

IELTS ACADEMIC READING

Passage 1 Collection

*A complete practice book of real and adapted IELTS Reading
Passage 1 texts, with full-text passages, original question types,
and a complete answer key*

IB

60

PASSAGES

807

QUESTIONS

Full

ANSWER KEY

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Introduction

Welcome to the IELTS Academic Reading — Passage 1 Collection, a practice book bringing together 60 real and adapted Passage 1 texts in one place, compiled and prepared by Teacher Ibodullaev Bekhruz.

Each passage is presented exactly as it would appear in the exam — the full text with paragraph labels, and every original question type (Matching Headings, Matching Information, Multiple Choice, Summary Completion, and more) — with blank spaces and answer circles left for you to write your own answers, just like on test day.

At the back of the book you will find a complete Answer Key for every passage, with the exact sentence from the passage that supports each correct answer, so you can check your work and understand exactly why an answer is correct.

How to use this book:

- Work through one passage at a time under timed conditions (about 20 minutes each).
- Write your answers directly in the circles and blanks provided.
- Check your answers against the key at the back, and read the supporting evidence for anything you got wrong.
- Revisit difficult passages after a few days to reinforce what you have learned.

Good luck with your preparation, and follow [**@bekhruzposts**](#) for more free IELTS materials.

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Passage 1

Ambergris – What Is It and Where Does It Come From?

Read the text and answer questions 1–13

In the ancient world, the waxy grey substance we now refer to as ambergris was highly prized for its medicinal properties, and was widely used as a spice, which was believed to be an aphrodisiac when added to food or wine. Ambergris itself is pleasantly aromatic, especially when warmed, and it was also highly valued as a fixing agent in the making of perfume, since it enabled a scent to retain its fragrance for much longer than might otherwise have been possible. Most ambergris was found in the form of lumps floating on the surface of the sea, or washed up on the shores of tropical and temperate oceans. At one time, ambergris was worth its weight in gold, but there was much confusion about its origins.

Ambergris was known to the Arabs as 'ambar' and was originally called amber in the West in the Middle Ages. This eventually led to further confusion in the popular mind between ambergris and true amber, the mineral known to mineralogists as succinite, which is actually fossilised tree resin, and generally yellow in colour. Both substances were rare and costly, and both were associated with the sea, largely because for Europeans the most common source of amber was the shores of the Baltic. In Chapter 92 of *Moby Dick*, the American writer Herman Melville pours scorn on those who believed the two substances to be the same:

"Though the word ambergris is but the French compound for grey amber, yet the two substances are quite distinct. For amber, though at times found on the sea-coast, is also dug up in some far inland soils, whereas ambergris is never found except upon the sea. Besides, amber is a hard, transparent, brittle, odorless substance, used for mouth-pieces to pipes, for beads and ornaments; but ambergris is soft, waxy, and so highly fragrant that it is largely used in perfumery."

Moby Dick was published in 1851, by which time the mystery of the origins of ambergris had been resolved by the scientific community. In 1783, the botanist Joseph Banks, who had accompanied Captain James Cook on his voyages of discovery in the Pacific, presented a paper to the Royal Society of London by the German physician Dr. Franz Xavier Schwediawer in which it was conclusively proved that ambergris came from sperm whales. In this, he was confirming an observation made in the 13th century by the great Venetian traveller Marco Polo who, while on the island of Socotra in the Indian Ocean, had witnessed a sperm whale vomiting up ambergris. But whereas Marco Polo imagined that the whale had swallowed the lump in the depths of the sea, Schwediawer showed that the origin of the material was inside the whale itself.

The sperm whale is the largest of the odontocetes, or toothed whales. Males can grow up to 20 meters in length. Melville described the sperm whale as the 'king of whales', and his novel *Moby Dick* is based on the pursuit of one such creature. Sperm whales are renowned for their ability to dive to great depths, possibly as far as 3,000 metres below the surface, and for remaining underwater for periods of two hours or more in pursuit of their favourite prey, the giant squid.

It is from the problems the whales have in digesting the beaks of such creatures that ambergris has its origins. The beak is sharp and irritates the whale's lower intestine, which responds by producing a black, foul-smelling liquid. It is not clear to scientists whether this secretion should be considered a normal response by the whale's digestive system or a pathological one, but from time to time large quantities of the liquid are vomited up by the whale. Once outside the whale's body and exposed to air, the substance hardens, acquiring the waxy, greyish and pleasantly aromatic characteristics of ambergris. Often the beaks of squid are still found embedded in lumps of ambergris, some of which can weigh several hundred kilograms. Melville took some delight in contrasting the origins of ambergris with the high value placed upon it by refined society:

"Who would think, then, that such fine ladies and gentlemen should regale themselves with an essence found in the inglorious bowels of a sick whale!"

Sperm whales were ruthlessly pursued by commercial whalers in the 19th and 20th centuries. In 1963–64 alone, almost 30,000 individuals were killed, and only the imposition of a ban on the hunting of sperm whales in 1984 saved the species from extinction. Ambergris was by far the most valuable product to be extracted during the processing of the whales' carcasses, and over 90 per cent of the annual worldwide total was acquired in this way, as a by-product of commercial whaling. However, even before the ban on hunting sperm whales was imposed, the 1972 Marine Mammal Protection Act had prohibited trade in ambergris. Just as petroleum and plastic products were replacing other natural products of whaling, so ambergris was supplanted in the making of perfume by other materials, some natural and some synthetic in origin. Nevertheless, it is possible that, as sperm whale populations recover to their former numbers in the wild, so the sight of lumps of ambergris washed ashore along the tide line will once again become a familiar one to beachcombers the world over.

Questions 1–6

According to the information in Reading Passage 1, classify the following facts as referring to:

CLASSIFICATION

- A ambergris only
- B amber only
- C both ambergris and amber
- D neither ambergris nor amber

Q	Statement	
1	very expensive	
2	a food flavouring	
3	used as currency	
4	sweet-smelling	
5	referred to by Herman Melville	
6	can be seen through	

Questions 7–9

Complete the flow-chart below. Choose **NO MORE THAN TWO WORDS** from the passage for each answer.

How ambergris is formed

Ambergris is formed in whales because of problems digesting the ⑦ _____ of giant squid.

↓

Black liquid is produced and is ⑧ _____ from time to time.

↓

The liquid ⑨ _____ on contact with the air.

Questions 10–13

Do the following statements agree with the information given in Reading Passage 1? In boxes 10–13, write

TRUE if the statement agrees with the information

FALSE if the statement contradicts the information

NOT GIVEN if there is no information on this

- ⑩ In the 20th century, most of the world's ambergris came from processing dead whales.

- ⑪ The value of ambergris has increased recently.
- ⑫ Ambergris remains an important ingredient in perfume.
- ⑬ New uses have recently been found for ambergris.

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Passage 2

A Look at the History of Poetry

Tracing the earliest forms of poetry to the modern era

Unlike other literary forms that we can date to precise texts and time periods, it is a challenge to pinpoint the earliest work of poetry. In one form or another, poetry has been around for thousands of years. Indeed, poetry as an art form may even predate literacy. Many ancient works, such as the Vedas (1700 – 1200 BCE), were composed in poetic form to aid memorization and oral transmission of information in prehistoric and ancient societies. Poetry appears among the earliest records of most literate cultures, with poetic fragments found engraved on ancient stones.

The lengthy adventure poem known as an epic might be considered as the first type of poetry. *The Epic of Gilgamesh* is often cited as one of the earliest epic works, dating back to the 18th century BCE. Consisting of poems written in an ancient language, it's a text that was discovered through many different versions during archaeological excavations. Other examples of early epic poems include the *Mahabharata* and the *Ramayana*, the latter of which has become an important narrative in both Hindu and Buddhist mythology throughout regions of Asia.

A list of the most notable works of epic poetry – at least in the Western world – would have to include the *Iliad* and the *Odyssey*, both works of Greek mythology that have been attributed by academics to the Greek poet Homer. The *Iliad* recounts the battle between Achilles, a famed warrior, and King Agamemnon during the Trojan War. A companion of sorts to the *Iliad*, the *Odyssey* depicts the long journey of its hero, Odysseus, after the conflict is over.

Sonnets were a form of poetry that came into being in the 13th century. The rhyme scheme of a sonnet varies depending upon whether it is Italian or English. The Italian poet Petrarca, after whom the Petrarchan sonnet is named, is perhaps one of the most well-known early writers of the sonnet, in which he specialized. Following his work in the 13th century, other poets created variations of the sonnet, but it became widely known as an English poetic form through the work of William Shakespeare in the 16th century. However, its popularity declined in the Restoration period (1660 – 1689).

The Restoration Period in England also saw the rise of literary elites, such as John Dryden. Perhaps one of the best-known names of English literary criticism, translation, drama and poetry, Dryden made his name as a poet with his poems making fun of important figures or genres. The form of these mock-heroic poems, and the use of poetic form for satire, did not end with Dryden. Alexander Pope carried on Dryden's tradition of using poetry for comic purposes.

The 19th century saw the emergence of many poets of importance. The so-called English Romantics included figures like William Blake, Samuel Taylor Coleridge, William Wordsworth and John Keats. During the same period, American poetry produced what are known as the fireside poets. Writers like Henry Wadsworth Longfellow, Oliver Wendell Holmes, and William Cullen Bryant belonged to this group. In other circles, Walt Whitman revolutionized the 19th-century American spirit with his 'Leaves of Grass', while most of Emily Dickinson's unconventional poems, written in the mid-to-late 1800s, were published only after her death.

During the 20th century, the waves of change brought about by world events influenced poetry, resulting in works by poets with distinct voices who came to enjoy global circulation. Around the turn of the 20th century, Whitman continued to edit and alter his 1855 edition of 'Leaves of Grass', which introduced a new period – and a variety of forms – for poetry. The extraordinary modernist style of Gertrude Stein overwhelmed American readers who bought her work 'Tender Buttons' in 1914. Those same readers were to be surprised again a short time later by T.S. Eliot's employment of ancient languages and obscure references to other poetical texts in his famous poem, 'The Waste Land', written in 1927.

Another example of modernist poetry is the hugely important work of Claude McKay. An African-American poet born in Jamaica who emigrated to the US in 1912, he carried his homeland into his distinctly American poetic voice. Writing on war, racism and memories connected with his birthplace, McKay wrote notable poems such as 'If We Must Die' and 'The Tropics in New York'.

The 20th century also witnessed a number of poets winning the Nobel Prize, from the United States to India. Rabindranath Tagore, who resisted pressure from the English language used by the rulers of the country at that time and wrote solely in his mother tongue, Bengali, won this esteemed award in 1913. He was followed by Irish poets William Butler Yeats in 1923 and Seamus Heaney in 1995.

Questions 1–6

Do the following statements agree with the information given in Reading Passage 1?

*Write **TRUE** if the statement agrees with the information, **FALSE** if the statement contradicts the information, or **NOT GIVEN** if there is no information on this.*

- 1 Nearly all early literate cultures had poetry in some form.
- 2 The *Ramayana* is of little significance in Asia.
- 3 The *Odyssey* was discovered many years later than the *Iliad*.

- 4 Petrarcha is associated with a particular form of poetry.
- 5 Shakespeare's sonnets were some of the finest ever written.
- 6 Dryden attempted to amuse people with his poetry.

Questions 7–13

Complete the notes below.

Choose **ONE WORD ONLY** from the passage for each answer.

A Look at the History of Poetry

19th-century poetry

- English poets such as Keats and Blake were known as the 7 _____ poets.
- The 8 _____ poets in America included names such as Longfellow and Bryant.

20th-century poetry

- A variety of new poetic forms followed the revision by 9 _____ of his earlier work.
- 'The Waste Land's' use of old 10 _____ took those who read it by surprise.
- Many of Claude McKay's poems were about his homeland of 11 _____.
- In 1913, Tagore won the Nobel Prize with poems written in 12 _____.
- Two 13 _____ writers won the Nobel Prize for their poetry in the 20th century.

Passage 3

Ahead of its Time

A chance discovery in New Zealand has challenged the country's recorded history

One October afternoon, a young New Zealander, Sam Tobin, called his dogs and went for a walk down to the nearby Ruamahanga River. Having been very high for days, the river had at last fallen, and Tobin was eager to see what changes the floods had brought. The family farm borders the river, and a four-metre-high flood bank testifies to its natural tendency to flood.

Tobin stepped out onto a broad shoulder of river sand, where he noticed what he initially took to be a whitish rock, lit by the sun. Then, getting closer, he realised it was a bone. Such a thing was not uncommon in these parts - he had often come across bone fragments, or even whole skeletons, of cows and sheep. But as he scraped aside the stones, he realised it was a human bone, something quite new in his experience. As he picked it up, he saw it was a skull, discoloured with age.

Tobin replaced the skull and hurried home to tell his mother what the river had delivered to their doorstep. It would prove to be a spectacular find, setting in motion an investigation by some of the country's most respected specialists, and ultimately challenging our most firmly held assertions about the human settlement of New Zealand.

The police were immediately called, but despite a thorough search could find nothing that might shed light on the identity of the Ruamahanga skull, or the circumstances of its sudden appearance. The skull was then taken north to be examined by forensic pathologist Dr Ferris, at Auckland Hospital. Despite being hampered by its damaged and incomplete condition - the jawbone and lower left portion of the cranium were missing - Dr Ferris determined that the skull was that of a female. He then consulted with a colleague, Dr Koelmeyer, who believed that the deterioration of the bone placed the time of death 'before living memory' and, most significantly as it would turn out, the skull appeared to be European in origin.

Wellington-based forensic anthropologist Dr Watt also examined the skull, and suggested it belonged to a 40-45-year-old. He believed that it could be the remains of an old farm burial, but was not certain, and proposed the use of radiocarbon dating to make sure it wasn't a recent death. As a result, the Institute of Geological and Nuclear Sciences (GNS) in Lower Hutt was contacted and provided with a sample of bone that had originated in the top of the skull. In a little over three weeks the seemingly astonishing results from the GNS laboratory came back. Cutting through the bewildering complexity of the scientific analysis was a single line reading: conventional radiocarbon age approximately 296 years. This was staggering, for the skull was about 200 years older than Dr Koelmeyer had believed.

Of course, a skull of this age wasn't particularly unusual in New Zealand. The Maori people have been living in the country for at least 800 years and scientists frequently come across human remains of considerable age. The fascinating question, however, was how a skull of this race, let alone this gender, had reached these remote islands in the South Pacific at such a time, long before the arrival of the explorer Captain Cook in 1769, and perhaps even before the very first European landfall - the fleeting visit of the Dutch explorer Tasman in 1642 - neither of whom had women among their crews.

The first known European women in the Pacific came with a doomed colonising venture which sailed from Peru in 1595 under the command of Spanish captain Mendana. However, it is unlikely the Ruamahanga skull originated from this expedition because no evidence of Mendana's ships has ever been found in New Zealand, while a team of archaeologists working in the Solomon Islands in 1970 did discover the remains of European vessels dating from the 16th century.

Two centuries were to pass before the first recorded European females arrived in New Zealand, both having escaped from prison in Australia. Kathleen Hagerty and Charlotte Edgar are known to have reached the country in 1806. How then do we account for the Ruamahanga skull, which appears to be about 100 years older than that? It is impossible to say with certainty, but the most likely explanation is that a Spanish or Portuguese trading-ship was washed onto these wild shores as a result of a shipwreck and a woman got ashore. Implausible, perhaps, but the Ruamahanga skull, today resting in the Wellington Museum, could be the kind of concrete evidence that demands such a drastic reevaluation of history.

Questions 1–4

Do the following statements agree with the information given in Reading Passage 1?

*Write **TRUE** if the statement agrees with the information, **FALSE** if the statement contradicts the information, or **NOT GIVEN** if there is no information on this.*

- 1 The Ruamahanga River often floods.
- 2 When Tobin first found the object in the river, he mistook it for something else.
- 3 Tobin could not decide what part of the body the bone came from.
- 4 Tobin's mother was surprised that the skull caused debate among specialists.

Questions 5–9

Complete the flow-chart below.

*Choose **NO MORE THAN TWO WORDS AND/OR A NUMBER** from the passage for each answer.*

The events after the river flooded

- Tobin found a human skull.
- The (5) _____ were initially involved in trying to explain the presence of the skull.
- Dr Ferris believed the skull belonged to a female.
- Dr Koelmeyer suggested it was a (6) _____ skull.
- Dr Watt recommended (7) _____ to establish the skull's age.
- A bone (8) _____ was sent to the GNS.
- The age of the skull was about (9) _____ years.

Questions 10–13

Complete the notes below.

Choose **ONE WORD ONLY** from the passage for each answer.

Problem of the skull's origins

- Old bones common in NZ – Maori living there for 800 years – Ruamahanga skull surprising because of its age, (10) _____ and gender.
- Mendana expedition – possible source of skull, but probably did not visit NZ. Evidence of this expedition found elsewhere by (11) _____.
- New Zealand – first European explorer arrived in 1642. Hagerty and Edgar arrived in 1806 from (12) _____ where they had been imprisoned.
- Possible solution – Ruamahanga skull may have reached NZ in 17th century after a (13) _____.

Passage 4

An Early Cultural Tourist

In the 15th century, the Italian merchant Cyriacus of Ancona journeyed in search of the Mediterranean's classical past, laying the groundwork for today's cultural holidays

Today we take it for granted that we travel around the world to admire the monuments of the past. We prepare for such trips by reading about what we are going to see, set out on the journey with a good idea of how we will get there and where we will stay, and have a sense of what we will encounter on location. Cyriacus of Ancona (1391–1452), the first cultural tourist since antiquity, lacked these advantages when, in the first half of the 15th century, he sailed around the Mediterranean in search of the remains of Greek and Roman civilisations.

Cyriacus first became fascinated by ancient monuments while walking in his home city Ancona and looking at the marble arch, erected in AD 115, to the Roman Emperor Trajan. He suddenly saw the structure in a new light. He no longer saw it as just a familiar and generally overlooked landmark, but as a doorway to the wonders of ancient imperial Rome. Not many people of Cyriacus's time were interested in historical travel, they generally ignored old buildings and structures, or worse, dismantled them for their building materials.

Cyriacus decided to see the world for himself and to record details of whatever other antiquities remained to be discovered. His training as a merchant did not prepare him for this vocation; he did not know ancient languages, history or art. However, he set out to solve these failings, first by learning Latin at the age of 30 and then adding ancient Greek. Having done this, he then set off on voyages around the Mediterranean to find, investigate and understand ancient cultures from their buildings, sculptures and inscriptions. Thus he became the first archaeologist and cultural tourist, predating other antiquarians by some 200 years.

Travel in the 15th century, however, was anything but simple or enjoyable. Overland journeys by foot or mule along bad roads, under constant threat from bandits, were bad; voyages by seas were even worse. When the weather cooperated, sailing went relatively smoothly, ships proceeded along coasts from one recognizable landmark to another. However, when there was no wind the ship did not move. Strong winds were no friends either, they drenched the ship with lashing waves and blew it off course. Water swamped the deck, splashed into the cabins and soaked mattresses, clothes and food. Remarkably, Cyriacus never complained about the miseries of travel. Optimistic by nature, he endured such hardships unafraid and saw opportunities where other people saw setbacks.

Among many of the important records made by Cyriacus was his crucial documenting, in 1431, of the remains of Cyzicus, an ancient Roman city that had relied on commerce for its financial success. He hired a local person to take him to the site and then had to work out for himself the significance of the ruins he was looking at because there was no guidebook on ancient architecture to help him. Cyzicus had been a splendid city in its prime. Unfortunately, the area was highly seismic and in AD 123 the city was so devastated by a major earthquake that, when the Roman Emperor Hadrian visited it the following year, he was so saddened that he decided to subsidise a campaign to reconstruct Cyzicus. He made a substantial donation for a new temple to the Roman god Jupiter. Cyriacus thought the ruined city was awe-inspiring. He found the remains of the temple and examined it in great detail, looking for clues in ancient texts to help him understand what he was seeing. He sketched the great doorway adorned with carved foliage and mythological characters. Cyriacus's account of this temple is the only record of this building as in the following centuries it was entirely stripped of all its stonework and all that remains is its base.

Cyriacus also visited mainland Greece, in 1436, when no one went to Greece in order to see the country's ancient ruins. One of his destinations was the sanctuary of Delphi. The ancient Greeks considered Delphi as being situated in the most beautiful spot in Greece. When Cyriacus arrived at the site of Delphi, however, he found war, earthquakes and avalanches had all but obliterated its ruins. Determined to find any ancient traces, Cyriacus spent six days walking all over the areas, peering at odd stone blocks sticking out of the ground, running his hands over inscriptions to trace fragments of words, and trying to puzzle out the few surviving structural remains. Climbing uphill towards the rocks that tower over the site, he came upon a theatre built into the slope. Soon after his visit, the site was buried by a rockslide and was not seen again until archaeologists began to excavate the area systematically in the late 19th century.

Cyriacus had hoped to visit Egypt and Ethiopia but he never got there. However, in his life he did record for posterity countless ancient monuments around the Mediterranean, paving the way for future archaeologists and cultural tourists.

Questions 1–8

Do the following statements agree with the information given in Reading Passage 1?

*Write **TRUE** if the statement agrees with the information, **FALSE** if the statement contradicts the information, or **NOT GIVEN** if there is no information on this.*

- 1 Cyriacus was unable to research his journeys before he left.
- 2 Cyriacus was inspired to begin travelling by a Roman ruin in his city.
- 3 The Roman Emperor Trajan built the city of Ancona.

- ④ Respect for ancient architecture was widespread in the 15th century.
- ⑤ Cyriacus's experience as a merchant gave him skills he needed to investigate the ancient world.
- ⑥ Before leaving on his journey, Cyriacus studied ancient languages.
- ⑦ Travelling by sea in the 15th century was easier than travelling on land.
- ⑧ Cyriacus tried to make his fellow sea travelers more comfortable.

Questions 9–16

Complete the notes below.

Choose **ONE WORD ONLY** from the passage for each answer.

Ancient Roman and Greek sites visited by Cyriacus

The city of Cyzicus

- The wealth of the city had come from ⑨ _____.
- ⑩ _____ to the ancient city ruins not available when visited by Cyriacus.
- The city was destroyed by a powerful ⑪ _____ in AD 123.
- A year later Emperor Hadrian supported a ⑫ _____ to rebuild the city.
- A temple to Jupiter was built – helped by an especially large ⑬ _____ from Emperor Hadrian.
- Cyriacus found part of the temple, which was built in the time of Emperor Hadrian, and made drawings of the ⑭ _____ to the temple and its decorative carvings.

The sanctuary of Delphi

- By the 15th century Delphi had almost disappeared due to natural disasters and ⑮ _____.
- Cyriacus found a ⑯ _____ above Delphi.

Passage 5

An Important Language Development

Cuneiform, the world's first known system of handwriting, and how a 19th-century British officer eventually decoded it

Cuneiform, the world's first known system of handwriting, originated some 6,000 years ago in Sumer in what is now southern Iraq. It was most often inscribed on palm-sized, rectangular clay tablets measuring several centimetres across, although occasionally larger tablets or cylinders were used. Clay was an excellent medium for writing. Other surfaces which have been employed — for example, parchment, papyrus and paper — are not long-lasting and are easily destroyed by fire and water. But clay has proved to be resistant to those particular kinds of damage.

The word "cuneiform" actually refers to the marks or signs inscribed in the clay. The original cuneiform signs consisted of a series of lines — triangular, vertical, diagonal and horizontal. Sumerian writers would impress these lines into the wet clay with a stylus — a long, thin, pointed instrument which looked somewhat like a pen. Oddly, the signs were often almost too small to see with the naked eye. Cuneiform signs were used for the writing of at least a dozen languages. This is similar to how the Latin alphabet is used today for writing English, French, Spanish and German, for example.

Before the development of cuneiform, tokens were used by the Sumerians to record certain information. For example, they might take small stones and use them as tokens or representations of something else, like a goat. A number of tokens, then, might mean a herd of goats. These tokens might then be placed in a cloth container and provided to a buyer as a receipt for a transaction, perhaps five tokens for five animals. It was not that different from what we do today when we buy some bread and the clerk gives us back a piece of paper with numbers on it to confirm the exchange.

By the 4th century BCE, the Sumerians had adapted this system to a form of writing. They began putting tokens in a container resembling an envelope, and now made of clay instead of cloth. They then stamped the outside to indicate the number and type of tokens inside. A person could then "read" what was stamped on the container and know what was inside. Gradually, Sumerians developed symbols for words. When first developed, each symbol looked like the concrete thing it represented. For example, an image which resembled the drawing of a sheep meant just that. Then another level of abstraction was introduced when symbols were developed for intangible ideas such as "female" or "hot" or "God". Cuneiform, in other words, evolved from a way used primarily to track and store information into a way to represent the world symbolically. Over the centuries, the marks became ever more abstract, finally evolving into signs that looked nothing like what they

referred to, just as the letters "h-o-u-s-e" have no visual connection to the place we live in. At this last stage in the evolution of cuneiform, the signs took the form of triangles, which became common cuneiform signs.

As the marks became more abstract, the system became more efficient because there were fewer marks a "reader" needed to learn. But cuneiform also became more complex because society itself was becoming more complex, so there were more ideas and concepts that needed to be expressed. However, most linguists and historians agree cuneiform developed primarily as a tool for accounting. Of the cuneiform tablets that have been discovered, excavated and translated, about 75 percent contain this type of practical information, rather than artistic or imaginative work.

Cuneiform writing was used for thousands of years, but it eventually ceased to be used in everyday life. In fact, it died out and remained unintelligible for almost 2,000 years. In the late 19th century, a British army officer, Henry Rawlinson, discovered cuneiform inscriptions which had been carved in the surface of rocks in the Behistun mountains in what is present-day Iran. Rawlinson made impressions of the marks on large pieces of paper, as he balanced dangerously on the surrounding rocks.

Rawlinson took his copies home to Britain and studied them for years to determine what each line stood for, and what each group of symbols meant. He found that in the writing on those particular rocks every word was repeated three times in three languages: Old Persian, Elamite and Babylonian. Since the meanings in these languages were already known to linguists, he could thus translate the cuneiform. Eventually, he fully decoded the cuneiform marks and he discovered that they described the life of Darius, a king of the Persian Empire in the 5th century BCE.

Questions 1–5

Do the following statements agree with the information given in Reading Passage 1?

Write TRUE, FALSE or NOT GIVEN.

- 1 Cuneiform tablets were produced in different shapes and sizes.
- 2 When Sumerian writers marked on the clay tablets, the tablets were dry.
- 3 Cuneiform was often difficult to read because of its size.
- 4 A number of languages adopted cuneiform.
- 5 Cuneiform signs can be found in some modern alphabets.

Questions 6–13

Complete the notes below.

Choose **ONE WORD ONLY** from the passage for each answer.

The development and translation of cuneiform

- Before cuneiform, tokens, for example ⑥ _____, were often used.
- The first tokens were kept in containers made of ⑦ _____.
- Tokens were used as a ⑧ _____ to give when selling something.
- By the 4th century BCE, tokens were put in a container that looked like a clay ⑨ _____.
- At first, signs looked like what they indicated, for example ⑩ _____.
- Eventually, cuneiform signs shaped like ⑪ _____ were developed.
- According to experts, cuneiform was mainly used for ⑫ _____.
- In the 19th century, Rawlinson copied inscriptions onto ⑬ _____.

Passage 6

Arctic Warming

Why the Arctic is heating up faster than anywhere else on the planet, and what that means for the rest of us

Before the 19th century, the balance between incoming energy from the sun and outgoing radiation from the earth kept the planet's average temperature relatively stable. Only natural processes, such as volcanic eruptions and variations in solar output, caused periods of relative warming and cooling. Today, however, thanks to human-driven activities such as the burning of fossil fuels, industrial processes, deforestation and agriculture, global carbon dioxide levels are rising rapidly, which has led to an average global temperature increase of approximately 1°C since 1900.

This number may seem small, but the Arctic region has actually warmed by about 2°C in this time. This disparity in warming between the poles and the tropics is a phenomenon known as polar amplification. In just the last 30 years, areas of Arctic sea ice larger than Norway, Sweden, and Denmark combined have melted away. In the documentary *Under Thin Ice*, diver and underwater explorer Jill Heinerth travels to the Arctic regions of Greenland and Northern Canada to document what is happening above and below this rapidly disappearing ice. "The Arctic is changing faster than anywhere else on the planet," says Heinerth. "And it's really evident here."

There is a scientific explanation for why this is occurring. Fresh snow is the brightest natural surface on earth, as it reflects nearly 85% of solar radiation that hits it. Oceans, on the other hand, are the darkest natural surface on earth, and they only reflect about 10% of sunlight, absorbing the remaining 90%. In the winter, the Arctic Ocean is covered in ice, acting as an insulating blanket for the ocean below. In the springtime, this ice melts and exposes the dark ocean underneath, which leads to warming of the region, and this warming melts even more ice, resulting in a positive feedback loop. This process of polar amplification is stronger in the Arctic than it is in Antarctica, and this is because the Arctic is primarily an ocean covered in sea ice, while Antarctica is an elevated landmass that is covered in thicker layers of ice and snow. In fact, the average temperature of Antarctica has not changed in the last several decades, despite an increase in the concentration of greenhouse gasses. In the autumn, large storms in the open ocean create more waves that cause the Arctic waters to mix with warmer waters, which increases the heat absorbed in the summer, making ice formation more difficult again in the autumn. These increased temperatures have also spread over the land. The Arctic land that was once covered in snow during the summer has shrunk in size significantly, and this darker land absorbs more heat than before, further warming the Arctic and the rest of the planet.

However, this melted water does not simply remain on the surface to freeze again in the fall. Instead, it sinks down into the ice sheet through large holes called moulins, which lubricates the ice sheet and causes glaciers to move more rapidly and release more icebergs into the ocean. The level of melting of the Greenland ice sheet is a major indicator of potential sea level rise; if it completely melts, global sea levels could rise by as much as 6 metres, endangering coastal cities and small island nations while making dangerous weather events, such as floods and hurricanes, even more deadly.

The most troubling effect of this feedback loop is the release of methane into the atmosphere. The Arctic stores an enormous amount of carbon, mostly in the soil, which is permanently stored in perennially frozen land called permafrost. Humans release around 9 billion tonnes of carbon dioxide per year through industrial activities. It is estimated that an astounding 14,000 billion tonnes of carbon exist in permafrost soil alone, or the equivalent of 155 years worth of fossil fuel combustion. As this permafrost melts, plants and animals that were frozen under the ice begin to decay, and if there is no restriction on oxygen, this releases mostly carbon dioxide. However, if the soil is waterlogged, fermentation occurs and this discharges a gas called methane. Although it only lingers in the atmosphere for a fraction of the time, methane traps heat in the atmosphere around 23 times more efficiently than carbon dioxide.

As the situation nears its tipping point, a global campaign is underway to actually remove carbon dioxide from the atmosphere manually, using techniques such as Direct Air Capture, a process of capturing carbon dioxide directly from the air surrounding us, as opposed to direct capture from point sources, such as factories, and sequestering this carbon directly or utilizing it for other purposes. This is what plants do naturally, but Direct Air Capture does it much more rapidly and requires less land space. In addition, it delivers the carbon dioxide in a pure, condensed form that can easily be stored in the ground or reused. The success of such efforts will likely spell the difference between a prosperous and sustainable future for mankind and a descent into bleak, climate-driven chaos.

Questions 1–6

Complete the notes below.

*Choose **NO MORE THAN TWO WORDS** from the passage for each answer.*

The Melting of Sea Ice

Background

- Before industrialization, only ① ————— affected the climate

- ② _____ is a term that refers to the difference between polar warming and tropical warming.

Explanation

- Snow ③ _____ most of the sunlight that falls on it
- Ice melts in the ④ _____, causing the region to warm

Consequences

- Islands and ⑤ _____ are at risk if Greenland ice sheet disappears
- Extreme weather could become more dangerous
- Risk of ⑥ _____ leakage into the atmosphere is a worry.

Questions 7–13

Do the following statements agree with the information given in Reading Passage 1?

*Write **TRUE** if the statement agrees with the information, **FALSE** if the statement contradicts the information, or **NOT GIVEN** if there is no information on this.*

- ⑦ Rising levels of greenhouse gasses have increased temperatures in the Antarctic.
- ⑧ Increase in wave action accelerates the melting of sea ice in the summer.
- ⑨ Coastal cities and small island countries may disappear if the Greenland ice sheet melts completely.
- ⑩ The darker surface exposed as a result of melting ice absorbs less heat.
- ⑪ The amount of water in soil determines whether carbon dioxide or methane is released when permafrost thaws.
- ⑫ The most effective way to address Arctic warming is Direct Air Capture technology.
- ⑬ Direct Air Capture technique captures carbon dioxide more efficiently than plants and trees.

Passage 7

Australia's Airborne Dentists

How the Royal Flying Doctor Service brings dental care to Australia's most remote communities

Australians living or travelling in rural and remote areas can face particular difficulties when they need medical care. Hundreds of kilometres from major cities and many hours by road from the closest hospital or clinic, some rural Australians do not have easy access to doctors, nurses and dentists.

Organisations such as the Royal Flying Doctor Service (RFDS) have been established to bring health services to outback Australian communities. The RFDS provides free medical care to people who live, work or travel in remote and regional parts of Australia. This non-profit organisation is the oldest and largest airborne health service of its kind in the world, and since 1928 it has used small aircraft to provide health services to some of Australia's most far-away communities.

In recent years, the RFDS has also started to fly dentists to regional Australia. As well as offering mobile dental clinics, the RFDS offers a range of preventative and educational services. Looking after the teeth of people in remote areas presents special challenges. These include providing care to disparate communities with no established dental facilities, and dealing with higher incidences of other diseases which are linked to, or caused by, poor dental health.

People in remote areas have very infrequent visits by health staff. RFDS dentists might only visit a community once every few months, or sometimes once per year. Because of infrequent dentist visits, patients in these areas often need to put up with their dental problems before they can get treatment; consequently, people in remote areas are more likely to have tooth decay (the blackening and deterioration of teeth) and develop gum and other mouth diseases.

In some locations that the RFDS visits, there are no suitable dental facilities, so dentists have to bring everything with them. This includes drills, dentists' chairs, portable X-ray machines, and computers for keeping track of patients' treatments. Equipment can weigh up to 100 kilograms, and since the small planes that transport dentists have limited space, dentists cannot always bring everything that they need.

While dentists in town or city centres can specialise in certain types of treatment, RFDS dentists need to be 'all-rounders'. They need to be able to do all kinds of dental procedures, as they don't have the ability to refer patients to more specialised dentists. Even with their broad experience, there are some services that are particularly challenging for RFDS

dentists. For example, dentures (artificial teeth) can be very difficult to provide, as they need to be the right shape and size for the patient, and this requires many visits over a long period of time. As a result, it is not practical to make dentures available.

Some chronic illnesses are more common in remote communities than in the rest of Australia. These illnesses can in turn lead to a lowered resistance to infection, including gum and other oral infections. As a result, people in outback Australian communities are more likely to experience oral health problems than city folk, and this poses extra challenges for both the dentists and the doctors of the RFDS.

Because there aren't a lot of dental services in remote areas, people living in these areas also receive less education about good dental hygiene than their city counterparts do. Australians in very remote communities might not be aware of things that people in cities take for granted, such as the importance of daily tooth brushing. Also, basic dental hygiene items such as toothpaste and toothbrushes can be more expensive in outback areas. Many people are on low incomes, meaning they have extra difficulty affording these products. If this is the case, the RFDS supplies these.

As well as treating patients, RFDS dentists try to focus on preventative oral health and educate their patients on good oral hygiene, such as tooth brushing and flossing. The RFDS also provides mouthguards for young sports players. Playing contact sports, such as rugby league or Australian rules football, can damage young people's teeth, so mouthguards provide protection which prevents accidental injuries.

Adding fluoride to water supplies has been proven to reduce the incidence of tooth decay in many parts of the world. City dwellers in Australia use water supplies that have been fluoridated, and their rates of tooth decay are lower because of this. In remote areas, it is not practical to fluoridate drinking water supplies, and so people living in these areas are more subject to tooth decay. As a result, it is particularly important that people living in areas without fluoridated water pay special attention to regular brushing of their teeth with fluoridated toothpaste.

Despite many challenges, the RFDS continues to offer much-needed dental and medical support. Its presence in isolated communities greatly improves the quality of dental health, and supports important oral hygiene and health initiatives.

Questions 1–7

Do the following statements agree with the information given in Reading Passage 1?

Write **TRUE**, **FALSE** or **NOT GIVEN**.

- ① Many of the RFDS doctors work as volunteers.

- ② RFDS dentists make trips to outback communities each month.
- ③ RFDS dentists are accompanied on their journeys to remote areas by a dental nurse.
- ④ RFDS dentists must provide a wide range of dental services.
- ⑤ Rural dental patients are more informed about oral hygiene than urban patients.
- ⑥ RFDS dentists educate patients about good eating habits.
- ⑦ Urban Australians generally have better teeth because their water is treated.

Questions 8–13

Complete the notes below.

Choose **ONE WORD ONLY** from the passage for each answer.

Challenges faced by RFDS dentists

- need to bring equipment including ⑧ _____ for records
- aircraft used to carry equipment have restricted ⑨ _____
- problems offering some services, e.g., fitting ⑩ _____
- people in remote areas are more likely to have infection in their mouth

Products supplied by RFDS dentists if necessary, RFDS provides:

- ⑪ _____ and ⑫ _____ for regular use
- ⑬ _____ to protect teeth of sports players

Passage 8

Bondi

Bondi is Australia's most famous beach, but how did this come about?

Australians have not always valued Sydney's Bondi Beach. Beaches attracted little attention in the first century of British colonial settlement, when it was the country's desert interior that stirred the settlers' imagination. The visual image of Australia was based on stories of explorers searching in the dry red centre for an inland sea, of men driving cattle vast distances, or shearing sheep in hot dusty sheds. It was a cultural landscape developed in hardship and tragedy.

Before the British settlement, there had been a strong traditional connection to the coast among Aboriginal people, the indigenous inhabitants of Australia, and evidence of this connection can still be seen. In 1788 Governor Phillip, who established the first British settlement on the eastern coast of Australia, estimated that there were approximately 1500 Aboriginal people living within a 10-mile radius of Sydney Harbour. But by the 1820s the local Aboriginal population had largely disappeared as a result of contact with the diseases brought by the settlers. In the minds of the new settlers, Aboriginal culture would come to be predominantly associated with the inland desert. The cultural associations that formed around Australia's beach landscape would, in turn, be almost entirely European in nature. Compared with the 'ancient' and bicultural mythology of Australia's interior, its beach culture would be depicted as being wholly modern and Western.

At the end of the 19th century, the beach emerged in art and literature as an alternative cultural landscape to the mythology of the interior. The two settings were similar in that they shared an idea of purity because of their association with nature, but otherwise they represented vastly different qualities. The interior of the country represented hard work and suffering, while the coast evoked images of health, leisure and egalitarianism.

In Britain, the association of beaches with health received royal blessing in 1783 when a prominent member of the royal family attended the health spas at Brighton Beach on the English coast, and built a pavilion there. With its entertainment, hotels and weekend villas, Brighton became the model for beaches throughout Britain's colonies. Reflecting the English resort's influence, seven Sydney beach suburbs still have a Brighton Street. Sydney's beach suburbs, rising to prominence in the late 19th century, shared Brighton's association with health. The perception of the city's beaches as places of purity and well-being was strengthened by contrast with the increasingly polluted city centre.

By the end of the 19th century, Sydney's beaches had begun to differ in their socio-economic characteristics. The beach in the suburb of Coogee was the first ocean beach to become accessible by tram from the city. Promoted as a garden suburb, Coogee became a

recreational meeting place for the city's middle class. The suburb of Manly, with its beach and esplanade, copied the British resort model, and attracted the city's wealthy class living in the northern suburbs as well as holiday-makers from farms in the countryside.

The suburb of Bondi was developed after Coogee and Manly and took on a character distinct from the city's other beach suburbs. With the extension of the tramline to Bondi in the 1890s, a number of substantial properties were built there. Property developers, expecting continued middle-class development, subdivided the area into house-sized plots. However, a sudden demand for housing in the first decade of the 20th century saw the plans for houses put aside, and investors built blocks of flats instead. A treeless plain of red-roofed apartments began to spread its way across Bondi. Built with cheap local bricks, the size and quality of the apartments were generally more modest than those in neighbouring suburbs, and quickly attracted a population of rent-paying, working-class tenants.

With tram access extended all the way to the beach, Bondi also became the most accessible ocean beach to people living in the inner city, who were mostly working class. The pleasures of the sun, sea and surf were not only for the wealthy, but free to all. The Australian notion of beaches as places of social equality became established, and Bondi exemplified this precisely.

Bondi's identity as a largely working-class suburb began to change in the 1950s. The suburb and its beach first came to international attention in 1954 when Queen Elizabeth of England and her husband Prince Philip attended a surf carnival there. Since then, Bondi has become one of Sydney's most used sites for large-scale public events. The film industry quickly built on the fame that Bondi beach acquired as a result of the surf carnival, and since then it has been the location of many Australian films. With the expanding use of Bondi as a media setting, the suburb became home to a community of artists as well as film and television workers. Since the 1970s Bondi has become the almost inevitable backdrop for any artistic or commercial project in Sydney that requires a beach setting. Bondi is now famous simply for being famous.

In the preparations for the 2000 Olympic Games, Bondi was nominated to host the 10,000-seat beach volleyball stadium. Some people in the Bondi community did not support the proposal because of concerns for the environment, but the opposition was dismissed by the Olympics Organising Committee. There were good practical reasons for the Committee to support Bondi's hosting of the event, but the main reason the Committee insisted the stadium be built there was the international expectation that Bondi, as Australia's best known beach, would be the stadium's scenic backdrop.

Questions 1–6

Do the following statements agree with the information given in Reading Passage 1?

*Write **TRUE** if the statement agrees with the information, **FALSE** if the statement contradicts the information, or **NOT GIVEN** if there is no information on this.*

- 1 Bondi beach has been popular since the arrival of the first British settlers in Australia.
- 2 Aborigines in the centre of Australia had a different culture from those on the coast.
- 3 Contemporary Australian beach culture reflects Aboriginal traditions.
- 4 At the end of the 19th century, some parts of Australia were associated with a difficult life.
- 5 Sydney's beach suburbs were influenced by a seaside town in England.
- 6 At the end of the 19th century, Sydney's beaches were regarded as unhealthy places to be.

Questions 7–13

Answer the questions below.

*Choose **NO MORE THAN THREE WORDS AND/OR A NUMBER** from the passage for each answer.*

- 7 What form of public transport could people use to get to the beach at the end of the 19th century?
- 8 Which beachside suburb was popular with people from rural areas?
- 9 Which suburb helped to meet the needs of Sydney's accommodation shortage?
- 10 When did British royalty visit Bondi?
- 11 Which industry is Bondi now associated with?
- 12 What sporting event at the 2000 Olympic Games was held at Bondi?
- 13 What did people think could be damaged by building a stadium on Bondi Beach?

Passage 9

Chili Peppers

How a fiery South American fruit conquered the world's cuisines

Chili peppers, the fruit of the capsicum plant, are believed to have originated in what is now Bolivia in South America, home to dozens of wild species that are probably the ancestors of each and every chili variety today, from mild green peppers to fiery hot naga jolokia peppers. The chili pepper contains a heat-generating ingredient known as capsaicin, which has long been known to affect human taste buds, nerve cells and nasal membranes. People have been spicing their food with chilies for at least 8,000 years. At first they used wild chilies, adding them to potatoes, grain and corn, says Linda Perry, archaeobotanist at the Smithsonian Institution, Washington DC in the US. She has found traces of chilies on ancient stones used to grind grains, and in vessels used in food preparation from the Bahamas to southern Peru.

Based on her examination of ancient pieces of pottery, Ms. Perry concludes that people in the Americas began cultivating chilies more than 6,000 years ago. 'Chilies were domesticated early and spread very quickly, just because people liked them,' she says. The chilies, it would seem, made otherwise quite bland food more enjoyable to eat. Other researchers, such as Jennifer Billing and Paul Sherman of Cornell University in the US, argue that people learned early on that chilies could reduce food spoilage. And some scholars point to medical uses: ancient Mayan Indians in Central America incorporated chilies into medicinal preparations for treating wounds, stomach problems and earaches.

Whatever the benefits, chilies spread around the world at astonishing speed, thanks in part to Christopher Columbus, who is credited with being the first European to reach America in 1492, and claiming the new land for Spain. When Columbus took chilies back to Spain, they were initially not popular, but soon became widely accepted throughout Europe. The Portuguese first encountered chilies in Brazil and carried them, along with tobacco and cotton, to Africa. By 1550, chilies were being cultivated all over the world, including in India, Japan and China.

When people call chilies hot, they are not just speaking metaphorically. Capsaicin stimulates the neural sensors in the tongue and skin that also detect rising temperatures. As far as these neurons and the brain are concerned, when someone eats a hot chili, the effect is the same as exposure to fire. With enough heat, adrenaline flows and the heart pumps faster. This reaction, according to some physiologists, is part of what makes peppers so stimulating and enjoyable to eat.

The scale that scientists use to describe a chili's heat was developed in 1912 by Wilbur Scoville, a chemist from Detroit in the US. He diluted a chili extract in sugar water until the heat was no longer detectable to a panel of trained tasters; that threshold is the basis of the Scoville rating. A sweet bell pepper, for instance, measures zero on the Scoville Heat Unit (SHU) scale, while a typical hot jalapeno chili pepper from Mexico falls between 2,500 and 8,000 SHUs. Last year, the naga jolokia chili, grown in India, was rated at one million SHUs.

But why would chilies have evolved to produce the varying degrees of heat of capsaicin? Chili peppers, like other fruit, lure birds and other animals to eat them and disperse their seeds. But chilies also attract predators, largely rodents like packrats and cactus mice, that crush seeds and make germination impossible. In order to deter animals from eating their seeds, many plants produce toxic or foul-tasting chemicals, but these are usually found in the plant's leaves and roots as well as its fruit. In chilies, however, capsaicin is found just in the fruit, and becomes much stronger with ripening. Joshua Tewksbury, an ecologist at the University of Washington in the US, suspected that at least one role that capsaicin performs is to protect the chili seeds from rodents.

To test his theory, Tewksbury grew plants that produced hot, spicy chilies as well as plants that yielded mild chilies (though both were from the same species). When he offered the fruits of those plants to laboratory rats and mice, they ate the mild chilies but avoided the ones with a strong concentration of capsaicin. He later found that birds do not seem to mind eating even the hottest chilies and, in fact, that the capsaicin had the strange effect of retarding birds' digestion, which helps some seeds germinate, possibly by softening the seed coat. Even so, Tewksbury didn't believe that deterring rodents and slowing bird digestion were enough to explain why the spicy heat of chilies evolved. Instead, he thinks that a chili's heat protects it from disease that a certain fungus can trigger, the primary reason chili seeds die prior to being dispersed. In looking at chili populations that contain both hot, spicy, and mild plants, Tewksbury has found that the more capsaicin, the less fungal infection. 'Capsaicin demonstrates the incredible elegance of evolution,' says Tewksbury. The capsaicin discourages rodents and deters microbes — and humans harness this ability when they use chilies to preserve food — but capsaicin doesn't prevent birds from eating chili fruit and spreading seeds. Once in a while, the complex, often conflicting demands that nature places on complex traits result in a truly elegant solution. This is one of those times.

Questions 1–6

Do the following statements agree with the information given in the passage?

Write **TRUE** if the statement agrees with the information, **FALSE** if the statement contradicts the information, or **NOT GIVEN** if there is no information on this.

- ① Archaeological evidence from pots has helped determine when people first began to grow chilies for food.
- ② Linda Perry thinks mankind began using chilies in cooking largely because they kept food from spoiling.
- ③ Christopher Columbus is said to have enjoyed the new experience of eating food spiced with chilies.
- ④ Explorers from Portugal were introduced to chilies in Africa.
- ⑤ The feeling of heat while eating chilies is a purely psychological effect rather than a physical one.
- ⑥ A chemist called Scoville created a heat scale based on the collective judgement of a group of individuals.

Questions 7–13

Complete the notes below.

Choose **NO MORE THAN TWO WORDS** from the passage for each answer.

The role of capsaicin

Chili seeds and capsaicin

- certain birds and other animals eat chili fruit and spread the seeds
- some animals destroy the seeds, preventing ⑦ _____
- unlike many other plants, chilies contain an unpleasant chemical only in their ⑧ _____
- the ⑨ _____ of the chili causes an increase in the capsaicin

Tewksbury's experiments to test effects of capsaicin

- chilies offered to rodents and birds in laboratory conditions
- rodents avoided chilies with a high ⑩ _____ of capsaicin
- birds do not mind eating capsaicin
- capsaicin slows the digestion in birds

- this may make the ⑪ _____ of the seed softer
- another role of capsaicin is in reducing infection caused by a ⑫ _____
- Tewksbury considers the role of capsaicin in chilies to be an example in nature of the beauty of ⑬ _____

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Passage 10

Companies Try Out a New Marketing Practice

Philip Kotler describes a business trend of the 1980s and '90s in his book, Marketing Management

Traditional marketing identifies target markets and satisfies consumer needs. In recent years, however, some have questioned whether this marketing concept is an appropriate philosophy in an age of environmental deterioration, resource shortages, explosive population growth, world hunger and poverty, and neglected social services. Are companies that do an excellent job of satisfying consumer wants necessarily acting in the best interests of consumers and society in the long run?

Let us first consider two examples of the marketing concept in action. The fast-food hamburger industry offers tasty but unhealthy food. The hamburgers have a high fat content, and the restaurants promote fries and pies, two products high in starch and fat. The products are wrapped in convenient packaging, which leads to much waste. In satisfying consumer wants, these restaurants may be hurting consumer health and causing environmental problems. The soft drinks industry has catered to our modern desire for convenience by increasing the share of single-use disposable bottles. However, the single-use bottle represents a great waste of resources; approximately 17 bottles are necessary where formerly one reusable bottle made 17 trips before it was damaged. In addition, many single-use bottles are not biodegradable, and these bottles often litter the environment.

These situations call for a new concept that enlarges the marketing concept. A suggested name for this new concept is 'the societal marketing concept'. It holds that the organization's task is to determine the needs, wants, and interests of target markets and to deliver the desired satisfactions more effectively and efficiently than competitors, in a way that preserves or enhances the consumer's and the society's well-being. Marketers need to build social and ethical considerations into their marketing practices. They must juggle the often conflicting criteria of company profits, consumer satisfaction, and public interest. Two pioneers in implementing the societal marketing concept are Ben & Jerry's and The Body Shop. But even they have begun to experience some difficulty.

In 1978, Ben Cohen and Jerry Greenfield formed a superpremium ice cream company called Ben & Jerry's Homemade. Sixteen years after Ben & Jerry's Scoop Shop opened in Burlington, Vermont, USA, sales climbed to US\$150 million. The company now has more than 600 employees and 10 franchisees. Their popularity rests on Ben & Jerry's creating innovative flavors such as Rainforest Crunch. Customers also know that 7.5% of Ben & Jerry's pre-tax profits are donated to a variety of social and environmental causes. Ben & Jerry's was additionally noted for its egalitarian employer-employee relations: the

company used to have a policy in which the highest-paid employee made no more than five times the lowest-paid worker. Business realities eventually forced the company to modify this ratio to seven to one.

In 1993, facing stiff competition in the super premium ice cream category, the company suffered its first loss since going public in 1984. A search began for a new, well-qualified Chief Executive Officer (CEO) to lead the company through its next stage of financial growth. To achieve this, the founders felt they had to raise the compensation level, yet Ben & Jerry's still conducted the CEO search in their own iconoclastic style. An essay contest entitled 'Yo, I'm Your CEO' attracted applications from 20,000 CEO hopefuls. Despite Ben & Jerry's unusual approach to finding a CEO, in the end the company made a rather conventional choice.

In 1976, Anita Roddick opened The Body Shop in Brighton, UK, in a tiny store selling beauty products in small plastic bottles. Now there are over 1,100 stores in 46 countries and The Body Shop's annual sales growth rate has been between 60% and 100%, with sales reaching US\$250 million in 1993. The company manufactures and sells cosmetics based on natural ingredients in simple and appealing recyclable bottles. The ingredients are largely plant-based and are often sourced from developing countries. Products are usually formulated without animal testing, and The Body Shop donates a percentage of profits each year to animal rights groups, homeless shelters, Amnesty International, Save the Rainforest, and other social causes. The outspoken Roddick routinely spoke out against the mainstream beauty industry and promoted her company's brand of socially responsible capitalism. However, going out on a limb means facing closer scrutiny. In 1992, the company filed a libel suit over a television report that challenged its image as a company with a social conscience. While The Body Shop won the suit in 1993, it is still besieged by critics who allege that the company does not live up to its reputation, and its profits have slipped slightly as copycat competitors enter the natural products market.

These companies are practising cause-related marketing, a version of the societal marketing concept, and it is one major factor in their success. Several companies use cause-related marketing on a limited scale, for example, by contributing to a specific charity based on consumer purchases during a limited period. Companies run these cause-related marketing campaigns for several purposes: to enhance corporate image, thwart negative publicity, pacify consumer groups, launch a new product or brand, broaden their customer base, and generate incremental sales. However, some critics complain that cause-related marketing exploits the cause's constituency and may make consumers feel that they have fulfilled their philanthropic duties by purchasing a product instead of donating directly.

Questions 1–4

Choose the correct letter, **A**, **B**, **C** or **D**.

- 1 What point is made about traditional marketing in the first paragraph?
 - A it can be influenced by population growth.
 - B it considers the long-term needs of society.
 - C it may be unsuited to a time of global problems.
 - D it leads to environmental destruction.
- 2 According to the writer, disposable bottles were introduced in the soft drinks industry
 - A to meet consumer demand.
 - B to reduce damage to bottles.
 - C because they are biodegradable.
 - D because they are more efficient.
- 3 In the sixth paragraph, what does the writer give as the main reason for criticisms of The Body Shop?
 - A The company's profits have decreased.
 - B The company has lost money to other businesses.
 - C The company is less concerned with social issues than it appears to be.
 - D The company uses fewer natural ingredients than it says it does.
- 4 In the seventh paragraph, what objection is made to cause-related marketing?
 - A It is practised in a restricted way.
 - B It is sometimes aimed at only one charity for a short time.
 - C It is practised in order to improve the company's image.
 - D It is likely to discourage people from donating straight to charity.

Questions 5–7

Which **THREE** of the following are part of the societal marketing concept?

Choose **THREE** correct answers, **A–G**.

OPTIONS

- A Assessing the requirements of customers
- B Imitating competitor practices
- C Informing consumers about products
- D Considering community welfare issues
- E Promoting items which are profitable
- F Managing differing requirements in competitive welfare
- G Involving managers in selling the products

Answer: 5 6 7

Questions 8–10

Which *THREE* of the following does the writer suggest were the reason for the success of Ben & Jerry's?

Choose **THREE** correct answers, **A–G**.

OPTIONS

- A Product names were very popular.
- B New tastes were introduced.
- C Funds were given to worthy organisations.
- D The company organized social events.
- E All workers were treated equally.
- F Fairness was a factor in salary levels.
- G Some shops were staffed by volunteers.

Answer: 8 9 10

Questions 11–13

Which *THREE* things show that *The Body Shop* is an example of the societal marketing practice?

Choose **THREE** correct answers, **A–G**.

OPTIONS

- A** Products are sold in small plastic bottles.
- B** Products are made from natural ingredients.
- C** Social and environmental organisations are consulted on company practice.
- D** Some of the income is given to social organisations.
- E** The product containers can be recycled.
- F** They have trained other companies in environmentally friendly practices.
- G** The company is disapproved of by the mainstream beauty industry.

Answer: **11** **12** **13**

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Passage 11

Computer Games for Preschoolers

Nintendo's Research and Design Process

Designing computer games for young children is a daunting task for game producers, who, for a long time, have concentrated on more "hard core" game fans. This article chronicles the design process and research involved in creating Nintendo DS games for preschool gamers.

After speaking with our producers who have a keen interest in designing for the DS, we finally agreed on three key goals for our project. First, to understand the range of physical and cognitive abilities of preschoolers in the context of handheld system game play; second, to understand how preschool gamers interact with the DS, specifically how they control the different forms of play and game mechanics offered by the games presently on the market for this platform; third, to understand the expectations of preschoolers' parents concerning the handheld systems as well as the purchase and play contexts within which game play occurs. The research team decided that in-home ethnographies with preschoolers and their families would yield a comprehensive database with which to give our producers more information and insights, so we started by conducting 26 in-home ethnographies in three markets across the United States: an East coast urban/suburban area, a West coast urban/suburban area, and a Midwest suburban/rural area.

The subjects in this study included 15 girls and 11 boys ranging from 3 years and 3 months old to 5 years and 11 months old. Also, because previous research had shown the effects of older siblings on game play (demonstrated, for example, by more advanced motor coordination when using a computer mouse), households were selected to have a combination of preschoolers with and without older siblings. In order to understand both "experienced" and "new" preschool users of the platform, we divided the sample so that 13 families owned at least one Nintendo DS and the others did not. For those households that did not own a DS, one was brought to the interview for the kid to play. This allowed us to see both the instinctive and intuitive movements of the new players (and of the more experienced players when playing new games), as well as the learned movements of the more experienced players. Each of those interviews took about 60 to 120 minutes and included the preschooler, at least one parent, and often siblings and another caregiver.

Three kinds of information were collected after each interview. From any older siblings and the parents that were available, we gathered data about: the buying decisions surrounding game systems in the household, the family's typical game play patterns, levels of parental moderation with regard to computer gaming, and the most favorite games played by family members. We could also understand the ideology of gaming in these homes because of these in-home interviews: what types of spaces were used for game play,

how the systems were installed, where the handheld play occurred in the house (as well as on-the-go play), and the number and type of games and game systems owned. Most importantly, we gathered the game-playing information for every single child.

Before carrying out the interviews, the research team had closely discussed with the in-house game producers to create a list of game mechanics and problems tied to preschoolers' motor and cognitive capabilities that were critical for them to understand prior to writing the games. These ranged from general dexterity issues related to game controllers to the effectiveness of in-game instructions to specific mechanics in current games that the producers were interested in implementing for future preschool titles. During the interviews, the moderator gave specific guidance to the preschooler through a series of games, so that he or she could observe the interaction and probe both the preschooler and his or her parents on feelings, attitudes, and frustrations that arose in the different circumstances.

If the subject in the experiment had previous exposure to the DS system, he or she was first asked to play his or her favorite game on the machine. This gave the researchers information about current level of gaming skill related to the complexity of the chosen one, allowing them to see the child playing a game with mechanics he or she was already familiar with. Across the 26 preschoolers, the Nintendo DS selections were very broad, including New Super Mario Bros, Sonic Rush, Nintendogs, and Tony Hawk's Proving Ground. The interviewer observed the child play, noting preferences for game mechanic and motor interactions with the device as well as the complexity level each game mechanic was for the tested subject. The researchers asked all of the preschoolers to play a specific game chosen in consultation with our producers, The Little Mermaid: Ariel's Undersea Adventure. The game was chosen for two major reasons. First, it was one of the few games on the market with characters that appeal to this young age group. Second, it incorporated a large variety of mechanics that highlighted the uniqueness of the DS platform, including using the microphone for blowing or singing.

The findings from this initial experiment were extensive. After reviewing the outcomes and discussing the implications for the game design with our internal game production team, we then outlined the design requirements and presented the findings to a firm specializing in game design. We worked closely with those experts to set the game design for the two preschool-targeted DS games under development based on what we had gathered.

As the two DS games went into the development process, a formative research course of action was set up. Whenever we developed new game mechanics, we brought preschoolers into our in-house utility lab to test the mechanics and to evaluate both their simplicity, and whether they were engaging. We tested either alpha or beta versions of different elements of the game, in addition to looking at overarching game structure. Once a full version of the

DS game was ready, we went back into the field test with a dozen preschoolers and their parents to make sure that each of the game elements worked for the children, and that the overall objective of the game was understandable and the process was enjoyable for players. We also collected parents' feedback on whether they thought the game was appropriate, engaging, and worth the purchase.

Questions 1–5

Complete the sentences below.

*Choose **ONE WORD ONLY** from the passage for each answer.*

Exploratory Research Project

Main Objectives

- Determine the relevant ① _____ in the context. Observe how preschoolers manage playing. Investigate attitudes of ② _____ towards games.

Subjects

- 26 children from different US ③ _____. Age range: 3 years and 3 months to 5 years and 11 months. Some children have older ④ _____. Equal number of new and ⑤ _____ players. Some households have Nintendo DS and some don't.

Length of Interview

1–2 hours

Questions 6–9

Do the following statements agree with the information given in Reading Passage 1?

*Write **TRUE** if the statement agrees with the information, **FALSE** if the statement contradicts the information, or **NOT GIVEN** if there is no information on this.*

- ⑥ One area of research is how far mothers and fathers controlled children's playing after school.
- ⑦ The researchers are allowed a free access to the subject's houses.
- ⑧ The researchers regarded The Little Mermaid: Ariel's Undersea Adventure as likely appeal to preschoolers.

- 9 The Little Mermaid: Ariel's Undersea Adventure is operated entirely by hand controls.

Questions 10–13

Complete the flow-chart below.

Choose **NO MORE THAN TWO WORDS** from the passage for each answer.

Using the results of the study

Presentation of design requirements to a specialist 10 _____

↓

Testing the mechanics of two new games in the Nintendo lab (assess 11 _____ and interest)

↓

A 12 _____ of the games trialled with twelve children

↓

Collection of 13 _____ from parents

Passage 12

Deep Sea Discovery

How a chance dive revealed hidden red light on the reef

A Nico Michiels is an ecologist from the University of Tübingen in Germany who spends part of each year in Egypt, where he dives in the Red Sea, observing fish life and gathering data on its coral reefs. In September 2007 he decided to find out how far red light could penetrate the ocean depths. Seawater absorbs different colours at different depths, and as an experienced diver, Michiels was aware that red light is extinguished not far below the surface whereas blue-green light penetrates deeper. To find out the depth at which red disappeared in this particular ocean, however, he attached a special plastic filter to his dive mask which was designed to block out all colours except red. Then he began to descend. In theory, once he reached about 15 metres, he should have been plunged into darkness. Instead, something totally unexpected happened. Sure enough, 20 metres down it was as dark as night. 'All the fish disappeared. With no light from the surface, they were effectively black and had become invisible,' he says. 'But it didn't stay black for long. Then I saw a group of goby fish with bright red eyes lit up against the background. After that red spots began to show up all over the reef.'

B Even with the red filter removed, Michiels could pick them out without much trouble once his eyes grew accustomed to the gloom. It seems strange that no diver or researcher had spotted all this red before, but as Michiels points out, no one saw it because no one expected to see it. On that one dive Michiels discovered three fish species with prominent red markings, and has found many others since.

C But how can fish appear red where there's no red light? Ordinary red pigments look red because they reflect red light while absorbing all other wavelengths. At 20 metres down, there had to be some other explanation for the red Michiels was seeing. He suspected fluorescence. Fluorescent pigments behave differently from ordinary ones: they receive incoming light of one wavelength, for example blue, and emit light of a longer wavelength, in this case red. On the reef in the Red Sea during daytime, the most likely explanation was that the predominantly blue and green wavelengths at depth triggered the emission of fluorescent red in the fish.

D With only a week left in Egypt, and lacking the equipment to confirm that the fish were fluorescent, Michiels photographed as many of them as he could. Then once back in Germany, he bought an assortment of tropical fish and installed them in his lab. Here he confirmed that the fish did indeed fluoresce. In most of the fish he looked at, the fluorescence could be traced to specialised pigment cells that lie in the skin beneath the scales. These cells contain 'guanine crystals', which scatter light to give fish their silvery

sheen. However, Michiels says they are still not sure exactly what is fluorescing. 'It's not the crystals themselves. It's probably a fluorescent protein built into the crystals, and we have a suspicion that it might be made by bacteria.'

E Intrigued, Michiels began a systematic search for red fluorescence in reef fish. He and his colleagues, Nils Anthes and Dennis Sprenger, have identified some 50 species with red fluorescence. The most common markings tend to be on the body towards the head and to a lesser extent around the eyes, and then the fins. To Michiels, the distribution of these markings is one of the strongest indications that red fluorescence has a very particular function: communication with other members of the species. According to several recent studies, a whole range of animals employ fluorescence as a natural highlighter to boost the visibility of body parts they use to signal, for example to ward off enemies. In reef fish, the red tends to be confined to parts of the body used to signal, suggesting these markings serve a similar function. But instead of highlighting an existing colour, the fluorescence gives the fish a colour that otherwise wouldn't exist. For example, fish commonly use eye rings to signal that they are present and their direction of gaze, and Michiels suspects that red-eyed gobies use signals to indicate their location and keep their group together.

F Red light, whatever its source, doesn't travel far through water, which suggests signals are intended to be private, seen only by nearby fish of the right species. There are several lines of evidence to support this, says Michiels. And closely related species do not have completely identical markings, which suggests they might be important in species recognition.

G Michiels suspects red fluorescence has another important role for some reef fish: helping them blend in. During his first dive with the red filter, he noticed corals glow a dark but faint red too. Against this irregular red background, a fish that glows red all over would be hard to distinguish. More compelling for Michiels is the case of the scorpionfish, which lies perfectly still until food swims past which it then sucks in.

H Yet if red plays any part in a fish's life then it must be able to see it. Fish that live in a world dominated by blue-green light are assumed to have eyes tuned to those wavelengths, and most marine fish that have been studied are thought incapable of seeing red. One exception is the seahorse, whose eyes are sensitive to red. As for the other fish, it remains to be seen.

Questions 1–6

Do the following statements agree with the information given in the passage?

*Write **TRUE** if the statement agrees with the information, **FALSE** if the statement contradicts the information, or **NOT GIVEN** if there is no information on this.*

- ① During his 2007 dive, Michiels expected to encounter total darkness at about 15 metres.
- ② Michiels could see the red markings on fish without the aid of the red filter.
- ③ Other divers had assumed they would see fish with red markings.
- ④ All the fish with red markings that Michiels found during his diving expeditions came from the Red Sea.
- ⑤ Michiels first thought of the possibility that fish could fluoresce while he was in Germany.
- ⑥ Michiels remains uncertain as to what creates fluorescence in fish.

Questions 7–13

Complete the notes below.

Write **ONE WORD ONLY** from the passage for each answer.

Michiels's findings

Michiels has observed:

- – 50 types of fish with red fluorescence in total; markings mainly near the

⑦ _____

Some of Michiels's beliefs are that:

- – Red fluorescence is used specifically for ⑧ _____ purposes.
- – Fish, like some animals, use fluorescence to keep ⑨ _____ away.
- – Gobies depend on red fluorescence to show their ⑩ _____.
- – There are variations in the markings of fish among those ⑪ _____ which are very similar.

Other benefits of red fluorescence:

- – Fish cannot easily be seen near backgrounds of ⑫ _____ which give off a red light.
- – Helps some fish catch their prey.

The ability to see red amongst fish:

- – The only fish proven to have this ability is the 13 _____.

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Passage 13

Diagnosing Bridges

Microwave sensors that constantly monitor bridge condition may soon replace risky visual inspections

A Most road and rail bridges are only inspected visually, if at all. Every few months, engineers have to clamber over the structure in an attempt to find problems before the bridge shows obvious signs of damage. Technologies developed at Los Alamos National Laboratory, New Mexico, and Texas A&M University may replace these surveys with microwave sensors that constantly monitor the condition of bridges.

B "The device uses microwaves to measure the distance between the sensor and the bridge, much like radar does," says Albert Migliori, a Los Alamos physicist. "Any load on the bridge - such as traffic - induces displacements, which change that distance as the bridge moves up and down." By monitoring these movements over several minutes, the researchers can find out how the bridge resonates. Changes in its behavior can give an early warning of damage.

C The Interstate 40 bridge over the Rio Grande River in Albuquerque provided the researchers with a rare opportunity to test their ideas. Chuck Farrar, an engineer at Los Alamos, explains: "The New Mexico authorities decided to raze this bridge and replace it. We were able to mount instruments on it, test it under various load conditions and even inflict damage just before it was demolished." In the 1960s and 1970s, 2500 similar bridges were built in the US. They have two steel girders supporting the load in each section. Highway experts know that this design is "fracture critical" because a failure in either girder would cause the bridge to fail.

D After setting up the microwave dish on the ground below the bridge, the Los Alamos team installed conventional accelerometers at several points along across it and while subjecting it to pounding from a "shaker", which delivered precise punches to a specific point on the road.

E "We then created damage that we hoped would simulate fatigue cracks that can occur in steel girders," says Farrar. They first cut a slot about 60 centimeters long in the middle of one girder. They then extended the cut until it reached the bottom of the girder and finally, they cut across the flange - the bottom of the girder's "I" shape.

F The initial, crude analysis of the bridge's behavior, based on the frequency at which the bridge resonates, did not indicate that anything was wrong until the flange was damaged. But later the data were reanalyzed with algorithms that took into account changes in the mode shapes of the structure - shapes that the structure takes on when excited at a

particular frequency. These more sophisticated algorithms, which were developed by Norris Stubbs at Texas A&M University, successfully identified and located the damage caused by the initial cut.

G "When any structure vibrates, the energy is distributed throughout with some points not moving, while others vibrate strongly at various frequencies," says Stubbs. "My algorithms use pattern recognition to detect changes in the distribution of this energy. NASA already uses Stubbs' method to check the behavior of the body flap that slows space shuttles down after they land.

H A commercial system based on the Los Alamos hardware is now available, complete with the Stubbs algorithms, from the Quatro Corporation in Albuquerque for about \$100,000.

I Tim Darling, another Los Alamos physicist working on the microwave interferometer with Migliori, says that as the electronics become cheaper, a microwave inspection system will eventually be applied to most large bridges in the US. "In a decade I would like to see a battery or solar-powered package mounted under the bridge, scanning it every day to detect changes," he says.

Questions 1–5

The reading Passage has nine paragraphs, **A–I**. Which paragraph contains the following information?

Write the correct letter, **A–I**.

Q	Statement	
1	The professional team put pressure to test the motion of the bridge.	
2	Engineers apply knife to the bridge to excite cracks.	
3	A precious chance of experiment to certificate ideas.	
4	The popular application of the microwave inspection system within a decade.	
5	How the microwave works.	

Questions 6–9

Choose the correct letter, **A, B, C** or **D**.

- 6 What is the responsibility of engineers in order to prevent the damage of the bridge before the invention of the microwave sensors?
- A they have to climb over the bridge.
 - B They have to regularly check the bridge.
 - C They have to inspect the condition of the bridge through monitors.
 - D They have to employ others to help them check the bridge.
- 7 What did the device take advantage of the microwaves to do?
- A to calculate the distance
 - B to induce displacements
 - C to change the distance
 - D to give an advanced warning
- 8 Why did highway experts think the design as “fracture critical”?
- A Engineers failed to take several tests according to different conditions.
 - B Engineers failed to install conventional accelerometers.
 - C The supporting part would probably make the bridge fall down.
 - D No cars drove past the bridge.
- 9 What was the achievement of Norris Stubbs' complicated algorithms?
- A the identification and location of the damage.
 - B the movement of some points.
 - C the renounce of the bridge.
 - D the distribution of the energy.

Questions 10–13

Complete the sentences below.

*Choose **NO MORE THAN THREE WORDS** from the passage for each answer.*

- The weight of the 2500 bridges is sustained by 10 _____ in every sector.
- 11 _____ were set up by the Los Alamos team in order to test the movement of the bridge.
- In order to cause break, the Los Alamos team decided to make a 12 _____ at first step.
- The 13 _____ in the bottom of the bridge resembles "I" shape.

Passage 14

The Inventor of the Periodic Table – Dmitri Ivanovich Mendeleev

Mendeleev's wish to find a better way of organizing chemistry led to the creation of his periodic table, one of the most iconic symbols in science

Dmitri Ivanovich Mendeleev was born on February 8, 1834 in Verkhnie Aremzyani, in the Russian province of Siberia. His father, a graduate of Saint Petersburg's Main Pedagogical Institute, died when Mendeleev was just 13. At age 16, Mendeleev relocated to Saint Petersburg, which was then Russia's capital city. He won a place at his father's old college, where he initially trained as a teacher, in part because the director of the Institute had known his father. However, he went on to achieve worldwide fame as a chemist.

By the time he was 20, Mendeleev was already having research papers published. However, he was troubled with various health issues and was often so ill with tuberculosis that he was forced to work from his bed. His uncontrollable temper made him unpopular with some of the staff and his classmates, but he still graduated as the top student in his year. In 1855, he got a job in Simferopol, Crimea, but soon returned to Saint Petersburg, where he worked towards a Master's degree in chemistry. He gained his Master's in 1856.

A few years later, he was given the opportunity to go to western Europe to pursue chemical enquiry. He spent most of 1859 and 1860 in Heidelberg, Germany. Here he had the good fortune to work briefly with renowned German chemist Robert Bunsen at Heidelberg University, before setting up a laboratory in his own apartment.

In 1860, Mendeleev attended the first ever international chemistry conference, held in Karlsruhe, Germany. Much of the event was spent discussing the need to standardize chemistry, and this played a key role in Mendeleev's eventual development of his periodic table of the elements.

By the time he returned to Saint Petersburg in 1861, this time to work at the Technical Institute, Mendeleev had become even more passionate about chemistry. He was concerned that Russia was trailing behind Germany in this field. He thought improved Russian-language chemistry textbooks were necessary, and was determined to do something about it. In just 61 days, the 27-year-old chemist wrote his 500-page Organic Chemistry, which put him at the forefront of Russian chemical education.

Mendeleev was a charismatic lecturer and held a number of academic positions until, in 1867, aged just 33, he was awarded the Chair of General Chemistry at the University of Saint Petersburg. In this prestigious position he continued pushing to improve chemistry

in Russia, publishing *The Principles of Chemistry* in 1869. The popularity of this work in Russia and elsewhere led to the publication of translations into three languages: English, French and German.

At this time, chemistry was a patchwork of observations and discoveries. Mendeleev was certain that better, more fundamental principles could be found. This was his mindset when, in 1869, he began writing a second volume of his book *The Principles of Chemistry*. At the heart of chemistry were hydrogen, oxygen and all its other elements. What, wondered Mendeleev, could they reveal if he could find some way of organizing them logically?

He wrote the names of the 65 known elements on cards - one element on each card - and then wrote the fundamental properties of each element, including atomic weight, on its card. He saw that atomic weight was important in some way - the behavior of the elements seemed to repeat as their atomic weights increased - but he could not see the pattern. Convinced that he was close to making a significant discovery, Mendeleev moved the cards about for hours until finally he fell asleep at his desk. When he awoke, he found that his subconscious mind had done his work for him. He now knew the pattern the elements fell into. He later wrote, 'In a dream I saw a table where all the elements fell into place as required. Awakening, I immediately wrote it down on a piece of paper.'

Two weeks later, he published a paper entitled *The Relation between the Properties and Atomic Weights of the Elements*. The periodic table had been released to the scientific world. As with many scientific discoveries, there is a time when a concept becomes ripe for discovery, and this was the case in 1869 with the periodic table. Lothar Meyer, for example, had proposed a rough periodic table in 1864 and by 1868 had devised one that was very similar to Mendeleev's, but he did not publish it until 1870.

Mendeleev was successful because he not only showed how the elements could be organized, but he used his periodic table to predict the existence of eight new elements and also to propose that some of the elements, whose behavior did not agree with what he predicted, must have had their atomic weights measured incorrectly. It turned out that chemists had measured some atomic weights incorrectly. Mendeleev was right. Scientists everywhere started to pay attention to his periodic table. And on the discovery of new elements, as per his prediction, Mendeleev's fame and scientific reputation were further enhanced.

In 1905, the British Royal Society gave him its highest honor, the Copley Medal, for his achievements, and in the same year he was elected to the Royal Swedish Academy of Sciences. Element 101 is named Mendelevium in his honor. Dmitri Mendeleev died in Saint Petersburg, on February 2, 1907.

Questions 1–8

Complete the notes below.

Choose **ONE WORD ONLY** from the passage for each answer.

Dmitri Ivanovich Mendeleev

Mendeleev's early life

- he studied to become a ① _____ in St Petersburg.
- he often had to work in bed when he was suffering from ② _____.

Mendeleev's career

- went to a ③ _____ in 1860, which inspired his work on the periodic table.
- 1861 — he wrote Organic Chemistry, having identified a need for better ④ _____ in Russian.
- several ⑤ _____ of The Principles of Chemistry were published.

Mendeleev's work on the periodic table

- he used cards to make a note of the atomic weight and other ⑥ _____ of the elements.
- when asleep, he subconsciously discovered a ⑦ _____ which organised the elements in a table.
- 1869 — Mendeleev's periodic table was made public.
- Mendeleev's reputation grew after he made ⑧ _____ about further elements.

Questions 9–16

Do the following statements agree with the information given in Reading Passage 1?

Write **TRUE** if the statement agrees with the information, **FALSE** if the statement contradicts the information, or **NOT GIVEN** if there is no information on this.

- ⑨ Dmitri Mendeleev was the first member of his family to receive a college education.

- ⑩ While he was studying in Saint Petersburg, Mendeleev often failed to control his anger.
- ⑪ During his time at Heidelberg University, Mendeleev published a paper with Robert Bunsen.
- ⑫ Mendeleev worried that Germany was more successful than Russia in the field of chemistry.
- ⑬ It took Mendeleev less than a year to write the second volume of his book The Principles of Chemistry.
- ⑭ Mendeleev was the first scientist to suggest the organisation of the element in a table.
- ⑮ Mendeleev's paper on the periodic table received a positive reception from the scientific community.
- ⑯ In his lifetime, Mendeleev failed to receive any awards for his work in chemistry.

Passage 15

Dyes and Fabric Dyeing

Putting color into clothes

When primitive people began using their hands to be creative, they began to add color to their lives. They used natural materials like ochre to stain animal hides, decorate shells and feathers, and paint on the walls of caves. Ochre is a naturally occurring brownish yellow earth containing iron oxide. Scientists have been able to date the black, white, yellow and reddish pigments made from ochre, used in cave paintings, to over 15,000 BCE. With the development of fixed settlements and agriculture, around 7,000 to 2,000 BCE, people began to produce fabrics, and used natural substances such as ochre to color them.

Natural dyes, or dyes made from substances found in nature, can be broken down into two categories: substantive and adjective. Substantive, or direct dyes, become fixed to fibers without the aid of any other chemicals or additives. Adjective dyes require a mordant (usually a metal salt) which acts as a fixative and prevents the color from washing out or being bleached by sunlight. Most natural dyes are adjective dyes, and require the application of a mordant solution to the fibers at some point in the dyeing process.

Historically, three natural fibers were used in making fabrics: wool, silk and cotton. Wool fabric remains have been found in Europe dating back to 2,000 BCE. It was a common medieval fabric worn in both dyed and natural colors and was processed by both professional manufacturers and by people in their own homes. Silk was imported from China to Europe, and in the 14th to 16th centuries major silk manufacturing centers were set up in France, Spain and Italy. These silk production centers also became centers of dye technology, as most silk was dyed and required the highest quality dyes available. Cotton was considered a luxury fabric in Europe, as it was imported all the way from India and was dyed before it was shipped. Cotton was also valued because of the brightness and colorfastness of the dyes used to color it.

Dyes that gave fabrics a good bright color and were able to withstand washing and exposure to sunlight without losing their color were highly prized. The names of some of these valued and traded colors are still familiar today. The color known as Tyrian purple, for example, originated in the Mediterranean 2,000 years ago, and cochineal is the name given to the red dye from Latin America. Both of these colors came from animals; the mucus found in certain species of shellfish produced the deep rich Tyrian purple and cochineal was extracted from insects. Two other natural sources of color were saffron and indigo. Saffron, the base of yellow dyes, comes from the flowers of a particular kind of crocus which is thought to have first been cultivated on Crete in the eastern Mediterranean. The leaves of a plant native to India were used to produce indigo, which was the main source of the color blue.

As societies developed over the centuries the demand for dyes and dyed fabric grew, and by the 17th century a world-wide shipping and trading network was in place, allowing dyestuffs from all parts of the world to be brought to Europe. This meant that numerous dyestuffs could be blended to create a variety of colors for the rich and powerful. Fiber dyeing in the lower classes was a bit more restrictive. Without the money to buy exotic imported dyes, clothing in the countryside tended to be black, brown, grey and tan. Country people had some resources they could use to get a wider range of colors. They had always used local plants as food, and many of these plants were also used as medicines and in some cases as sources of dyes. Home dyers used any plants they could find that would give a good color. People who picked blackberries to make jam soon recognized this wild fruit as a source for a blue dye. Washing beehives in preparation for making mead (a popular drink containing honey) yielded a liquid that could be used as a yellow dye. The mosses which grow in many parts of Europe were used to produce green dye.

With the tremendous rise of interest in chemistry in the mid-19th century, several important innovations in dyeing came about. WH Perkin, a student of celebrated European scientist Wilhelm von Hofmann, accidentally discovered the first synthetic dye, later called mauve. The color was so popular that Perkin was able to open a factory of his own and went on to develop more synthetic dye colors. Synthetic dye production grew in Europe, and hardly a year passed until the end of the century without a new synthetic dye being patented.

Eventually, the old natural dyes lost popularity in favor of the newer synthetic ones and now the use of natural dyes on a commercial scale only exists in a few remote areas where people have either little access to synthetic dyes or a vested interest in retaining their ancient dyeing customs.

Questions 1–7

Do the following statements agree with the information given in Reading Passage 1?

Write **TRUE**, **FALSE** or **NOT GIVEN**.

- 1 Ochre was used in paintings before it was used in fabric dyes.
- 2 Natural dyes that need a mordant are rare.
- 3 In medieval times people sometimes wore fabric made of undyed wool.
- 4 Silk has always been more expensive than cotton and wool.
- 5 Cotton imported from India was dyed upon arrival in Europe.
- 6 Perkin became more famous than his teacher, von Hofmann.
- 7 Very few synthetic dyes were produced in Europe in the second half of the 19th century.

- 8 Today the commercial production of natural dyes is limited to a small number of isolated communities.

Questions 9–13

Complete the notes below.

Choose **ONE WORD ONLY** from the passage for each answer.

The history of some dyes — Highly valued dyes

- saffron (a yellow dye) — initially made from crocuses found on the island of Crete
- indigo (a blue dye) — made from plant leaves from India
- Tyrian purple — mucus from 9 — found in the Mediterranean
- Cochineal (a red dye) — made from 10 — found in Latin America

17th century Europe — dyes made by country people

- 11 — had two uses: jam and blue dye
- Liquid from cleaning 12 — had two uses: making mead and yellow dye

19th century Europe

- Progress in study of 13 — led to synthetic dyes

Passage 16

Endangered Chocolate

Cacao farming faces disease, deforestation and a broken supply chain

A The cacao tree, once native to the equatorial American forest, has some exotic traits for a plant. Slender and shrubby, the cacao has adapted to life close to the leaf littered forest floor. Its large leaves droop down, away from the sun. Cacao doesn't flower, as most plants do, at the tips of its outer and uppermost branches. Instead, its sweet white buds hang from the trunk and along a few fat branches which form where leaves drop off. These tiny flowers transform into pulp-filled pods almost the size of rugby balls. The low-hanging pods contain the bitter-tasting magical seeds.

B Somehow, more than 2,000 years ago, ancient humans in Mesoamerica discovered the secret of these beans. If you scoop them from the pod with their pulp, let them ferment and dry in the sun, then roast them over a gentle fire, something extraordinary happens: they become chocolaty. And if you then grind and press the beans, which are half-cocoa butter or more, you will obtain a rich crumbly, chestnut brown paste — chocolate at its most pure and simple.

C The Maya and Aztecs revered this chocolate, which they frothed up with water and spices to make bracing concoctions. It was an edible treasure, offered up to their gods, used as money and hoarded like gold. Long after Spanish explorers introduced the beverage to Europe in the sixteenth century, chocolate retained an aura of aristocratic luxury. In 1753, the Swedish botanist Carolus Linnaeus gave the cacao tree genus the name *Theobroma*, which means 'food of the gods'.

D In the last 200 years, the bean has been thoroughly democratized — transformed from an elite drink into ubiquitous candy bars, cocoa powders and confections. Today chocolate is becoming more popular worldwide, with new markets opening up in Eastern Europe and Asia. This is both good news and bad, because although farmers are producing record numbers of the cacao bean, this is not enough, some researchers worry, to keep pace with global demand. Cacao is also facing some alarming problems.

E Philippe Petithuguenin, head of the cacao program at the Centre for International Cooperation in Development-Oriented Agricultural Research (CIRAD) in France, recently addressed a seminar in the Dominican Republic. He displayed a map of the world revealing a narrow band within 18° north and south of the equator, where cacao grows. In the four centuries since the Spanish first happened upon cacao, it has been planted all around this hot humid tropical belt — from South America and the Caribbean to West Africa, East Asia, and New Guinea and Vanuatu in the Pacific.

F Today 70% of all chocolate beans come from West Africa and Central Africa. In many parts, growers practice so-called pioneer farming. They strip patches of forest of all but the tallest canopy trees and then they put in cacao, using temporary plantings of banana to shade the cacao while it's young. With luck, groves like this may produce annual yields of 50 to 60 pods per tree for 25 to 30 years. But eventually, pests, pathogens and soil exhaustion take their toll and yields diminish. Then the growers move on and clear a new forest patch — unless farmers of other crops get there first. 'You cannot keep cutting the tropical forest, because the forest itself is endangered,' said Petithuguenin. 'World demand for chocolate increases by 3% a year on average. With a lack of land for new plantings in tropical forests, how do you meet that?'

G Many farmers have a more imminent worry: outrunning disease. Cacao, especially when grown in plantations, is at the mercy of many afflictions, mostly rotting diseases caused by various species of fungi which cover the pods in fungus or kill the trees. These fungi and other diseases spoil more than a quarter of the world's yearly harvest and can devastate entire cacao-growing regions.

H One such disease, witches' broom, devastated the cacao plantations in the Bahia region of Brazil. Brazil was the third largest producer of cacao beans but in the 1980s the yields fell by 75%. According to Petithuguenin, 'if a truly devastating disease like witches' broom reached West Africa (the world's largest producer), it could be catastrophic.' If another producer had the misfortune to falter now, the ripples would be felt the world over. In the United States, for example, imported cacao is the linchpin of an \$8.6 billion domestic chocolate industry that in turn supports the nation's dairy and nut industries; 20% of all dairy products in the US go into confectionery.

I Today research is being carried out to try to address this problem by establishing disease-resistant plants. However, even the best plants are useless if there isn't anywhere to grow them. Typically, farmers who grow cacao get a pittance for their beans compared with the profits reaped by the rest of the chocolate business. Most are at the mercy of local middlemen who buy the beans then sell them for a much higher price to the chocolate manufacturers. If the situation is to improve for farmers, these people need to be removed from the process. But the economics of cacao is rapidly changing because of the diminishing supply of beans. Some companies have realized that they need to work more closely with the farmers to ensure that sustainable farming practices are used. They need to replant areas and create a buffer for the forest, to have ground cover, shrubs and small trees as well as the canopy trees. Then the soil will be more robust and more productive. They also need to empower the farmers by guaranteeing them a higher price for their beans so that they will be encouraged to grow cacao and can maintain their way of life.

Questions 1–3

Choose the correct letter, **A**, **B**, **C** or **D**.

- 1 The flowers of the cacao plant appear
- A at the end of its top branches.
 - B along all of its branches.
 - C mainly on its trunk.
 - D close to its leaves.
- 2 In Africa, banana trees are planted with the cacao plants in order to
- A replace the largest trees.
 - B protect the new plants.
 - C provide an extra crop.
 - D help improve soil quality.
- 3 In paragraph H, what is the writer referring to when he says 'the ripples would be felt the world over'?
- A the impact a collapse in chocolate production could have on other industries.
 - B the possibility of disease spreading to other crops.
 - C the effects of the economy on world chocolate growers.
 - D the link between Brazilian growers and African growers.

Questions 4–9

The reading passage has nine paragraphs, **A–I**.

Which paragraph contains the following information?

NB You may use any letter more than once.

Q	Statement	
4	a list of the cacao growing areas	
5	an example of how disease has affected one cacao growing region	
6	details of an ancient chocolate drink	
7	a brief summary of how the chocolate industry has changed in modern times	
8	the typical lifespan and crop size of a cacao plantation	
9	a reference to the scientific identification of the cacao plant	

Questions 10–13

Complete the notes below.

Choose **NO MORE THAN TWO WORDS** from the passage for each answer.

WAYS OF DEALING WITH THE PLANT'S PROBLEMS

- Need to find plants which are not affected by (10) _____.
- Chocolate producers need to work directly with farmers instead of (11) _____.
- Need to encourage farmers to use (12) _____ methods to grow cacao plants.
- Make sure farmers receive some of the (13) _____ made by the chocolate industry.

Passage 17

Experience versus Speed

Certain mental functions slow down with age, but the brain compensates in ways that can keep seniors as sharp as youngsters

Jake, aged 16, has a terrific relationship with his grandmother Rita, who is 70. They live close by, and they even take a Spanish class together twice a week at a local college. After class, they sometimes stop at a cafe for a snack. On one occasion, Rita tells Jake, 'I think it's great how fast you pick up new grammar. It takes me a lot longer.' Jake replies, 'Yeah, but you don't seem to make as many silly mistakes on the quizzes as I do. How do you do that?'

In that moment, Rita and Jake stumbled across an interesting set of differences between older and younger minds. Popular psychology says that as people age their brains 'slow down'. The implication, of course, is that elderly men and women are not as mentally agile as middle-aged adults or even teenagers. However, although certain brain functions such as perception and reaction time do indeed take longer, that slowing down does not necessarily undermine mental sharpness. Indeed, evidence shows that older people are just as mentally fit as younger people because their brains compensate for some kinds of declines in creative ways that young minds do not exploit.

Just as people's bodies age at different rates, so do their minds. As adults advance in age, the perception of sights, sounds and smells takes a bit longer, and laying down new information into memory becomes more difficult. The ability to retrieve memories also quickly slides and it is sometimes harder to concentrate and maintain attention.

On the other hand, the ageing brain can create significant benefits by tapping into its extensive hoard of accumulated knowledge and experience. The biggest trick that older brains employ is to use both hemispheres simultaneously to handle tasks for which younger brains rely predominantly on one side. Electronic images taken by cognitive scientists at the University of Michigan, for example, have demonstrated that even when doing basic recognition or memorization exercises, seniors exploit the left and right side of the brain more extensively than men and women who are decades younger. Drawing on both sides of the brain gives them a tactical edge, even if the speed of each hemisphere's process is slower.

In another experiment, Michael Falkenstein of the University of Dortmund in Germany found that when elders were presented with new computer exercises they paused longer before reacting and took longer to complete the tasks, yet they made 50% fewer errors, probably because of their more deliberate pace.

One analogy for these results might be the question of who can type a paragraph 'better': a 16-year-old who glides along at 60 words per minute but has to double back to correct a number of mistakes or a 70-year-old who strikes keys at only 40 words per minute but spends less time fixing errors? In the end, if 'better' is defined as completing a clean paragraph, both people may end up taking the same amount of time.

Computerized tests support the notion that accuracy can offset speed. In one so-called distraction exercise, subjects were told to look at a screen, wait for an arrow that pointed in a certain direction to appear, and then use a mouse to click on the arrow as soon as it appeared on the screen. Just before the correct symbol appeared, however, the computer displayed numerous other arrows aimed in various other directions. Although younger subjects cut through the confusion faster when the correct arrow suddenly popped up, they more frequently clicked on incorrect arrows in their haste.

Older test takers are equally capable of other tasks that do not depend on speed, such as language comprehension and processing. In these cases, however, the elders utilize the brain's available resources in a different way. Neurologists at Northwest University came to this conclusion after analyzing 50 people ranging from age 23 to 78. The subjects had to lie down in a magnetic resonance imaging (MRI) machine and concentrate on two different lists of printed words posted side by side in front of them. By looking at the lists, they were to find pairs of words that were similar in either meaning or spelling.

The eldest participants did just as well on the tests as the youngest did, and yet the MRI scans indicated that in the elders' brains, the areas which are responsible for language recognition and interpretation were much less active. The researchers did find that the older people had more activity in brain regions responsible for attentiveness. Darren Gleitman, who headed the study, concluded that older brains solved the problems just as effectively but by different means.

Questions 1–3

Choose the correct answer, **A**, **B**, **C** or **D**.

- ① The conversation between Jake and Rita is used to give an example of
- A** the way we learn languages.
 - B** the changes that occur in our brains over time.
 - C** the fact that it is easier to learn a language at a young age.
 - D** the importance of young and old people doing things together.

- 2 In paragraph six, what point is the analogy used to illustrate?
- A Working faster is better than working slower.
 - B Accuracy is less important than speed.
 - C Accuracy can improve over time.
 - D Working faster does not always save time.
- 3 In the computerized distraction exercises, the subjects had to
- A react to a particular symbol on the screen.
 - B type a text as quickly as possible.
 - C move an arrow in different directions around the screen.
 - D click on every arrow that appeared on the screen.

Questions 4–7

Complete each sentence with the correct ending, A–F, below. Write the correct letter in each box.

- | | |
|---|---------|
| 4 According to popular psychology | 4 _____ |
| 5 Researchers at the University of Michigan showed that | 5 _____ |
| 6 Michael Falkenstein discovered that | 6 _____ |
| 7 Scientists at Northwest University concluded that | 7 _____ |

List of Endings

- A the older we get the harder it is to concentrate for any length of time.
- B seniors take longer to complete tasks but with greater accuracy.
- C old people use both parts of their brain more than young people.
- D older people use their brains differently but achieve the same result.
- E the speed of our brain decreases with age.
- F older people do not cope well with new technology.

Questions 8–12

Complete the summary below.

Choose **NO MORE THAN ONE WORD** from the passage for each answer.

- People's bodies and 8 _____ grow older at varying stages.
- As we age our senses take longer to process information and our aptitude for recalling 9 _____ also decreases.

- However, older people's brains do have several advantages. Firstly, they can call upon both the ⑩ _____ and ⑪ _____ which is already stored in their brain.
- Secondly, although the ⑫ _____ of each side of their brain is reduced, they are able to use both sides at once.

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Passage 18

Flying the Coast

The development of an air service on the west coast of New Zealand's South Island

Cut off from the rest of the country by a range of mountains, the west coast of New Zealand's South Island — or the 'Coast' as it is commonly known — was the country's 'wild west frontier'. But unlike Fiordland to the south, which was and still is an uninhabitable wilderness, the Coast in the 1930s was not only habitable, it was also potentially rich. Settlers hunted and fished, logged, milled and mined. They farmed where they managed to clear the forest and drain the swamps. It was pure survival at times. The inescapable isolation was not so much because of the great distances that travellers had to cover, but rather due to the topography of the place — the mountains, gorges, glaciers, rivers and headlands — which necessitated long detours and careful timing with regard to weather and tides.

Bridges were few and far between in the early years, and even ferry crossings were often impossible after heavy rains. Each river had its attendant ferryman or woman whose attention a traveller would attract by firing a rifle shot. It was the kind of country where one would greatly benefit from a pair of wings.

Maurice Buckley, a World War I pilot, was the first to give Coasters, as the residents of the region were called, such wings, by establishing the Arrow Aviation Company in 1923. That year he bought an Avro biplane on the east coast, which he transported across the country by rail, wings off, before reassembling it in a local garage. When he opened for business the following year, the colourful Avro became a familiar sight, and Coasters queued up for joyrides. For the first major flight, Buckley invited Dr Teichelmann, a local mountaineer, to join him. They flew over the Franz Josef Glacier and landed at Okarito. Afterwards, Teichelmann wrote about how extraordinary it was to look at the world from the air, 'like taking the roof off the house and watching the performances from above.'

Next came an aviator named Bert Mercer, who made a reconnaissance flight to the Coast in August 1933 and started Air Travel (NZ) the following year. Mercer himself chose a DH83 Fox Moth. By comparison with the regular, open-air aircraft of the day, the Fox Moth was a plane that offered considerable luxury, housing four passengers in an enclosed forward area fully protected from the weather. Mercer opened for business in December 1934, picking up the airline's first passengers and, on the last day of that year, commenced a regular delivery of mail, carrying 73½ lb to Haast and Okuru. From that day on, the Fox Moth became a much-anticipated sight on the coast.

Mercer got on with everyone and won their respect by anticipating, then meeting their needs. One of those was setting up the first aerial shipping route to help transport a kind of small fish known as whitebait. Starting in 1935 Mercer would put the plane down where there was no airstrip, instead using remote beaches such as the one at the mouth of the Paringa River, to collect the whitebait and whisk them off to the right train and waiting city markets in perfectly fresh condition. Mercer relied on his senses — what he could see and hear — to navigate, flying around the weather and contours of the land. Although often warned to do so by aviation authorities, he refused to develop the skills necessary to navigate the plane 'blind', using just its instruments on the console in front of him. The old habits were too hard to change.

With the outbreak of World War II, Mercer's aircraft were considered so essential to the remote Coast that they were not militarised. In fact, the business continued to grow in the early years, thanks in large part to a government-issued subsidy, which allowed him to expand into neighbouring areas. Despite the war in far-off lands, life on the Coast was business as usual. The settlers were always in need of mail and transportation. In time, though, this presented Mercer with a pressing issue: with so many now joining the Air Force, he no longer had enough pilots. In 1942 he wrote in his diary, 'I am back to where I started eight years ago — on my own.'

The only solution to keep the airline going was to pack as much into every plane as possible and make every flight count. But some of Mercer's newly formed team objected to the amount of cargo they had to carry, which for a small rural airline was a fact of life. One man, Norm Suttie, left the airline after a few months in protest about carrying more than was appropriate for the aircraft. This marked another decline in the airline's fortunes. When Bert Mercer died in 1944, the airline was taken over by Fred Lucas, a man who shared Mercer's pioneering spirit. Under Lucas's leadership the newly formed West Coast Airways saw another decade of profitable returns. But in the following decade, times changed fast. Helicopters were soon found to be ideal machines for the Coast terrain, and quickly took over the vast majority of the local air transport business.

Questions 1–6

Do the following statements agree with the information given in the passage?

*Write **TRUE** if the statement agrees with the information, **FALSE** if the statement contradicts the information, or **NOT GIVEN** if there is no information on this.*

- 1 In the 1930s, the Coast and Fiordland had populations of a similar size.
- 2 Most settlers on the Coast were migrants from overseas.
- 3 The Coast's geographical features made moving around the region difficult.

- 4 The first bridges to be built on the Coast were swept away by floods.
- 5 Maurice Buckley flew his Avro biplane to the Coast in 1923.
- 6 Coasters were unwilling to fly at first.

Questions 7–13

Complete the notes below.

Choose **ONE WORD ONLY** from the passage for each answer.

Bert Mercer and aviation on the Coast

Early Years

- Mercer set up Air Travel (NZ) in 1934
- the Fox Moth was noted for its 7 ————— compared to other planes
- in 1934 Mercer's company started to transport 8 ————— and passengers
- from 1935 planes landed on 9 ————— to pick up fresh produce

World War II

- the airline expanded at first because it got a 10 ————— from the state
- there was a shortage of 11 ————— by 1942

Final Years

- there were disputes at the airline about the quantity of 12 ————— in each plane
- 1950s: 13 ————— became popular and the airline suffered

Passage 19

Gobekli Tepe: A Remarkable Discovery

A prehistoric monument in Turkey is rewriting the story of how civilisation began

Between roughly 10,000 and 11,000 BC, in a remote corner of Mesopotamia, now in modern-day Turkey, a people who pre-dated the invention of the wheel and lacked even basic technology such as metal tools, built an enormous complex of stone pillars called Gobekli Tepe (Potbelly Hill). To achieve this, they shaped blocks of limestone into pillars at a stone quarry. They then moved each pillar, weighing between 11 and 22 tons, a distance of 100 to 500 metres from the quarry to the Gobekli Tepe site. There, the pillars were arranged in rings, which usually consisted of eight upright pillars. Each pillar comprised two stones placed together in the form of a 'T'. Typically, six of these pillars, connected by low walls, made up the circumference of each ring and two taller pillars were situated in the centre. The tallest pillars reached almost five metres in height, and the largest rings were almost 20 metres in diameter.

To date, nearly 200 pillars have been found by archaeologists at the site. A menagerie of animals, including bulls, foxes, snakes, spiders, donkeys, gazelles and lions had been carved in relief on the stones. There are also strange depictions of human-like forms on the stones. Experts have noted that art from other sites around the world created at roughly the same time typically depicts animals in herds, while at Gobekli Tepe, the images are of individual fiercely-looking beasts.

Although archaeologists had been aware since the 1960s that there were old limestone slabs at Gobekli Tepe, these were for years generally dismissed as nothing more than a jumble of broken stones of fairly recent origin and uncertain purpose, and the site was essentially ignored. Then in 1994, German archaeologist Klaus Schmidt took a closer look and realised that the unusually rounded appearance of the hill was something that 'only man could have created' and archaeologists began studying the site more carefully.

In addition to traditional excavation, which has uncovered a treasure trove of flint tools as well as four rings of pillars, geomagnetic surveys were also conducted. The surveys revealed 16 additional rings buried in the hill that had been left undisturbed. It appears that every now and then, ancient builders filled in a ring, pillars and all, with dirt, gravel, flint and old tools. They then erected a new circle over or near the old, buried one.

There are differing opinions about the purpose of Gobekli Tepe. The stones display prominent depictions of vultures, which are birds commonly associated with the dead, and it is possible that Gobekli Tepe was built as an ancient cemetery. However, some people say that the great age of the stones and the lack of any writing on them make interpreting the builders' motivations virtually impossible.

Of the many discoveries made at the site, perhaps the most intriguing is that the builders do not appear to have become settled before constructing the monuments. The indications are, in fact, that they were nomadic hunter-gatherers. This is remarkable because, up until this discovery, most experts believed that in order to build massive structures like those found at Gobekli Tepe, a society needed to have the organisation and resources that could only come about through people living in a settled agricultural community. The evidence found at Gobekli Tepe, however, appears to seriously challenge this conventional wisdom.

In more than 20 years of digging at the site, archaeologists have found no evidence of a well-established ancient settlement such as houses, hearths, cooking sites, or rubbish dumps. However, they have found 100,000 fragments of the bones of wild game animals like boar, gazelle and deer, which strongly suggests the site was used by hunters. And, unlike the dry landscape found around Gobekli Tepe today, 11,000 years ago the region was a paradise that would have provided ample natural nourishment. It would have been full of fruit and nuts, all readily available to be collected by gatherers without the need for farming.

All of the evidence suggests that these vast monuments at Gobekli Tepe were not built after a settled way of life had been adopted, but that the process of construction itself eventually caused the builders to abandon hunting and gathering. It seems that a large number of the enormous population required to build the complex eventually chose to remain nearby. Only after the monuments were finished did they adopt a more permanent lifestyle. They began cultivating wild grasses in order to feed themselves and in this way started basic agricultural practices. In fact, the original strains of the modern wheat that people farm today have been traced to an ancient village only 32 kilometres from Gobekli Tepe. In addition, there are indications that the first pigs raised in domestic conditions were kept at a village about 100 kilometres away.

Questions 1–8

Do the following statements agree with the information given in Reading Passage 1?

*Write **TRUE** if the statement agrees with the information, **FALSE** if the statement contradicts the information, or **NOT GIVEN** if there is no information on this.*

- ① The builders of Gobekli Tepe used tools made of iron.
- ② Pillars were carved from stone at a quarry and then taken to Gobekli Tepe.
- ③ The arrangement of the pillars in rings is thought to relate to the seasons of the year.
- ④ Specially trained workers carved the animals on the stones.
- ⑤ The stone carvings of creatures at Gobekli Tepe are similar to those found at other sites from the same period.

- 6 Until the 1990s, archaeologists believed that Gobekli Tepe site was of little importance.
- 7 Klaus Schmidt thought that hill at Gobekli Tepe was formed naturally.
- 8 Archaeologists had to get permission before they could start studying the Gobekli Tepe site.

Questions 9–16

Complete the notes below.

Choose **ONE WORD ONLY** from the passage for each answer.

Gobekli Tepe

Purpose of Gobekli Tepe

- the kind of bird shown on many stones suggests the site may have been a 9 _____.
- it is impossible to know why the site was constructed because there's no 10 _____ on the stones.

Discoveries made at the site

- no remains of houses have been found to indicate a permanent 11 _____.
- many pieces of animal 12 _____ have been found indicating an abundance of wild animals.

What we have learned about Gobekli Tepe and the area

- Gobekli Tepe was in a region where 13 _____ as well as fruit were easy to find.
- after the 14 _____ of Gobekli Tepe many people chose to remain in the region.
- an early form of 15 _____ was grown near Gobekli Tepe.
- domesticated 16 _____ were kept at a place nearby.

Passage 20

Going Nowhere Fast

Why the cars that get us around our cities have scarcely changed in over a century - and what might finally replace them

THIS is ludicrous! We can talk to people anywhere in the world or fly to meet them in a few hours, we can even send probes to other planets. But when it comes to getting around our cities, we depend on systems that have scarcely changed since the days of Gottlieb Daimler.

In recent years, the pollution belched out by millions of vehicles has dominated the debate about transport. The problem has even persuaded California - that home of car culture - to curb traffic growth. But no matter how green they become, cars are unlikely to get us around crowded cities any faster. And persuading people to use trains and buses will always be an uphill struggle. Cars, after all, are popular for very good reasons, as anyone with small children or heavy shopping knows.

So politicians should be trying to lure people out of their cars, not forcing them out. There's certainly no shortage of alternatives. Perhaps the most attractive is the concept known as personal rapid transit (PRT), independently invented in the US and Europe in the 1950s.

The idea is to go to one of many stations and hop into a computer-controlled car, which can whisk you to your destination along a network of guideways. You wouldn't have to share your space with strangers, and with no traffic lights, pedestrians or parked cars to slow things down, PRT guideways can carry far more traffic, nonstop, than any inner-city road.

It's a wonderful vision, but the odds are stacked against PRT for a number of reasons. The first cars ran on existing roads, and it was only after they became popular - and after governments started earning revenue from them - that a road network designed specifically for motor vehicles was built. With PRT, the infrastructure would have to come first - and that would cost megabucks. What's more, any transport system that threatened the car's dominance would be up against all those with a stake in maintaining the status quo, from private car owners to manufacturers and oil multinationals. Even if PRTs were spectacularly successful in trials, it might not make much difference. Superior technology doesn't always triumph, as the VHS versus Betamax and Windows versus Apple Mac battles showed.

But 'dual-mode' systems might just succeed where PRT seems doomed to fail. The Danish RUF system envisaged by Palle Jensen, for example, resembles PRT but with one key difference: vehicles have wheels as well as a slot allowing them to travel on a monorail, so

they can drive off the rail onto a normal road. Once on a road, the occupant would take over from the computer, and the RUF vehicle - the term comes from a Danish saying meaning to "go fast" - would become an electric car.

Build a fast network of guideways in a busy city centre and people would have a strong incentive not just to use public RUF vehicles, but also to buy their own dual-mode vehicle. Commuters could drive onto the guideway, sit back and read as they are chauffeured into the city. At work, they would jump out, leaving their vehicles to park themselves. Unlike PRT, such a system could grow organically, as each network would serve a large area around it and people nearby could buy into it. And a dual-mode system might even win the support of car manufacturers, who could easily switch to producing dual-mode vehicles.

Of course, creating a new transport system will not be cheap or easy. But unlike adding a dedicated bus lane here or extending the underground railway there, an innovative system such as Jensen's could transform cities.

And it's not just a matter of saving a few minutes a day. According to the Red Cross, more than 30 million people have died in road accidents in the past century - three times the number killed in the First World War - and the annual death toll is rising. And what's more, the Red Cross believes road accidents will become the third biggest cause of death and disability by 2020, ahead of diseases such as AIDS and tuberculosis. Surely, we can find a better way to get around?

Questions 1–6

Do the following statements agree with the information given in Reading Passage 1?

*Write **TRUE** if the statement agrees with the information, **FALSE** if the statement contradicts the information, or **NOT GIVEN** if there is no information on this.*

- ① City transport developed slower than other means of communication.
- ② The pollution caused by city transport has been largely ignored.
- ③ Most states in America have taken actions to reduce vehicle growth.
- ④ Public transport is particularly difficult to use on steep hills.
- ⑤ Private cars are much more convenient for those who tend to buy a lot of things during shopping.
- ⑥ Government should impose compulsory restrictions on car use.

Questions 7–12

Classify the following descriptions as referring to

A PRT only **B** RUF only **C** both PRT and RUF

Write the correct letter, **A**, **B** or **C**.

Q	Statement	
7	It is likely to be resisted by both individuals and manufacturers.	
8	It can run at high speed in cities.	
9	It is not necessary to share with the general public.	
10	It is always controlled by a computer.	
11	It can run on existing roads.	
12	It can be bought by private buyers.	

Question 13

Choose **THREE** letters, **A–G**.

Which **THREE** of the following are advantages of the new transport system?

OPTIONS

- A** economy
- B** space
- C** low pollution
- D** suitability for families
- E** speed
- F** safety
- G** suitability for children

Answer: 13

Passage 21

How Consumers Decide

Professor John Maule from the University of Leeds describes new research into the way that consumers choose a product

Consumers are creatures of habit: they buy the same products time and time again, and such is their familiarity with big brands and the colours and logos that represent them, that they can register a brand they like with barely any conscious thought process. The packaging of consumer products is, therefore, a crucial vehicle for delivering the brand and the product into our shopping baskets.

Having said this, understanding how consumers make decisions, and the crucial role of packaging in this process has been a neglected area of research so far. This is surprising given that organisations invest huge amounts of money in developing packaging that they believe is effective - especially at the retail level. Our Centre for Decision Research at Leeds University's Business School, in collaboration with Faraday Packaging, is now undertaking work in this area. It has already led to some important findings that challenge the ways in which organisations think about consumer choice.

The research has focused on two fundamental types of thinking. On the one hand, there's 'heuristic processing', which involves very shallow thought and is based on very simple rules: 1) buy what you recognize, 2) choose what you did last time, or 3) choose what a trusted source suggests. This requires comparatively little effort, and involves looking at - and thinking about - only a small amount of the product information and packaging. One can do this with little or no conscious thought.

On the other hand, 'systematic processing' involves much deeper levels of thought. When people choose goods in this way, they engage in quite detailed analytical thinking - taking account of the product information, including its price, its perceived quality and so on. This form of thinking, which is both analytical and conscious, involves much more mental effort.

The role of packaging is likely to be very different for each of these types of decision making. Under heuristic processing, for example, consumers may simply need to be able to distinguish the pack from those of competitors since they are choosing on the basis of what they usually do. Under these circumstances, the simple perceptual features of the pack may be critical - so that we can quickly discriminate what we choose from the other products on offer. Under systematic processing, however, product-related information may be more important, so the pack has to provide this in an easily identifiable form.

Consumers will want to be able to compare the product with its competitors, so that they can determine which option is better for them. A crucial role of packaging in this situation is to communicate the characteristics of the product, highlighting its advantages over possible competitors.

So, when are people likely to use a particular type of thinking? First, we know that people are cognitive misers; in other words, they are economical with their thinking because it requires some effort from them. Essentially, people only engage in effort-demanding systematic processing when the situation justifies it, for example when they are not tired or distracted and when the purchase is important to them.

Second, people have an upper limit to the amount of information they can absorb. If we present too much, therefore, they will become confused. This, in turn, is likely to lead them to disengage and choose something else.

Third, people often lack the knowledge or experience needed, so will not be able to deal with things they do not already understand, such as the ingredients of food products, for example.

And fourth, people vary in the extent to which they enjoy thinking. Our research has differentiated between people with a high need for thinking - who routinely engage in analytical thinking - and those low in the need for cognition, who prefer to use very simple forms of thinking.

This work has an important impact on packaging in that what makes packaging effective is likely to vary according to the type of processing strategy that consumers use when choosing between products. You need to understand how consumers are selecting your products if you are to develop packaging that is relevant. Furthermore, testing the effectiveness of your packaging can be ineffective if the methods you are employing concern one form of thinking (e.g. a focus group involving analytical thinking) but your consumers are purchasing in the other mode (i.e. the heuristic, shallow form of thinking).

For the packaging industry, it is important that retailers identify their key goals. Sustaining a consumer's commitment to a product may involve packaging that is distinctive at the heuristic level (if the consumers can recognize the product, they will buy it) but without encouraging consumers to engage in systematic processing (prompting deeper level thinking that would include making comparisons with other products).

Conversely, getting consumers to change brands may involve developing packaging that includes information that does stimulate systematic processing and thus encourages consumers to challenge their usual choice of product. Our work is investigating these issues, and the implications they have for developing effective packaging.

Questions 1–6

Do the following statements agree with the information given in the passage?

Write **TRUE** if the statement agrees with the information, **FALSE** if the statement contradicts the information, or **NOT GIVEN** if there is no information on this.

- 1 Little research has been done on the link between packaging and consumers choosing a product.
- 2 A person who buys what another person recommends is using heuristic thinking.
- 3 Heuristic processing requires more energy than systematic processing.
- 4 The concept of heuristic processing was thought up by Dr Maule's team.
- 5 A consumer who considers how much a product costs, is using systematic processing.
- 6 For heuristic processing, packaging must be similar to other products.

Questions 7–8

Choose the correct answer, **A**, **B**, **C** or **D**.

- 7 When trying to determine how effective packaging is, testing can be made 'ineffective' if
 - A you rely upon a very narrow focus group.
 - B your consumers use only heuristic thinking.
 - C the chosen consumers use only shallow thinking.
 - D your tests do not match the consumers' thinking type.
- 8 If a retailer wants consumers to change brands their packaging needs to be
 - A informative.
 - B distinctive.
 - C familiar.
 - D colourful.

Questions 9–13

Complete the summary below.

Choose **NO MORE THAN TWO WORDS** from the passage for each answer.

Comparing competition

- For consumers who want to compare products, it is important that your packaging stresses the 9 _____ of your product.

- We know that people only use systematic processing if the ⑩ _____ makes it necessary or desirable.
- We also know that too much ⑪ _____ could make consumers choose another product.
- Furthermore, consumers may not fully understand details such as the ⑫ _____ of a product.
- While some people like using systematic processing, others like to think in a ⑬ _____ way.

Passage 22

How Tennis Rackets Have Changed

From natural gut to co-polyester — the quiet science behind every professional's racket

In 2016, the British professional tennis player Andy Murray was ranked as the world's number one. It was an incredible achievement by any standard – made even more remarkable by the fact that he did this during a period considered to be one of the strongest in the sport's history, competing against the likes of Rafael Nadal, Roger Federer and Novak Djokovic, to name just a few. Yet five years previously, he had been regarded as a talented outsider who entered but never won the major tournaments.

Of the changes that account for this transformation, one was visible and widely publicised: in 2011, Murray invited former number one player Ivan Lendl onto his coaching team – a valuable addition that had a visible impact on the player's playing style. Another change was so subtle as to pass more or less unnoticed. Like many players, Murray has long preferred a racket that consists of two types of string: one for the mains (verticals) and another for the crosses (horizontal). While he continued to use natural string in the crosses, in 2012 he switched to a synthetic string for the mains. A small change, perhaps, but its importance should not be underestimated.

The modification that Murray made is just one of a number of options available to players looking to tweak their rackets in order to improve their games. 'Touring professionals have their rackets customised to their specific needs,' says Colin Triplow, a UK-based professional racket stringer. 'It's a highly important part of performance maximisation.' Consequently, the specific rackets used by the world's elite are not actually readily available to the public; rather, each racket is individually made to suit the player who uses it. Take the US professional tennis players Mike and Bob Bryan, for example: 'We're very particular with our racket specifications,' they say. 'All our rackets are sent from our manufacturer to Tampa, Florida, where our frames go through a ... thorough customisation process.' They explain how they have adjusted not only racket length, but even experimented with different kinds of paint. The rackets they use now weigh more than the average model and also have a denser string pattern (i.e. more crosses and mains).

The primary reason for these modifications is simple: as the line between winning and losing becomes thinner and thinner, even these slight changes become more and more important. As a result, players and their teams are becoming increasingly creative with the modifications to their rackets as they look to maximise their competitive advantage.

Racket modifications mainly date back to the 1970s, when the amateur German tennis player Werner Fischer started playing with the so-called spaghetti-strung racket. It created a string bed that generated so much topspin that it was quickly banned by the

International Tennis Federation. However, within a decade or two, racket modification became a regularity. Today it is, in many ways, an aspect of the game that is equal in significance to nutrition or training.

Modifications can be divided into two categories: those to the string bed and those to the racket frame. The former is far more common than the latter: the choice of the strings and the tension with which they are installed is something that nearly all professional players experiment with. They will continually change it depending on various factors including the court surface, climatic conditions, and game styles. Some will even change it depending on how they feel at the time.

At one time, all tennis rackets were strung with natural gut made from the outer layer of sheep or cow intestines. This all changed in the early 1990s with the development of synthetic strings that were cheaper and more durable. They are made from three materials: nylon (relatively durable and affordable), Kevlar (too stiff to be used alone) or co-polyester (polyester combined with additives that enhance its performance). Even so, many professional players continue to use a 'hybrid set-up', where a combination of both synthetic and natural strings are used.

Of the synthetics, co-polyester is by far the most widely used. It's a perfect fit for the style of tennis now played, where players tend to battle it out from the back of the court rather than coming to the net. Studies indicate that the average spin from a co-polyester string is 25% greater than that from natural string or other synthetics. In a sense, the development of co-polyester strings has revolutionised the game.

However, many players go beyond these basic adjustments to the strings and make changes to the racket frame itself. For example, much of the serving power of US professional player Pete Sampras was attributed to the addition of four to five lead weights onto his rackets, and today many professionals have the weight adjusted during the manufacturing process.

Other changes to the frame involve the handle. Players have individual preferences for the shape of the handle and some will have the handle of one racket moulded onto the frame of a different racket. Other players make different changes. The professional Portuguese player Gonalo Oliveira replaced the original grips of his rackets with something thinner because they had previously felt uncomfortable to hold.

Racket customisation and modification have pushed the standards of the game to greater levels that few could have anticipated in the days of natural strings and heavy, wooden frames, and it's exciting to see what further developments there will be in the future.

Do the following statements agree with the information given in Reading Passage 1?

Write **TRUE** if the statement agrees with the information, **FALSE** if it contradicts it, or **NOT GIVEN** if there is no information on this.

- 1 People had expected Andy Murray to become the world's top tennis player for at least five years before 2016.
- 2 The change that Andy Murray made to his rackets attracted a lot of attention.
- 3 Most of the world's top players take a professional racket stringer on tour with them.
- 4 Mike and Bob Bryan use rackets that are light in comparison to the majority of rackets.
- 5 Werner Fischer played with a spaghetti-strung racket that he designed himself.
- 6 The weather can affect how professional players adjust the strings on their rackets.
- 7 It was believed that the change Pete Sampras made to his rackets contributed to his strong serve.

Questions 8–13

Complete the notes below.

Choose **ONE WORD ONLY** from the passage for each answer.

THE TENNIS RACKET AND HOW IT HAS CHANGED

- Mike and Bob Bryan made changes to the types of 8 ——— used on their racket frames.
- Players were not allowed to use the spaghetti-strung racket because of the amount of 9 ——— it created.
- Changes to rackets can be regarded as being as important as players' diets or the 10 ——— they do.
- All rackets used to have natural strings made from the 11 ——— of animals.
- Pete Sampras had metal 12 ——— put into the frames of his rackets.
- Gonalo Oliveira changed the 13 ——— on his racket handles.

Passage 23

Katherine Mansfield

Katherine Mansfield was a modernist writer of short fiction who was born and brought up in New Zealand

Katherine Mansfield Beauchamp Murry was born in 1888, into a prominent family in Wellington, New Zealand. She became one of New Zealand's best-known writers, using the pen name of Katherine Mansfield. The daughter of a banker, and born into a middle-class family, she was also a first cousin of Countess Elizabeth von Arnim, a distinguished novelist in her time. Mansfield had two older sisters and a younger brother. Her father, Harold Beauchamp, went on to become the chairman of the Bank of New Zealand. In 1893, the Mansfield family moved to Karori, a suburb of Wellington, where Mansfield would spend the happiest years of her childhood; she later used her memories of this time as an inspiration for her Prelude story.

Her first published stories appeared in the High School Reporter and the Wellington Girls' High School magazine in 1898 and 1899. In 1902, she developed strong feelings for a musician who played the cello, Arnold Trowell, although her feelings were not, for the most part, returned. Mansfield herself was an accomplished cellist, having received lessons from Trowell's father. Mansfield wrote in her journals of feeling isolated to some extent in New Zealand, and, in general terms of her interest in the Maori people (New Zealand's native people), who were often portrayed in a sympathetic light in her later stories, such as How Pearl Button Was Kidnapped.

She moved to London in 1903, where she attended Queen's College, along with her two sisters. Mansfield recommenced playing the cello, an occupation that she believed, during her time at Queen's, she would take up professionally. She also began contributing to the college newspaper, with such a dedication to it that she eventually became its editor. She was particularly interested in the works of the French writers of this period and in the 19th-century British writer, Oscar Wilde, and she was appreciated amongst fellow students at Queen's for her lively and charismatic approach to life and work. She met fellow writer Ida Baker, a South African, at the college, and the pair became lifelong friends. Mansfield did not actively support the suffragette movement in the UK. Women in New Zealand had gained the right to vote in 1893.

Mansfield first began journeying into other parts of Europe in the period 1903–1906, mainly to Belgium and Germany. After finishing her schooling in England, she returned to her New Zealand home in 1906, only then beginning to write short stories in a serious way. She had several works published in Australia in a magazine called Native Companion,

which was her first paid writing work, and by this time she had her mind set on becoming a professional writer. It was also the first occasion on which she used the pseudonym “K. Mansfield”.

Mansfield rapidly grew discontented with the provincial New Zealand lifestyle, and with her family. Two years later she headed back again to London. Her father sent her an annual subsidy of £100 for the rest of her life. In later years, she would express both admiration and disdain for New Zealand in her journals.

In 1911, Mansfield met John Middleton Murry, the Oxford scholar and editor of the literary magazine *Rhythm*. They were later to marry in 1918. Mansfield became a co-editor of *Rhythm*, which was subsequently called *The Blue Review*, in which more of her works were published. She and Murry lived in various houses in England and briefly in Paris. *The Blue Review* failed to gain enough readers and was no longer published. Their attempt to set up as writers in Paris was cut short by Murry's bankruptcy, which resulted from the failure of this and other journals. Life back in England meant frequently changed addresses and very limited funds.

Between 1915 and 1918, Mansfield moved between England and Bandol, France. She and Murry developed close contact with other well-known writers of the time such as DH Lawrence, Bertrand Russell and Aldous Huxley. By October 1918 Mansfield had become seriously ill; she had been diagnosed with tuberculosis and was advised to enter a sanatorium. She could no longer spend winters in London. In the autumn of 1918 she was so ill that she decided to go to Ospedaletti in Italy. It was the publication of *Bliss and Other Stories* in 1920 that was to solidify Mansfield's reputation as a writer.

Mansfield also spent time in Menton, France, as the tenant of her father's cousin at “The Villa Isola Bella”. There she wrote eight stories including *Miss Brill* and *The Daughter of the Late Colonel*, the latter of which she pronounced to be “...the only story that satisfies me to any extent”.

Mansfield produced a great deal of work in the final years of her life, and much of her prose and poetry remained unpublished at her death in 1923. After her death, her husband, Murry, took on the task of editing and publishing her works. His efforts resulted in two additional volumes of short stories, *The Doves' Nest* and *Something Childish*, published in 1923 and 1924 respectively, the publication of her *Poems* as well as a collection of critical writings (*Novels and Novelists*) and a number of editions of Mansfield's previously unpublished letters and journals.

Questions 1–6

Do the following statements agree with the information given in the passage?

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Write **TRUE** if the statement agrees with the information, **FALSE** if the statement contradicts the information, or **NOT GIVEN** if there is no information on this.

- ① The name Katherine Mansfield, that appears on the writer's book, was exactly the same as her origin name.
- ② Mansfield won a prize for a story she wrote for the High School Reporter.
- ③ How Pearl Button Was Kidnapped portrayed Maori people in a favorable way.
- ④ When Mansfield was at Queen's College, she planned to be a professional writer.
- ⑤ Mansfield was unpopular with the other students at Queen's College.
- ⑥ In London, Mansfield showed little interest in politics.

Questions 7–13

Complete the notes below.

Choose **ONE WORD AND/OR A NUMBER** from the passage for each answer.

Katherine Mansfield's adult years

- ⑦ _____
 - moved from England back to New Zealand
 - first paid writing work was in a publication based in ⑧ _____
 - her ⑨ _____ and the New Zealand way of life made her feel dissatisfied
- 1908
 - returned to London
- 1911–1919
 - Met John Middleton Murry in 1911
 - ⑩ _____ prevented Mansfield and Murry from staying together in Paris
 - spent time with distinguished ⑪ _____
 - from 1916, tuberculosis restricted the time she spent in London
- 1920

- her ⑫ ————— was consolidated when Bliss and Other Stories was published
- wrote several stories at “Villa Isola Bella”
- 1923–1924
- Mansfield's ⑬ ————— published more of her works after her death

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Passage 24

Living in a Noisy World

How noise pollution is affecting hearing, health and quality of life around the globe

People living in a typical urban environment experience a wide range of sounds on a daily basis. In its Guidelines for Community Noise, the World Health Organisation (WHO) declared, 'Worldwide, noise-induced hearing impairment is the most prevalent irreversible occupational hazard, and it is estimated that 120 million people worldwide now have disabling hearing difficulties.'

The growing noise pollution problem has many different sources. Booming population growth and the loss of rural land to urban sprawl both play a role. Other factors include the inability of authorities in many parts of the world to implement noise-reducing legislation; the electronic nature of our age, which encourages many noisy gadgets; the rising number of vehicles on streets; and busier airports.

Sound intensity is measured in decibels (dB); the unit A-weighted dB (dBA) is used to indicate how humans hear a given sound. Zero dBA is considered the point at which a person begins to hear sound. A soft whisper at three feet equals 30 dBA, a busy freeway at 50 feet is around 80 dBA, and a chainsaw can reach 110 dBA.

Mark Stephenson, a senior research audiologist at the National Institute for Occupational Safety and Health (NIOSH), says his agency's definition of hazardous noise is sound that exceeds 85 dBA, meaning the average noise exposure measured over a typical eight-hour work day.

In the United States, about 30 million workers are exposed to hazardous sound levels on the job, according to NIOSH. Noise in US industry is an extremely difficult problem to monitor, acknowledges Craig Moulton, an industrial hygienist for the Occupational Safety and Health Administration (OSHA). Still, he says, 'OSHA does require that any employer with staff overexposed to noise safeguard those employees against the harmful impact of noise.'

'For many people in the United States, noise has drastically affected the quality of their lives,' says Arline Bronzaft, chair of the Noise Committee of the New York City Council of the Environment, and a psychologist who has done pioneering research on the effects of noise on children's reading ability. 'My daughter lives near La Guardia airport in New York City, and she can't open a window or enjoy her backyard in the summer because of the airplane noise.'

The United States is not the only country where noise pollution is affecting the quality of life. In Japan, noise pollution caused by public loudspeaker announcements and other forms of city noise has forced many Tokyo citizens to wear earplugs as they go about their daily lives. In Europe, about 65% of the population is exposed to ambient sound at levels above 55 dBA, while about 17% is exposed to levels above 65 dBA, according to the European Environment Agency.

Numerous scientific studies over the years have confirmed that exposure to certain levels of sound can damage hearing. NIOSH studies from the mid to late 1990s show that 90% of coal miners have hearing impairment by age 52 – compared to 9% of the general population. NIOSH research also reveals that by age twenty-five, the average carpenter's hearing is equivalent to that of a fifty-year-old male who hasn't been exposed to noise.

In 2001 researchers from the Center for Disease Control and Prevention reported that, based on audiometric testing of 5,249 children, an estimated 12.5% of American children have noise-induced hearing threshold shifts – or dulled hearing – in one or both ears. Most children with dulled hearing have only limited hearing damage, but continued exposure to excessive noise can lead to difficulties with high-frequency sound discrimination.

The effects of sound don't stop with the ears. The non-auditory effects of noise were noted as far back as 1930 in a study published by E L Smith and D L Laird in the Journal of the Acoustical Society of America. The results showed that exposure to noise caused stomach problems in healthy human beings.

Noise has also been shown to affect learning ability. In 1975 Bronzaft collaborated on a study of children in a school near a train track that showed how exposure to noise affects reading ability. Half of the students in the study were in classrooms facing the train track and the other half were in classrooms in the school's quieter section. The findings were that students on the quieter side performed better on reading tests.

Bronzaft and the school principal persuaded the school board to have acoustical tiles installed in the classrooms adjacent to the tracks. The Transit Authority also undertook work on the track near the school in order to reduce noise levels. A follow-up study in 1981 found that children's reading scores improved after these interventions were put in place.

Anti-noise activists say that Europe and several countries in Asia are more advanced than the United States in terms of combating noise. 'Population pressure has prompted Europe to move more quickly on the noise issue,' says Ken Hume, a principal lecturer in human physiology at the Manchester Metropolitan University in England. European cities with at least 250,000 people are developing noise maps of those cities to help leaders determine

noise pollution strategies. Paris has already prepared its first noise maps. The map data will be fed into computer models that will help test the sound impact of new road layouts or buildings before construction begins.

Bronzaft stresses that governments worldwide need to direct more financial resources towards studies investigating the effects of noise, and do a better job coordinating their noise pollution efforts, so they can establish health and environmental policies based on solid scientific research.

Questions 1–4

Complete the sentences below.

*Choose **NO MORE THAN TWO WORDS AND/OR A NUMBER** from the passage for each answer.*

- Experts consider noise levels to be harmful if they go above ① _____.
- Children with ‘dulled hearing’ may eventually have trouble distinguishing ② _____ sounds.
- Early research into physical effects other than hearing problems revealed that adults may suffer from ③ _____ if exposed to noise.
- Some European cities are creating ④ _____ in order to help governments develop noise policies.

Questions 5–10

Look at the following statements and the list of people and organisations below.

*Match each statement with the correct person or organisation, **A–E**. You may use any letter more than once.*

LIST OF PEOPLE AND ORGANISATIONS

- A** World Health Organisation (WHO)
- B** National Institute for Occupational Safety and Health (NIOSH)
- C** Craig Moulton
- D** Arline Bronzaft
- E** Center for Disease Control and Prevention

Q	Statement	
5	A significant proportion of American children have suffered some hearing loss as a result of exposure to noise.	
6	Noise in our environment can affect the kind of lifestyle we have.	
7	American companies are expected to protect workers from the effects of excessive noise.	
8	Authorities should increase funding for noise pollution research.	
9	Permanent hearing loss caused by noise is the most commonly occurring danger in the global workplace today.	
10	Workers in certain occupations in the US have high rates of hearing damage.	

Questions 11–13

Choose the correct letter, **A, B, C or D**.

- 11 As a result of the findings from Bronzaft's train track study
- A the school moved the classrooms located near the track.
 - B the track was treated to make it less noisy.
 - C the classrooms facing the track were no longer used.
 - D the amount of traffic on the track was reduced.
- 12 According to the passage, European countries are examining how noise level will be affected by
- A population distribution.
 - B the size of cities.
 - C street design.
 - D the purpose of tower blocks.
- 13 Which of the following is the most suitable title for Reading Passage 1?
- A Effects of noise on workers' job performance
 - B The cities most effective at reducing noise
 - C Educating workers about noise
 - D Living in a noisy world

Passage 25

Manhattan's Skyscrapers Race to the Top

The rise of New York's skyline, 1890–1960s

During the late 19th and 20th centuries, New York City and the skyline of Manhattan, the most famous of the city's five boroughs, or districts, were radically transformed. New technology, stronger building materials, and abundant labor allowed the towers of Manhattan to rise – eventually becoming the tallest in the world.

The race to the sky began in 1854, when inventor Elisha Otis demonstrated his elevator safety brake to curious crowds in New York. No matter how many times a man cut through cables holding the elevator platform high above gasping onlookers, Otis would descend just a few inches before safely stopping. Along with Otis's elevator, other innovations laid the groundwork for the future. In 1855 Henry Bessemer patented his process for producing a new kind of strong and low-cost steel. Plate glass, produced in volume from the 1850s, and electric lighting in the 1880s also contributed to the rise of the skyscraper.

Technology alone did not create the Manhattan skyline. Other essential factors included an increasing population, soaring property values, rivalry, geology, vanity, and greed. In 1871, when Chicago developers built the first so-called 'skyscraper', New York entrepreneurs responded – but they had a geological advantage. While Chicago was built on muddy soil, Manhattan stood on solid granite. And because Manhattan was surrounded by water, there was little room for ambitious new buildings to grow unless they looked upward.

The first tall buildings to rise in New York City appeared in the late 1800s. Completed in 1890, the 94-meter New York World Building was the tallest in the world until the Manhattan Life Insurance Building surpassed it four years later, setting the fast pace and ambitious character of future Manhattan skyscrapers.

These buildings were striking achievements, but perhaps the most stunning was the Woolworth Building, completed in 1912. At 222 meters, the Woolworth became the world's tallest building. Upon its completion, this architectural wonder was soon as much a symbol of New York City as the famous Statue of Liberty. It promised freedom and very big dreams.

As parts of New York began to resemble man-made mountain ranges, concerns over the nature of skyscrapers grew. From the sidewalks, streets felt increasingly like canyons. To counteract this, from 1916, city authorities issued zoning laws that stipulated that tall buildings must become narrower at higher levels – causing Manhattan's skyscrapers to resemble multi-layered wedding cakes.

In 1928, construction began on one of New York's most iconic structures: the Chrysler Building. Walter Chrysler, a motor-industry leader, incorporated several unique features into his building's design. Much of the building's distinctive ornamentation resembled features of the automobile from which Chrysler made his fortune. Upon completion, the Chrysler building was the world's tallest, at 319 meters high.

The Chrysler Building's time at the top would be short. Just 11 months after the Chrysler's completion in May 1930, the 102-story Empire State Building became the world's tallest building. It was opened by the U.S. president to great acclaim. At 381 meters, the tower matched the ambition of the New York financier John J. Raskob, who commissioned the legendary skyscraper.

Unfortunately, the American economy crashed as the Empire State Building rose at record speed. Not even the building's famous appearance in the 1933 movie King Kong could boost its fortunes. Unable to find tenants, Raskob's magnificent structure was mockingly called the 'Empty State Building.' It took World War II to fill the building with government agencies. Even then, a B-25 bomber almost brought it down, crashing – accidentally – between the 78th and 80th floors in dense fog on July 28, 1945. Five years later, the Empire State Building finally turned a profit. The building itself was an architectural marvel, remaining the world's tallest skyscraper for 40 years.

In the years following World War II, architecture in New York transformed again. Instead of decorative designs, buildings started to favor a modern look with a sleek design and also gleaming surfaces. Completed in 1952, the United Nations Building was constructed in this style. Designed by the Swiss architect Le Corbusier, the slim tower was like nothing the city had seen before. The trend toward modern lines continued into the 1950s. By the time the Seagram Building was completed in 1958, rectilinear office buildings – with their straight up-and-down walls – were to become typical. With its unique bronze exterior, it was not the tallest building in New York, but it became a model for designing other office towers built from the 1960s onward.

Le Corbusier is often cited as the villain behind this move towards less elegant architecture, yet when he visited New York in 1935 and witnessed the physical impact of skyscrapers for the first time, he said: 'The skyscrapers of New York are romantic, a gesture of pride... but the street has been killed...'. Today his observation hardly seems true: Manhattan's skyscrapers failed to kill the street. In fact, New York City remains one of the world's great walking cities, its avenues and streets lined by towering giants born of a golden age of opportunity and ambition.

Questions 1–8

Do the following statements agree with the information given in Reading Passage 1?

Write **TRUE** if the statement agrees with the information, **FALSE** if the statement contradicts the information, or **NOT GIVEN** if there is no information on this.

- 1 Construction of tall buildings in New York City started in the twentieth century.
- 2 An important force in the development of Manhattan's skyscrapers was the increasing price of land.
- 3 Chicago was a better place to build skyscrapers than New York.
- 4 The New York World Building took four years to complete.
- 5 In 1912, the Woolworth Building was visited as often as the Statue of Liberty.
- 6 In its early years, the Empire State Building was a commercial failure.
- 7 The Empire State Building was the first skyscraper to be featured in a film.
- 8 Le Corbusier is frequently considered responsible for changing New York's architecture.

Questions 9–16

Complete the table below.

Choose **ONE WORD ONLY** from the passage for each answer.

Manhattan's Skyscrapers Race to the Top

1854

- Elisha Otis introduced New Yorkers to a new invention – a 9 ————— for elevators.

1855

- Henry Bessemer created a way of making 10 ————— that was sturdy and inexpensive.

1916

- New 11 ————— required that the upper floors of buildings were smaller in area than the lower floors.

1928

- Some of the decorations on the Chrysler Building looked like 12 ————— parts.

1945

- An aeroplane lost in (13) _____ hit the Empire State Building.

After 1945

- New buildings featured straight lines as well as shiny (14) _____.

1958

- The shape of the Seagram Building was a (15) _____ for other office blocks in New York.

Today

- Despite its tall buildings, New York is still an ideal place for (16) _____.

Passage 26

Moving Continents

From a 1596 hunch to the theory of plate tectonics

A The belief that continents have not always been fixed in their present positions was first suggested as early as 1596 by the Dutch map maker Abraham Ortelius. Ortelius suggested that the Americas were torn away from Europe and Africa by earthquakes and floods, and went on to say that the vestiges of the rupture reveal themselves if someone brings forward a map of the world and considers carefully the coasts of the three continents. However, it was not until 1912 that the idea of moving continents was seriously considered as a full-blown scientific theory, when the theory of continental drift was introduced in two articles by a 32-year-old German meteorologist named Alfred Lothar Wegener. He contended that all the present continents used to form one supercontinent, which he called Pangaea, and that this began to split apart around 200 million years ago. The parts drifted across the Earth, eventually breaking into the various smaller continents that exist today.

B Wegener's theory was based in part on what appeared to him to be the remarkable fit of the South American and African continents, first noted by Ortelius three centuries earlier. Wegener was also intrigued by the occurrences of unusual geologic structures and of fossils — of both plants and animals — found on the matching coastlines of South America and Africa, which are now widely separated by the Atlantic Ocean. He reasoned that it was physically impossible for most of these creatures to have swum or have been transported across the vast ocean. To him, the presence of identical fossils along the coastal parts of Africa and South America was the most compelling evidence that the two continents were once joined.

C In Wegener's mind, the drifting of continents after the break-up of Pangaea explained not only the matching fossil occurrences but also the evidence of dramatic climate change on some continents. For example, indications that there were once glaciers in regions of South Africa led to the conclusion that this land must once have been situated much further away from the tropics. Other mismatches of geology and present-day climate included the occurrence of coal — the product of ancient temperate or tropical forests — in Spitzbergen, an island that is now close to the North Pole.

D At the time Wegener introduced his theory, the scientific community firmly believed the continents and oceans to be permanent features on the Earth's surface. Not surprisingly, his proposal was not well received, even though it seemed to agree with the scientific information available at the time. A fatal weakness in Wegener's theory was that it could not satisfactorily answer the most fundamental question raised by his critics: what kind of forces could be strong enough to move such large masses of solid rock over such great distances?

E After Wegener's death in 1930, new evidence from ocean floor exploration and other studies rekindled interest in his theory, ultimately leading, in the early 1960s, to the development of the theory of plate tectonics. This scientific concept has revolutionised our understanding of the dynamic planet upon which we live. The theory states that the Earth's outermost layer is fragmented into a dozen or more plates of various sizes that are moving relative to one another as they ride on top of hotter, more mobile material.

F The theory has provided explanations to questions that scientists had speculated upon for centuries — such as why earthquakes and volcanic eruptions occur in very specific areas around the world, and how and why mountain ranges like the Alps and Himalayas formed.

Questions 1–6

Answer the questions below.

*Choose **NO MORE THAN THREE WORDS** from the passage for each answer.*

In Ortelius's opinion, what caused continents to be separated?

1 _____

In the theory of continental drift, what was given the name 'Pangaea'?

2 _____

Apart from shape and geological structures, what similar phenomena are found in South America and Africa?

3 _____

What do indications of glaciers in South Africa and coal in Spitzbergen provide evidence of?

4 _____

According to the 1960s theory, what does the top layer of the Earth consist of?

5 _____

What natural phenomena may have been created by movement on the Earth's surface?

6 _____

Questions 7–12

The reading passage has six paragraphs, A–F.

Which paragraph contains the following information?

NB *You may use any letter more than once.*

Q	Statement	
7	how later developments revived consideration of the theory of continental drift	
8	evidence that some areas used to be situated in much warmer or cooler seasons	
9	something that Wegener's theory could not explain	
10	a reason why living organisms can't have moved from one continent to another	
11	an apparent contradiction in the reaction to the theory of continental drift	
12	an outline of a theory that explains how continental drift occurs	

Passage 27

Museum Blockbuster

Do blockbuster exhibitions really pay off for museums, galleries and science centres?

A Since the 1980s, the term 'blockbuster' has become the fashionable word for special spectacular museum, art gallery or science centre exhibitions. These exhibitions have the ability to attract large crowds and often large corporate sponsors. Here is one of some existing definitions of blockbuster: put by Elsen (1984), a blockbuster is a '... large scale loan exhibition that people who normally don't go to museums will stand in line for hours to see ...' James Rosenfield, writing in *Direct Marketing* in 1993, has described a successful blockbuster exhibition as a '... triumph of both curatorial and marketing skills ...' My own definition for blockbuster is 'a popular, high profile exhibition on display for a limited period, that attracts the general public, who are prepared to both stand in line and pay a fee in order to partake in the exhibition.' What both Elsen and Rosenfield omit in their descriptions of blockbuster, is that people are prepared to pay a fee to see a blockbuster, and that the term blockbuster can just as easily apply to a movie or a museum exhibition.

B Merely naming an exhibition or movie a blockbuster however, does not make it a blockbuster. The term can only apply when the item in question has had an overwhelmingly successful response from the public. However, in literature from both the UK and USA the other words that also start to appear in descriptions of blockbuster are 'less scholarly', 'non-elitist' and 'populist'. Detractors argue that blockbusters are designed to appeal to the lowest common denominator, while others extol the virtues of encouraging scholars to cooperate on projects, and to provide exhibitions that cater for a broad selection of the community rather than an elite sector.

C Maintaining and increasing visitor levels is paramount in the new museology. This requires continued product development. Not only the creation or hiring of blockbuster exhibitions, but regular exhibition changes and innovations. In addition, the visiting publics have become customers rather than visitors, and the skills that are valued in museums, science centres and galleries to keep the new customers coming through the door have changed. High on the list of requirements are commercial, business, marketing and entrepreneurial skills. Curators are now administrators. Being a director of an art gallery no longer requires an Art Degree. As succinctly summarised in the *Economist* in 1994 'business nous and public relation skills' were essential requirements for a director, and the ability to compete with other museums to stage travelling exhibitions which draw huge crowds.

D The new museology has resulted in the convergence of museums, the heritage industry, and tourism, profit-making and pleasure-giving. This has given rise to much debate about the appropriateness of adapting the activities of institutions so that they more closely

reflect the priorities of the market place and whether it is appropriate to see museums primarily as tourist attractions. At many institutions you can now hold office functions in the display areas, or have dinner with the dinosaurs. Whatever commentators may think, managers of museums, art galleries and science centres worldwide are looking for artful ways to blend culture and commerce, and blockbuster exhibitions are at the top of the list. But while blockbusters are all part of the new museology, there is proof that you don't need a museum, science centre or art gallery to benefit from the drawing power of a blockbuster or to stage a blockbuster.

E But do blockbusters held in public institutions really create a surplus to fund other activities? If the bottom line is profit, then according to the accounting records of many major museums and galleries, blockbusters do make money. For some museums overseas, it may be the money that they need to update parts of their collections or to repair buildings that are in need of attention. For others in Australia, it may be the opportunity to illustrate that they are attempting to pay their way, by recovering part of their operating costs, or funding other operating activities with off-budget revenue. This makes the economic rationalists cheerful. However, not all exhibitions that are hailed to be blockbusters will be blockbusters, and some will not make money. It is also unlikely that the accounting systems of most institutions will recognise the real cost of either creating or hiring a blockbuster.

F Blockbusters require large capital expenditure, and draw on resources across all branches of an organisation; however, the costs don't end there. There is a Human Resource Management cost in addition to a measurable 'real' dollar cost. Receiving a touring exhibition involves large expenditure as well, and draws resources from across functional management structures in project management style. Everyone from a general labourer to a building servicing unit, the front of house, technical, promotion, education and administration staff, are required to perform additional tasks. Furthermore, as an increasing number of institutions in Australia try their hand at increasing visitor numbers, memberships (and therefore revenue), by staging blockbuster exhibitions, it may be less likely that blockbusters will continue to provide a surplus to subsidise other activities due to the competitive nature of the market. There are only so many consumer dollars to go around, and visitors will need to choose between blockbuster products.

G Unfortunately, when the bottom-line is the most important objective to the mounting of blockbuster exhibitions, this same objective can be hard to maintain. Creating, mounting or hiring blockbusters is exhausting for staff, with the real costs throughout an institution difficult to calculate. Although the direct aims may be financial, creating or hiring a blockbuster has many positive spin-offs; by raising their profile through a popular blockbuster exhibition, a museum will be seen in a more favorable light at budget time. Blockbusters mean crowds, and crowds are good for the local economy, providing

increased employment for shops, hotels, restaurants, the transport industry and retailers. Blockbusters expose staff to the vagaries and pressures of the market place, and may lead to creative excellence. Either the success or failure of a blockbuster may highlight the need for managers and policy makers to rethink their strategies. However, the new museology and the apparent trend towards blockbusters make it likely that museums, art galleries and particularly science centres will be seen as part of the entertainment and tourism industry, rather than as cultural icons deserving of government and philanthropic support.

H Perhaps the best pathway to take is one that balances both blockbusters and regular exhibitions. However, this easy middle ground may only work if you have enough space, and have alternate sources of funding to continue to support the regular less exciting fare. Perhaps the advice should be to make sure that your regular activities and exhibitions are more enticing, and find out what your local community wants from you.

Questions 1–4

The reading passage has eight paragraphs, **A–H**.

Which paragraph contains the following information?

NB You may use any letter more than once.

Q	Statement	
1	a reason for changing the exhibition programs	
2	the time people have to wait in a queue in order to enjoy exhibitions	
3	terms people used when referring to blockbuster	
4	there was some controversy over confining target groups of blockbuster	

Questions 5–8

Complete the following summary of the paragraphs of Reading Passage.

Using **NO MORE THAN THREE WORDS** from the Reading Passage for each answer.

MUSEUMS AND THE NEW MUSEOLOGY

- Instead of being visitors, people turned out to be **5** —————, who require the creation or hiring of blockbuster exhibitions as well as regular exhibition changes and innovations.
- Business nous and **6** —————, succinctly summarised in a magazine, are not only important factors for directors, but also an ability to attract a crowd of audiences.

- 7 _____ has contributed to the linking of museums, the heritage industry, tourism, profit-making and pleasure-giving.
- There occurs some controversy over whether it is proper to consider museums mainly as 8 _____.

Questions 9–10

The list below gives some advantages of blockbusters.

Which **TWO** advantages are mentioned by the writer of the text?

Choose **TWO** correct answers, **A–E**.

OPTIONS

- A** To offer sufficient money to repair architectures.
- B** To maintain and increase visitor levels.
- C** Presenting the mixture in the culture and commerce of art galleries and science centres worldwide.
- D** Being beneficial for the development of local business.
- E** Being beneficial for the directors.

Answer: 9 10

Questions 11–13

The list below gives some disadvantages of blockbusters.

Which **THREE** disadvantages are mentioned by the writer of the text?

Choose **THREE** correct answers, **A–F**.

OPTIONS

- A** People left hesitated to choose exhibitions.
- B** Workers has become tired of workloads.
- C** The content has become more entertaining rather than cultural.
- D** General labourers are required to perform additional tasks
- E** Huge amounts of capital invested in specialists.
- F** Exposing staff to the fantasies and pressures of the market place.

Answer: 11 12 13

Passage 28

New Zealand's Early Crafts and Traditions

How Polynesian settlers adapted their craftsmanship to a new land

The first groups of people to discover New Zealand came from Polynesia. Exactly when these explorers arrived has often been a matter of debate, but today the general understanding is that it was during the 13th century that their canoes eventually landed on New Zealand's shores. In some ways the new country must have seemed like an ideal place to settle: the land was fertile, and thick forests provided firewood, shelter and building materials. Still, life would have been challenging for the different Polynesian tribes, who had to adapt to a new environment. The tribes only began to refer to themselves as Maori, meaning 'ordinary people', when Europeans in search of new opportunities began arriving in the 18th century. To the Maori, of course, the European settlers and sailors were not 'ordinary', but very strange.

It was not only a knowledge of canoe-building and navigation that the Polynesians brought to New Zealand. They were also skilled craftsmen. There is archaeological evidence that the tools they produced were of high quality and would have enabled tribes to plant and harvest crops. Craftsmen were also occupied with making weapons such as knives and axes, which were used for both construction and fighting. Interestingly, some crafts that had once been popular in Polynesian islands were no longer done in New Zealand, although researchers are unsure why. Pottery is an example of this, despite the fact that the clay needed to make pots and bowls could easily be found in the new country.

The Maori word whakairo can be translated as 'decorative work' — this can refer to bone, wood and greenstone carving. Although Maori carvers were influenced by their Polynesian heritage, they developed their own style, including the curved patterns and spirals inspired by New Zealand plants. The same term can also apply to weaving; the crafting of, for example, woven baskets and mats all required knowledge and skill. Carving greenstone, or pounamu as it is called in Maori, was a long process, requiring great patience. Further, because of this mineral's rarity, any greenstone object, such as a piece of jewellery or cutting blade, was a prized possession. For that reason, it was the few people of high status rather than low-ranking members of a tribe who would possess such objects.

As New Zealand had no native mammals except for bats, dolphins and whales, Maori largely had to depend on plants to provide material for their clothing, including their cloaks. Weavers experimented with the inner bark of the houhere, the lacebark tree, but found it unsuitable. But the dried-out leaves and fibres of the flax plant provided a solution. Once a cloak had been woven from flax, it could be decorated. Borders might be dyed black or red, for example. In the case of superior ones made for chiefs or the more important members of a tribe, feathers from kiwi, pigeons or other native birds might be

attached. All flax cloaks were rectangular in shape, so had no sleeves, and neither was a hood a feature of this garment. Short cloaks were fastened around a person's neck, and came only to the waist.

Pins made of bone, wood or greenstone allowed longer cloaks to be secured at the shoulder; these were a type that were often used for ceremonial occasions. Of course, the construction of the cloaks was influenced by the plant material available to Maori weavers. This meant that cloaks were loose-fitting, and while they protected wearers from New Zealand's strong sunshine, they were not useful during the winter months. A cloak made from fur or wool could provide insulation from the cold, but not so a cloak made of flax.

The warriors of a tribe required a different kind of cloak to help protect them. To create these special cloaks, the tough fibres of the mountain cabbage tree were used instead. It is not clear to researchers what the entire process involved, but they believe the fibres were left to soak in water over a period of time in order to soften them and make them easier to weave together. Later, once the whole cloak had been constructed, it would be dyed black. To do this, Maori weavers covered it in a special kind of mud they had collected from riverbeds. This was rich in iron due to New Zealand's volcanic landscape. The particular advantage of these cloaks was that the tough cabbage tree fibres they were woven from could reduce the impact of spear tips during a fight with enemy tribes. It is fortunate that some cloaks from the 1800s still survive and can provide us with further insight into the materials and construction techniques that Maori craftsmen used.

Questions 1–6

Do the following statements agree with the information given in Reading Passage 1?

*Write **TRUE** if the statement agrees with the information, **FALSE** if the statement contradicts the information, or **NOT GIVEN** if there is no information on this.*

- 1 It is now widely thought that humans reached New Zealand in the 13th century.
- 2 The first Europeans to come to New Zealand were keen to trade with Maori.
- 3 Members of Maori tribes were responsible for either tool- or weapon-making.
- 4 A craft that the Maori once practised in New Zealand was making pottery.
- 5 Weaving baskets and mats was seen as a form of decorative work.
- 6 It used to be common for everyone in a Maori tribe to wear greenstone jewellery.

Questions 7–13

Complete the notes below.

Choose **NO MORE THAN TWO WORDS** from the passage for each answer.

Māori cloaks

flax cloaks

warrior cloaks

methods of
construction

Māori made flax cloaks by:

- weaving leaves and fibres
- sometimes adding

⑦ _____ to the better
cloaks

appearance

Flax cloaks were:

- rectangular in shape
- designed without a

⑧ _____

- tied at either the wearer's neck or

their ⑨ _____

good/bad points

Flax cloaks offered no

⑩ _____ during winter

Weavers had to use ⑪ _____
to make cabbage tree fibres less stiff

Mud containing ⑫ _____ was
used to make the cloaks look black

⑬ _____ could not easily go
through the cloaks' tough fibres

Passage 29

Obesity: Do Citizens of Developed Countries Really Have a Choice?

According to researchers, we may not have much control over what and how much we eat

According to researchers, we may not have much control over what and how much we eat. Numerous recent studies show that Americans are fatter than ever before. According to the OECD*, US citizens are more likely to be obese than those in any other nation - though other highly developed countries are following the trend. The figures raise the question of what cultural shift has occurred to make Americans eat so enthusiastically.

Marion Nestle, a nutritionist, argues that the desire to reach for the next hamburger may be more commercially driven than people like to admit. Nestle attributes part of the blame for ever-expanding waists to the food industry - from fertiliser manufacturers to farmers to restaurants - which needs to promote the consumption of ever greater quantities to remain competitive, increase sales, and keep shareholders happy. Even after accounting for imports and exports, Nestle writes, America produces far more food than its citizens need. The US food industry provides a daily average of 3,800 calories per person, twice an individual's requirement.

With such an oversupply on hand, the food industry has resorted to whatever tools it can to sell its products to the American public. Soft-drink companies use their economic power to get vending machines into schools, chocolate-bar makers flood us with advertising, and industry lobby groups help write nutritional guidelines - all of which get Americans eating and drinking more.

The tactics work. According to the US Department of Agriculture, each American now consumes an average of 86kg of red meat, poultry, and fish each year, up 5.5kg from 20 years ago. More surprising still, the average American uses 68kg of sweeteners such as refined sugar and corn syrup each year, up from 5.5kg in 1910. According to the Center for Disease Control and Prevention (CDCP) in Atlanta, Georgia, that results in more than 30 per cent of men and women in America being classified as obese, more than twice as many as 40 years ago. That gain in weight has cut across all age groups, income strata, and racial, professional and regional divisions, though some groups are affected more than others. The overweight can now be found everywhere.

The current politically correct acceptance of obesity overlooks one important fact. According to Professor Gortmaker, cardiovascular disease and hypertension are just two of the conditions that researchers have connected with obesity, and people need to be aware of the health risk. And it is not a marginal problem. The total cost in direct expenses, such as medical bills, and indirect expenses, such as lost wages due to obesity-related problems,

is close to \$100 billion a year, according to the US government's National Institutes of Health (NIH). An additional \$33 billion is spent on weight-loss products and services. But Americans continue to eat more, and do little or nothing about it. According to the NIH, only 15 per cent of Americans get the recommended amount of vigorous activity each week, and a quarter reported no physical activity at all outside of work.

Gortmaker classifies obesity as a 'chronic health condition' similar to asthma. However, he says, asthma sufferers usually seek professional medical help, while those suffering from obesity rarely do unless they are experiencing serious problems.

The condition is defined in terms of Body Mass Index (BMI), a measure of body fat that compares height to weight. A BMI of 20-25 is considered normal, 25-30 overweight and above 30 obese.

Some economists accept obesity and its risks as the price of progress. Darius Lakdawalla locates the roots of American weight gain in technological change. He argues that agricultural technology has brought down the cost of food, and because it is less costly, Americans eat more. 'It's becoming easier to consume lots of calories with very small amounts of money and very small amounts of time,' says Lakdawalla.

At the same time, Philipson points out that technological advances in the work place have made Americans' lives less strenuous, causing people to burn fewer calories than they once did. He has plotted the rise in America's weight using data stretching back to the American Civil War. He found that the average BMI rose steadily and then accelerated after the Second World War. A 35-year-old male had a BMI of just over 23 in 1864. It edged up to 23 and three quarters by 1944, and then shot up to 26 by 1991. The obesity epidemic can be traced to the way the US economy gets children to consume more and also to be more sedentary. If we sit children in front of the television, and we feed them at fast food outlets, and we provide sweetened drink machines at school, do children have a choice?

Geoffrey Sobel, a sociologist, says that marriage can have the same effect on women. 'When couples marry, they consume a high proportion of their calories together,' says Sobel. 'The new household diet is typically the man's diet. Features of the feminine diet like salad and yoghurt are reduced.'

Where exercise was once the natural by-product of work and life, it is now a luxury that must be paid for either in money or time or both. In poor countries, body weight generally increases with wealth. But in developed economies, the very rich are often actually thinner because they can afford to exercise. Is the upward trend in body mass a side-effect of economic development - the price we pay for enjoying technology?

Look at the following findings (Questions 1–8) and the list of researchers/research organizations below.

Match each statement with the correct researcher/research organization, **A–I**.

Q	Statement	
1	Business competition drives food consumption.	
2	Cheaper food production has led to increased food consumption.	
3	Improved technology makes work less physically demanding.	
4	Obesity is a serious illness.	
5	Many Americans are physically active only during their working hours.	
6	Obese people often consider medical treatment to be unnecessary.	
7	Obesity is more prevalent in the US than anywhere else.	
8	The number of obese Americans has doubled in 40 years.	

List of People

A Organization for Economic Cooperation and Development

B Marion Nestle

C US Department of Agriculture

D Center for Disease Control and Prevention

E Gortmaker

F National Institutes of Health

G Darius Lakdawalla

H Phillipson

I Geoffrey Sobel

Questions 9–13

Do the following statements agree with the claims of the writer in Reading Passage 1?

Write **YES** if the statement agrees with the claims of the writer, **NO** if the statement contradicts the claims of the writer, or **NOT GIVEN** if it is impossible to say what the writer thinks about this.

- 9 The US produces a large food surplus.
- 10 Large companies can impose their marketing on schools.
- 11 Since 1970, the use of sweeteners by the average American has decreased slightly.
- 12 Direct and indirect medical expenses resulting from obesity are the same.
- 13 These days, exercising is time-consuming and costs money.

Passage 30

Stonehenge

Unravelling the mysteries of England's most iconic prehistoric monument

For centuries, historians and archaeologists have puzzled over the many mysteries of Stonehenge, a prehistoric monument that took an estimated 1,500 years to erect. Located on Salisbury Plain in southern England, it is comprised of roughly 100 massive upright stones placed in a circular layout.

Archaeologists believe England's most iconic prehistoric ruin was built in several stages, with the earliest constructed 5,000 or more years ago. First, Neolithic Britons used primitive tools, which may have been fashioned out of deer antlers, to dig a massive circular ditch and bank, or henge. Deep pits dating back to that era and located within the circle may have once held a ring of timber posts, according to some scholars.

Several hundred years later, it is thought, Stonehenge's builders hoisted an estimated 80 bluestones, 43 of which remain today, into standing positions and placed them in either a horseshoe or circular formation. These stones have been traced all the way to the Preseli Hills in Wales, some 300 kilometers from Stonehenge. How, then, did prehistoric builders without sophisticated tools or engineering haul these boulders, which weigh up to four tons, over such a great distance?

According to one long-standing theory among archaeologists, Stonehenge's builders fashioned sledges and rollers out of tree trunks to lug the bluestones from the Preseli Hills. They then transferred the boulders onto rafts and floated them first along the Welsh coast and then up the River Avon toward Salisbury Plain; alternatively, they may have towed each stone with a fleet of vessels. More recent archaeological hypotheses have them transporting the bluestones with supersized wicker baskets on a combination of ball bearings and long grooved planks, hauled by oxen.

As early as the 1970s, geologists have been adding their voices to the debate over how Stonehenge came into being. Challenging the classic image of industrious builders pushing, carting, rolling or hauling giant stones from faraway Wales, some have suggested that it was glaciers, not humans, that carried the bluestones to Salisbury Plain. Most archaeologists have remained skeptical about this theory, however, wondering how the forces of nature could possibly have delivered the exact complete the circle.

The third phase of construction took place around 2000 BCE. At this point, sandstone slabs – known as 'sarsens' – were arranged into an outer crescent or ring; some were assembled into the iconic three-pieced structures called trilithons that stand tall in the centre of Stonehenge. Some of 50 stones are now visible on the site, which may once have

contained many more. Radiocarbon dating has revealed that work continued at Stonehenge until roughly 1600 BCE, with the bluestones in particular being repositioned multiple times.

But who were the builders of Stonehenge? In the 17th century, archaeologist John Aubrey made the claim that Stonehenge was the work of druids, who had important religious, judicial and political roles in Celtic society. This theory was widely popularized by the antiquarian William Stukeley, who had unearthed primitive graves at the site. Even today, people who identify as modern druids continue to gather at Stonehenge for the summer solstice. However, in the mid-twentieth century, radiocarbon dating demonstrated that Stonehenge stood more than 1,000 years before the Celts inhabited the region.

Many modern historians and archaeologists now agree that several distinct tribes of people contributed to Stonehenge, each undertaking a different phase of its construction. Bones, tools and other artefacts found on the site seem to support this hypothesis. The first stage was achieved by Neolithic agrarians who were likely to have been indigenous to the British Isles. Later, it is believed, groups with advanced tools and a more communal way of life left their mark on the site. Some believe that they were immigrants from the European continent, while others maintain that they were probably native Britons, descended from the original builders.

If the facts surrounding the architects and construction of Stonehenge remain shadowy at best, the purpose of the striking monument is even more of a mystery. While there is consensus among the majority of modern scholars that Stonehenge once served the function of burial ground, they have yet to determine what other purposes it had.

In the 1960s, the astronomer Gerald Hawkins suggested that the cluster of megalithic stones operated as a form of calendar, with different points corresponding to astrological phenomena such as solstices, equinoxes and eclipses occurring at different times of the year. While his theory has received a considerable amount of attention over the decades, critics maintain that Stonehenge's builders probably lacked the knowledge necessary to predict such events or that England's dense cloud cover would have obscured their view of the skies.

More recently, signs of illness and injury in the human remains unearthed at Stonehenge led a group of British archaeologists to speculate that it was considered a place of healing, perhaps because bluestones were thought to have curative powers.

Questions 1–8

Complete the notes below.

Choose **NO MORE THAN TWO WORDS** from the passage for each answer.

Compiled by @bekhruzposts

Stonehenge

Construction Stage 1

- the ditch and henge were dug, possibly using tools made from ① _____
- ② _____ may have been arranged in deep pits inside the circle.

Stage 2

bluestones from the Preseli Hills were placed in standing position — theories about the transportation of the bluestones:

- archaeological* — builders used ③ _____ to make sledges and rollers;
④ _____ pulled them on giant baskets.
- geological* — they were brought from Wales by ⑤ _____.

Stage 3

sandstone slabs were arranged into an outer crescent or ring

Builders

- theory arose in the 17th century that its builders were Celtic ⑥ _____.

Purpose

- many experts agree it has been used as a ⑦ _____ site; in the 1960s, it was suggested that it worked as a kind of ⑧ _____.

Questions 9–16

Do the following statements agree with the information given in Reading Passage 1?

Write **TRUE** if the statement agrees with the information, **FALSE** if the statement contradicts the information, or **NOT GIVEN** if there is no information on this.

- ⑨ The ditch and bank at Stonehenge were already in place when the bluestones arrived.
- ⑩ During the third phase of construction, sandstone slabs were placed in both the outer areas and the middle of the Stonehenge site.

- ⑪ There is scientific proof that the bluestones stood in the same spot until approximately 1600 BCE.
- ⑫ John Aubrey's claim about Stonehenge was supported by twentieth-century findings.
- ⑬ Objects discovered at Stonehenge seem to indicate that it was constructed by a number of different groups of people.
- ⑭ It is accepted that the builders who used advanced tools to work on Stonehenge came from the British Isles.
- ⑮ Criticisms of Gerald Hawkins' theory about Stonehenge has come mainly from other astronomers.
- ⑯ The theory that people came to Stonehenge to cure their illnesses and injuries has been accepted by archaeologists worldwide.

Passage 31

Sydney Opera House

The story behind one of the world's most recognisable buildings

A Sydney Opera House is an example of late modern architecture; it is admired internationally and treasured by the people of Australia.

B In 1956 the Premier of New South Wales, Australia, announced an international competition for the design of an opera house for Sydney. It attracted more than 200 entries from around the world and was won by Jørn Utzon, a relatively little-known architect from Denmark. The story goes that during the judging of the competition, one judge, American architect Eero Saarinen, arrived in Sydney after the other three judges had started assessing the entries. He looked through their rejected entries and stopped at the Utzon design, declaring it to be outstanding.

C It was Utzon's life and travels that had shaped his design for the Sydney Opera House. Though he had never visited the site, he used his maritime background to study naval charts of Sydney Harbour. His early exposure to shipbuilding provided the inspiration for the design of the roof of the Sydney Opera House, which is a series of curved 'shells' that look like the sails of a sailing ship billowing on the wind. From his travels to Mexico, he had the idea of placing his building on a wide horizontal platform.

D Construction of the platform began in 1959, and throughout the early 1960s Utzon amended his original designs in order to develop a way to build the large 'shells' that cover the two main halls. The construction of the roof brought together some of the world's best engineers and craftsmen, devising innovative techniques to create a major visual impact in accordance with Utzon's vision. The design was one of the first examples of the use of computer-aided design for complex shapes.

E Although Utzon had spectacular plans for the interior, he was unable to realise them. Cost overruns contributed to criticism of the project and, after a change of government, the Minister of Works began questioning Utzon's schedules and cost estimates. Payments to Utzon were stopped and he was forced to withdraw as chief architect in 1966. Following his resignation, there were protests through the streets led by prominent architect Harry Seidler and others, demanding Utzon be reinstated as architect. However, Utzon was not reinstated and left Australia in 1966. He never returned and new architects were appointed to complete the building in his absence. The original cost estimate for the Opera House was \$7 million, with the completion date set at 26 January 1963. However, the Opera House was not formally completed until 1973, having cost \$102 million.

F Since its opening in 1973, Sydney Opera House has earned a reputation as a world-class performing arts centre, and become a symbol of both Sydney and Australia. Situated at Bennelong Point on Sydney Harbour, it consists of a series of large precast 'shells' made of concrete, each composed of sections of a sphere of 75.2 metres radius, forming the roofs of the structure, set on a monumental platform. The building is 183 metres long and 120 metres wide at its widest point. It is supported on 588 concrete piers which are sunk approximately 25 metres below sea level.

G Although the roof structures are commonly referred to as 'shells', they are precast concrete panels supported by concrete ribs. The 'shells' are covered with 1,056,000 white and cream-coloured tiles manufactured in a factory in Sweden that generally produced stoneware tiles for the paper-mill industry. The design solution and construction of the shell structure took eight years to complete, and the development of the special ceramic tiles took over three years. Apart from the tiles covering the 'shells', the building's exterior is mostly clad with granite quarried in Australia.

H Contrary to its name, Sydney Opera House includes multiple performance venues. It is among the busiest performing arts centres in the world, holding over 1,500 performances each year. It hosts a large number of performing arts companies, including the four resident companies: Opera Australia; the Australian Ballet; the Sydney Theatre Company and the Sydney Symphony Orchestra.

I With its grand setting and cathedral-like atmosphere, the Concert Hall is Sydney Opera House's most prestigious performance space. The largest of all Sydney Opera House interior venues, it delivers outstanding acoustics thanks to its high ceiling and wood panelling. There is a sizeable outdoor forecourt from which people ascend to the main entrance. The steps, which lead up from the forecourt to the main performance venues, are nearly 100 metres wide.

J In 1999, Utzon was re-engaged to develop a set of design principles to act as a guide for future changes to the building. All of this design work he did from his base in Europe. These principles help to ensure that the building's architectural integrity is maintained. The first alteration to the exterior of the building was the addition of a new colonnade, which shades nine large glass openings into the previously solid exterior wall. This Utzon-led project, completed in 2006, enabled theatre patrons to see the harbour for the first time from the theatre foyers. The design also incorporates the first public lift and interior escalators to assist less mobile patrons.

K Since 2007, the cultural, heritage and architectural importance of Sydney Opera House has been protected by its inclusion on the World Heritage List.

Questions 1–7

Do the following statements agree with the information given in the passage?

Write **TRUE** if the statement agrees with the information, **FALSE** if the statement contradicts the information, or **NOT GIVEN** if there is no information on this.

- 1 Utzon was famous for his work before he designed the Opera House.
- 2 Utzon's design was favoured by the four judges of the competition from the beginning.
- 3 Utzon's knowledge of boats gave him the idea for parts of the Opera House.
- 4 Utzon was impressed by the opera houses he had seen in Mexico.
- 5 Utzon changed his designs in the 1960s after construction began.
- 6 Seidler defended Utzon's role as architect.
- 7 Utzon went back to Australia in 1973 for the opening of the Opera House.

Questions 8–13

Complete the notes below.

Choose **ONE WORD AND/OR A NUMBER** from the passage for each answer.

Sydney Opera House

Final cost

- – \$ 8 _____

Construction

- – a large platform acting as a base for the building
- – concrete panels used to make 'shells', which are covered in tiles
- – over a million tiles from 9 _____
- – 10 _____ from Australia covering the outside walls

Use

- – more than 1,500 performances annually
- – 11 _____ performing arts companies have their home base at the Opera House

Outside

- – a large ⑫ _____ at the foot of a wide staircase

Alterations

- – a colonnade was added in 2006
- – openings made the ⑬ _____ visible from foyers

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Passage 32

Temperature Rising

The impact of global warming on the tourism industry

The tourism industry depends to a large extent on good weather. The past few years have brought intense hurricanes, typhoons, torrential rains, heat waves and drought. According to the World Meteorological Organization, 2004 was the fourth hottest year on record. And while global warming, defined as human-driven climate change traced to the burning of fossil fuels such as oil, coal and natural gas, cannot be blamed for single-handedly wreaking such havoc, the scientific consensus is that it may intensify some weather events. For example, meteorologists attribute the back-to-back hurricanes in 2004 to a 50- to 70-year natural cycle of increased hurricane activity. But hurricanes tend to become even more formidable on a warmer globe. Bill MacCracken, a scientist at the Climate Institute in Washington DC, likens the present climate to a pot of slowly boiling water on a stove: if you turn up the flame, you get more bubbles, but you can't attribute one particular bubble to the increase in heat. He says, 'We're slowly going to be getting more extreme weather, but we can't say that any particular type of weather is due to the changing climate.'

Tourism companies worldwide are starting to feel the financial effects of weather change. In Scotland higher winter temperatures have led several ski areas to diversify into summer leisure pursuits such as golf, and paragliding in the winter. A similar scenario is taking place in the European Alps. Robert Mills, of the United States Tourism Association, reports that in the upper Midwest of the United States, cooler summers and warmer winters have resulted in resort owners having to restrict outdoor sports such as swimming in lakes and ice fishing. And in the west of the United States, intense forest fires in recent years have limited public access to national and state parks. For the tourism industry, which generated US\$578 billion in 2002, or seven per cent of the world's economy, the costs of extreme weather are high. Last year was the most expensive on record for the insurance side of the industry. 'Weather and tourism go hand in hand,' says Tim Warren, an adventure business consultant based in California. 'If the weather's not co-operating, the impacts on profits are very real.'

One of the most impacted areas is the diving industry, whose very survival depends on the health of coral reefs. The Global Coral Reef Monitoring Network reports that 20 per cent of the world's reefs have been effectively destroyed by factors such as pollution, overfishing, diseases, and bleaching - the last directly related to rise in water temperatures that essentially heats coral until it sheds the algae that feed it, leaving it colourless. Even Australia's Great Barrier Reef, with its 400 species of coral spanning 1,240 miles, hasn't been spared. Scientists estimate that bleaching has affected significant areas of the Great

Barrier Reef. 'We hear tourists say, "this doesn't look like the postcard or as it did three years ago",' says John Rumney, manager of Undersea Explorer, a dive and research vessel that operates out of Queensland in Australia.

The ski industry has also felt the effects of climate change. A 2003 report by Brendan Stark, of the Global Environment Programme (GEP), predicted that downhill skiing could disappear altogether at low-altitude ski resorts such as Kitzbuhel in Austria. Snowfall in the New England region of the United States has decreased by 15 per cent since 1953 and, perhaps more significantly, there are seven fewer days per annum with snow on the ground here than there were fifty years ago. While there is no concrete evidence that the warming in the New England is due to the emission of greenhouse gases, many scientists in the United States believe it is consistent with global warming. The ski industry has made great efforts to downplay the impact of climate change. However, less snow and more variable temperatures have had disastrous consequences for many smaller ski operators, who can only open for business when snow is on the ground. According to Michael Berry, president of the Denver-based National Ski Areas Association, 'the reason why some small ski resorts aren't around anymore is because they decided they could not afford to install snow-making systems.' The best snowmaking in the world won't save the industry if current trends continue.

The Intergovernmental Panel on Climate Change, made up of some 2,000 scientists, estimates a rise of between two- and 10.4-degrees Fahrenheit by 2100. As a result, the GEP forecasts that the ski and tourism industry in Switzerland, for example, could eventually lose up to US\$1.6 billion annually. The outlook for other travel destinations is not much brighter. Scientists feel polar bears will die out within the next hundred years or so. James Wilson, of the Planet Travel Club, believes that in just 20 to 30 years, if people have not already had the opportunity to see the snows of Mt Kilimanjaro in Tanzania or visit the low-lying Maldivian Islands of the Indian Ocean, it will be too late because they will have gone too. Kyoto's climate treaty, signed by more than 100 countries, required industrialised nations to reduce greenhouse gases by at least five per cent. It is a step towards slowing the trend, but clearly much larger strides are necessary if the travel industry hopes to become less affected by the impact of global warming.

Questions 1–4

Do the following statements agree with the information given in Reading Passage 1?

*Write **TRUE** if the statement agrees with the information, **FALSE** if the statement contradicts the information, or **NOT GIVEN** if there is no information on this.*

- 1 In Europe, some ski resorts now offer summer activities during the winter months.
- 2 In parts of the United States, fires have prevented tourists from visiting some park areas.

- 3 The financial consequences of global warming on the tourism industry in 2002 was US\$578 billion.
- 4 Global warming has caused the closure of some Australian ski resorts.

Questions 5–8

Complete the sentences below.

Choose **NO MORE THAN TWO WORDS AND/OR A NUMBER** from the passage for each answer.

- The 5 _____ is the area of tourism which is suffering most as a result of coral reef damage.
- Warmer water ultimately changes the appearance of coral to the point where it becomes 6 _____.
- International scientists predict that global temperatures will have increased by at least 7 _____ degrees Fahrenheit by the next century.
- Experts believe that this century will see the extinction of 8 _____.

Questions 9–13

Look at the following statements (Questions 9–13) and the list of people below.

Match each statement with the correct researcher, **A–G**. **NB** You may use any letter more than once.

List of People

- A** Bill MacCracken
B Robert Mills
C Tim Warren
D John Rumney
E Brendan Stark
F Michael Berry
G James Wilson

Q	Statement	
9	The technology required to extend the skiing season is too expensive for some businesses.	

10	It is difficult to know exactly which weather conditions are caused by global warming.	
11	Tourists have observed changes in the environment over time.	
12	The change in the climate will mean that some locations near sea level will no longer exist.	
13	Temperature changes have forced holiday complexes to cut down on certain types of activities.	

Passage 33

The Ascension Island Experiment

How Charles Darwin's bold idea turned a barren volcanic rock into a self-sustaining cloud forest

The tiny tropical island of Ascension in the South Atlantic Ocean has an area of only 98 square kilometres. It is located 1,600 km from the coast of Africa and 2,250 km from South America and is certainly difficult for an ordinary commercial traveler to reach even today. Ascension sits on top of what scientists call the mid-Atlantic ridge. This is a chain of underwater volcanoes which are still in the process of forming as the ocean floor is slowly pulled apart by massive geological forces. A million years ago, molten magma from beneath the ocean floor burst explosively above the waves and thus, a new island was born which was to become Ascension. Because the island occupies a 'hot spot' on the ridge, its volcano is especially active to this day.

The island was discovered by a Portuguese seafarer, João da Nova Castella, in 1501, but was given the name Ascension Island two years later by a second Portuguese explorer, Alfonso de Albuquerque. At that time, Ascension was little more than a barren volcanic rock, largely treeless, with no indigenous people at all and no animals to speak of. Its coastal regions, however, supported a few small, underdeveloped plants, while the central peaks were essentially bald.

Because of its location, by the early 19th century, Ascension had become a vital strategic base for the British military. It was a thriving waystation providing much needed provisions for British ships.

However, the big problem that impeded further expansion of this outpost was the supply of fresh water. Ascension was extremely arid and experienced strong trade winds from southern Africa. Due to the lack of trees, the little rain that did fall swiftly disappeared in the hot environment.

But Charles Darwin, the British naturalist who is widely regarded as the father of the theory of evolution, devised a bold plan for the barren island. In 1836, the young Darwin was finishing a five-year scientific expedition around the world, reaching strange places like Ascension, where no other naturalist had gone before. Darwin was aboard the British Naval vessel HMS Beagle when it called in at Ascension. Darwin later admitted that his expectations of Ascension were low, and indeed, it cannot have appeared to be a very promising place. But arriving on Ascension, Darwin became excited by what he considered to be its potential. Everywhere, he encountered bright red volcanic cones and rugged black lava, evidence of the violent geological forces that must have formed the island. Although it remained quite barren, there was evidence of human presence, including a few sheep,

goats, cows and horses that had been introduced by the Navy. Being the inventive scientist that he was, Darwin was intrigued by what he saw and began to plan how this place of wild desolation could be transformed.

Only a few years after Darwin's return to England, he laid out his imaginative scheme for Ascension to his good friend, Joseph Hooker, also a respected botanist, who ultimately implemented the plan. In 1839 Hooker set off on his own four-year adventure by sea, an ambitious journey around the icy polar region of Antarctica aboard HMS Erebus and HMS Terror. Mirroring Darwin's journey, the ships called in at Ascension on their way home in 1843.

In 1847 Hooker advised the Royal Navy to set in motion Darwin's elaborate plan. With the help of British scientists, shipments of trees were to be sent to Ascension. These would provide shade, helping to reduce evaporation after rain. And the ultimate results of the introduction of trees would be even more important because the retention of water would create fertile soil, encouraging the growth of even more vegetation of all kinds.

So, beginning in 1850 and continuing year after year, ships started to arrive at Ascension, each depositing trees selected from special gardens maintained by scientists in Europe, South Africa and South America. Soon, on the highest peak of 859m, great changes became apparent. By the late 1870s, a variety of trees including fast-growing eucalyptus, Norfolk Island pine, bamboo and banana trees had all taken root and begun to thrive.

Today, Green Mountain — as the highest peak is now known — is sometimes described as a 'cloud forest'. The trees capture sea mist, creating moisture in the centre of an essentially dry island. This forest is a fully functioning, but in fact totally artificial ecosystem. When naturally occurring, ecosystems develop over millions of years through the process of evolution. By contrast, the Green Mountain cloud forest was created in a matter of decades. In effect, what Darwin, Hooker and the Navy achieved in an effort to make Ascension Island more habitable was the world's first experiment in creating a self-sustaining and self-reproducing forest.

Could the Ascension experiment have more far-reaching consequences? Dr Dave Wilkinson, an ecologist at Liverpool John Moores University in the UK, who has extensively studied Ascension and its history, first visited there in 2003. 'I remember thinking, this is really weird,' he said. 'There were all kinds of plants that don't belong together in nature, growing side by side. I only later found out about Darwin, Hooker and everything that had happened,' he later recalled. Wilkinson has suggested that the principles upon which the Ascension experiment was based could one day be used to transform the planet Mars to make it sustainable for colonies from Earth.

Questions 1–8

Do the following statements agree with the information given in Reading Passage 1?

Write **TRUE** if the statement agrees with the information, **FALSE** if the statement contradicts the information, or **NOT GIVEN** if there is no information on this.

- 1 People can visit Ascension easily today.
- 2 The mid-Atlantic ridge continues to experience volcanic activity.
- 3 João da Nova Castella was the first to call his discovery Ascension Island.
- 4 When the island was discovered, its center contained more life than the area near the sea.
- 5 The Portuguese military wanted to use the island as a base.
- 6 In the 19th century, Ascension Island acted as an important port for naval supplies.
- 7 The British imported fresh water to Ascension Island from southern Africa.
- 8 When it rained on Ascension Island, the water quickly dried up.

Questions 9–16

Complete the notes below.

Choose **ONE WORD ONLY** from the passage for each answer.

The Ascension Island experiment

Darwin's plan

- before his 1836 visit, Darwin had limited 9 ————— of what he might find
- after his arrival, Darwin saw the 10 ————— of the island and so devised a plan

Joseph Hooker's implementation

- on an expedition to explore the area around 11 —————, Hooker visited Ascension
- trees were planted on Ascension to lessen the 12 ————— of rainwater
- from the 1850s, plants were brought to Ascension from 13 ————— around the world

Results of the plan today

- the cloud forest traps moisture (which is) contained in the ⑭ _____ which blows off the ocean
- the cloud forest is a successful man-made ⑮ _____
- According to Dr Dave Wilkinson, the ideas learned from Ascension may be applicable to life in future human ⑯ _____ on Mars

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Passage 34

A review of Nigel Townson's *The British at Play*

A critical look at a social history of sport in Britain, from football hooliganism to the business of sport

An estimated three million Britons take part in some sort of sporting activity every week. Globally, around four billion people - over half the world's entire population - watch at least part of major events like the Olympic Games. Sport is big business. In fact, it is the UK's 11th largest industry, employing over 400,000 people.

But these figures don't get to the heart of the social power and significance of sport in the modern world. It is a powerful social force in Britain, as in many other cultures. Friends and colleagues regularly discuss sport, and it is one of relatively few topics that are acceptable when initiating social interaction with strangers. Expressions from sport have passed into general use: we talk about 'team players' in situations that have nothing to do with sport, and the word 'goal', meaning an objective, probably evolved from its meaning in sport.

Why is sport so important in society? *The British at Play - a Social History of British Sport from 1600 to the Present*, by Nigel Townson, sets out to answer that question by examining the connections between sport and social class, gender, violence, commercialism, race and even our sense of national identity.

The British at Play explains these complex issues simply and straightforwardly. For example, it highlights the way in which sport contributes to the creation of 'in groups', most notably the supporters of particular football teams. Such informal associations define themselves by their loyalty to their own group and opposition to others, the 'out groups', and in an extreme form, this opposition leads to the phenomenon of football hooliganism. The author handles the issue well, showing what is wrong with the well-known stereotypes of soccer hooligans. He argues that media coverage of fan behaviour helps to create a climate in which hooliganism occurs. And when trouble does break out, the media sensationalises and exaggerates it, with the result that an atmosphere of panic builds up in the country.

Several of the topics relate to social changes in Britain in recent decades. Women are entering fields of activity which would have been closed to them just a generation ago - as football commentators, producers of sports programmes for radio and television, editors of sports magazines. This greater visibility of women highlights the weakening of the traditional view that sport is mainly for men.

The worldwide health-and-fitness boom has to some extent been driven by our growing wish to have a 'perfect' body shape. And that desire has been encouraged, if not created, by the emphasis in sport on images of 'ideal' male and female bodies.

Sport fits in well with the global TV world of beautiful and perfectly muscled young people, exercising or playing sports, dressed in the latest fashionable sports gear. Sport images in the media do not depend on the written word, just on strong images, reaching out directly to the emotions of the viewer - the perfect medium for advertisers.

So sport becomes big money, attracting more and more commercial interest and investment. Some international TV companies depend on the popularity of sport for their survival. Football clubs turn themselves into businesses, raising money by selling merchandise and by selling their shares on the stock market.

The British at Play is an excellent work, of great value and interest to a wide range of audiences, but if I have one criticism of it, it is this: despite its title, the book is not about play. It does not ask the most basic question of all - why do people do sport? Why is it so popular? The book did not, for me, go far enough in transmitting the power, the energy, the passion, the emotion and the joy of sport. The social power of sport ultimately rests on this psychological and physical appeal - the way it involves the whole person, the way in which it allows us to play.

Questions 1–8

Complete the summary below.

Choose **NO MORE THAN TWO WORDS AND/OR A NUMBER** from the passage for each answer.

Summary

- Sport is of major importance around the world, with around ① _____ participants in the UK.
- It has great social significance. Unlike many other subjects, it is sometimes used to start conversations between ② _____, and it is the source of a number of everyday ③ _____.
- The writer examines the way in which groups are formed whose members show ④ _____ to each other.
- Their ⑤ _____ to 'out groups' can lead to violence.

- The writer rejects the common (6) _____ that are used to describe football hooligans.
- Instead, he focuses on (7) _____ in the media, and shows how this can create a public sense of (8) _____.

Questions 9–13

Complete each sentence with the correct ending, A–G, below. Write the correct letter in each box.

- | | | |
|----|---|------------|
| 9 | The change in personnel in sports activities | (9) _____ |
| 10 | Growing interest in health and fitness | (10) _____ |
| 11 | Advertising related to sport | (11) _____ |
| 12 | The greater role of business in sport | (12) _____ |
| 13 | The only weakness of <i>The British at Play</i> | (13) _____ |

List of Endings

- A is unlikely to continue in the same form.
- B lies in how the appeal of sport is explained.
- C shows that sport is no longer seen as a mainly masculine activity.
- D makes considerable use of pictures of sportsmen and women.
- E has resulted in large salaries for players.
- F is partly caused by a focus on the physical characteristics of sportsmen and women.
- G has led to changes in the activities of some football clubs.

Passage 35

The Burgess Shale Fossils

Fauna vanished with a whimper, not a bang — a century-old fossil bed keeps overturning what we thought we knew

Some discoveries are so unusual it takes decades and sometimes even centuries to understand their full significance. One such discovery is the fossil bed known as the Burgess Shale, formed over 500 million years ago. It was discovered in the Canadian Rockies over a century ago and was popularised in 1989 in a book, *Wonderful Life*, by Stephen Jay Gould, an American palaeontologist.

The Burgess Shale fossils were created at a time when the future Canadian land mass was situated near the Earth's equator. The creatures were preserved when an entire marine ecosystem was buried in mud that eventually hardened and became exposed hundreds of millions of years later in an outcrop of the Rocky Mountains. American palaeontologist Charles Walcott, following reports of fabulous fossils finds by construction workers on Canadian railways who were digging in the mountains in the late 19th century, is said to have tripped over a block of shale in 1909 that revealed the area's remarkable supply of specimens. It has long been believed that the curious fauna that lived there vanished in a series of extinctions because the fossil record ends abruptly, but that no longer appears to be the case.

The Burgess Shale began to form soon after a period of time known as the Cambrian explosion, when most major groups of complex animals arose over a surprisingly short period. Before 560m years ago, most living things were either individual cells or simple colonies of cells. Then, and for reasons that remain a mystery, life massively diversified and became ever more complex as the rate of evolution increased. An unusual feature of the Burgess Shale is that it is one of the earliest fossil beds to contain impressions of the soft body parts alongside the remains of bones and shells, which is highly unusual.

Although the fossil bed was discovered on a mountain, these animals originally existed below an ocean, the bed of which was later pushed up to create the Rockies. Nobody knows exactly why they were so well preserved. One possibility is that the creatures were buried quickly and in conditions that were hostile to the bacteria that cause decomposition of soft body parts.

Those that first worked on the Burgess Shale, unearthing 65,000 specimens over a 14-year period up to 1924, assumed that the fossils came from extinct members of groups of animals in existence today. This turned out to be misleading because many of the creatures are so unusual that they are still difficult to classify.

One such example is Opabinia, a creature that grew to about 8cm (3 inches), had five eyes, a body that was a series of lobes, a tail in the shape of a fan and that ate using a long proboscis. The proboscis had a set of grasping claws on the end, with which it grabbed food and stuffed it into its mouth. Nectocaris, meanwhile, could be mistaken for a leech, with fins and tentacles. Weirdest of all was Hallucigenia, described by palaeontologist Simon Conway Morris when he re-examined Walcott's specimens in 1979. With its multiplicity of spines and tentacles, little about Hallucigenia made sense. Like an abstract painting, its orientation is a mystery at first, making it difficult to work out which way up it went, which hole food went into, and which hole food came out of.

Palaeontologists had long thought that many of the Burgess Shale animals were examples of experiments in evolution. In other words, entirely new forms of life that did not survive or lead to other groups or species. Hallucigenia, ironically, turned out to be the exception that proved the rule. It is now thought to be an ancestor of the modern group of arthropods, which includes everything from flies and butterflies to centipedes and crabs.

Now another misconception has been quashed. Writing in Nature, Peter Van Roy of Yale University and his colleagues suggest that the sudden absence of such crazy soft-bodied fossils does not indicate a mass extinction, but merely an end to the unusual local circumstances that caused the creatures to be preserved. In an area of the Atlas Mountains of Morocco, Van Roy's team of researchers have found another diverse (and sometimes bizarre) assemblage of soft-bodied organisms from a period after the Burgess Shale was formed. One discovery includes something that may be a stalked barnacle. This suggests that the evolution of such complex life went on uninterrupted. For its part, the Burgess Shale continues to produce an astonishing array of indefinable creatures faster than palaeontologists can examine them. The world still has plenty to learn about this wonderful life.

Questions 1–5

Do the following statements agree with the information given in the passage?

*Write **TRUE** if the statement agrees with the information, **FALSE** if the statement contradicts the information, or **NOT GIVEN** if there is no information on this.*

- ① The Burgess Shale became widely known to the public because of Gould's book.
- ② Charles Walcott had to get permission from Canadian authorities to gain access to the fossil site.
- ③ The Burgess Shale includes impressions of soft and hard body parts.
- ④ The Burgess Shale creatures were land animals.
- ⑤ Researchers now believe that Hallucigenia is unrelated to any modern creature.

Questions 6–9

Complete the notes below.

Choose **NO MORE THAN TWO WORDS AND/OR A NUMBER** from the passage for each answer.

Formation

- Burgess Shale was formed following a time called the ⑥ _____.

Discovery and Investigation in the twentieth century

- discovered in 1909
- Charles Walcott learned of the fossil finds from people building a ⑦ _____.
- the first work on Burgess Shale was undertaken at the start of the century
- a researcher looked at Burgess Shale findings again in ⑧ _____.

Recent theories

- Peter Van Roy
- believes that discoveries in Morocco show that the ⑨ _____ of complex life forms continued.

Questions 10–13

Complete the table below.

Choose **ONE WORD ONLY** from the passage for each answer.

Burgess Shale Creatures

Opabinia

- five eyes
- tail resembling a ⑩ _____
- claws used to hold ⑪ _____

Nectocaris

- looked like a 12 _____
- fins
- tentacles

Hallucigenia

- spines used to 13 _____
- tentacles

Passage 36

The Clipper Races

An era of competition between cargo ships, as merchants raced to bring the year's first tea harvest home

During the seventeenth and eighteenth centuries, the British East India Company had the monopoly on trade with China and India. This meant that because no rival could legally import tea or other goods from these countries at this time, the company was rarely in a hurry to transport its merchandise. Instead, its priority was to minimise costs by carrying as much as possible on each ship. This meant that its ships – known as East Indiamen – were enormous, strong and very slow.

By 1800, the average East Indiaman could carry 1,200 tons of merchandise. The trading pattern for China tea usually meant the East Indiamen set sail from Britain in January, sailed round the Cape of Good Hope at the southernmost tip of Africa, and arrived in China in September. There they would load up that year's tea harvest, set off again and, depending on the wind and weather, aim to arrive back by the following September, so even with favourable sailing conditions, the round trip lasted almost two years, and if anything went wrong it could take a lot longer.

However, by 1834 the company had lost its trading monopoly, and tea had become a freely traded item. Having no more use for its great ships, the company sold them off, and many were bought by merchants or their captains, who continued to plough the seas between Britain and China. But now that tea could be traded freely, a few smart sailors began to realise that whoever brought each new harvest of tea to Britain first, stood to make the most money.

This was partly because if you were home first, you could sell your shipment of tea before your competitors even arrived, and partly because consumers in Britain in the nineteenth century believed that the fresher and earlier-picked the tea, the better the resulting drink. Tea traders now needed faster, sleeker ships to bring their precious cargo back. Nevertheless, in Britain this idea only caught on slowly, and while the 1840s saw a few faster ships launched, for the time being many merchants remained satisfied with the slow but reliable East Indiamen.

In fact, it was the Americans who pioneered the first clipper ships. These vessels were fast and slender, with a narrow hull that was deeper at the back than at the front, and masses of sails on tall masts. They earned their name from the way that they 'clipped off' journey time. British merchants resolved to build their own clippers to rival the Americans, and the first British tea clipper, *Stornaway*, was built in Aberdeen in 1850. More tea clippers were

designed and built in Britain throughout the 1850s and 1860s; they had a narrower beam than their American equivalents, making them less powerful during storms, but faster in calmer weather.

There was a great spirit of competition between the British and American ships plying the tea trade, but to begin with the Americans had the edge. Then in 1851 a British ship owner, Richard Green, built the aptly named clipper Challenger with the stated intention of beating the American ships. Loaded with tea, Challenger left China for London in 1852 at the same time as the American clipper Challenge, a much larger older ship, already greatly admired for its speed. Large sums were bet on which would complete the journey first. In the event, the British ship beat its rival to London by two days, amid much jubilation. From then on, such international races grew in popularity.

After 1855, American participation in the British tea trade gradually stopped. But even without the Anglo-American rivalry, the competitive spirit continued. It was really ignited when new ports were opened up for trade in China. These included Fouchow, which was much closer to the tea-producing areas than Canton, the port used previously. As a result, tea could be loaded on board earlier and fresher, and the clippers could set off in late May or early June – sometimes not even taking time to fill out the official paperwork – racing back to Britain whatever the difficulties.

They sped down through the South China Sea and into the Indian Ocean, then raced to get round the southernmost tip of Africa at the Cape of Good Hope. Then it was north across the vast Atlantic, past the Azores, through the English Channel and into the estuary of the River Thames. Once there, they would be towed by tugs, up the river and into the docks.

The cargo of the winning ship could earn a premium of up to sixpence per pound – and so the captain and crew were rewarded by the owners of the cargo. But the races were about more than just money: the crews, about 40 men on each clipper, were expert sailors, proud of their ships, and they delighted in competing against each other. Without their enthusiasm, the races would never have happened, since getting home as fast as possible required the crew to be totally dedicated and to sacrifice much of their rest for the duration of the race.

Questions 1–6

Do the following statements agree with the information given in the passage?

*Write **TRUE** if the statement agrees with the information, **FALSE** if the statement contradicts the information, or **NOT GIVEN** if there is no information on this.*

- ① In the seventeenth and eighteenth centuries, the British East India Company faced a lot of competition.

- 2 Before 1800, cargo size was the most important consideration for the East India Company.
- 3 At best, voyages of the East Indiamen to China and back took nearly two years to complete.
- 4 Before 1834, voyages to and from China were considered to be highly dangerous.
- 5 After 1834, the ships which had served the East India Company stopped being used for commercial purposes.
- 6 In the nineteenth century, British drinkers preferred tea made from mature leaves to that made from younger leaves.

Questions 7–13

Complete the notes below.

Choose **ONE WORD ONLY** from the passage for each answer.

Clipper races

The ships

- Clipper ships were first used for trading by American merchants.
- The ships were remarkable for the number of 7 ————— they had.
- The performance of British tea clippers was particularly affected when there were 8 ————— at sea.

The races

- It was in a ship called 9 ————— that the British first competed successfully against the Americans.
- Richard Green's ship arrived two days ahead of its competitor.
- Competition increased when additional Chinese trading 10 ————— were established.
- Merchants were occasionally in such a hurry that they failed to complete the 11 ————— before leaving China.
- At the end of their journey, the ships needed the help of 12 —————.

The rewards

- The crews were motivated by both 13 ————— and their enthusiasm for the competition.

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Passage 37

The Development of Plastics

From billiard balls to biodegradable bottles: a century of invention

A The first plastics were developed as a substitute for natural rubber. Chemically, rubber is a polymer — a compound containing large molecules that are formed by the bonding of many smaller, simpler units, repeated over and over again. The same bonding principle, polymerization, is the basis of the creation of a huge range of plastics by the chemical industry. The first plastic was developed as a result of a competition in the USA. In the 1860s, \$10,000 was offered to anybody who could replace ivory, supplies of which were declining, with something equally good as a material for making billiard balls. The prize was won by John Wesley Hyatt, with a material called celluloid. Celluloid was made by dissolving cellulose, a carbohydrate obtained from plants, in a solution of camphor dissolved in ethanol. This new material rapidly found other applications in the manufacture of everyday products such as knife handles and detachable collars and cuffs. But perhaps the best-known celluloid product was photographic film, without which the film industry could never have taken off at the end of the 19th century. Celluloid can be repeatedly softened and reshaped by heat, and is known as a thermoplastic.

B In 1907, Leo Baekeland (1863–1944), a Belgian chemist working in the USA, invented a different kind of plastic, by causing phenol and formaldehyde to react together. Baekeland called it Bakelite, and it was the first of the thermosets, plastic that can be cast and moulded while hot, but cannot be softened by heat and reshaped once they have set. Bakelite was a good insulator, and was resistant to water and acid. With these properties it was soon being used in the manufacture of electrical switches as well as a variety of domestic items.

C As the century went on, the range of newly developed plastic increased. Chemists began looking for other small molecules that could be strung together to make polymers. In the 1930s, chemists in Britain discovered that the gas ethylene would polymerize under heat and pressure to form a thermoplastic they called polythene. Polypropylene followed in the 1950s. Both are used to make bottles, pipes and plastic bags. A small change in the starting material — replacing a hydrogen atom in ethylene with a chlorine atom — produced rigid PVC (polyvinyl chloride), a fireproof plastic suitable for drains and gutters. By adding certain chemicals, a soft form of PVC can be produced, suitable as a substitute for rubber in items such as waterproof clothing. A closely related plastic is Teflon or PTFE (polytetrafluoroethylene). It produces very little friction, making it ideal for products such as non-stick frying pans.

D Polystyrene, a hard, clear material like glass, was developed during the 1930s in Germany, and its applications included food containers and toys. Expanded polystyrene is rigid and is widely used in packaging and insulation. Polyurethane, developed in the same country, was commonly produced as a foam which was very useful in the production of insulating materials.

E In the 1930s, the first of the man-made fibres was created — nylon. Its inventor was a chemist called Wallace Carothers (1896–1937), who worked for the Du Pont company in the USA. He found that under the right conditions two particular chemicals would form a polymer that could be pumped out through holes and then stretched to form long glossy threads that could be woven like silk. Its first use was to make parachutes for the US armed forces in World War II. In the postwar years, it completely replaced silk in the manufacture of stockings. Many other synthetic fibres joined nylon, including Orion, Acrilan, and Terylene. Today most garments are made of a blend of natural fibres, such as cotton and wool, and man-made fibres that make fabrics easier to look after.

F Despite its enormous usefulness, plastic has its drawbacks. In fact one of its great strengths, its indestructibility, is its greatest disadvantage. Beaches all over the world, even on the remotest island, are littered with plastic bottles that nothing can destroy. Nor is it very easy to recycle plastics, as different types of plastic are often found in the same items and call for different treatments.

G Plastics can be made biodegradable by incorporating into their structure a material such as starch, which is attacked by bacteria and causes the plastic to fall apart. Other materials can be incorporated that gradually decay in sunlight — although bottles made of such materials have to be stored in the dark, to ensure they do not disintegrate before they have been used.

Questions 1–7

Complete the table below.

Choose **NO MORE THAN THREE WORDS** from the passage for each answer.

Early plastics

Name of plastic	Date of invention	Original region	Property	Common use
Celluloid	1860s	US	—	① _____

Name of plastic	Date of invention	Original region	Property	Common use
② _____	1907	US	Can be cast and moulded but cannot be softened by heat	③ _____
Polythene	1930	④ _____		household items and car parts
Rigid PVC	—	—	⑤ _____	—
Polystyrene	1930s	Germany	⑥ _____	Food container
Polyurethanes	—	Germany	⑦ _____ foams	Adhesives, coatings and insulation

Questions 8–13

Do the following statements agree with the information given in the passage?

Write **TRUE** if the statement agrees with the information, **FALSE** if the statement contradicts the information, or **NOT GIVEN** if there is no information on this.

- ⑧ The chemical structure of rubber is very different to that of plastics.
- ⑨ John Wesley Hyatt was an industrial chemist.
- ⑩ Celluloid and Bakelite react in the same way to heat.
- ⑪ If an object is made of several plastics, these prove hard to break down and reuse.
- ⑫ Adding starch to plastic makes it more durable.
- ⑬ Containers which are designed to decompose need particular storage conditions.

Passage 38

The Early History of Scott and Bowne's Cod Liver Oil

How an unpleasant-tasting fish oil became one of the best-known patent medicines in the world

Cod liver oil is a type of fish oil used today for general health purposes, but northern European fishing communities used cod liver for centuries before the doctors and chemists of 19th-century Europe began to take an interest. Its manufacture was simple: the livers of the codfish were left for some days, then oil was taken from them. The oil grew darker according to how long the livers were left, resulting in three grades of oil: pale, light brown and dark brown.

Ludovicus Josephus de Jongh of the Netherlands produced the first extensive chemical analysis of cod liver oil in 1843. His studies of the three grades of oil led him to conclude that the light-brown oil was the healthiest. He attributed this superiority to the larger quantities of iodine, phosphate of chalk and volatile acids found in it.

In 1846, de Jongh traveled to Norway to obtain the purest oil available. By the 1850s, 'Dr. de Jongh's Light Brown Cod Liver Oil' was marketed throughout Europe and exported to the United States. Each bottle had de Jongh's signature and stamped seal on it - a blue codfish on a red shield - guaranteeing that the product was 'put to the test of chemical analysis'. Advertising emphasized de Jongh's credentials as a doctor and chemist, and included testimonials from other men of science and medicine.

However, even the most enthusiastic supporters of cod liver oil admitted that the highly disagreeable taste and smell presented a significant obstacle to its use. De Jongh believed the problem of the oil's unpleasant taste and smell could be overcome with a little perseverance or, failing that, by following it with some fruit or biscuit, or glass of wine. But his recommendations appear not to have worked well. It was often combined with coffee, although a few people recommended taking the oil with tomato ketchup.

In 1873, Alfred B. Scott came to New York and, along with partner Samuel W. Bowne, began experimenting to produce a more pleasant preparation of cod liver. Three years later they established the firm of Scott and Bowne, and began marketing their product as Scott's Emulsion. Though not a doctor or pharmacist by training, Scott had the eye for opportunity that was necessary for achievement in business. Advertising, the two men believed, would propel their product to success. And so it did: by the 1890s Scott and Bowne had factories in five European countries, and were selling their emulsion throughout the Americas, Europe and Asia.

Scott got his oil for Scott's Emulsion directly from the Lofoten Islands in Norway, the world center of cod fishery - located above the Arctic Circle. The codfish streamed to the islands in early January to lay their eggs, and by the end of April were gone. The Gulf

Stream, the Arctic waters, and the Norwegian fjords combined to create a perfect breeding group for the codfish and an unequaled fishing industry for the fishermen.

Scott and Bowne's first trademark, registered in 1879, included the initials P.P.P. and three words - 'Perfect, Permanent, Palatable'. The mark reflected that Scott's Emulsion was a perfect formula, a permanent emulsion (that is, one in which the ingredients would not separate), and most importantly, that it had a pleasant taste. 'Palatable as milk' became a key phrase in Scott's advertising.

A man with a fish on his back first appeared on Scott's Emulsion around 1884 and became Scott and Bowne's trademark in 1890. As Scott told it, he saw this fisherman with his record-breaking catch while on business in Norway. A photographer was quickly found to record the scene. Later, the photo was faithfully reproduced as a drawing, and registered as the company's trademark. In the drawing, the man stoops forward, glances out from under the brim of his hat, legs tensed under the weight of his load. A thick rope, wrapped round his waist, shoulders and hands, secures the load on his back - a huge fish with gaping mouth and glassy yellow eye, its tail sweeping the floor. The common codfish is recognizable by the brown and amber spots all over its body, the light stripe down its side, and the three dorsal fins. The words 'SCOTT'S EMULSION' appears in the title of the picture.

Trade cards and booklets featured the fisherman and his catch along with the words 'Scene taken from life on the coast of Norway' and 'The Codfish, weighing 156 pounds, was caught off the coast of Norway'. The realistic image, a direct reference to the natural source of the medicine, served as a reassurance of quality in a market that contained some impure, unsafe products.

By the 1900s, 'the man with the fish' was famous. His image appeared on countless boxes and bottles of a cod-liver-oil preparation. It was printed in full colour on advertising trade cards, booklets, and posters distributed around the globe, and in one instance painted several stories high on the side of a building. The man with the fish endures today, a testament to the persistence of an age-old tradition, even as scientific and commercial interest in cod liver oil has risen and fallen.

Questions 1–8

Do the following statements agree with the information given in Reading Passage 1?

*Write **TRUE** if the statement agrees with the information, **FALSE** if the statement contradicts the information, or **NOT GIVEN** if there is no information on this.*

- ① In the manufacturing process, cod livers left for the longest time produced the lightest type of oil.

- 2 A Dutch scientist called de Jongh suggested why one grade of cod liver oil was particularly healthy.
- 3 De Jongh was both a researcher in, and a supplier of, cod liver oil.
- 4 Many scientists tried to find a solution to the bad smell of cod liver oil.
- 5 The experimental methods of Scott and Bowne were much better than de Jongh's.
- 6 Scott was a trained chemist as well as a businessman.
- 7 Scott found a new location for cod fishing.
- 8 Cod around the Lofoten Islands could be caught during the first four months of the year.

Questions 9–16

Complete the notes below.

Choose **ONE WORD ONLY** from the passage for each answer.

Two versions of cod liver oil

De Jongh's version — to deal with the taste of the oil

- de Jongh suggested drinking some 9 _____ after taking it.
- others frequently added it to 10 _____.

Scott and Bowne's version — trademarks and advertising

- one slogan compared the product with 11 _____; the image of the man with a fish contains:
- a man with a 12 _____ around his body.
- a fish whose 13 _____ is touching the ground, and which has multi-coloured 14 _____ on it.
- suggested the 15 _____ of the product to customers.
- appeared in large-scale on a 16 _____.

Passage 39

The Extinction of Cave Bear

A chemical detective story reveals why Europe's largest bear vanished

Herve Bocherens, an evolutionary biologist at the University of Tübingen, Germany, says his colleagues find his research methods a little crude. He dissolves 30,000-year-old animal bones in hydrochloric acid, which is strong enough to burn through metal, soaks the bone solution, cooks it at about 200 degrees Fahrenheit and freeze-dries it, until what's left is a speck of powder weighing less than one one-hundredth of an ounce. The method may be harsh, but the yield is precious — the biography of a cave bear as told through its chemical components. Bocherens is at the forefront of research on the bear, a European species that died out 25,000 years ago. People have been excavating cave bear remains for hundreds of years — in the Middle Ages, the massive skulls were attributed to dragons — but the past decade has seen a burst of discoveries about how the bears lived and why they became extinct. An abundance of bones has been found from Spain to Romania in caves where the animals once hibernated. 'Caves are good places to preserve bones, and cave bears had the good sense to die there,' Bocherens says.

Along with mammoths, lions and woolly rhinos, cave bears (*Ursus spelaeus*) were once among Europe's most impressive creatures. Males weighed up to 1,500 pounds, 50 per cent more than the largest modern grizzly bears. Cave bears had wider heads than today's bears, and powerful shoulders and forelimbs. But the relationship between humans and cave bears has been mysterious. Were humans prey for the bears, or predators? Were bears the object of worship or fear? Initially cave bears shared the continent of Europe more than 100,000 years ago with Neanderthals, a primitive species of humans. Modern humans arrived in Europe about 40,000 years ago, and were soon aware of the bears. They painted images of the animals on cave walls, and carved their likeness in fragments of mammoth tusk. The walls of France's Chauvet cave are painted with lions, hyenas and bears; dating back 32,000 years these are perhaps the oldest paintings in the world. The painters weren't the cave's only occupants: the floor is covered with 150 cave bear skeletons, and its soft clay still holds paw prints. Most dramatically, a cave bear skull was perched on a stone slab in the center of one chamber, placed deliberately by some long gone cave inhabitant. 'There's no way to tell if it was just curiosity that made someone put a skull on the rock, or if it had religious significance,' says Bocherens.

Another discovery, hundreds of miles to the east of Chauvet, would shed light on the relationship between cave bears and humans. The Swabian Jura is a limestone plateau in southwestern Germany that is riddled with caves. A steel gate guards the Hohle Fels cave from vandals and curiosity-seekers. Floodlights in the cave's main chamber illuminate the

ceiling, vaulted like a cathedral above 5,000 square feet of floor space. Long ago, as shown by the bones and tools that archaeologists have found, cave bears and human beings sought shelter here from winter weather.

In 2000, University of Tübingen paleobiologist Susanne Munzel unearthed a bear vertebra with a tiny triangular piece of flint embedded in it. The flint was likely a broken spear point, hard evidence of a successful bear hunt 29,000 years ago. Munzel also found bear bones that had clearly been scraped by stone tools. Evidence for bears being a food source for early humans was shown by cut marks on skulls and leg bones where the flesh had been cut away. Many of the bones were from baby bears, perhaps caught while hibernating.

Cave bears disappeared not long after humans spread throughout Europe, and one theory is that hunting led to the bears' extinction. Herve Bocherens' test tubes may hold the clues. Running his white powder through a mass spectrometer, he identifies different isotopes, or chemical forms, of elements such as carbon and nitrogen that reflect what the bears were eating and how quickly they grew. After studying hundreds of bones from dozens of sites in Europe, Bocherens has found that cave bears ate mainly plants. This would have made bears particularly vulnerable to the last Ice Age, which began around 30,000 years ago. This Ice Age shortened or eliminated growing seasons, and altered the distributions of plant species across Europe. Cave bears began to move from their old territories, according to a DNA analysis of teeth found near the Danube River. The cave bear population there was relatively stable, with the same genetic patterns showing up generation after generation. But about 28,000 years ago, newcomers with different DNA patterns arrived — a possible sign of hungry bears suddenly on the move.

But changes in the climate can't be solely to blame for the bears' extinction. According to the latest study, by Erik Trinkaus and his colleagues at the Max Planck Institute, cave bear populations began a long, slow decline 50,000 years ago — well before the climate began to change. The new study supports a different explanation for the cave bears' demise. As Neanderthals, and then a growing population of modern humans moved into the caves of Europe, cave bears had fewer safe places to hibernate. An acute housing shortage may have been the final blow for these magnificent beasts.

Questions 1–6

Do the following statements agree with the information given in Reading Passage 1?

*Write **TRUE** if the statement agrees with the information, **FALSE** if the statement contradicts the information, or **NOT GIVEN** if there is no information on this.*

- 1 Bocherens' findings on cave bears involve measuring the length of complete bones.
- 2 Bocherens was the first person to conduct analytical research on cave bears in a laboratory.

- 3 Cave bears have been extinct for 25,000 years.
- 4 At one time people thought the excavated remains of bears were those of dragons.
- 5 Modern grizzly bears are similar in size to cave bears.
- 6 Neanderthals understood cave bear behaviour better than modern humans did.

Questions 7–13

Complete the notes below.

Choose **ONE WORD ONLY** from the passage for each answer.

Research into the existence and disappearance of cave bears

Evidence from the Chauvet cave in France

- Paintings on the walls include bears.
- Over a hundred 7 _____ of cave bears together with paw prints were found on the floor of the cave.
- The 8 _____ of a cave bear was found in a prominent position in the cave.

Evidence from the Hohle Fels cave in the Swabian Jura in Germany

- Bones and tools found in the cave indicate it provided 9 _____ for both bears and humans.

Susanne Munzel's findings

- A spear point in the vertebra of a bear indicates that a hunt had taken place.
- Marks on the bones of bears indicate the removal of flesh.

Herve Bocherens' findings

- Isotopes from the bones indicate that 10 _____ were the cave bears' preferred food.

Factors leading to the disappearance of the cave bear

- Climate change
- The Ice Age changed vegetation growth in 11 _____.

- Bears were forced to migrate.
- Support for the above explanation comes from a study of changes in the DNA in the (12) _____ of the population of cave bears near the Danube River at this time.
- Human population expansion
- Competition from Neanderthals and modern humans for caves may have reduced opportunities for cave bears to (13) _____.

Passage 40

The Giant Panda

DNA testing finally settles a century-old scientific argument about what kind of animal a panda really is

For more than 100 years, scientists have argued over exactly what a panda is. Now, finally, with the help of DNA testing, the panda has been admitted to the ursidae (bear) family, and the spectacled bear of South America has been confirmed as its closest living relative.

In 1869, French Jesuit missionary Pere David first described the giant panda to western science. With just a pelt and reported sighting to go on, he classified it as a bear. However, the following year, zoologist Alphonse Milne Edwards dissected the first specimen and concluded that it had more in common with the red panda, a member of the raccoon family. For more than a century, scientists quarreled over whether the giant panda belonged to the bear family, the raccoon family or a separate family of its own.

They had good reason to be confused. The giant panda shares many physical characteristics with the red panda. Both have evolved to feed on bamboo, grasping and eating it in the same way, with similar teeth, skulls and forepaws. They also both have a distinctive cry, which they use to communicate with others in their group.

In the mid-1980s there were several studies involving DNA comparisons between the species. The first investigations linked the giant panda with bears, but in 1991 further tests contradicted these findings and placed it in the raccoon family with the red panda. By the year 2000, approximately twelve studies had been completed, and all except two placed the panda in the bear family. The data from these two studies was reanalysed by other researchers, who finally concluded that the giant panda was indeed a bear.

Today, there are eight species of bear. Along with dogs, their closest relatives, cats, raccoons and weasels, they belong to the order Carnivora, a group of meat-eating predators that evolved some 57 million years ago. The ancestors of modern bears split from this group about 34 million years ago, and today the panda is our oldest living bear, followed by the spectacled bear. Both are survivors of an ancient lineage dating back 18 million years. The rest - the brown, black, polar, Asiatic black, sloth and sun bears - are relatively modern, dating back four to five million years.

Researchers have found that the spectacled bear and the panda have several physical features in common. The spectacled bear's muzzle is comparatively short and it has blunt molar teeth and large jaw muscles, which are good for grinding fibrous vegetation - vegetation such as bamboo. Indeed, scientists in Venezuela have found that bamboo makes up 70% of the diet of some spectacled bear populations. For most spectacled bears, however, the bromeliad, a tropical plant with fleshy leaves, is their main food source. Most

species of bromeliad grow in trees, and spectacled bears therefore have to be adept tree climbers because they spend their lives foraging for these plants, as well as fruits, in the cloud forest of the Andes.

The giant panda's diet is famously dull, with bamboo representing 99% of its intake. This is rather strange given that its physiology is typical of a carnivore and it has no special adaptation for digesting cellulose, the main constituent of plant cell walls. A panda manages to digest only about 17% of the bamboo it eats (a deer living on grass achieves 80% efficiency). It typically feeds for 14 hours a day, consuming 20 kg or more of bamboo. Unable to store fat effectively, it continues eating in the bitterly cold winter, at a time when many other bears hibernate.

With such a specialized diet, the giant panda has evolved a sixth digit, a prehensile elongated wrist bone called the radial sesamoid. They use this 'false thumb' to roll bamboo leaves into fat, cigar-shaped wads which they then sever using their powerful jaws. They feed mainly on the ground but are capable of climbing trees as well. The spectacled bear is a more frequent climber and will even climb spiky cacti plants to reach fruit at the top. They also construct tree nests to act as a bed as well as a platform to feed from fruit-laden branches.

Very occasionally, the giant panda supplements its diet with meat, which it scavenges. Spectacled bears eat carrion, too, and some have been known to kill small calves. Spectacled bears are highly adaptable and are found in a wide range of habitats including rainforest, dry forest and coastal scrub desert. In contrast, the giant panda lives at an altitude of between 1,200 and 3,500 metres in mountain forests that are characterized by dense stands of bamboo.

There have been many theories as to why the panda has such a distinctive coat, but the most convincing argument is that of George Schaller, one of the first western scientists to study wild pandas. He believes the contrasting coat may help prevent close encounters with other pandas. 'In pandas, a stare is a threat,' Schaller says. 'The eye patches enlarge the panda's small, dark eyes tenfold, making the stare more powerful. A staring panda will hold its head low, so presenting the eye patches. To show lack of aggressive intent, a panda will avert its head, cover its eye patches with its paws or hide its face.' Interestingly, the spectacled bear is the only other bear with comparably obvious markings around the eye.

Questions 1–8

Classify the following characteristics as belonging to

A the giant panda **B** the spectacled bear **C** both the giant panda and the spectacled bear

Write the correct letter, A, B or C.

- 1 an extra thumb on each paw
A the giant panda
B the spectacled bear
C both the giant panda and the spectacled bear
- 2 a tendency to sleep in trees
A the giant panda
B the spectacled bear
C both the giant panda and the spectacled bear
- 3 their species originated 18 million years ago
A the giant panda
B the spectacled bear
C both the giant panda and the spectacled bear
- 4 the ability to adjust to different environments
A the giant panda
B the spectacled bear
C both the giant panda and the spectacled bear
- 5 the use of noises to socialize with each other
A the giant panda
B the spectacled bear
C both the giant panda and the spectacled bear
- 6 the ability to climb trees
A the giant panda
B the spectacled bear
C both the giant panda and the spectacled bear
- 7 the eating of meat
A the giant panda
B the spectacled bear
C both the giant panda and the spectacled bear
- 8 a similarity to a type of raccoon
A the giant panda
B the spectacled bear
C both the giant panda and the spectacled bear

Questions 9–13

Complete the sentences below.

*Choose **NO MORE THAN TWO WORDS** from the passage for each answer.*

- The panda's digestive system is that of a 9 _____.

- The giant panda must eat constantly because it can only ⑩ _____ a small amount of bamboo.
- In winter, giant pandas cannot ⑪ _____ because of their feeding habits.
- Spectacled bears build ⑫ _____ to help reach their food.
- Giant pandas may use their ⑬ _____ to threaten other pandas.

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Passage 41

The History and Production of Chocolate

From a sacred Mayan drink to a global industry — tracing chocolate's journey and how it is made today

Early history in Central America

The climate of tropical rain forests combines great quantities of rainfall with high year-round temperatures and humidity. This results in the perfect conditions for the cultivation of the cacao tree plant from whose bean chocolate is derived. This tree has grown in the wild in Central America for about 10,000 years. The Olmec Indians, whose civilization flourished in what is now southern Mexico from 1500 to 300 BC, are believed to have been the first to grow cocoa beans as a domestic crop.

The cacao tree was worshipped from about 600 AD by the Mayan people of Central America and southern Mexico, who believed it to be of divine origin. Cacao, which the early explorers renamed as cocoa, is actually a Mayan word meaning 'god food'. The Mayans brewed a spicy, bitter-sweet drink by pounding the cocoa beans with maize and chilli peppers and letting the mixture ferment. This drink was reserved for use in traditional ceremonies as well as for drinking by the wealthy and by priests, who also ate a porridge made from the beans.

In about 1200 AD the Aztecs of central Mexico were introduced to cocoa beans and came to highly prize them, but because they lived further north at more arid, higher altitudes not suitable for the cultivation of the cacao tree, they had to acquire the beans through trade or by force. The Aztecs used cocoa beans as currency and taxes were paid in the beans to their rulers. In fact, the Aztec royal treasury consisted not of precious metals but of cocoa beans. The Aztecs also enjoyed a beverage made from the beans, and this again featured prominently in rituals and as a luxury available only to the very wealthy. The Aztecs called this drink 'xocolatl'. The Spanish explorers found this word almost impossible to pronounce and so corrupted it to 'chocolat', which the English subsequently changed to chocolate.

Introduction to Europe and North America

Chocolate was brought back to Spain in the early 16th century by Spanish explorers, who at first found the Aztec drink distasteful because of its bitterness. However, they learned to improve its taste by mixing the ground roasted beans with sugar and vanilla. Chocolate factories were established in Spain and by the early 17th century, chocolate powder was being exported to other parts of Europe. The Spanish kept the source of the drink – the beans – a secret for many years, so successfully, in fact, that when English pirates boarded

a Spanish ship they thought carried great treasure and only found a load of cocoa beans, they burned the ship in frustration. If only they had known that chocolate at that time was extremely valuable.

Within a few years, the cocoa beverage produced in Spain was sought after throughout Europe. The first chocolate house in England opened in 1657, followed rapidly by many others. These establishments were used as clubs where wealthy people could meet to talk to their friends and to discuss business over a cup of chocolate. The first recorded instance of chocolate being eaten in solid form occurred when a baker in England began adding cocoa powder to cakes in the mid-1600s. Then in 1828, a Dutch chemist, Johannes Van Houten, invented a method of extracting the bitter-tasting cocoa butter from the roasted beans. His aim was to make the cocoa drink smoother and more palatable, however, unknowingly he paved the way for the chocolate bar that we consume today.

In the early 18th century, certain companies owned by prominent English families including the Frys, Rowntrees and Cadburys, began making chocolate. Many of these chocolate factory owners were noticed for their enlightened treatment of employees, providing not just jobs but facilities such as housing and schools. In fact, the Cadbury company constructed a large purpose-built town near Birmingham. And interestingly, Milton Hershey, founder of perhaps the most famous US chocolate-making company of its day, also established a town in Pennsylvania in 1900.

The production of chocolate

Producing chocolate is a time-consuming and complicated process. After being harvested and separated from the pods which contain them, the beans are allowed to ferment and then dried either in the sun (which produces the best chocolate) or artificially, using specially designed machines (which creates an inferior chocolate). After that the beans are roasted according to various formulas which affect the flavour of the final product.

The beans are then lightly crushed, the outer layer of shell discarded, and the internal kernel or 'nib' retained. At this point, the nibs, which contain high quantities of a fat known as cocoa butter, may be treated to develop their flavour and colour. The nibs are finely ground, and in the grinding process heat is produced which liquefies the product into a substance called the 'liquor'. This is cooled and allowed to solidify. After this stage, the chocolate is made into any of a multitude of solid forms or is ground into cocoa powder, depending on the ultimate intended use.

Questions 1–5

Do the following statements agree with the information given in the passage?

Write **TRUE** if the statement agrees with the information, **FALSE** if the statement contradicts the information, or **NOT GIVEN** if there is no information on this.

- ① A climate with seasonal variation in temperature is ideal for growing the cacao tree.
- ② It is likely that the Olmec Indians were the first people to cultivate the cacao trees.
- ③ The location of the Aztec homeland meant that they had to obtain cocoa beans from elsewhere.
- ④ The Aztec rulers demanded heavy taxes from their subjects.
- ⑤ Drinks made from cocoa beans were widely consumed by ordinary Aztecs.

Questions 6–10

Complete the notes below.

Choose **ONE WORD ONLY** from the passage for each answer.

The history of chocolate

Introduction to Europe and North America

- Spanish explorers introduced chocolate drink to Europe in 16th century
- ingredients added to reduce ⑥ _____
- chocolate houses established in 17th century in England for social, ⑦ _____ and other activities

Developments in the production of chocolate

- Englishman who was a ⑧ _____ creates first known edible chocolate product in 1600s
- Dutch scientist invents process leading to creation of the chocolate ⑨ _____
- many English producers provide variety of facilities for ⑩ _____
- Cadbury (in England) and Hershey (in America) each establish a town

Questions 11–13

Complete the flow chart below.

Choose **ONE WORD ONLY** from the passage for each answer.

The chocolate manufacturing process

- After being separated from their pods, cocoa beans are ⑪ _____ naturally or mechanically
- Beans then ⑫ _____ to create distinctive tastes
- Nibs are separated from the shells
- Heat generated during the grinding of the nibs produces ⑬ _____. Further treatment produces desired final chocolate product

Passage 42

The History of Lighting

From tallow candles to LEDs — how artificial light transformed daily life

We forget how painfully dark the world once was. A candle – a good candle – provides barely a hundredth of the illumination of a single 100 watt light bulb, and yet candles provided most of the world's artificial illumination before the arrival of electricity. Candles were of two types – tallow and wax. Tallow, made from animal fat, had the great advantage that, as it could be made at home as a by-product of any animal which also supplied food to eat, it was plentiful and cheap. But tallow candles melted extremely quickly and had to be frequently replaced, they burned with an uneven light, and also they smelled very unpleasant. Far superior in all these respects were candles made of beeswax. These gave a steadier light, but they cost about four times as much, and tended to be used only on special occasions.

For the well-to-do in many large cities, gas was an additional option from about 1820. Mostly, however, it was used in businesses and for street lighting, and didn't become common in homes until closer to the middle of the century. Gas had many drawbacks. Those who worked in gas-supplied offices or visited gas-lit theatres, for example, often complained of headaches and nausea. To minimise that problem, gas lights were sometimes erected outside windows to light the interior. If installed indoors, they tended to blacken ceilings, discolour fabrics, corrode metal and leave greasy marks on every horizontal surface. Gas also needed some care in use. Most gas-supply companies reduced gas flow through their pipes during the day when demand was low. So anyone lighting a gas jet during the day had to open the tap wide to get a cleaner flame.

The advent of electric lighting in the late 19th century revolutionised artificial light, but its adoption was gradual. Thomas Edison's successful development of a long-lasting incandescent light bulb in 1879 marked a turning point, yet electricity networks were slow to spread. Initially, electric lighting was a luxury, installed first in public buildings, wealthy homes, and commercial districts. Unlike gas, electric light was clean, bright, consistent, and much safer – it produced no fumes, left no soot, and posed a far lower fire risk. However, early electric systems were not standardised; voltages varied from city to city, and direct current (DC) competed with alternating current (AC) in what became known as the “War of the Currents.” By the early 20th century, with the expansion of power grids and falling costs, electric light became the dominant form of illumination across the developed world, eventually rendering gas and candles obsolete for everyday use.

The improvement in lighting technology had profound social and economic effects. Longer hours of useful light extended the working day, boosted industrial productivity, and made night-time leisure activities more practical and popular. Streets became safer, reading after

dark ceased to be a strain on the eyes, and public spaces like libraries, shops, and restaurants could operate into the evening. Historians often link the spread of reliable artificial light to broader patterns of modernisation, including shifts in sleep patterns, increased literacy, and the growth of a vibrant night-time culture in cities.

Today, lighting technology continues to evolve. Fluorescent tubes, halogen bulbs, and now light-emitting diodes (LEDs) have each brought gains in efficiency, longevity, and environmental sustainability. Smart lighting systems allow for personalised control over colour and intensity, while advances in design integrate light into architecture in ever more seamless ways. Yet the fundamental human desire that drove the history of lighting – to push back the darkness – remains unchanged.

Questions 1–5

Complete the table below.

Choose **NO MORE THAN TWO WORDS** from the passage for each answer.

Type of Lighting	Key Advantages	Key Disadvantages
Tallow Candle	Plentiful and ① _____	Melted quickly, uneven light, badly ② _____
Beeswax Candle	③ _____ light	Expensive, used for ④ _____
Gas Lighting	Available from c.1820 for the wealthy	Caused health issues, eroded ⑤ _____

Questions 6–10

Do the following statements agree with the information given in Reading Passage 1?

Write **TRUE** if the statement agrees with the information, **FALSE** if it contradicts it, or **NOT GIVEN** if there is no information on this.

- ⑥ Candles provided more light than early 100-watt light bulbs.
- ⑦ Beeswax candles were commonly used for everyday lighting in most households.
- ⑧ Gas lighting was commonly used in homes before it was used in theatres.
- ⑨ Early electric light bulbs were developed by Thomas Edison.
- ⑩ Electric lighting immediately replaced gas lighting in all cities.

Questions 11–13

Choose the correct letter, **A**, **B**, **C** or **D**.

- 11 The main reason tallow candles were widely used was that they were
- A very bright.
 - B easy to make at home.
 - C long-lasting.
 - D pleasant-smelling.
- 12 One way people tried to reduce the negative health effects of gas lighting was to
- A use it only in winter.
 - B place lights outside windows.
 - C burn candles at the same time.
 - D avoid using it in theatres.
- 13 What is the main idea of the final section, "Modern lighting and beyond"?
- A Lighting technology has stopped improving.
 - B The basic purpose of lighting has changed completely.
 - C Advances continue, but the human need for light remains.
 - D LEDs are the final and perfect form of artificial light.

Passage 43

The History of Pencil

How a lightning strike in 16th-century England led to one of the world's most enduring writing tools

A The beginning of the story of pencils started with a lightning. Graphite, the main material for producing pencil, was discovered in 1564 in Borrowdale in England when a lightning struck a local tree during a thunder. Local people found out that the black substance spotted at the root of the unlucky tree was different from burning ash of wood. It was soft, thus left marks everywhere. Chemistry was barely out of its infancy at the time, so people mistook it for lead, equally black but much heavier. It was soon put to use by locals in marking their sheep for signs of ownership and calculation.

B Britain turns out to be the major country where mines of graphite can be detected and developed. Even so, the first pencil was invented elsewhere. As graphite is soft, it requires some form of encasement. In Italy, graphite sticks were initially wrapped in string or sheepskin for stability, becoming perhaps the very first pencil in the world. Then around 1560, an Italian couple made what are likely the first blueprints for the modern, wood-encased carpentry pencil — their version was a flat, oval, more compact type of pencil. Their concept involved the hollowing out of a stick of juniper wood. Shortly thereafter in 1662, a superior technique was discovered by German people: two wooden halves were carved, a graphite stick inserted, and the halves then glued together — essentially the same method in use to this day. The news of usefulness of these early pencils spread far and wide, attracting the attention of artists all over the known world.

C Although graphite core in pencils is still referred to as lead, modern pencils do not contain lead as the "lead" of the pencil is actually a mix of finely ground graphite and clay powders. This mixture is important because the amount of clay content added to the graphite depends on intended pencil hardness, and the amount of time spent on grinding the mixture determines the quality of the lead. The more clay you put in, the higher hardness the core has. Many pencils across the world, and almost all in Europe, are graded on the European system. This system of naming used B for black and H for hard; a pencil's grade was described by a sequence or successive Hs or Bs such as BB and BBB for successively softer leads, and HH and HHH for successively harder ones. The standard writing pencil is graded HB.

D In England, pencils continued to be made from whole sawn graphite. But with the mass production of pencils, they are getting drastically more popular in many countries with each passing decade. As demands rise, appetite for graphite soars. According to the United States Geological Survey (USGS), world production of natural graphite in 2012 was 1,100,000 tonnes, of which the following major exporters are: China, India, Brazil, North

Korea and Canada. When the value of graphite was realised, the mines were taken over by the government and guarded. One of its chief uses during the reign of Elizabeth I in the second half of the 16th century was as moulds for the manufacture of cannon balls. Graphite was transported from Keswick to London in armed stagecoaches. In 1751 an Act of Parliament was passed making it an offence to steal or receive "wad". This crime was punishable by hard labour or transportation.

E That the United States did not use pencils in the outer space till they spent \$1000 to make a pencil to use in zero gravity conditions is in fact a fiction. It is widely known that astronauts in Russia used grease pencils, which don't have breakage problems. But it is also a fact that their counterparts in the United States used pencils in the outer space before real zero-gravity pencil was invented. They preferred mechanical pencils, which produced fine lines, much clearer than the smudgy lines left by the grease pencils the Russians favoured. But the lead tips of these mechanical pencils broke often. That bit of graphite floating around the space capsule could get into someone's eye, or even find its way into machinery or electronics, causing an electrical short or other problems. But despite the fact that the Americans did invent zero-gravity pencils later, they stuck to mechanical pencils for many years.

F Against the backcloth of a digitalized world, the prospect of pencils seems bleak. In reality, it does not. The application of pencils has by now become so widespread that they can be seen everywhere, such as classrooms, meeting rooms and art rooms, etc. A spectrum of users are likely to continue to use it into the future: students to do math works, artists to draw on sketch pads, waiters or waitresses to mark on order boards, make-up professional to apply to faces, and architects to produce blueprints. The possibilities seem limitless.

Questions 1–6

Complete the sentences below.

*Choose **ONE WORD ONLY** from the passage for each answer.*

- Graphite was found under a (1) _____ in Borrowdale.
- Ancient people used graphite to sign possession and number of (2) _____.
- The first pencil was graphite wrapped in (3) _____ or animal skin.
- In the eighteenth century, the (4) _____ protected the mines when the value of graphite was realised.

- During the reign of Elizabeth I, people were condemnable if they ⑤ _____ or received the "wad".
- Russian astronauts preferred ⑥ _____ pencils to write in outer space.

Questions 7–13

Do the following statements agree with the information given in the passage?

Write **TRUE**, **FALSE** or **NOT GIVEN**.

- ⑦ Italy is probably the first country in the world to make pencils.
- ⑧ Germany used various kinds of wood to make pencils.
- ⑨ Graphite makes a pencil harder and sharper.
- ⑩ Pencils are not produced any more since the reign of Elizabeth I.
- ⑪ Pencil was used during the first American space expedition.
- ⑫ American astronauts did not replace mechanical pencils immediately after the zero-gravity pencils were invented.
- ⑬ Pencils are unlikely to be used in the future.

The History of the Biro

How a frustrated journalist's search for a better pen led to a decades-long race across three continents

① **A** One chilly autumn morning in 1945, five thousand shoppers crowded the pavements outside Gimbels Department Store in New York City. The day before, Gimbels had taken out a full-page newspaper advertisement in the New York Times, announcing the sale of the first ballpoint pens in the United States. The new writing instrument was heralded as "fantastic ... miraculous ... guaranteed to write for two years without refilling!" Within six hours, Gimbels had sold its entire stock of ten thousand ballpoints at \$12.50 each - approximately \$130 at today's prices.

② **B** In fact, this 'new' pen was not new after all, and was just the latest development in a long search for the best way to deliver ink to paper. In 1884 Lewis Waterman had patented the fountain pen, giving him the sole rights to manufacture it. This marked a significant leap forward in writing technology, but fountain pens soon became notorious for leaking. In 1888, a leather tanner named John Loud devised and patented the first "rolling-pointed marker pen" for marking leather. Loud's design contained a reservoir of ink in a cartridge and a rotating ball point that was constantly bathed on one side with ink. Loud's pen was never manufactured, however, and over the next five decades, 350 additional patents were issued for similar ball-type pens, though none advanced beyond the design stage. Each had their own faults, but the major difficulty was the ink: if the ink was thin, the pens leaked, and if it was too thick, they clogged. Depending on the climate or air temperature, sometimes the pens would do both.

③ **C** Almost fifty years later, Ladislav and Georg Biro, two Hungarian brothers, came up with a solution to this problem. In 1935 Ladislav Biro was working as a journalist, editing a small newspaper. He found himself becoming more and more frustrated by the amount of time he wasted filling fountain pens with ink and cleaning up ink smudges. What's more, the sharp tip of his fountain pen often scratched or tore through the thin newsprint paper. Ladislav and Georg (a chemist) set about making models of new pen designs and creating better inks to use in them. Ladislav had observed that the type of ink used in newspaper printing dried rapidly, leaving the paper dry and smudge-free. He was determined to construct a pen using the same type of ink. However, the thicker ink would not flow from a regular pen nib so he had to develop a new type of point. Biro came up with the idea of fitting his pen with a tiny ball bearing in its tip. As the pen moved along the paper, the ball bearing rotated and picked up ink from the ink cartridge which it delivered to the paper.

④ **D** The first Biro pen, like the designs that had gone before it, relied on gravity for the ink to flow to the ball bearing at the tip. This meant that the pens only worked when they were held straight up, and even then, the ink flow was sometimes too heavy, leaving big smudges of ink on the paper. The Biro brothers had a rethink and eventually devised a new design, which relied on capillary action rather than gravity to feed the ink. This meant that the ink could flow more smoothly to the tip and the pen could be held at an angle rather than straight up. In 1938, as World War II broke out, the Biro brothers fled to Argentina, where they applied for a patent for their pen and established their first factory.

⑤ **E** The Biros' pen soon came to the attention of American fighter pilots, who needed a new kind of pen to use at high altitudes. Apparently, it was ideal for pilots as it did not leak like the fountain pen and did not have to be refilled frequently. The United States Department of War contacted several American companies, asking them to manufacture a similar writing instrument in the U.S. Thus, fortune smiled on the Biro brothers in May 1945, when the American company 'Eversharp' paid them \$500,000 for the exclusive manufacturing and marketing rights of the Biro ballpoint for the North American market. Eversharp were slow to put their pen into production, however, and this delay ultimately cost them their competitive advantage.

⑥ **F** Meanwhile, in June 1945 an American named Milton Reynolds stumbled upon the Biro pen while on vacation in Buenos Aires. Immediately seeing its commercial potential, he bought several pens and returned to Chicago, where he discovered that Loud's original 1888 patent had long since expired. This meant that the ballpoint was now in the public domain, and he, therefore, wasted no time making a copy based on the Biro design. Establishing his pen company with just \$26,000, Reynolds quickly set up a factory with 300 workers who began production on 6th October 1945, stamping out pens from precious scraps of aluminum that hadn't been used during the war for military equipment or weapons. Just 23 days later, it was Reynolds' ballpoint pen that caused the stampede at Gimbels Department Store. Following the ballpoint's debut in New York City, Eversharp challenged Reynolds in the law courts, but lost the case because the Biro brothers had failed to secure a U.S. patent on their invention.

Questions 1–6

Reading Passage 1 has six paragraphs, A–F.

Choose the correct heading for each paragraph from the list of headings below. Write the correct number, i–ix, in the circle before each paragraph.

- i Fountain pens are history
- ii Fame at last for the Biro brothers

- iii A holiday helps bring the biro to America
- iv A second design and a new country
- v War halts progress
- vi Dissatisfaction leads to a new invention
- vii Big claims bring big crowds
- viii A government request brings a change of ownership
- ix Many patents and many problems

Questions 7–9

Choose the correct answer, A, B, C or D.

- 7 The problem with the ballpoint pens invented between 1888 and 1935 was that
 - A they cost a great deal of money to manufacture.
 - B the technology to manufacture them did not exist.
 - C they could not write on ordinary paper.
 - D they were affected by weather conditions.
- 8 The design of the Biro brothers' first pen
 - A was similar to previous pens.
 - B was based on capillary action.
 - C worked with heavy or light inks.
 - D worked when slanted slightly.
- 9 Milton Reynolds was able to copy the Biro brothers' design because
 - A the Biro brothers' original patent was out of date.
 - B it was legal to copy other designs at the time.
 - C they did not have a patent for North America.
 - D the Biro brothers gave him permission.

Questions 10–12

Answer the questions below.

Choose **NO MORE THAN TWO WORDS AND/OR A NUMBER** from the passage for each answer.

- 10 What material was the first ballpoint pen designed to write on?
- 11 Where did the Biro brothers open their first factory?
- 12 In what year did the first American biro factory begin production?

Passage 45

The History of the Picnic

How an 18th-century French indoor pastime evolved into the outdoor meal we know today

Picnics became popular in France in the 18th century. A favourite pastime of the upper classes, they were initially purely indoor affairs, held at home or in hired rooms. It was assumed that invited guests would contribute - they could either bring a dish or pay a share of the overall cost. These picnics were often marked by conversation and wit and, as such, were associated with intellectual refinement. At larger gatherings, there was also music, with the result that 'picnic' came to mean the same as a ball or party. Picnics were, however, nothing if not adaptable. When the first restaurants appeared, picnickers quickly began going to them and having their picnics there.

The French Revolution in 1789 changed everything, with many wealthy French aristocrats fleeing abroad. Some went to Austria, others to Prussia or even America, but the majority chose to go to England. Settling mainly in London, some were now in need of money, but they did their best to maintain their old way of life - and in doing so, introduced the picnic to England. This led to several important developments. The first, and shortest-lived, was that the picnic became decidedly less refined and noisier. This was thanks to a group of around 200 wealthy young people who loved France and French culture and, in late 1801, founded the 'Pic Nic Society'. Held in hired rooms on Tottenham Street in London, the society's gatherings were intended to impress. Every member was required to bring a costly dish and something to drink. As The Times newspaper reported, each member tried to outdo the others in luxury and expense. After dinner, the principal entertainment was always a play. The productions were, admittedly, quite basic: the stage was small and crudely built, unlike in a theatre, and the actors were never professionals. But even so, they were wildly popular.

Another development in England in the 1800s was more profound. Picnics were adopted by the emerging middle classes and moved outside. What caused this change is unclear, but the most likely explanation is that the English simply applied a fashionable French word to a pre-existing practice, without being aware of what it meant. One of the results was that music ceased to be provided at picnics, which became just rather plain, open-air meals to which people were invited by a host. Another was that it became both more 'genteel' and - thanks to the idealisation of the countryside - more innocent.

In the decades which followed, this form of picnic found its way to the United States. As was only to be expected, it remained a refined activity most popular with the middle classes in cities; but unlike in England, its setting was associated more with a flight from civilisation than with childlike simplicity. Though still tinged with innocence, paintings of

picnics by well-known American artists - such as those by Winslow Homer - tended, therefore, to contain fewer gently rolling fields than shown by English artists and instead had more gnarled trees and rough mountains.

Picnics in the open air were less well received elsewhere, however. Though the restoration of the French monarchy was accompanied by the return of picnicking to France, their indoor version still predominated, especially among the aristocracy. But the legacy of revolutionary equality in France also opened up the picnic there to others lower down the social scale. By the mid-19th century, even members of the working classes in France were starting to adopt the practice. When outdoor picnics eventually did begin to gain ground in France, they were regarded with considered suspicion. Perhaps because of a growing reaction in France against the Romantic idealisation of nature, they were not seen as innocent and wholesome, as they were in England and the United States.

In general, not until the early 20th century did the outdoor picnic prevail over the indoor kind. Particularly in England, the development of different forms of transport and the acceleration of social change made the countryside accessible to a far greater proportion of the population. Before long, the popularity of the picnic had grown so much that specialist picnic baskets were being produced for the mass market and their contents were becoming more standardised.

Over the last 100 years, picnics have undergone still further changes - largely as a result of the international availability of a huge variety of food to be transported over long distances. Picnics will undoubtedly continue to evolve in the future. Shifting patterns of trade will almost certainly continue to change the foods we pack for a picnic. But whatever happens, one thing is certain: as long as there are friends with whom to share a meal, there will be few things so pleasant as a picnic lunch.

Questions 1–8

Do the following statements agree with the information given in Reading Passage 1?

*Write **TRUE** if the statement agrees with the information, **FALSE** if the statement contradicts the information, or **NOT GIVEN** if there is no information on this.*

- ① The first picnics in 18th-century France were held in parks.
- ② People who attended early French picnics were expected to bring food or help with the expense.
- ③ The hosts of early French picnics often asked their guests to play music.
- ④ The picnic was brought to England by aristocrats escaping the French Revolution.
- ⑤ Many complaints were received about the noise at the first English picnics.

- 6 People who wanted to join the Pic Nic Society had to have the approval of existing members.
- 7 The picnics held by the Pic Nic Society were extravagant events.
- 8 The picnics given by the Pic Nic Society took place in theatres.

Questions 9–16

Complete the notes below.

Choose **ONE WORD ONLY** from the passage for each answer.

Changes in picnics

England

- in the 1800s, picnics began to be held 9 _____.
- picnics no longer usually included 10 _____ for entertainment.

In the USA

- picnics were a favourite activity for people living in 11 _____.
- American 12 _____ of picnics were different from those by English artists.

Return to France

- indoor picnics were still preferred by members of the 13 _____.

20th century onwards

- new kinds of 14 _____ meant that picnics in the countryside became more common.
- 15 _____ designed especially for use by picnickers were widely sold.
- there is now greater 16 _____ in the food that is used for picnics.

Passage 46

The Life of Louis Braille

Teacher and advocate for the blind, Louis Braille designed the coding system, based on raised dots, by which the blind can read through touch

Louis Braille was born in Coupvray, France on January 4, 1809, the only child of Louis and Constance Braille. His father made leather saddles and harnesses for farmers in the area. At the age of three, while playing in his father's shop, young Louis was struck in the eye by a tool. Within weeks of the accident, the infection that had developed in his eyes took away his sight completely. Although there were few options for blind people at that time, Braille's parents wanted their son to be educated. He attended school in their village with sighted children and followed the lessons through listening alone. He was an excellent student, mostly because of his exceptional memory.

In 1819, Braille received a scholarship that enabled him to attend the Institut National des Jeunes Aveugles (National Institute for Blind Youth) in Paris, one of the first schools of its kind in the world. At that time, the school had only a handful of books for the blind, and these were massive volumes with raised letters of the standard alphabet pressed into the paper. Reading them was slow and laborious, and writing was nearly impossible. Moreover, the books were extremely expensive to produce, and the library contained no more than fourteen such volumes.

While at the institute, Braille learned of a communication system developed by Charles Barbier, a French army captain. Barbier had invented 'night writing' — a code of raised dots and dashes that soldiers could use to read and send messages in complete darkness without speaking. The system was based on grids of twelve dots, but it proved too complex for military use. Braille immediately saw the potential for the blind. He spent the next several years simplifying Barbier's system, reducing it to a six-dot cell (two columns of three dots each). This six-dot cell could produce sixty-four different combinations, enough to represent the letters of the alphabet, punctuation marks, numbers, and even musical notation.

By 1824, when Braille was just fifteen years old, he had completed the basic framework of his system. He later published a book explaining his method, titled *Method of Writing Words, Music, and Plain Songs by Means of Dots, for the Use of the Blind and Arranged for Them*. Despite its brilliance, the system was not immediately accepted. Many sighted teachers at the institute feared that Braille's code would isolate blind students from the sighted world. Instead, they continued to teach raised print letters. In 1839, Braille published further refinements, including symbols for mathematics and music, but the institute's directors remained resistant.

Tragically, Braille's own health began to decline. He had suffered from tuberculosis since his early twenties, and the disease gradually weakened him. In 1844, during a public exhibition, Braille demonstrated his system successfully, and the audience applauded enthusiastically. However, official recognition remained elusive. In 1852, at the age of 43, Louis Braille died from his illness. He never witnessed the widespread adoption of his invention.

After his death, Braille's system gradually gained acceptance. In 1854, one year after his passing, the Institut National officially adopted the Braille code. By the 1860s, schools for the blind across Europe and North America were teaching it. Over the ensuing decades, Braille has been adapted into nearly every major language on earth. Today, millions of blind and visually impaired people around the world read and write using the six-dot system that one determined teenager invented over two centuries ago. Louis Braille's legacy is not merely a code — it is the gift of literacy, independence, and dignity.

Questions 1–5

Do the following statements agree with the information given in the passage?

Write **TRUE**, **FALSE** or **NOT GIVEN**.

- ① Louis Braille was the oldest child in his family.
- ② He lost his sight immediately after the accident in his father's workshop.
- ③ Braille attended a school for blind children before going to Paris.
- ④ Charles Barbier originally designed his code for military purposes.
- ⑤ The Institut National officially adopted the Braille system before Braille died.

Questions 6–9

Complete the sentences below.

Use **NO MORE THAN TWO WORDS** from the passage for each answer.

- Braille's father produced leather ⑥ _____ for local farmers.
- Before Braille's system, books for the blind used raised letters of the ⑦ _____.
- Barbier's 'night writing' was based on grids of ⑧ _____ dots.
- Braille published his first complete system when he was only ⑨ _____ years old.

Questions 10–13

Choose the correct letter, **A**, **B**, **C** or **D**.

- 10 Why did Braille succeed as a student in his village school?
- A He had excellent teachers.
 - B He possessed an extraordinary memory.
 - C He used special books for the blind.
 - D His parents helped him at home.
- 11 What was a major disadvantage of raised-print books?
- A They were too small to read.
 - B They could only be used during the day.
 - C They were slow to read and very expensive.
 - D They did not include musical notation.
- 12 Why did some teachers initially reject Braille's system?
- A They thought it was too difficult to learn.
 - B They feared it would separate blind people from the sighted world.
 - C They had invented their own system for reading.
 - D They believed blind people should not learn to read.
- 13 What is the author's main purpose in writing this passage?
- A To describe the history of blind education in France.
 - B To compare different reading systems for the blind.
 - C To explain the life and achievements of Louis Braille.
 - D To argue that Braille was not recognised during his lifetime.

Passage 47

The National Parks of America

How a 19th-century vision for protecting the natural world grew into one of the country's most influential institutions

America's national parks are one of the country's most remarkable and influential initiatives. The concept of the national park originated in America, though it was quickly replicated by similar parks in Canada, Finland, New Zealand, Australia and elsewhere. The first American national park was Yellowstone, created in 1872. Yellowstone comprises some 3,500 square miles of wilderness and today attracts more than four million visitors a year. Since 1872, nearly 60 additional national parks have been established in the USA, incorporating almost every type of natural environment, from mountains to deserts, lakes, rivers and coastlines. They are administered by the National Parks Service (NPS), which also manages significant monuments and historic structures such as the Statue of Liberty, the Gettysburg Museum and Ford's Theatre, where President Lincoln was assassinated.

The creation of the national parks came in response to a growing attachment to the natural world that was developing in American society. In 1832, the painter George Catlin dreamed of creating 'a nation's park' to preserve the grasslands in the state of South Dakota. Catlin failed to see such a park established in his lifetime, but his vision lived on. Four years later the writer Ralph Waldo Emerson published an important essay entitled 'Nature' in which he argued that natural landscapes should be protected rather than exploited for agriculture and commerce. The campaigner Frederick Olmsted went further, claiming that every American, not only the wealthy and powerful, had the right to access places of natural beauty. The creation of a public park in the Yosemite Valley was directly the result of his work.

The development of the national parks was a process that continued through the late 19th century and the early 20th century. The establishment of the first park at Yellowstone in 1872 was perhaps the defining moment in this process. The Yellowstone national park not only preserved the headwaters of the Mississippi, Colorado and Snake rivers, but also provided a sanctuary for countless endangered species by imposing a ban on hunting in all its forms. Today such an initiative would be commonplace but in these earliest days of the conservation movement, many Americans still regarded it as their right to shoot or trap wild animals.

Yellowstone was followed by Sequoia (1890), Mount Rainier (1899), Crater Lake (1902) and Wind Cave (1903), among many other parks. But while the creation of additional parks enjoyed wide popular support, there were frequent disagreements about their purpose. For some, the primary goal was the protection of the natural world, while for others the parks were about providing public access to nature. These two goals were at

times contradictory and led to a variety of competing voices and factions. In recognition of the difficulties of running these vast wilderness areas, their management was for many years placed in the hands of the military, rather than the federal or state government.

The late 19th century was also a time when railroads were being constructed across America. Increasingly, these ran through even the most remote areas, where many parks were located, making tourism possible in these locations for the very first time. While this was welcomed by many park supporters, others were concerned that visitors should appreciate the parks in particular ways. Nathaniel Langford was instrumental in setting up Yellowstone and saw the park as an ideal opportunity for the education of ordinary citizens in all aspects of the natural world and environmental conservation, which was his number one priority.

In the early 20th century, the first really significant development was the setting up of the NPS in 1916. Now for the first time a single body was charged with administering the network and managing the numerous groups and interests that had a stake in it. Stephen Mather was appointed as the first Superintendent of Parks and under his leadership seven new parks were established, including Grand Canyon, Hawaii and Mount McKinley. However, arguably even more important was Mather's transformation of the hitherto poorly managed finances of the parks.

The NPS generally saw the primary role of the parks as being environmental protection. For this reason, most human activities were regulated – farming and forestry were certainly not allowed. However, the controversial decision to permit the construction of dams in national parks was made on the basis that the resulting supplies of water and hydroelectric power would be in the public interest. Certainly, visitor numbers were unaffected as park admissions increased every year. Then in the 1930s the number of visitors expanded dramatically, largely because mass-produced automobiles became affordable, allowing ever more city-dwellers to reach the park network. With this, the national parks assumed a pre-eminent position in the social fabric of America, a legacy that continues to the present day.

Questions 1–6

Do the following statements agree with the information given in the passage?

*Write **TRUE** if the statement agrees with the information, **FALSE** if the statement contradicts the information, or **NOT GIVEN** if there is no information on this.*

- 1 America copied the idea of national parks from other countries.
- 2 Yellowstone is the most frequently visited national park in the USA.
- 3 The NPS is responsible for certain buildings as well as natural landscapes.

- ④ Catlin was criticised by some people for proposing 'a nation's park'.
- ⑤ Emerson believed that farms and businesses should be allowed to use land freely.
- ⑥ Olmsted helped establish a park in the Yosemite Valley.

Questions 7–13

Complete the notes.

*Write **ONE WORD ONLY** from the text in each gap.*

Development of the National Parks

Late 19th Century

- in Yellowstone Park, every type of ⑦ _____ was prohibited
- Yellowstone and other parks were administered by the ⑧ _____
- the development of a railroad system encouraged new kinds of ⑨ _____
- Langford believed that public ⑩ _____ was the most important role of the parks

Early 20th Century

- Mather greatly improved the ⑪ _____ of the parks
- the NPS allowed some ⑫ _____ to be built in national parks
- in the 1930s, the popularity of the parks increased with the availability of ⑬ _____

Passage 48

The Origins of Laughter

While joking and wit are uniquely human inventions, laughter certainly is not

While joking and wit are uniquely human inventions, laughter certainly is not. Other creatures, including chimpanzees, gorillas and even rats, laugh. The fact that they laugh suggests that laughter has been around for a lot longer than we have.

There is no doubt that laughing typically involves groups of people. "Laughter evolved as a signal to others - it almost disappears when we are alone," says Robert Provine, a neuroscientist at the University of Maryland. Provine found that most laughter comes as a polite reaction to everyday remarks such as "see you later", rather than anything particularly funny. And the way we laugh depends on the company we're keeping. Men tend to laugh longer and harder when they are with other men, perhaps as a way of bonding. Women tend to laugh more and at a higher pitch when men are present, possibly indicating flirtation or even submission.

To find the origins of laughter, Provine believes we need to look at play. He points out that the masters of laughing are children, and nowhere is their talent more obvious than in the boisterous antics, and the original context is play. Well-known primate watchers, including Dian Fossey and Jane Goodall, have long argued that chimps laugh while at play. The sound they produce is known as a pant laugh. It seems obvious when you watch their behavior - they even have the same ticklish spots as we do. But after removing the context, the parallel between human laughter and a chimp's characteristic pant laugh is not so clear. When Provine played a tape of the pant laughs to 119 of his students, for example, only two guessed correctly what it was.

These findings underline how chimp and human laughter vary. When we laugh the sound is usually produced by chopping up a single exhalation into a series of shorter ones, with one sound produced on each inward and outward breath. The question is: does this pant laughter have the same source as our own laughter? New research lends weight to the idea that it does. The findings come from Elke Zimmerman, head of the Institute for Zoology in Germany, who compared the sounds made by babies and chimpanzees in response to tickling during the first year of their life. Using sound spectrographs to reveal the pitch and intensity of vocalizations, she discovered that chimp and human baby laughter follow broadly the same pattern. Zimmerman believes the closeness of baby laughter to chimp laughter supports the idea that laughter was around long before humans arrived on the scene. What started simply as a modification of breathing associated with enjoyable and playful interactions has acquired a symbolic meaning as an indicator of pleasure.

Pinpointing when laughter developed is another matter. Humans and chimps share a common ancestor that lived perhaps 8 million years ago, but animals might have been laughing long before that. More distantly related primates, including gorillas, laugh, and anecdotal evidence suggests that other social mammals can do too. Scientists are currently testing such stories with a comparative analysis of just how common laughter is among animals. So far, though, the most compelling evidence for laughter beyond primates comes from research done by Jaak Panksepp from Bowling Green State University, Ohio, into the ultrasonic chirps produced by rats during play and in response to tickling. All this still doesn't answer the question of why we laugh at all. One idea is that laughter and tickling originated as a way of sealing the relationship between mother and child. Another is that the reflex response to tickling is protective, alerting us to the presence of crawling creatures that might harm us or compelling us to defend the parts of our bodies that are most vulnerable in hand-to-hand combat. But the idea that has gained the most popularity in recent years is that laughter in response to tickling is a way for two individuals to signal and test their trust in one another. This hypothesis starts from the observation that although a little tickle can be enjoyable, if it goes on too long it can be torture. By engaging in a bout of tickling, we put ourselves at the mercy of another individual, and laughing is what makes it a reliable signal of trust, according to Tom Flamson, a laughter researcher at the University of California, Los Angeles. "Even in rats, laughter, tickle, play and trust are linked. Rats chirp a lot when they play," says Flamson. "These chirps can be aroused by tickling. And they get bonded to us as a result, which certainly seems like a show of trust." We'll never know which animal laughed the first laugh, or why. But we can be sure it wasn't in response to a prehistoric joke. The funny thing is that while the origins of laughter are probably quite serious, we owe human laughter and our language-based humor to the same unique skill. While other animals pant, we alone can control our breath well enough to produce the sound of laughter. Without that control there would also be no speech - and no jokes to endure.

Questions 1–6

Look at the following research findings (Questions 1–6) and the list of people below.

Match each finding with the correct person, **A**, **B**, **C** or **D**. **NB** You may use any letter more than once.

List of People

- A** Provine
- B** Zimmerman
- C** Panksepp
- D** Flamson

Q	Statement	Compiled by @bekhruzposts
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1	Babies and some animals produce laughter which sounds similar.	
2	Primates are not the only animals who produce laughter.	
3	Laughter can be used to show that we feel safe and secure with others.	
4	Most human laughter is not a response to a humorous situation.	
5	Animal laughter evolved before human laughter.	
6	Laughter is a social activity.	

Questions 7–10

Complete the summary using the list of words, **A–K**, below.

Some scientists believe that laughter first developed out of (7) _____. Research has revealed that human and chimp laughter may have the same (8) _____. Scientists have long been aware that (9) _____ laugh, but it now appears that laughter might be more widespread than once thought. Although the reasons why humans started to laugh are still unknown, it seems that laughter may result from the (10) _____ we feel with another person.

List of Words

A combat	B chirps	C pitch	D origins	E play	F rats
G primates	H confidence	I fear	J babies	K tickling	

Questions 11–13

Do the following statements agree with the information given in Reading Passage 1?

Write **TRUE** if the statement agrees with the information, **FALSE** if the statement contradicts the information, or **NOT GIVEN** if there is no information on this.

- (11) Both men and women laugh more when they are with members of the same sex.
- (12) Primates lack sufficient breath control to be able to produce laughs the way humans do.
- (13) Chimpanzees produce laughter in a wider range of situations than rats do.

Passage 49

The Origins of Weather Forecasting

How Admiral Robert FitzRoy brought forecasting to the British public

The world's first public weather forecast appeared in *The Times* newspaper over 150 years ago: the beginning of a service that gives everyone something to comment on. The pioneering forecast on August 1, 1861, was introduced with a deceptively simple heading: 'General weather probable during next two days.' But those few words started an ongoing controversy that still continues today — after all, everyone remembers the forecasts that go wrong, but rarely the ones that are right.

The first forecast was the idea of Admiral Robert FitzRoy, head of the newly founded Meteorological Department, and one of Britain's greatest but least-known heroes. It was a brave undertaking, because in those days predicting the weather had gained a reputation similar to astrology or fortune-telling. The few people who attempted it had become a national joke when their forecasts inevitably went wrong.

FitzRoy was different, though. His background was faultless, most famously his captaining of the ship HMS Beagle on Charles Darwin's voyage around the world, which eventually led to the publication of Darwin's well-known book *On the Origin of Species*. When FitzRoy retired from active service as an admiral in Britain's navy, he was looking for a new direction. He subsequently secured a job as chief statistician at the Meteorological Department, part of the Board of Trade. Parliament had voted to set up and fund the Meteorological Department on June 30, 1854, to chart the safest sailing routes across the Atlantic Ocean. However, the British Government was not impressed with the idea of collecting weather reports and establishing a weather forecast, and the suggestion by one Member of Parliament for this was met with laughter from the other MPs.

FitzRoy was given a staff of three clerks in a small office in London, but he soon grew bored with the tedium of transcribing thousands of ships' reports. Instead, he focused on a much more pressing issue — the huge loss of life at sea from storms. This problem came to a head in 1859, when gale-force winds sank the ship named Royal Charter, with the loss of 430 lives, within sight of the shore. Even in those days when ships regularly sank in storms, the loss of the Royal Charter was considered a national tragedy. FitzRoy stirred up feelings even further when he wrote a letter to *The Times* newspaper explaining that the storm had been entirely predictable.

That same year, a total of 1,645 lives were lost off the British coast, and FitzRoy wrote repeatedly to *The Times* pushing for a storm-warning service for shipping. This was just the right moment for a national forecasting service. Knowledge of the weather was

improving, barometer readings gave warning of approaching storms, and the invention of the electric telegraph and Morse code in 1844 gave instant communication for the first time.

So in February 1861, FitzRoy pressed ahead with a storm-warning service. Previous weather reports had been unreliable, but FitzRoy's innovation was to issue ships with standardised instruments which allowed for accurate weather reports. At exactly the same time each day the information was sent to him by telegraph, and he was able to draw up more detailed weather maps. The approach of an incoming storm could be recognised, and from London, FitzRoy could telegraph warnings to ports on the coast, where an ingenious system of flags was hoisted up high for passing ships to read. This represented the biggest advance in shipping safety after the introduction of the lifeboat, and in subsequent years the number of lives lost around Britain fell by about a third. FitzRoy became a hero to the fishing and maritime fleets. Encouraged by his storm-warning success, FitzRoy then started a public weather forecasting service. As he had a close connection with *The Times*, this newspaper introduced his daily forecast in August 1861.

It attracted huge attention, including from Queen Victoria, who regularly sent messengers to FitzRoy's office to get a forecast for the sea crossing over the Solent to an island which she liked to visit off the south coast of England. He was also popular with ordinary fishermen, who were happy not to be out at sea in bad weather. The owners of fishing vessels, however, were not supportive of FitzRoy's forecasts, which often caused delays for them when bad weather was forecast.

Nevertheless, weather forecasting at that time was very difficult, and one mistake could result in much bad publicity. It wasn't long before some forecasts turned out wrong, and the letters pages of *The Times* were full of heated correspondence.

It is indisputable that FitzRoy was way ahead of his time, but it is only in recent years that he has been recognized as a towering figure in meteorology, with the honour of an area of sea being given his name in the shipping forecast (a radio broadcast of weather reports for the seas around Britain). His extraordinary determination to get forecasting established cost him dearly though, and he died prematurely. But his legacy lives on, because late in his life he published a book about weather forecasting which was recognized by other scientists to be well ahead of its time.

Questions 1–5

Do the following statements agree with the information given in the passage?

*Write **TRUE** if the statement agrees with the information, **FALSE** if the statement contradicts the information, or **NOT GIVEN** if there is no information on this.*

- ① In Robert FitzRoy's time, weather forecasting was a respected profession.
- ② When FitzRoy retired from the navy, he had already had a successful career at sea.
- ③ FitzRoy faced competition from other applicants for the chief statistician's job.
- ④ The British Parliament supported the idea of regular weather forecasts in 1854.
- ⑤ Progress in technology made it a good time for publicly available weather forecasting to begin.

Questions 6–13

Complete the notes below.

Choose **ONE WORD ONLY** from the passage for each answer.

February 1861: Storm warning service

- – FitzRoy received weather reports produced by ships using reliable ⑥ _____
- – simultaneous weather reports were taken daily
- – storm warnings were telegraphed to ⑦ _____ along the shore
- – signals for ships were displayed using raised ⑧ _____
- – the storm warning service was the greatest development in sea safety since the ⑨ _____ was introduced

August 1861: Public weather forecasting service

- – very popular in *The Times* newspaper
- – frequently used by the English Queen for safe trips to a favourite ⑩ _____
- – FitzRoy's forecasts were popular with fishermen but not with boat ⑪ _____
- – In modern times FitzRoy's ⑫ _____ is used for a marine area
- – FitzRoy produced a ⑬ _____ which was respected by the scientific community

Passage 50

The Pyramid of Cestius

A 2000-year-old pyramid in the city of Rome has been restored by archaeologists

Though Rome draws tourists from around the world to its many impressive sites, one notable monument there has never attracted nearly as much interest: The Pyramid of Cestius. But why would there be a pyramid in Italy? After the Roman conquest of Egypt in 30 B.C, Egyptian architectural style became the fashion in Rome.

Though obelisks and other monuments inspired by Egypt's great pyramids sprung up around the city, only two actual pyramids are known to have been built. The only one left standing, the pyramid of Cestius, was designed as the burial pyramid for a Roman politician named Caius Cestius, who ordered that the building work be completed within a period of 330 days.

Construction took place at some point between 18 B.C. and 12 B.C. Cestius' pyramid had a layer of white Carrara marble on the outside, and was constructed from brick held together by a basic kind of cement on the inside. One of the things that strikes you when you look at the pyramid is how steep it is, so that the shape of Cestius' pyramid is quite unlike that of typical Egyptian ones. This is a difference that could have been the result of inaccurate information sent back to Rome by soldiers who saw the pyramids in person in Egypt. Alternatively, Roman builders could have drawn inspiration from the pyramids in Nubia, a region located in what is today northern Sudan and southern Egypt.

At the time of its construction, since there was a strict Roman law prohibiting the placement of tombs within the city itself, the pyramid of Cestius would have stood in the countryside. Rome grew enormously during the next two centuries, and by the 3rd century A.D., the pyramid would have been surrounded by buildings. We also know that in the 3rd century A.D., the pyramid of Cestius was hidden behind a high wall on the orders of Emperor Aurelian; this probably helped it survive throughout the centuries to come, even as other ancient monuments disappeared.

By the Middle Ages, the pyramid was covered in vegetation and thick dirt, and popular myth had developed that it might be a tomb for one of the twin brothers Romulus and Remus, who were regarded as the men who had established the city of Rome. Cestius' actual tomb within the pyramid and the inscription identifying the pyramid as his, weren't rediscovered until the 1660's when the pyramid underwent restoration. During excavations, when trees and plants were cleared away, two marble bases were found in front of the pyramid, as well as fragments of bronze statues that had once stood on them on either side of the pyramid. The people employed to excavate the pyramid did not find the urn that would have contained Cestius' remains, but they did come across a tunnel. It

was quite possible, therefore, that robbers had at some earlier time removed the contents of the tomb. But while some of the features of Cestius' tomb no longer exist, at least the pyramid itself has survived.

Today, the foundations of Cestius' pyramid rest below street level near an intersection with heavy traffic, so that passing tourists and residents could easily fail to notice its full height of 119 feet. Across the intersection is the Piramide station, located on line B of the Rome Metro. In 2011, the Japanese clothing company entrepreneur Yuzo Yagi, president of Yagi Tsusho Ltd, announced his intention to help the Italian government pay for an ambitious renovation of the pyramid of Cestius. 'It's an act of gratitude', he later told journalists. "Our company has grown thanks to Italy." Work began at the site shortly after Yagi signed an official agreement with the Special Superintendency for the Archaeological Heritage of Rome, and was completed ahead of schedule thanks to his 2-million-euro contribution.

As archaeologist Leonardo Guarnieri explained to journalists, officials are now conducting tours of the newly renovated pyramid twice a month by reservation. Visitors who take advantage of the tour can make their way through a narrow corridor in order to enter the burial chamber itself. It is within these walls that they can admire the frescoes: watercolour paintings typical of the time. In the chamber, it is possible to make out four frescoes of the winged Roman goddess of war, Victoria, a figure from Roman legend, as well as of a series of vases, the type that would have been used for special rituals and purification purposes. We know from the writings of earlier visitors that there used to be more here, but the majority have disappeared over time.

Only one problem remains now that the restoration is complete. The white exterior of the pyramid of Cestius will have to be cleaned every few months to remove the layer of urban pollution. A team of free-climbers will be employed to do the job, in order to avoid placing builder's scaffolding around the newly welcoming monument.

Questions 1–7

Do the following statements agree with the information given in Reading Passage 1?

*Write **TRUE** if the statement agrees with the information, **FALSE** if the statement contradicts the information, or **NOT GIVEN** if there is no information on this.*

- ① The pyramid of Cestius has always been one of Rome's most popular tourist attractions.
- ② The construction of the pyramid was completed before Cestius' death.
- ③ In the Middle Ages, people thought an original founder of Rome was buried in the pyramid of Cestius.
- ④ Today the height of the pyramid is something that tourists and residents immediately notice.

- 5 Japanese businessman Yuzo Yagi was an admirer of both Italian and Egyptian architecture.
- 6 The restoration of the pyramid of Cestius, which was funded by Yuzo Yagi, finished earlier than expected.
- 7 Most of the original frescoes inside Cestius' tomb have survived to this day.

Questions 8–13

Complete the notes below.

Choose **ONE WORD ONLY** from the passage for each answer.

History of the pyramid of Cestius

Construction of Cestius' pyramid

- It was made from 8 ————— marble and cement.
- Its 9 ————— is different to the pyramids found in Egypt.
- It was originally built in the 10 ————— as building tombs in the city was forbidden.
- Emperor Aurelian ordered that a wall had to be built around it.

Restoration of Cestius' pyramid in the 1660s

- In the 1660s, some broken 11 ————— were found next to it.
- The 12 ————— inside the tomb suggests that robbers had been there.
- The frescoes show mythological scenes, and images of vases.

Restoration of Cestius' pyramid today

- A Japanese businessman paid for its restoration.
- Climbers are helping to get rid of signs of 13 —————.

Passage 51

The Seed Hunters

With a quarter of the world's plants set to vanish within the next 50 years, scientists are working against the clock to preserve the earth's botanical heritage

They travel the four corners of the globe, scouring jungles, forests and savannahs. But they're not looking for ancient artefacts, lost treasure or undiscovered tombs - just seed pods. It may lack the romantic allure of archaeology, or the whiff of danger that accompanies going after big game, but seed hunting is an increasingly serious business. Some seek seeds for profit - hunters in the employ of biotechnology firms, pharmaceutical companies and private corporations on the lookout for species that will yield the drugs or crops of the future. Others collect to conserve, working to halt the sad slide into extinction facing so many plant species.

Among the pioneers of this botanical treasure hunt was John Tradescant, an English royal gardener who brought back plants and seeds from his journeys abroad in the early 1600s. Later, the English botanist Sir Joseph Banks who was the first director of the Royal Botanic Gardens at Kew and travelled with Captain James Cook on his voyages near the end of the 18th century - was so driven to expand his collections that he sent botanists around the world at his own expense.

Those heady days of exploration and discovery may be over, but they have been replaced by a pressing need to preserve our natural history for the future. This modern mission drives hunters such as Dr Michiel van Slageren. He and three other seed hunters work at the Millennium Seed Bank, which is an £80 million international conservation project that aims to protect the world's most endangered wild plant species.

The group's headquarters are in a modern glass-and-concrete structure on a 200-hectare estate at Wakehurst Place in the West Sussex countryside in the UK. Within its underground vaults are 260 million dry seeds from 220 different countries, all stored at 20°C to survive for centuries. These stored seeds can be used in various ways, including the restoration of damaged environments, or in scientific research to find new benefits to society - in medicine, agriculture or local industry. Among the 5,100 species represented at the seed bank are virtually all of Britain's 1,400 native seed-bearing plants, the most complete such collection of any country's flora.

Overseen by the Royal Botanic Gardens, the Millennium Seed Bank is the world's largest wild-plant depository with over 24,000 species in its collection. The reason is simple: thanks to humanity's efforts, an estimated 25 per cent of the world's plants are on the verge of extinction and may vanish within 50 years. We're currently responsible for habitat destruction on an unprecedented scale and, during the past 400 years, plant species'

extinction rates have been about 70 times greater than those indicated by the geological record as being 'normal'. Experts predict that during the next 50 years a further one billion hectares of wilderness will be converted to farmland in developing countries alone. The implications of this loss are enormous. Besides providing staple crops for human consumption, plants are a source of many medicines, and the principal supply of fuel and building materials in many parts of the world. They also protect and improve the quality of the soil and help regulate the climate. Yet, across the globe, plant species are being driven to extinction before their potential benefits are discovered.

The World Conservation Union has listed 5,714 threatened plant species worldwide, but it admits this is only scratching the surface. With only four per cent of the world's described plants having been evaluated, the true number of threatened species is sure to be much higher. In the UK alone, 300 wild plant species are classified as endangered. The Millennium Seed Bank aims to ensure that even if a plant becomes extinct in the wild, it won't be lost for ever.

Seed banks are an 'insurance policy' to protect the world's heritage for the future, explains Dr Paul Smith, another Royal Botanic Gardens seed hunter. "Seed conservation techniques were originally developed by farmers," he says. "Storage is the basis of what we do, conserving seeds until we can use them - just as in farming." Smith says there's no reason why any plant species should become extinct, given today's technology. But he admits the biggest challenge is finding, naming and categorising all the world's plants. And someone has to gather these seeds before it is too late. "There aren't a lot of people out there doing this," he says. "The key is to know the flora from a particular area, and that knowledge takes years to acquire."

When in the field, Royal Botanic Gardens collectors are guided by a target list, generally provided by the host country, of the seeds they need. Priorities are determined by economic importance, endangered status and distribution - although they will deviate from the list if the chance arises to gather an unknown plant they've stumbled upon.

Questions 1–6

Complete the summary below.

*Choose **NO MORE THAN THREE WORDS** from the passage for each answer.*

Seed Hunting

- Apart from the pressing need to collect seeds in an effort to protect certain plants from (1) _____, seed hunting also makes an important contribution to the development of potential (2) _____.
- Botanists and gardeners were some of the (3) _____ in the hunt for seeds. One of these, called (4) _____, was so keen that he personally provided funding for trips to collect seeds.
- Present-day seed hunters employed by the Millennium Seed Bank bring back seeds which are then stored in the (5) _____ beneath the bank's headquarters at Wakehurst Place. These seeds play a vital role in the re-construction of (6) _____ and also in various fields of scientific research.

Questions 7–12

Do the following statements agree with the information given in Reading Passage 1?

*Write **TRUE** if the statement agrees with the information, **FALSE** if the statement contradicts the information, or **NOT GIVEN** if there is no information on this.*

- (7) The reasons for collecting seeds have changed over the years.
- (8) The Millennium Seed Bank was the first wild plant repository to be set up.
- (9) A major threat to plant species is posed by the increasing spread of farmland.
- (10) The World Conservation Union has been restricted in its work by lack of resources.
- (11) The methods of seed conservation resemble those used in agriculture.
- (12) Seed banks will only be successful if the technology improves.

Question 13

*Choose **TWO** letters, **A–E**.*

*Which **TWO** of the following uses of plants are mentioned by the writer?*

OPTIONS

- A** to improve air quality
- B** as a source of energy
- C** to enhance our understanding of nature
- D** to provide food
- E** to discover more about species that have died out

Answer: 13

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Passage 52

The Story of Silk

The history of the world's most luxurious fabric, from ancient China to the present day

Silk is a fine, smooth material produced from the cocoons – soft protective shells – that are made by mulberry silkworms (insect larvae). Legend has it that it was Lei Tzu, wife of the Yellow Emperor, ruler of China in about 3000 BC, who discovered silkworms. One account of the story goes that as she was taking a walk in her husband's gardens, she discovered that silkworms were responsible for the destruction of several mulberry trees. She collected a number of cocoons and sat down to have a rest. It just so happened that while she was sipping some tea, one of the cocoons that she had collected landed in the hot tea and started to unravel into a fine thread. Lei Tzu found that she could wind this thread around her fingers. Subsequently, she persuaded her husband to allow her to rear silkworms on a grove of mulberry trees. She also devised a special reel to draw the fibres from the cocoon into a single thread so that they would be strong enough to be woven into fabric. While it is unknown just how much of this is true, it is certainly known that silk cultivation has existed in China for several millennia.

Originally, silkworm farming was solely restricted to women, and it was they who were responsible for the growing, harvesting and weaving. Silk quickly grew into a symbol of status, and originally, only royalty were entitled to have clothes made of silk. The rules were gradually relaxed over the years until finally during the Qing Dynasty (1644–1911 AD), even peasants, the lowest caste, were also entitled to wear silk. Sometime during the Han Dynasty (206 BC–220 AD), silk was so prized that it was also used as a unit of currency. Government officials were paid their salary in silk, and farmers paid their taxes in grain and silk. Silk was also used as diplomatic gifts by the emperor. Fishing lines, bowstrings, musical instruments and paper were all made using silk. The earliest indication of silk paper being used was discovered in the tomb of a noble who is estimated to have died around 168 AD.

Demand for this exotic fabric eventually created the lucrative trade route now known as the Silk Road, taking silk westward and bringing gold, silver and wool to the East. It was named the Silk Road after its most precious commodity, which was considered to be worth more than gold. The Silk Road stretched over 6,000 kilometres from Eastern China to the Mediterranean Sea, following the Great Wall of China, climbing the Pamir mountain range, crossing modern-day Afghanistan and going on to the Middle East, with a major trading market in Damascus. From there, the merchandise was shipped across the Mediterranean Sea. Few merchants travelled the entire route; goods were handled mostly by a series of middlemen.

With the mulberry silkworm being native to China, the country was the world's sole producer of silk for many hundreds of years. The secret of silk-making eventually reached the rest of the world via the Byzantine Empire, which ruled over the Mediterranean region of southern Europe, North Africa and the Middle East during the period 330–1453 AD. According to another legend, monks working for the Byzantine emperor Justinian smuggled silkworm eggs to Constantinople (Istanbul in modern-day Turkey) in 550 AD, concealed inside hollow bamboo walking canes. The Byzantines were as secretive as the Chinese, however, and for many centuries the weaving and trading of silk fabric was a strict imperial monopoly. Then in the seventh century, the Arabs conquered Persia, capturing their magnificent silks in the process. Silk production thus spread through Africa, Sicily and Spain as the Arabs swept through these lands. Andalusia in southern Spain was Europe's main silk-producing centre in the tenth century. By the thirteenth century, however, Italy had become Europe's leader in silk production and export. Venetian merchants traded extensively in silk and encouraged silk growers to settle in Italy. Even now, silk processed in the province of Como in northern Italy enjoys an esteemed reputation.

The nineteenth century and industrialisation saw the downfall of the European silk industry. Cheaper Japanese silk, trade in which was greatly facilitated by the opening of the Suez Canal, was one of the many factors driving the trend. Then in the twentieth century, new manmade fibres, such as nylon, started to be used in what had traditionally been silk products, such as stockings and parachutes. The two world wars, which interrupted the supply of raw material from Japan, also stifled the European silk industry. After the Second World War, Japan's silk production was restored, with improved production and quality of raw silk. Japan went on to remain the world's biggest producer of raw silk, and practically the only major exporter of raw silk, until the 1970s. However, in more recent decades, China has gradually recaptured its position as the world's biggest producer and exporter of raw silk and silk yarn. Today, around 125,000 metric tons of silk are produced in the world, and almost two thirds of that production takes place in China.

Questions 1–9

Complete the notes below.

*Choose **ONE WORD ONLY** from the passage for each answer.*

The Story of Silk

Early silk production in China

- Around 3000 BC, according to legend: silkworm cocoon fell into emperor's wife's
① _____.
- emperor's wife invented a ② _____ to pull out silk fibres.
- Only ③ _____ were allowed to produce silk.
- Only ④ _____ were allowed to wear silk.
- Silk used as a form of ⑤ _____ — e.g. farmers' taxes consisted partly of silk.
- Silk used for many purposes — e.g. evidence found of ⑥ _____ made from silk around 168 AD.

Silk reaches rest of world

- Merchants use Silk Road to take silk westward and bring back ⑦ _____ and precious metals.
- 550 AD: ⑧ _____ hide silkworm eggs in canes and take them to Constantinople.
- Silk production spreads across Middle East and Europe.
- 20th century: ⑨ _____ and other manmade fibres cause decline in silk production.

Questions 10–13

Do the following statements agree with the information given in Reading Passage 1?

*Write **TRUE** if the statement agrees with the information, **FALSE** if the statement contradicts the information, or **NOT GIVEN** if there is no information on this.*

- ⑩ Gold was the most valuable material transported along the Silk Road.
- ⑪ Most tradesmen only went along certain sections of the Silk Road.
- ⑫ The Byzantines spread the practice of silk production across the West.
- ⑬ Silk yarn makes up the majority of silk currently exported from China.

Passage 53

The Terracotta Army

How a chance discovery by farmers digging a well in 1974 uncovered one of the most astonishing archaeological finds in modern history

It was a prolonged drought in 1974 that forced farmer Yang Zhifa and his five brothers to start digging a well in a small wooded area just outside their village in Lingtong County near the city of Xi'an in Northwest China, eventually reaching a depth of 15 metres. On March 29th, they found something much more remarkable than water – the terracotta head of a Chinese warrior along with a heavily-corroded bronze arrowhead. They did not understand the significance of what they had found at the time, but the brothers had stumbled across the first emperor of China's Terracotta Army.

When local archaeologist and museum curator Zhao Kangmin got a phone call a couple of weeks later about the farmers' discovery, he knew he needed to get down to the dig site as quickly as possible. The farmers weren't archaeologists, and he was worried they would unwittingly damage the site as they continued to dig for treasure. Zhao jumped on his bicycle and raced to the site. Once there, his worst fears were realized. The brothers had indeed continued to dig, and the site was now littered with hundreds of fragments of shattered terracotta. He told the farmers that they had to stop what they were doing immediately. They were reluctant to agree, but finally did so.

Zhao gathered the broken fragments together and loaded them onto a truck to transport them to the Lingtong County Museum where he worked. Over the course of the next few days, Zhao painstakingly put the fragments back together. When he was finished, two imposing, 1.78-metre-tall figures stood before him. This was an incredible find, but he wondered if it was just the tip of the iceberg. To find out, he assigned a small team to start work on excavating the site. It soon became clear, as more soil was cleared and more figures and fragments of figures emerged, that far from there being just a few figures buried underground, there were potentially hundreds.

Zhao was uncertain whether the government in Beijing would allocate resources to a large-scale excavation and decided to continue the work with a small team. However, when a journalist visiting the area heard about his work and published an article about the site, word travelled fast. The authorities were quick to recognise the importance of the find and ordered a full excavation of the site. Soon, the dig was teeming with archaeologists. It was quickly dawning on everyone that the farmers had stumbled upon one of the most astonishing archaeological finds in modern history. What they had found was an army cast in terracotta – made to defend the first emperor of China, Qin Shi Huang, in the afterlife.

Over the months that followed, four pits were uncovered. To the astonishment of everyone involved, what was unearthed was an entire army, complete with horses and chariots. The first pit revealed the main bulk of the emperor's army – six thousand warriors in total arranged in battle formation, and scattered around their feet a huge collection of swords. The second pit contained more infantry units, alongside cavalry units and chariots pulled by teams of terracotta horses. The third pit revealed high-ranking officers and an elaborate bronze war chariot. The fourth and final pit was empty. In total, the emperor's army numbered eight thousand strong, all lying undiscovered for 2,200 years, buried just a kilometre and a half away from the giant burial mound that houses the mausoleum of emperor Qin Shi Huang.

As more and more of the figures were painstakingly uncovered, one thing became clear – no two were the same. Built on an industrial scale in various workshops in the late 3rd century BC, each soldier's face had been individually finished off by hand to give it a unique look. Once cast and put together, the soldiers had then been painted to make them look as realistic as possible. Unfortunately, this paint quickly flaked off from the figures when they were unearthed, leading to criticism about the fact that insufficient environmental research had been undertaken before the figures were exposed to the elements.

A year after the discovery of the army, the Chinese government decided to open up the site to the public. A museum was built, and a protective roof was constructed over the now fully-exposed pits. It wasn't long before tourists began to trickle into the small provincial city of Xi'an, eager to see this amazing sight. Soon word spread around the world, and tourists flew in to see the army in all its glory. The Terracotta Army was officially recognised as a UNESCO World Heritage Site in 1987. Today, it is one of the most popular tourist attractions in China, and on the occasions when groups of the terracotta soldiers have been allowed out of the country to go on display in national museums around the world, they have attracted hundreds of thousands of visitors.

Questions 1–8

Complete the notes below.

*Choose **ONE WORD ONLY** from the passage for each answer.*

The Terracotta Army

Discovery and early excavation

- Yang Zhifa and his brothers were creating a (1) _____ when they made their discovery.
- Together with the terracotta head they found a metal (2) _____.
- Zhao Kangmin put the terracotta pieces into a (3) _____ and took them to the museum.
- a (4) _____ became aware of the site and wrote about it.
- the government sent more archaeologists to the site.

Objects in the four pits

- Pit one: warriors and numerous (5) _____.
- Pit two: warriors, chariots and (6) _____.
- Pit three: warriors and a chariot.
- Pit four: empty.

How the terracotta figures were made

- mass produced in a number of (7) _____.
- the (8) _____ of individual figures completed by hand to make them look different.

Questions 9–16

Do the following statements agree with the information given in Reading Passage 1?

*Write **TRUE** if the statement agrees with the information, **FALSE** if the statement contradicts the information, or **NOT GIVEN** if there is no information on this.*

- (9) Yang Zhifa and his brothers realised immediately that their discovery was important.
- (10) When Zhao Kangmin arrived at the site, he knew he had been right to be concerned.
- (11) Yang and his brothers carried out Zhao's order straight away.
- (12) After he had assembled two terracotta figures, Zhao sent other people to continue digging.
- (13) The government sent archaeologists from every part of China to work on the site.
- (14) The figures found in the third pit represented the leaders of the emperor's army.

- ⑮ Extensive research was carried out to discover how excavation would affect the paint on the figures.
- ⑯ Large groups of tourists need special permission to visit the site.

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Passage 54

The Trans-Australian Railway

How a single line across the Nullarbor Plain finally connected Perth to the rest of Australia

Up until the first part of the 20th century, communications across the Australian continent were relatively undeveloped. Perth, the capital of Western Australia, remained isolated from Australia's eastern states until the Trans-Australian Railway (TAR) was built across the arid area known as the Nullarbor (No Trees) Plain. During the 19th century, the only form of transport between the west coast and the eastern cities was a long and often rough sea voyage across the Great Australian Bight, while the first east-west crossing of the Nullarbor Plain on foot was achieved by the English explorer Edward Eyre and his indigenous Australian guide Wylie in 1841. In 1877 a telegraph line was built across the plain, and it was this man-made structure that a mining engineer called Arthur Richardson followed, when he set out in 1896 to be the first to ride a bicycle around the Australian continent.

The first railway in Western Australia had opened in 1871, and by 1896 there was a 313-mile line going eastwards through the inhospitable landscape as far as Kalgoorlie. This line was built alongside a pipeline carrying drinking water to the arid goldfields around Kalgoorlie. Meanwhile, in the south-east of the continent, a railway line from Port Augusta to Oodnadatta had opened in 1891. These early railways in both the east and the west were built to the narrow gauge of 3ft 6in.

In 1901, the self-governing colonies of Queensland, New South Wales, Victoria, Tasmania, South Australia and Western Australia got together to form a federation. One of the incentives held out to Western Australia to become a member had been the promise of a trans-continental railway linking Perth with the eastern cities. A survey of the route across the featureless Nullarbor Plain was completed in 1909, and construction started to join the two existing railways at each end in 1912. Despite the onset of the First World War, work progressed until the two lines finally converged, at a site in an area of desert, in 1917. The railway's claim to fame is that its route includes the longest stretch - 297 miles- of straight track in the world. Although the existing railways at each end had been built to the narrow gauge, the new 1,052 mile stretch was built to the standard gauge of 4ft 8in. These two breaks of gauge would mean a true trans-continental service did not exist until 1970.

Water supplies for steam engines on the TAR route were non-existent, as the railway did not cross a permanent watercourse along its entire length. To overcome this, the TAR line's original chief engineer had requested diesel engines, but his request was not granted, so steam haulage had to be supported by extra water tankers carried on each train. Australian-built, diesel electric engines eventually replaced steam haulage in 1951.

As well as being an important freight route, the new line saw the introduction of the Trans Australian Express, a service which commenced running in 1917 to serve isolated communities and settlements along its route. For much of its long-running life, this twice-weekly train consisted solely of sleeping cars; seated accommodation was only included from 1981 until the service terminated ten years later. The route of the train was extended following the conversion to standard gauge of its narrow-gauge sections in 1970, but facing increased competition from air travel it last ran in June 1991.

One long-lived curiosity of the trans-Australian network was the Tea and Sugar Train, which also started running in 1917. The train brought to communities along the route not only the necessities for everyday life, but also a butcher's shop, medical facilities and a cinema. It last ran in 1996.

Despite the withdrawal of the Trans-Australian Express between Adelaide and Perth in 1991, the Indian Pacific service has operated between Sydney on the east coast and Perth since 1970, the year that the TAR route was completely converted to standard gauge. This service is operated by Great Southern Rail. It runs twice a week between Adelaide and Perth, and once a week between Sydney and Perth. The journey time for the complete 2,704-mile journey is 66¾ hours westbound and 71¼ hours eastbound.

The Indian Pacific service calls by at the isolated outback railway settlements of Pimba, which has a population of around fifty, Kingoonya, which only has a hotel, and Tarcoola, which has a population of around forty and is a junction for the south-north line to Darwin. It then travels to Cook, which, despite having a population of fewer than ten, is a scheduled stop, before crossing the Nullarbor Plain in a dead straight line for 297 miles to Loongana, which is uninhabited and is a request stop, and Rawlinna, also a request stop serving Australia's largest sheep station. Civilisation is eventually reached at Kalgoorlie before the last leg of the journey westwards to East Perth Station. The train has carriages with seating, sleeping cars, a restaurant car, and wagons conveying passengers' motor cars.

Questions 1–8

Complete the table below.

*Choose **ONE WORD ONLY** from the passage for each answer.*

Early long-distance communications in Australia

19th century

- lack of public transport across the continent resulted in people travelling by

1841

- an explorer walked across the Nullarbor Plain with the help of a (2) _____

1877

- equipment for the (3) _____ was installed in the Nullarbor Plain

1896

- a new railway line in Western Australia followed the route of a (4) _____

1901

- the prospect of an interstate railway persuaded Western Australia to join a (5) _____

1917

- TAR was completed in a (6) _____ location
- the (7) _____ section of TAR was longer than that in any other country
- The new 4ft 8½ inch TAR track was a (8) _____ width

Questions 9–13

Do the following statements agree with the information given in Reading Passage 1?

*Write **TRUE** if the statement agrees with the information, **FALSE** if the statement contradicts the information, or **NOT GIVEN** if there is no information on this.*

- (9) The first chief engineer's recommendation for TAR trains was fully implemented.
- (10) The Trans-Australian Express was popular with both local residents and tourists.
- (11) Demand for Trans-Australian Express services eventually fell due to the availability of alternative transport.
- (12) The Tea and Sugar Train provided health services as well as food supplies.
- (13) The Indian Pacific service was introduced before the entire TAR route became a standard gauge.

Passage 55

Thomas Cole: American Nature Painter

America's first major landscape artist, and an early voice for the protection of the American wilderness

Thomas Cole (1801–1848) was America's first major landscape artist. Though he was born in England and spent his formative time in Europe, his identity was as an American nature painter. He was also an advocate and protector of untouched American land threatened by industrial development.

Cole's father, a small-time businessman, moved his family from place to place in northwestern England. In the process, Cole worked as a print shop apprentice, designing styles of cloth and later making reproductions of well-known paintings. In 1818, the family sailed to the United States, settling at first in the east coast city of Philadelphia. The teenage Cole stayed behind on the east coast as his parents traveled westward. From this point, his career as a largely self-taught artist began.

Cole began to travel and wherever he went — including, briefly, to the Caribbean — he sketched what he saw. He had little patience with cities and distanced himself from them when he could. At the Pennsylvania Academy of the Fine Arts, he became familiar with new American and European work. But within a few years, he felt ready to head to New York City, where he made lasting contacts in the still small art world, and was able to sell some paintings. And on an 1825 trip up the Hudson River, he found the subject that he would make his own: the American landscape, which he approached as both a mirror of reality and an object of fantasy.

Encouraged, Cole resolved to leave for Europe, a move that would let him see firsthand the art he had known mostly in reproduction. In terms of sales, London, the first stop, was a failure. No one noticed him, but he noticed a lot. He saw England suffering from chaos, with old social structures breaking down as a new industrial elite threatened a traditional wealthy rural class. And he saw artists responding to that. From England he went on to Italy, a more pleasant stay, though it was also suffering its own social issues. Outside Rome, he saw and painted countryside bathed in the hazy sunset light that he had seen and loved in 17th-century landscapes.

When Cole returned to America in 1832 he found it confusing and depressing. Though he had worked hard to leave his poor childhood behind through identification with a cultural elite, Cole found a new America: competitive, commercial, and uncivil — a land of uncontrolled and destructive greed. He was committed to the USA — he became a citizen in 1834 — but left New York City with his wife and children for the countryside. What most disturbed him was the government's seizure and destruction of natural forest lands. From

his house he could see forests being rapidly cut down. Increasingly, he used painting to protest against these developments. Cole chose wilderness, and used the rest of his life to paint it.

One of Cole's earliest pieces illustrating his dedication to the preservation of nature is *The Tree*, dated 1823. Done while he was living in the Caribbean, it's an ink drawing of a single tree. But it is no ordinary tree: although realistically drawn, it looks bizarre, almost resembling some kind of animal, swinging its arm-like branches and walking on its roots.

Cole's paintings from his Italian trip feature a different kind of countryside. These landscapes focused not on pure and untouched nature, but on the ruins of a long-dead culture. One of his most famous works is a series called *The Course of Empire* (1834–36). In five separate paintings, it narrates the stages that an unnamed fictional civilization passes through: from forest origins, to rural innocence, to proud flourishing high society, to destruction, to vanishing.

Returning to America, Cole settled on producing and selling landscape paintings. An early landscape, *View of the Round-Top* (1827), depicts thin clouds in the cool fresh air, a forested mountain, and a river winding in the distance. Paintings such as these were popular and sold quickly for high prices to New York's new wealthy elite.

A similar image that was far more effective is the single, large 1836 painting titled *View From Mount Holyoke (The Oxbow)*. From a bird's-eye height we look down at the spot where the Connecticut River, flowing around an island-like mass, creates a question-mark shape known as the Oxbow. The painting divides vertically into two distinct halves. To the left is a wild cluster of old trees and dense vegetation, which looks messy and threatening, but full of life; to the right, flat, treeless, square-cut fields which are cultivated and fertile, but feel bare and bland. In the foreground is a tiny self-portrait of Cole at work. He turns away from his canvas and looks right at us, as if to say: Here are the alternatives that you will face; you choose.

When he died suddenly at 47, in 1848, he was widely mourned. After Cole's death, respect for his life and work was shown by fellow artists in the form of paintings done in a similar style for just a few years. The most famous was Asher Brown Durand's 1849 *Kindred Spirits*, which illustrates Cole and the nature-poet William Bryant engaged in deep discussion in a rugged paradise of New York State forest. Yet, within a few years, Cole was all but forgotten, and it took more than a century before he was rediscovered.

Questions 1–7

Do the following statements agree with the information given in Reading Passage 1?

Write **TRUE**, **FALSE** or **NOT GIVEN**.

- 1 Thomas Cole's early work experience included creating copies of famous artwork.
- 2 Cole moved to the west of the USA while still a teenager.
- 3 Cole sold paintings in the Caribbean.
- 4 Cole remained in touch with other New York-based painters for a long time.
- 5 The first paintings that Cole sold in New York showed the Hudson River.
- 6 On Cole's trip to Europe, he stayed longer in Italy than in England.
- 7 Although Cole painted in the countryside, his family lived in New York City.

Questions 8–13

Complete the notes below.

*Choose **ONE WORD ONLY** from the passage for each answer.*

Cole's Work — Early work

- The Tree: a drawing that looks like a strange 8 _____

Italian landscapes

- Cole's paintings show 9 _____ from a past society instead of natural beauty
- The Course of Empire shows 10 _____ of an imaginary society

American landscapes

- View of the Round-Top shows a 11 _____ covered in trees
- The Oxbow shows wild vegetation on one side, and organised farmers' 12 _____ on the other
- Cole put himself in the picture, next to a 13 _____

Passage 56

Tickling and Laughter

Why does tickling, or even the thought of it, produce laughter?

A Tickling is the act of touching so as to cause laughter or twitching movements. Tickling may have been one of the first ways early humans communicated with each other and is a useful form of non-verbal communication, particular with babies and children. The word itself comes from the English of the Middle Ages when tickelen meant 'to touch lightly'.

B If you don't know whether you are ticklish, you'll have to ask someone else. Tickling is not included in the spectrum of pains and pleasures that we can inflict on ourselves and while we can stroke and scratch and hurt ourselves, the one thing we can't do effectively is tickle ourselves. And no-one knows why. It is a subject that has intrigued philosophers and scientists since antiquity. The ancient Greek philosophers Plato and Aristotle speculated about tickling and its purpose. The 19th-century British scientist Charles Darwin was the first to attempt to analyse this peculiar phenomenon, observing the involuntary spasm it seems to trigger in babies and primates, and he came to the conclusion that tickling was an ingredient in forming social bonds. In 1872 he noted that the key to the success of tickling is that 'the precise point to be tickled must not be known'. So, it is surprise, rather than tactile pressure, that is a key ingredient in successful tickling. Indeed, in people who are extremely suggestible, the threat of being tickled without even being touched is enough to induce hysterical laughter. This is as effective with adults as with children and provides a clue to the fact that tickling is not merely a physical sensation. Ticklishness is not something that diminishes with age, nor does anyone know why some people are more ticklish than others, and there are no distinctions to be made along gender lines. The whole thing is mysterious.

C Research has been done on animals on the relationship between tickling and laughter. Neuroscientists at Bowling Green University in Ohio in the USA have recently discovered that rats respond to being tickled with squealing, chirping sounds, increased excitement and little kicks - especially when tickled on the nape of the neck. Dogs may not respond quite as effusively, but it is common for tummy-tickling to trigger frantic hind leg action which appears to be a sign of pleasure. More controversial is the claim that Washoe, a female gorilla living in the primate facility at Washington Central University and trained in American sign language, frequently makes the sign for 'tickle me', suggesting that it is a pleasant sensation.

D For eminent neuroscientist Professor V S Ramachandran, head of the Department of Brain and Cognition at the University of California, laughter is the essential key to unlocking the mystery of tickling. 'Laughter' is a signal that the tickling is a false alarm, that there is nothing to really worry about, the subject is not really under attack,'

Ramachandran says. 'When someone tells you a joke, they take you along a path of expectation to a punchline which is a twist in the path. When the subject laughs at the punchline, it is a recognition that danger has been averted. The same applies to tickling.'

E Ramachandran has studied the response by children to tickling. He says: 'Most babies are ticklish. In evolutionary terms it may be that in humans, ticklishness is a leftover of childhood behaviour with some social benefits. But because there are so many layers to the human mind, people who do not consider themselves ticklish may be inhibited about laughing and exposing their vulnerability.' Another researcher, Christine Harris believes that there are two types of tickling. The lighter pressure results in the urge to scratch or rub, while the heavier provokes laughter. As to why some areas of the body appear to be more sensitive to tickling than others - the soles of the feet, the underarm area, the stomach and the neck are most commonly mentioned - Ramachandran suggests that 'these are areas that are not normally touched by other people so it is an indication that they are considered private space'. Other especially ticklish areas include the waist and ribs.

F The laughter response to the stimulus of tickling comes from the brain. Sarah Jayne Blakemore, a cognitive neuroscientist at London's University College, says the cerebellum, a more primitive part of the brain, dampens the tickle sensation if you try to tickle yourself, telling the cortex to ignore the feeling. To demonstrate the theory, she constructed a robotic tickle machine with a foam-tipped arm and operated by an unseen person. Blakemore used MRI scans which measure blood flow in the brain to compare cerebral activity when six volunteers tried to tickle themselves and when they were tickled by the machine. The part of the brain that registers touch reacted more strongly when the machine tickled them than when they tickled themselves. Recent studies suggest that reaction to tickling, like laughter, is innate. Children born deaf and blind react normally to being tickled. No specific studies have been conducted, however, on tickling in people suffering from autism.

Questions 1–5

Reading Passage 1 has six paragraphs, A–F. Which paragraph contains the following information?

Write the correct letter, A–F. NB You may use any letter more than once.

Q	Statement	
1	the parts of the human body which are sensitive to tickling	
2	the interest in tickling shown by scientists and thinkers throughout history	
3	the similarity between response to tickling and response to telling funny stories	
4	an experiment on tickling oneself	

5	a reason why some people do not believe they are ticklish	
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Questions 6–11

Look at the following claims (Questions 6–11) and the list of people below.

Match each claim with the correct person, **A, B, C** or **D**. **NB** You may use any letter more than once.

Q	Statement	
6	Laughter demonstrates that tickling is not a threat.	
7	Tickling strengthens relations between people.	
8	Different sorts of tickling cause different reactions.	
9	Tickling oneself results in a weaker sensation than being tickled by someone or something else.	
10	Effective tickling relies on not knowing where it will happen.	
11	Understanding laughter will allow us to understand tickling.	

List of People

A Charles Darwin

B Professor V S Ramachandran

C Christine Harris

D Sarah Jayne Blakemore

Questions 12 and 13

Complete the summary below.

Choose **NO MORE THAN TWO WORDS** from the passage for each answer.

- When people are tickled, the brain produces a laughter-response. If you attempt to tickle yourself, the part of the brain called the **12** _____ weakens the feeling you experience.
- A recent experiment testing why people do not laugh when they tickle themselves examined **13** _____ in the brain to assess brain activity.

Passage 57

Tunnelling under the Thames

The first tunnel ever to be built under a major river was the tunnel under London's River Thames

At the beginning of the 19th century, the port of London was the busiest in the world. Cargoes that had travelled thousands of miles and survived all the hazards of the sea were unloaded on the banks of the Thames, only for their owners to discover that the most frustrating portion of their journey lay ahead. Consignments intended for the southern parts of Britain had to be lifted onto horse carts, pulled through the docks and across London Bridge, built in the 12th century and as impractical as its early date implies. By 1820, London Bridge had become the centre of the world's largest traffic jam.

It was an intolerable situation, and it was clear that if private enterprise could build another crossing closer to the docks, there would be good money to be made in tolls paid by users. Another bridge was out of the question, as this would deny sailing ships access to the city centre and ambitious men turned their thoughts to tunnelling beneath the Thames instead. This was not such an obvious idea as it might appear. Although increasing demand for coal had meant a great many tunnels had been dug in mines in Britain, working methods remained primitive. Tunnels were dug by men with simple tools, by candlelight. However, in 1807, a group of businessmen set themselves up as the Thames Archway Company. Their ambition was to tunnel below the Thames, but there was little to guide them as there had been no previous attempt to do this. Their chief engineer was Richard Trevithick, designer of the world's first high-pressure steam engine. His men made progress at the beginning, but then things began to go disastrously wrong, with muddy soil pouring into the tunnel. Eventually, the Thames Archway Company had had enough. Its funds were exhausted, Trevithick was sick from exposure to the river water, and its efforts had proved only that a passage under the river exceeded the limits of contemporary mining technology.

At that time, the only machines used in mines were pumps. It took a man of genius to recognise that a different sort of machine was needed, a machine that could prevent the roof and walls of a tunnel from collapsing. This man was Marc Brunel, a Frenchman who had become one of the most prominent engineers in Britain. Not long after the failure of the Thames Archway Company, Brunel saw a rotten piece of wood lying on the riverbank. Examining the wood through a magnifying glass, he observed it was infested with something that looked like a worm. Brunel realised that as it tunnelled through the wood, it would push chewed wood into its mouth and digest it, then excrete a hard substance that lined the new tunnel. Brunel realised that the worm's digging technique could be adapted to produce a new way of tunnelling. His realisation led him to invent a device that has been used in one form or another in most major tunnels built since – the tunnelling shield.

It consisted of a heavy iron frame that could be pushed forward a few inches at a time. The front of the frame was made up of a series of iron frames that could be folded back to allow miners to dig the ground ahead. Behind these frames was a wall consisting of a series of iron plates pressed against the tunnel face and supported on a set of horizontal wooden planks, that would prevent the face from collapsing. It was a complex and rather cumbersome machine and not easy to use, but it seemed that it would protect the miners from the worst of the river's water. Brunel's team carefully examined earth samples taken from beneath the riverbed, and subsequently decided to dig the tunnel close to the muddy river bottom, where he could expect to find clay. This would be a more solid and safe substance to dig through than the sand that was found deeper down.

Brunel began work on his tunnel in 1825, but the problems of such an operation soon became apparent. Although the shield itself worked well, water began to drip into the tunnel. This was more of an annoyance than a danger while the pump was working, but this machine proved unreliable and sometimes failed altogether. When the pump broke down, work had to stop as the tunnel quickly flooded. There were occasions when the miners had to abandon their tools and flee for their lives. Even when Brunel's men were able to work, they had to run the constant risk of the pumps failing. They also complained of frequent headaches and dizziness, caused by the poor air quality. The air underground was dirty and stale, contaminated due to the lack of an adequate ventilation system. There were lighting problems too. Illuminating the tunnels by candlelight was a constant challenge. Lamps give off only a very weak glow, and there were a number of accidents because the miners could not see what they were doing. Lastly, a number of Brunel's miners walked off the job because they could not tolerate the excessive temperatures that developed in the cramped conditions underground.

Despite all these setbacks, the tunnel finally emerged on the opposite river bank on August 12, 1841. Brunel's triumph, however, was only partial. The small payment per person made by the thousands of visitors who flocked to see the marvel hardly paid even a penny per foot of the tunnel's construction costs. Brunel had gone bankrupt long before the project was completed, and the government loan he had required to complete the project had to be paid back with interest. As a result, there was not enough funding to make it accessible to horse-drawn vehicles, as intended. Instead, the passageways were filled with souvenir sellers and entertainers. In the end, the tunnel was closed two years later, used at night, before it was finally closed entirely and fell into dereliction for decades.

It was only when the underground railway came to London in the 1880s that the Thames Tunnel found and achieved a measure of real usefulness. It was bought in 1869 by the East London Railway, who found it to be in such excellent condition that it was immediately pressed into service as a route for passenger trains heading east. The tunnel became, and remains, part of the London Underground network.

Questions 1–8

Do the following statements agree with the information given in the passage?

Write **TRUE** if the statement agrees with the information, **FALSE** if the statement contradicts the information, or **NOT GIVEN** if there is no information on this.

- 1 In the early 19th century, the port of London was considered a safer destination than other ports.
- 2 London Bridge provided quick access for cargo being sent to southern Britain.
- 3 It was generally believed that a new river crossing would be profitable.
- 4 Building a second bridge crossing was initially considered to be the best solution.
- 5 It was believed that coal could be found under the River Thames.
- 6 The Thames Archway Company was the first group to try tunnelling below the Thames.
- 7 Some of Trevithick's men were injured during a mudslide at his tunnel.
- 8 The Thames Archway Company ran out of money to finance the tunnel project.

Questions 9–13

Complete the notes below.

Choose **ONE WORD ONLY** from the passage for each answer.

Marc Brunel's tunnel

Preparing to build the tunnel

- Brunel noticed how a kind of 9 ————— made its tunnels in wood
- Brunel created a device called a tunnelling shield, to protect people working under the river
- Brunel planned to build a shallow tunnel so the earth would have a higher content of 10 —————

Problems faced by miners

- There were frequent floods caused by mechanical breakdowns.
- The miners suffered from 11 ————— because of pollution in the tunnels.
- Lighting problems led to several 12 —————

- Some workers quit because of the high temperatures in the tunnel.

After the tunnel was finished

- The tunnel was finally completed in 1841.
- Brunel did not have enough money to repay his debt to the ⑬ _____.
- The tunnel was abandoned until the 1880s.

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Passage 58

What Exactly Is a Hoard?

Why people buried collections of coins and precious objects, and what their discovery tells archaeologists today

What exactly is a hoard? Broadly speaking, a hoard is a group of items kept together, perhaps gathered all at once or gradually over time. Even though a typical image of a hoard is a pot full of gold coins, it need not be a collection of metal objects. Hoards that have survived over the centuries are the ones which were either lost or deliberately thrown away. Many of those discovered have been split up, spent or melted down, leaving no traces behind.

People have been finding hoards since the practice of burying them began. In the past, they were dug up by farmers while working the soil and clearing land, or were exposed by fallen trees or eroded riverbanks. Some of the places where hoards have been found seem to have held particular significance throughout centuries. Bronze Age (around 3000 to 1200 BC) objects appear in later hoards, and the Romans in particular (Britain's rulers 43 to 410 AD) seem to have found ancient artefacts intriguing: they buried fossils and prehistoric weapons alongside their own items. In Britain's historic town centers today, the past is revealed when buildings are knocked down or rebuilt. In these urban areas, hoards are usually found by archaeologists excavating sites before they are redeveloped, whereas many recent discoveries in rural areas have been made by amateur metal-detector users.

Each newly found hoard raises questions. Who did it belong to? Why was it not recovered? The actual contents of hoards have much to tell us when studied. The act of hiding a group of objects in a pot in the ground or behind a wall often keeps them in good condition. In contrast to single objects that have been accidentally dropped on the ground. These are often later damaged as fields are ploughed. Hoarded objects may be rare survivals: things that would normally be melted down for recycling or coins that would have been recalled by the authorities had they not been hidden.

Archaeologists have come to realize, however, that the key to understanding a hoard is usually held not in the group of objects itself but in its context; that is, in the information held in the soil immediately around it and evidence of human activity in the wider landscape. Where the precise locations of hoards have been investigated further, their stories are enriched with detail. Although metal items may be thrown up to the surface during farming activities, archaeological remains are often waiting to be discovered below ground.

Why were hoards buried? Some smaller hoards were certainly accidental losses, or so-called 'purse hoards.' This may be the case for smaller groups of coins found together, sometimes still with the bags that had contained them. Larger hoards may have been emergency hoards hidden in times of conflict, when people who owned precious objects had to flee suddenly, or felt their homes were under threat.

A combination of conflict and economic insecurity may explain why so many hoards were buried in late third-century Britain. Instability in the Roman empire led to rapid inflation and the official coinage decreased in silver content, to the point where the coins had little intrinsic value. When the coinage was eventually reformed, older coins may have been unacceptable for official payments, so coins were discarded, or hoards were not recovered.

Concealing coins and valuable items would have been more common before ordinary people had access to banks. Savings would need to be kept secure and hidden, and many hoards clearly started off this way. The Beau Street hoard of over 17,500 coins found in Bath in 2007 is composed of eight lots of money that appear to have been sorted and then stored under the floor of a Roman building. The hoard could represent savings made by a wealthy individual or business, but the reason why the hoard was not recovered is uncertain. It is also important to remember that not all hoards found in the ground had been buried there: many hoards from Roman villas were hidden within walls that later collapsed.

There are other reasons for the burial of hoards. A strong theme that runs through prehistory is the practice of 'deliberate deposition' - that is, putting something underground or under water, which was perhaps a way of releasing it to the gods. There are no records to allow us ever to know the exact significance of prehistoric rituals but some later offerings are accompanied by legible inscriptions. Such offerings may have been made on one occasion or built up over a long period of time.

Seemingly unusual rites were often simply part of everyday life in the past. In Iron Age Britain (800 BC to the Roman Invasion), for instance, chosen objects were carefully placed in pits and ditches in settlements as well as at shrines, possibly marking the beginning or end of use of a building, or defending significant boundaries. In some societies, a public destruction or donation of valuable items enhances the status of the person giving them away. These ceremonies may be carried out when there is a change of leader, to create social cohesion in a time of uncertainty or to make significant events in the community - and some hoards may be the result of such ceremonies.

Questions 1–8

Do the following statements agree with the information given in Reading Passage 1?

Write **TRUE** if the statement agrees with the information, **FALSE** if the statement contradicts the information, or **NOT GIVEN** if there is no information on this.

- 1 People typically think of a hoard as consisting of money.
- 2 In the past, most hoards were discovered in wooded areas.
- 3 The Romans took little interest in objects from earlier civilizations.
- 4 Nowadays hoards in British towns and cities are generally discovered using metal detectors.
- 5 Items in hoards are often better preserved than single lost objects.
- 6 Archaeologists have realized that certain soil types cause less harm to buried objects.
- 7 The Beau Street hoard is the largest collection of coins discovered in the UK.
- 8 The precise meaning of prehistoric rituals has become clear following recent discoveries.

Questions 9–16

Complete the notes below.

Choose **ONE WORD ONLY** from the passage for each answer.

Why were hoards buried?

Accidental loss

- in 'purse hoards', the 9 _____ are often found alongside their contents.

Loss of value

- coins in late third-century Britain had less and less 10 _____ in them.

Greater security

- money found in Bath in 2007 may have been 11 _____ before being hidden.
- hoards from Roman villas were often placed in 12 _____.

Rituals

- 'deliberate deposition' in prehistory involved people placing objects in 13 _____ or below the ground.
- offerings are sometimes found with 14 _____.

Common rites

- in Iron Age Britain important ⑮ ————— were protected by burying objects.
- ceremonies in which items were destroyed or donated sometimes indicated a new ⑯ ————— within a community.

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Passage 59

What if Everything Had a Barcode?

A vast new database will let us catalogue every plant and animal on the planet, and identify them in seconds

Imagine going for a walk and spotting a wild flower. Its beauty and fragrance delight you, but the name eludes you. No problem. You whip out a hand-held scanner, about the size of a mobile phone, and pop a fragment of a leaf into the device. A few seconds, and the read-out tells you that you're looking at a pyramidal orchid. Satisfied, you continue on your way.

Sound far-fetched? Not at all. Scientists are currently creating a DNA barcode for every species of plant and animal on the planet. It won't be long before everyone, from experts to amateurs, will be able to scan the world's flora and fauna as if they were checking out groceries at a supermarket, to look up or confirm their identities.

There are numerous practical uses too. Such a device would let you scan fish at the fishmonger's to check if it's been labelled properly, work out exactly what is in your mixed vegetable dish, and confirm whether a piece of furniture really has come from a renewable forest, as the retailer claims. It would also assist forensic science teams, who could quickly identify the pollen on a suspect, to link him to a particular location; customs officials, in their efforts to prevent disease-carrying pests being taken across national borders; and environmental inspectors assessing water quality, who need to work out what microbes are lurking in a particular sample.

It was Professor Paul Hebert, a biologist from the University of Guelph in Canada, who came up with the idea of DNA barcoding the natural world. The inspiration came while he was walking up and down the aisles of a supermarket, marvelling at the ability of the store to keep track of all the lines stocked and sold using the thick and thin lines that make up a barcode. Could scientists, he wondered, exploit a barcode system to record the millions of species on earth via their DNA?

The compilation of a planetary inventory began more than 250 years ago, with the Swedish life classifier Carl Linnaeus. In 1758, he founded the science of taxonomy – a method of classifying living things – based on physical and behavioural characteristics. To date, scientists have classified about 1.7 million organisms, a small fraction of the total number of species, which has been estimated at anywhere between 5 and 30 million. But taxonomy is difficult and time-consuming. Many species, such as the different kinds of flies, look remarkably similar. Only an expert who has spent years examining a particular group can distinguish one from another. Even the experts can be stumped, however, when presented

with an egg, an embryo, a seedling or a root. The next problem is that we are running out of time in which to complete the inventory. The International Union for the Conservation of Nature estimates that a quarter of the world's mammals are threatened with extinction.

So, Hebert's idea centred on finding a fragment of DNA that would disclose the identity of a species without having to decode its entire genetic code. He envisaged a 'DNA barcode reader', similar to the scanners at retail checkouts. Outlining his idea in *Scientific American*, Hebert writes: 'An inspector at a busy seaport, a hiker on a mountain trail, or a scientist in a lab could insert a sample containing DNA – a snippet of whisker, say, or the leg of an insect – into the device, which would detect the sequence of nucleic acids in the barcode segment. This information would be instantly relayed to a reference database, a public library of DNA barcodes. Anyone, anywhere, could identify species.'

To create the barcode, Hebert proposed the use of a section of DNA, from the energy-producing units found in all cells. He selected a gene that gives rise to an enzyme known as CO1. This gene is small enough to be quickly and easily deciphered, but has sufficient variation for us to be able to tell most animal species apart. You and I, for instance, will have different versions of CO1, but they will be similar enough to show that we're both humans and not chimpanzees.

In 2003, Hebert and his team published their first results. They showed that the barcode system could identify the group an animal came from (for example, whether it was a vertebrate, a worm or an insect) and even the species when it was stored in the barcode library. After five more years of work, results indicate that animals can now be identified by their barcodes in 98 per cent of cases. Early results have confirmed the additional benefits of the new system: for example, caterpillars of the tropical butterfly *Astraptes fulgerator*, which was first recognised as a species in 1775, all look very similar, and were assumed to belong to a single species. Barcoding has shown there are 10 different kinds.

Of course, the value of the system depends on a comprehensive reference library of the DNA (CO1) barcodes of established species. The Barcode of Life Data (BOLD) system is an enormous international collaboration supported by 150 institutions in 45 countries. To date, it has compiled more than 500,000 records from 50,000 species. The consortium is hoping that the world's birds will be barcoded by 2011. 'People have watched birds for so long that they might think every different tweet has been heard, every different colour observed, but barcoding may prove otherwise,' says Professor Mark Stoeckle, professor of the human environment at Rockefeller University, New York, who works with Hebert. He estimates that out of the world's 10,000 bird species, DNA barcoding will distinguish at least 1,000 new ones.

Questions 1–3

Complete the notes below.

Choose **NO MORE THAN ONE WORD AND/OR A NUMBER** from the passage for each answer.

PROBLEMS WITH TAXONOMY

- only (1) _____ species have been classified so far
- difficult to distinguish between species of certain creatures, for example (2) _____
- possibility of a large number of species of (3) _____ dying out soon

Questions 4–8

Complete the flow-chart below.

Choose **NO MORE THAN THREE WORDS AND/OR A NUMBER** from the passage for each answer.

Hebert's system

decided to create a device called a (4) _____, like ones used in shops

↓

chose a (5) _____ that produces a substance called CO₁

↓

samples of CO₁ read by the device and matched with those kept in the (6) _____

↓

current results show that (7) _____ of animal species can be identified in this way

↓

results show different species being identified, eg of (8) _____

Questions 9–13

Do the following statements agree with the information given in Reading Passage 1?

Write **TRUE** if the statement agrees with the information, **FALSE** if it contradicts it, or **NOT GIVEN** if there is no information on this.

- (9) The writer believes that the barcode system will be widely used by the general public.
- (10) It is likely that the barcode device will show that many foods and goods have not been correctly described.
- (11) Hebert got the idea for DNA barcoding from someone who worked at a supermarket.

- ⑫ The number of organisations supporting Hebert's barcoding project is growing all the time.
- ⑬ A large number of new bird species have already been identified by the DNA barcode system.

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Passage 60

William Gilbert and Magnetism

The Elizabethan physician whose experiments on magnets laid the foundations of modern science

① **A** The 16th and 17th centuries saw two great pioneers of modern science: Galileo and Gilbert. The impact of their findings is eminent. Gilbert was the first modern scientist, also the accredited father of the science of electricity and magnetism, an Englishman of learning and a physician at the court of Elizabeth. Prior to him, all that was known of electricity and magnetism was what the ancients knew, nothing more than that the lodestone possessed magnetic properties and that amber and jet, when rubbed, would attract bits of paper or other substances of small specific gravity. However, he is less well known than he deserves.

② **B** Gilbert's birth pre-dated Galileo. Born in an eminent local family in Colchester County in the UK, on May 24, 1544, he went to grammar school, and then studied medicine at St John's College, Cambridge, graduating in 1573. Later he travelled in the continent and eventually settled down in London.

③ **C** He was a very successful and eminent doctor. All this culminated in his election to the president of the Royal Science Society. He was also appointed personal physician to the Queen (Elizabeth I), and later knighted by the Queen. He faithfully served her until her death. However, he didn't outlive the Queen for long and died on November 30, 1603, only a few months after his appointment as personal physician to King James.

④ **D** Gilbert was first interested in chemistry but later changed his focus due to the large portion of mysticism of alchemy involved (such as the transmutation of metal). He gradually developed his interest in physics after the great minds of the ancient, particularly about the knowledge the ancient Greeks had about lodestones, strange minerals with the power to attract iron. In the meantime, Britain became a major seafaring nation in 1588 when the Spanish Armada was defeated, opening the way to British settlement of America. British ships depended on the magnetic compass, yet no one understood why it worked. Did the Pole Star attract it, as Columbus once speculated; or was there a magnetic mountain at the pole, as described in Odyssey, which ships would never approach, because the sailors thought its pull would yank out all their iron nails and fittings? For nearly 20 years, William Gilbert conducted ingenious experiments to understand magnetism. His works include *On the Magnet*, *Magnetic Bodies*, and *the Great Magnet of the Earth*.

⑤ **E** Gilbert's discovery was so important to modern physics. He investigated the nature of magnetism and electricity. He even coined the word "electric". Though the early beliefs of magnetism were also largely entangled with superstitions such as that rubbing garlic on lodestone can neutralise its magnetism, one example being that sailors even believed the

smell of garlic would even interfere with the action of compass, which is why helmsmen were forbidden to eat it near a ship's compass. Gilbert also found that metals can be magnetised by rubbing materials such as fur, plastic or the like on them. He named the ends of a magnet "north pole" and "south pole". The magnetic poles can attract or repel, depending on polarity. In addition, however, ordinary iron is always attracted to a magnet. Though he started to study the relationship between magnetism and electricity, sadly he didn't complete it. His research of static electricity using amber and jet only demonstrated that objects with electrical charges can work like magnets attracting small pieces of paper and stuff. It is a French guy named du Fay that discovered that there are actually two electrical charges, positive and negative.

⑥ **F** He also questioned the traditional astronomical beliefs. Though a Copernican, he didn't express in his quintessential beliefs whether the earth is at the centre of the universe or in orbit around the sun. However, he believed that stars are not equidistant from the earth but have their own earth-like planets orbiting around them. The earth itself is like a giant magnet, which is also why compasses always point north. They spin on an axis that is aligned with the earth's polarity. He even likened the polarity of the magnet to the polarity of the earth and built an entire magnetic philosophy on this analogy. In his explanation, magnetism is the soul of the earth. Thus a perfectly spherical lodestone, when aligned with the earth's poles, would wobble all by itself in 24 hours. Further, he also believed that the sun and other stars wobble just like the earth does around a crystal core, and speculated that the moon might also be a magnet caused to orbit by its magnetic attraction to the earth. This was perhaps the first proposal that a force might cause a heavenly orbit.

⑦ **G** His research method was revolutionary in that he used experiments rather than pure logic and reasoning like the ancient Greek philosophers did. It was a new attitude towards scientific investigation. Until then, scientific experiments were not in fashion. It was because of this scientific attitude, together with his contribution to our knowledge of magnetism, that a unit of magneto motive force, also known as magnetic potential, was named Gilbert in his honour. His approach of careful observation and experimentation rather than the authoritative opinion or deductive philosophy of others had laid the very foundation for modern science.

Questions 1–7

Reading Passage 1 has seven paragraphs, A–G.

Choose the correct heading for each paragraph from the list of headings below. Write the correct number, i–ix, in the circle before each paragraph.

i Early years of Gilbert

- ii What was new about his scientific research method
- iii The development of chemistry
- iv Questioning traditional astronomy
- v Pioneers of the early science
- vi Professional and social recognition
- vii Becoming the president of the Royal Science Society
- viii The great works of Gilbert
- ix His discovery about magnetism
- x His change of focus

Questions 8–10

Do the following statements agree with the information given in the text?

Write **TRUE**, **FALSE** or **NOT GIVEN**.

- 8 He is less famous than he should be.
- 9 He was famous as a doctor before he was employed by the Queen.
- 10 He lost faith in the medical theories of his time.

Questions 11–13

Which **THREE** of the following are parts of Gilbert's discovery?

Choose **THREE** correct answers, **A–F**.

OPTIONS

- A Metal can be transformed into another.
- B Garlic can remove magnetism.
- C Metals can be magnetised.
- D Stars are at different distances from the earth.
- E The earth wobbles on its axis.
- F There are two charges of electricity.

Answer: 11 12 13

Answer Key & Explanations

All 60 Passages

Passage 1 — Ambergris – What Is It and Where Does It Come From?

1. C

2. A

3. D

4. A

5. C

6. B

7. beaks

problems digesting the ____ ~ para 6: "the problems the whales have in digesting the beaks of such creatures". Direct lift from the passage — answer: **beaks**.

8. vomited

is ____ from time to time ~ para 6: "from time to time large quantities of the liquid are vomited up by the whale". Answer: **vomited** (up).

9. hardens

____ on contact with the air ~ para 6: "once outside the whale's body and exposed to air, the substance hardens". Answer: **hardens**.

10. TRUE

most... came from processing dead whales ~ para 8: "over 90 per cent of the annual worldwide total was acquired... as a by-product of commercial whaling", and this whaling took place "in the 19th and 20th centuries". Matches the statement — answer: **TRUE**.

11. NOT GIVEN

The passage says ambergris "was worth its weight in gold" historically, but never compares that to *recent* value — no trend is given. Answer: **NOT GIVEN**.

12. FALSE

remains an important ingredient contradicts para 8: "ambergris was supplanted in the making of perfume by other materials, some natural and some synthetic" — it was *replaced*. Answer: **FALSE**.

13. NOT GIVEN

The passage only speculates ambergris might *wash ashore again* as whale populations recover — it never mentions any *new use* being found. Answer: **NOT GIVEN**.

Passage 2 — A Look at the History of Poetry

1. TRUE

‘Poetry appears among the earliest records of **most literate cultures**, with poetic fragments found engraved on ancient stones.’ Answer: **TRUE**.

2. FALSE

the opposite: the *Ramayana* has become **an important narrative in both Hindu and Buddhist mythology throughout regions of Asia**. Answer: **FALSE**.

3. NOT GIVEN

the passage says both works are **attributed by academics to the Greek poet Homer**, but never states **WHEN** either was discovered (discovery dates are only given for the Epic of Gilgamesh). Answer: **NOT GIVEN**.

4. TRUE

a sonnet form is literally named after him: ‘the Italian poet Petrarca, after whom **the Petrarchan sonnet is named**, is perhaps one of the most well-known early writers of the sonnet, in which he specialized.’ Answer: **TRUE**.

5. NOT GIVEN

the passage only says the sonnet **became widely known as an English poetic form through the work of William Shakespeare** — about popularity, not quality; his sonnets are never judged as among the finest. Answer: **NOT GIVEN**.

6. TRUE

‘Dryden made his name as a poet with **his poems making fun of important figures or genres**’, and the passage adds ‘the use of poetic form for satire’. Answer: **TRUE**.

7. Romantic

‘The so-called English **Romantics** included figures like William Blake...and John Keats.’ Answer: **Romantic**.

8. fireside

‘American poetry produced what are known as the **fireside** poets. Writers like Henry Wadsworth Longfellow...and William Cullen Bryant belonged to this group.’ Answer: **fireside**.

9. Whitman

‘**Whitman** continued to edit and alter his 1855 edition of ‘Leaves of Grass’, which introduced a new period...for poetry.’ Answer: **Whitman**.

10. languages

readers ‘were to be surprised...by T.S. Eliot’s employment of ancient **languages** and obscure references.’ Answer: **languages**.

11. Jamaica

‘An African-American poet born in **Jamaica** who emigrated to the US in 1912, he carried his homeland into his...poetic voice.’ Answer: **Jamaica**.

12. Bengali

‘wrote solely in his mother tongue, **Bengali**, won this esteemed award in 1913.’ Answer: **Bengali**.

13. Irish

‘He was followed by **Irish** poets William Butler Yeats in 1923 and Seamus Heaney in 1995.’ Answer: **Irish**.

Passage 3 — Ahead of its Time

1. TRUE

a four-metre-high flood bank testifies to its natural tendency to flood. Answer: **TRUE**.

2. TRUE

he noticed what he initially took to be a whitish rock, lit by the sun. Answer: **TRUE**.

3. FALSE

as he picked it up, he saw it was a skull. Answer: **FALSE**.

4. NOT GIVEN

the passage only says tobin hurried home to tell his mother, but does not describe her reaction to the later scientific debate. Answer: **NOT GIVEN**.

5. police

the police were immediately called. Answer: **police**.

6. European

the skull appeared to be european in origin. Answer: **European**.

7. radiocarbon dating

proposed the use of radiocarbon dating to make sure it wasn't a recent death. Answer: radiocarbon dating.

8. sample

GNS was contacted and provided with a sample of bone that had originated in the top of the skull. Answer: sample.

9. 296

conventional radiocarbon age approximately 296 years. Answer: 296.

10. race

how a skull of this race, let alone this gender, had reached these remote islands. Answer: race.

11. archaeologists

a team of archaeologists working in the solomon islands in 1970 did discover the remains of european vessels. Answer: archaeologists.

12. Australia

both having escaped from prison in australia. Answer: Australia.

13. shipwreck

a spanish or portuguese trading-ship was washed onto these wild shores as a result of a shipwreck. Answer: shipwreck.

Passage 4 — An Early Cultural Tourist

1. TRUE

‘Cyriacus of Ancona...the first cultural tourist since antiquity, **lacked these advantages**’ (the advantages of reading about a destination in advance). Answer: **TRUE**.

2. TRUE

‘Cyriacus first became fascinated by ancient monuments while...looking at the **marble arch, erected in AD 115, to the Roman Emperor Trajan.**’ Answer: **TRUE**.

3. NOT GIVEN

the passage only says an arch was ‘erected...to the Roman Emperor Trajan’ (dedicated to him) — it never says Trajan built the city itself. Answer: **NOT GIVEN**.

4. **FALSE**

the opposite: '**not many people** of Cyriacus's time were interested in historical travel, they generally ignored old buildings...or worse, dismantled them.' Answer: **FALSE**.

5. **FALSE**

the opposite: '**his training as a merchant did not prepare him for this vocation**; he did not know ancient languages, history or art.' Answer: **FALSE**.

6. **TRUE**

'he set out to solve these failings, first by **learning Latin at the age of 30 and then adding ancient Greek**. Having done this, he then set off on voyages.' Answer: **TRUE**.

7. **FALSE**

the opposite: 'overland journeys...were bad; **voyages by seas were even worse**.' Answer: **FALSE**.

8. **NOT GIVEN**

the passage describes his personal optimism during hardships, but never mentions him trying to help or comfort other travelers. Answer: **NOT GIVEN**.

9. **commerce**

'Cyzicus, an ancient Roman city that had relied on **commerce** for its financial success.' Answer: **commerce**.

10. **guidebook**

'there was no **guidebook** on ancient architecture to help him.' Answer: **guidebook**.

11. **earthquake**

'in AD 123 the city was so devastated by a major **earthquake**.' Answer: **earthquake**.

12. **campaign**

'he decided to subsidise a **campaign** to reconstruct Cyzicus.' Answer: **campaign**.

13. **donation**

'He made a substantial **donation** for a new temple to the Roman god Jupiter.' Answer: **donation**.

14. **doorway**

'He sketched the great **doorway** adorned with carved foliage and mythological characters.' Answer: **doorway**.

15. **war**

'he found **war**, earthquakes and avalanches had all but obliterated its ruins.' Answer: **war**.

16. theatre

'he came upon a **theatre** built into the slope.' Answer: **theatre**.

Passage 5 — An Important Language Development

1. TRUE

different shapes and sizes ≈ para 1: tablets were "palm-sized, rectangular... although occasionally **larger tablets or cylinders** were used" — different shapes (rectangular vs cylinder) and sizes. Answer: **TRUE**.

2. FALSE

tablets were dry contradicts para 2: writers "would impress these lines into the **wet clay** with a stylus". Answer: **FALSE**.

3. TRUE

Para 2: "the signs were often almost **too small to see with the naked eye**". Answer: **TRUE**.

4. TRUE

Para 2: "Cuneiform signs were used for the writing of **at least a dozen languages**." Answer: **TRUE**.

5. NOT GIVEN

Para 2 only draws a *functional* comparison — "similar to how the **Latin alphabet is used today** for writing English, French, Spanish and German" — meaning one script can serve many languages, like cuneiform did. It never claims cuneiform signs literally appear in a modern alphabet. Answer: **NOT GIVEN**.

6. stones

tokens, for example ____ ≈ para 3: "they might take small **stones** and use them as tokens". Answer: **stones**.

7. cloth

Para 3: tokens "might then be placed in a **cloth container**". Answer: **cloth**.

8. receipt

Para 3: tokens were "provided to a buyer as a **receipt** for a transaction". Answer: **receipt**.

9. envelope

Para 4: "a container **resembling an envelope**, and now made of clay". Answer: **envelope**.

10. sheep

Para 4: "an image which resembled the drawing of a **sheep** meant just that". Answer: **sheep**.

11. triangles

Para 4: "the signs took the form of **triangles**, which became common cuneiform signs". Answer: **triangles**.

12. accounting

Para 5: "linguists and historians agree cuneiform developed primarily as a tool for **accounting**". Answer: **accounting**.

13. paper

Para 6: "Rawlinson made impressions of the marks on large pieces of **paper**". Answer: **paper**.

Passage 6 — Arctic Warming

1. natural processes

only natural processes, such as volcanic eruptions and variations in solar output, caused periods of relative warming and cooling. Answer: **natural processes**.

2. polar amplification

this disparity in warming between the poles and the tropics is a phenomenon known as polar amplification. Answer: **polar amplification**.

3. reflects

fresh snow **reflects** nearly 85% of solar radiation that hits it. Answer: **reflects**.

4. springtime

in the springtime, this ice melts and exposes the dark ocean underneath, which leads to warming of the region. Answer: **springtime**.

5. coastal cities

if the ice sheet completely melts, sea levels could rise, **endangering coastal cities and small island nations**. Answer: **coastal cities**.

6. methane

the most troubling effect of this feedback loop is the release of methane into the atmosphere. Answer: **methane**.

7. FALSE

the average temperature of antarctica has not changed in the last several decades, despite an increase in the concentration of greenhouse gasses. Answer: **FALSE**.

8. **TRUE**

large storms in the open ocean create more waves that cause the arctic waters to mix with warmer waters, which increases the heat absorbed in the summer. Answer: **TRUE**.

9. **TRUE**

global sea levels could rise by as much as 6 metres, endangering coastal cities and small island nations. Answer: **TRUE**.

10. **FALSE**

this darker land absorbs more heat than before — not less. Answer: **FALSE**.

11. **TRUE**

without oxygen restriction this releases mostly carbon dioxide; however, if the soil is waterlogged, fermentation occurs and this discharges a gas called methane. Answer: **TRUE**.

12. **NOT GIVEN**

the passage only says a global campaign is underway to actually remove carbon dioxide from the atmosphere manually, using techniques such as direct air capture — not that it is the single most effective method. Answer: **NOT GIVEN**.

13. **TRUE**

this is what plants do naturally, but direct air capture does it much more rapidly and requires less land space. Answer: **TRUE**.

Passage 7 — Australia's Airborne Dentists

1. **NOT GIVEN**

work as volunteers — para 2 only says the RFDS is a non-profit providing "free medical care"; this describes the service, not how staff are employed or paid. No mention of volunteers anywhere. Answer: **NOT GIVEN**.

2. **FALSE**

each month contradicts para 4: RFDS dentists "might only visit a community once every few months, or sometimes once per year" — far less often than monthly. Answer: **FALSE**.

3. **NOT GIVEN**

a dental nurse — the passage never mentions who, if anyone, accompanies RFDS dentists on their trips. No information is given either way. Answer: **NOT GIVEN**.

4. **TRUE**

a wide range of dental services ≈ para 6: RFDS dentists "**need to be able to do all kinds of dental procedures, as they don't have the ability to refer patients to more specialised dentists**".

Answer: **TRUE**.

5. **FALSE**

more informed... than urban patients is the reverse of para 8: rural patients "**receive less education about good dental hygiene than their city counterparts do**". Answer: **FALSE**.

6. **NOT GIVEN**

good eating habits — para 9 only mentions education on "**tooth brushing and flossing**". "Such as" gives examples, not an exhaustive list, so this doesn't rule eating habits out — it just never comes up.

Answer: **NOT GIVEN**.

7. **TRUE**

better teeth because their water is treated ≈ para 10: "**City dwellers in Australia use water supplies that have been fluoridated, and their rates of tooth decay are lower because of this**".

Answer: **TRUE**.

8. **computers**

____ **for records** ≈ para 5: dentists bring "**computers for keeping track of patients' treatments**".

Answer: **computers**.

9. **space**

restricted ____ ≈ para 5: "**the small planes that transport dentists have limited space**" — restricted = limited. Answer: **space**.

10. **dentures**

problems... fitting ____ ≈ para 6: "**dentures (artificial teeth) can be very difficult to provide**".

Answer: **dentures**.

11. **toothpaste**

____ **and toothbrushes for regular use** ≈ para 8: "**toothpaste and toothbrushes can be more expensive in outback areas**" — the RFDS supplies these if patients can't afford them. Answer: **toothpaste**.

12. **toothbrushes**

toothpaste and ____ **for regular use** ≈ para 8: "**toothpaste and toothbrushes can be more expensive in outback areas**". Answer: **toothbrushes**.

13. mouthguards

_____ to protect teeth of sports players ≈ para 9: "The RFDS also provides mouthguards for young sports players" to prevent injuries in contact sports. Answer: **mouthguards**.

Passage 8 — Bondi

1. FALSE

the opposite: 'beaches attracted little attention in the first century of British colonial settlement.' Answer: **FALSE**.

2. NOT GIVEN

the passage says settlers later ASSOCIATED Aboriginal culture with the inland desert in their own minds, but never compares actual inland vs. coastal Aboriginal cultures. Answer: **NOT GIVEN**.

3. FALSE

the opposite: 'the cultural associations that formed around Australia's beach landscape would...be **almost entirely European in nature**.' Answer: **FALSE**.

4. TRUE

'the interior of the country represented **hard work and suffering**.' Answer: **TRUE**.

5. TRUE

'**Brighton became the model for beaches** throughout Britain's colonies...seven Sydney beach suburbs still have a Brighton Street.' Answer: **TRUE**.

6. FALSE

the opposite: Sydney's beach suburbs '**shared Brighton's association with health**' and were seen as places of purity and well-being. Answer: **FALSE**.

7. tram

'The beach in the suburb of Coogee was the first ocean beach to become accessible by **tram** from the city.' Answer: **tram**.

8. Manly

'The suburb of **Manly**...attracted...holiday-makers from farms in the countryside.' Answer: **Manly**.

9. Bondi

a sudden demand for housing saw investors build blocks of flats in **Bondi** instead of houses. Answer: **Bondi**.

10. 1954

‘in **1954** when Queen Elizabeth of England and her husband Prince Philip attended a surf carnival there.’
Answer: **1954**.

11. film industry

‘The **film industry** quickly built on the fame that Bondi beach acquired.’ Answer: **film industry**.

12. beach volleyball

‘Bondi was nominated to host the 10,000-seat **beach volleyball** stadium.’ Answer: **beach volleyball**.

13. the environment

‘Some people in the Bondi community did not support the proposal because of concerns for the **environment**.’ Answer: **the environment**.

Passage 9 — Chili Peppers

1. TRUE

Based on her examination of ancient pieces of pottery, Ms. Perry concludes that people in the Americas began cultivating chilies more than 6,000 years ago. Answer: **TRUE**.

2. FALSE

Perry's own view is that **chilies were domesticated early and spread very quickly, just because people liked them** — the food-spoilage theory belongs to different researchers (Billing and Sherman), not Perry. Answer: **FALSE**.

3. NOT GIVEN

The passage says Columbus brought chilies back to Spain, but never mentions whether he personally enjoyed eating them. Answer: **NOT GIVEN**.

4. FALSE

The Portuguese first encountered chilies in Brazil and carried them, along with tobacco and cotton, to Africa — they met chilies in Brazil, not Africa. Answer: **FALSE**.

5. FALSE

Capsaicin stimulates the neural sensors in the tongue and skin that also detect rising temperatures — a genuine physical/neural effect, not purely psychological. Answer: **FALSE**.

6. TRUE

He diluted a chili extract in sugar water until the heat was no longer detectable to a panel of trained tasters — a panel is a group of individuals. Answer: **TRUE**.

7. **germination**

that crush seeds and make germination impossible. Answer: **germination**.

8. **fruit**

capsaicin is found just in the fruit. Answer: **fruit**.

9. **ripening**

becomes much stronger with ripening. Answer: **ripening**.

10. **concentration**

they ate the mild chilies but avoided the ones with a strong concentration of capsaicin.
Answer: **concentration**.

11. **coat**

helps some seeds germinate, possibly by softening the seed coat. Answer: **coat**.

12. **fungus**

protects it from disease that a certain fungus can trigger. Answer: **fungus**.

13. **evolution**

'Capsaicin demonstrates the incredible elegance of evolution,' says Tewksbury. Answer: **evolution**.

Passage 10 — Companies Try Out a New Marketing Practice

1. **C**

the writer questions **whether this marketing concept is an appropriate philosophy in an age of environmental deterioration, resource shortages, explosive population growth, world hunger and poverty, and neglected social services**. Answer: C.

2. **A**

the soft drinks industry has catered to our modern desire for convenience by increasing the **share of single-use disposable bottles**. Answer: A.

3. **C**

it is still besieged by critics who allege that the company does not live up to its reputation.
Answer: C.

4. **D**

cause-related marketing **may make consumers feel that they have fulfilled their philanthropic duties by purchasing a product instead of donating directly**. Answer: D.

5. **A / D / F**

Correct three are **A, D, F**. Para 3: A ≈ **determine the needs, wants, and interests of target markets**; D ≈ **preserves or enhances the consumer's and the society's well-being**; F ≈ **juggle the often conflicting criteria of company profits, consumer satisfaction, and public interest**.

6. **A / D / F**

7. **A / D / F**

8. **B / C / F**

Correct three are **B, C, F**. Para 4: B ≈ **creating innovative flavors such as Rainforest Crunch**; C ≈ **7.5% of Ben & Jerry's pre-tax profits are donated to a variety of social and environmental causes**; F ≈ **the highest-paid employee made no more than five times the lowest-paid worker**.

9. **B / C / F**

10. **B / C / F**

11. **B / D / E**

Correct three are **B, D, E**. Para 6: B ≈ **cosmetics based on natural ingredients**; D ≈ **donates a percentage of profits each year to animal rights groups, homeless shelters, Amnesty International**; E ≈ **appealing recyclable bottles**.

12. **B / D / E**

13. **B / D / E**

Passage 11 — Computer Games for Preschoolers

1. **abilities**

'to understand the range of physical and cognitive **abilities** of preschoolers.' Answer: **abilities**.

2. **parents**

'to understand the expectations of preschoolers' **parents** concerning the handheld systems.' Answer: **parents**.

3. **markets**

'conducting 26 in-home ethnographies in three **markets** across the United States.' Answer: **markets**.

4. **siblings**

'the effects of older **siblings** on game play.' Answer: **siblings**.

5. **experienced**

'to understand both "**experienced**" and "new" preschool users.' Answer: **experienced**.

6. **NOT GIVEN**

the passage says researchers gathered data about '**levels of parental moderation with regard to computer gaming**' — but never specifies this related to time 'after school'. Answer: **NOT GIVEN**.

7. **NOT GIVEN**

the passage describes conducting '**26 in-home ethnographies**' but never characterises the nature or freedom of that access. Answer: **NOT GIVEN**.

8. **TRUE**

'it was one of the few games on the market with **characters that appeal to this young age group**.' Answer: **TRUE**.

9. **FALSE**

the opposite: it 'incorporated a large variety of mechanics...including **using the microphone for blowing or singing**' — not hand controls alone. Answer: **FALSE**.

10. **firm**

'presented the findings to a **firm** specializing in game design.' Answer: **firm**.

11. **simplicity**

'to test the mechanics and to evaluate both their **simplicity**, and whether they were engaging.' Answer: **simplicity**.

12. **field test**

'we went back into the **field test** with a dozen preschoolers and their parents.' Answer: **field test**.

13. **feedback**

'We also collected parents' **feedback** on whether they thought the game was appropriate, engaging, and worth the purchase.' Answer: **feedback**.

Passage 12 — Deep Sea Discovery

1. **TRUE**

Para A: **In theory, once he reached about 15 metres, he should have been plunged into darkness.** Answer: **TRUE**.

2. **TRUE**

Para B: **Even with the red filter removed, Michiels could pick them out without much trouble.** Answer: **TRUE**.

3. **FALSE**

Para B: **no one saw it because no one expected to see it** — the opposite of the statement. Answer: **FALSE**.

4. **FALSE**

Para D: back in Germany, **he bought an assortment of tropical fish and installed them in his lab** — some of the fish he studied were bought in Germany, not found diving in the Red Sea. Answer: **FALSE**.

5. **FALSE**

Para C: **He suspected fluorescence** while still processing the Red Sea dive; Para D then says "**With only a week left in Egypt**" he still lacked equipment to confirm it — so the idea came while still in Egypt, not Germany. Answer: **FALSE**.

6. **TRUE**

Para D: **they are still not sure exactly what is fluorescing... we have a suspicion that it might be made by bacteria**. Answer: **TRUE**.

7. **head**

Para E: **The most common markings tend to be on the body towards the head**. Answer: **head**.

8. **communication**

Para E: **red fluorescence has a very particular function: communication with other members of the species**. Answer: **communication**.

9. **enemies**

Para E: **a whole range of animals employ fluorescence... to ward off enemies**. Answer: **enemies**.

10. **location**

Para E: **Michiels suspects that red-eyed gobies use signals to indicate their location**. Answer: **location**.

11. **species**

Para F: **closely related species do not have completely identical markings**. Answer: **species**.

12. **corals**

Para G: **he noticed corals glow a dark but faint red too. Against this irregular red background, a fish that glows red all over would be hard to distinguish**. Answer: **corals**.

13. **seahorse**

Para H: **One exception is the seahorse, whose eyes are sensitive to red**. Answer: **seahorse**.

Passage 13 — Diagnosing Bridges

1. **D**

Para D: 'subjecting it to pounding from a "shaker", which delivered precise punches to a specific point on the road.' Answer: D.

2. **E**

Para E: 'they then extended the cut until it reached the bottom of the girder and finally, they cut across the flange.' Answer: E.

3. **C**

Para C: 'provided the researchers with a rare opportunity to test their ideas.' Answer: C.

4. **I**

'a microwave inspection system will eventually be applied to most large bridges in the us.' Answer: I.

5. **B**

Para B: 'the device uses microwaves to measure the distance between the sensor and the bridge, much like radar does.' Answer: B.

6. **B**

'every few months, engineers have to clamber over the structure in an attempt to find problems before the bridge shows obvious signs of damage.' Answer: B.

7. **A**

'the device uses microwaves to measure the distance between the sensor and the bridge.' Answer: A.

8. **C**

'a failure in either girder would cause the bridge to fail.' Answer: C.

9. **A**

'successfully identified and located the damage caused by the initial cut.' Answer: A.

10. **two steel girders**

'They have **two steel girders** supporting the load in each section.' Answer: **two steel girders**.

11. **conventional accelerometers**

'the Los Alamos team installed **conventional accelerometers** at several points along across it.' Answer: **conventional accelerometers**.

12. slot

'they first cut a **slot** about 60 centimeters long in the middle of one girder.' Answer: **slot**.

13. flange

'they cut across the **flange** - the bottom of the girder's "I" shape.' Answer: **flange**.

Passage 14 — The Inventor of the Periodic Table – Dmitri Ivanovich Mendeleev

1. teacher

'he won a place at his father's old college, where he initially trained as a **teacher**.' Answer: **teacher**.

2. tuberculosis

'he was often so ill with **tuberculosis** that he was forced to work from his bed.' Answer: **tuberculosis**.

3. conference

'Mendeleev attended the first ever international chemistry **conference**, held in Karlsruhe, Germany.' Answer: **conference**.

4. textbooks

'he thought improved Russian-language chemistry **textbooks** were necessary.' Answer: **textbooks**.

5. translations

'the publication of **translations** into three languages: English, French and German.' Answer: **translations**.

6. properties

'wrote the fundamental **properties** of each element, including atomic weight, on its card.' Answer: **properties**.

7. pattern

'he now knew the **pattern** the elements fell into.' Answer: **pattern**.

8. prediction

'he used his periodic table to **predict** the existence of eight new elements.' Answer: **prediction**.

9. FALSE

'his father, a **graduate of Saint Petersburg's Main Pedagogical Institute**, died when Mendeleev was just 13.' His father had already been to college. Answer: **FALSE**.

10. **TRUE**

'his uncontrollable temper made him unpopular with some of the staff and his classmates.' Answer: **TRUE**.

11. **NOT GIVEN**

he worked briefly with renowned German chemist Robert Bunsen at Heidelberg University — working together is mentioned, but no joint publication is stated. Answer: **NOT GIVEN**.

12. **TRUE**

'he was concerned that Russia was trailing behind Germany in this field.' Answer: **TRUE**.

13. **NOT GIVEN**

the passage states that **Organic Chemistry** (his earlier book) took 61 days to write, but never gives a timeframe for the **SECOND** volume of *The Principles of Chemistry*. Answer: **NOT GIVEN**.

14. **FALSE**

'Lothar Meyer, for example, had proposed a rough periodic table in 1864 and by 1868 had devised one that was very similar to Mendeleev's, but he did not publish it until 1870.' Answer: **FALSE**.

15. **TRUE**

'scientists everywhere started to pay attention to his periodic table. And...Mendeleev's fame and scientific reputation were further enhanced.' Answer: **TRUE**.

16. **FALSE**

'in 1905, the British Royal Society gave him its highest honor, the Copley Medal, for his achievements.' Answer: **FALSE**.

Passage 15 — Dyes and Fabric Dyeing

1. **TRUE**

to date the black, white, yellow and reddish pigments made from ochre, used in cave paintings to over 15,000 BCE, versus fabric production starting around 7,000–2,000 BCE. Answer: **TRUE**.

2. **FALSE**

most natural dyes are adjective dyes, and require the application of a mordant solution — the majority, not rare. Answer: **FALSE**.

3. **TRUE**

it was a common medieval fabric worn in both dyed and natural colors — natural colors means undyed. Answer: **TRUE**.

4. **NOT GIVEN**

Cotton is called a "luxury fabric" and silk "required the highest quality dyes", but their prices are never directly compared. Answer: **NOT GIVEN**.

5. **FALSE**

it was imported all the way from india and was dyed before it was shipped — before, not upon arrival. Answer: **FALSE**.

6. **NOT GIVEN**

Von Hofmann is called "celebrated" and Perkin's mauve was popular, but their fame is never compared. Answer: **NOT GIVEN**.

7. **FALSE**

hardly a year passed until the end of the century without a new synthetic dye being patented — almost every year, not very few. Answer: **FALSE**.

8. **TRUE**

only exists in a few remote areas where people have either little access to synthetic dyes. Answer: **TRUE**.

9. **shellfish**

the mucus found in certain species of shellfish produced the deep rich tyrian purple. Answer: **shellfish**.

10. **insects**

cochineal was extracted from insects. Answer: **insects**.

11. **blackberries**

people who picked blackberries to make jam soon recognized this wild fruit as a source for a blue dye. Answer: **blackberries**.

12. **beehives**

washing beehives in preparation for making mead. Answer: **beehives**.

13. **chemistry**

with the tremendous rise of interest in chemistry in the mid-19th century. Answer: **chemistry**.

Passage 16 — Endangered Chocolate

1. **C**

Para A: its sweet white buds **hang from the trunk and along a few fat branches**. Answer: **C**.

2. **B**

Para F: **using temporary plantings of banana to shade the cacao while it's young**. Answer: **B**.

3. **A**

Para H: imported cacao **is the linchpin of an \$8.6 billion domestic chocolate industry that in turn supports the nation's dairy and nut industries**. Answer: **A**.

4. **E**

Para E: cacao is planted **from South America and the Caribbean to West Africa, East Asia, and New Guinea and Vanuatu in the Pacific**. Answer: **E**.

5. **H**

Para H: witches' broom **devastated the cacao plantations in the Bahia region of Brazil**. Answer: **H**.

6. **C**

Para C: the Maya and Aztecs **frothed up with water and spices to make bracing concoctions**. Answer: **C**.

7. **D**

Para D: **the bean has been thoroughly democratized** — from an elite drink into everyday candy bars. Answer: **D**.

8. **F**

Para F: **annual yields of 50 to 60 pods per tree for 25 to 30 years**. Answer: **F**.

9. **C**

Para C: Linnaeus **gave the cacao tree genus the name Theobroma**. Answer: **C**.

10. **disease**

Para I: research is being carried out by **establishing disease-resistant plants**. Answer: **disease**.

11. **middlemen**

Para I: **at the mercy of local middlemen** who buy the beans then sell them for a higher price — "these people need to be removed from the process". Answer: **middlemen**.

12. **sustainable farming**

Para I: **sustainable farming practices are used**. Answer: **sustainable farming**.

13. profits

Para I: farmers get a pittance compared with the **profits reaped by the rest of the chocolate business**. Answer: **profits**.

Passage 17 — Experience versus Speed

1. B

the anecdote leads straight into **rita and jake stumbled across an interesting set of differences between older and younger minds**. Answer: **B**.

2. D

the typing analogy concludes that **both people may end up taking the same amount of time**. Answer: **D**.

3. A

subjects had to **wait for an arrow that pointed in a certain direction to appear, and then use a mouse to click on the arrow as soon as it appeared**. Answer: **A**.

4. E

4: popular psychology says that as people age their brains ‘slow down’. 5: seniors exploit the left and right side of the brain more extensively than men and women who are decades younger. 6: they paused longer before reacting and took longer to complete the tasks, yet they made 50% fewer errors. 7: older brains solved the problems just as effectively but by different means. Answers: **E / C / B / D**.

5. C

4: popular psychology says that as people age their brains ‘slow down’. 5: seniors exploit the left and right side of the brain more extensively than men and women who are decades younger. 6: they paused longer before reacting and took longer to complete the tasks, yet they made 50% fewer errors. 7: older brains solved the problems just as effectively but by different means. Answers: **E / C / B / D**.

6. B

4: popular psychology says that as people age their brains ‘slow down’. 5: seniors exploit the left and right side of the brain more extensively than men and women who are decades younger. 6: they paused longer before reacting and took longer to complete the tasks, yet they made 50% fewer errors. 7: older brains solved the problems just as effectively but by different means. Answers: **E / C / B / D**.

7. **D**

4: popular psychology says that as people age their brains 'slow down'. 5: seniors exploit the left and right side of the brain more extensively than men and women who are decades younger. 6: they paused longer before reacting and took longer to complete the tasks, yet they made 50% fewer errors. 7: older brains solved the problems just as effectively but by different means. Answers: E / C / B / D.

8. **minds**

just as people's bodies age at different rates, so do their minds. Answer: **minds**.

9. **memories**

the ability to retrieve memories also quickly slides. Answer: **memories**.

10. **knowledge**

the ageing brain taps into its extensive hoard of accumulated **knowledge and experience**. Answer: **knowledge**.

11. **experience**

knowledge and experience already stored in their brain. Answer: **experience**.

12. **speed**

'drawing on both sides of the brain gives them a tactical edge, even if **the speed of each hemisphere's process is slower**.' Answer: **speed**.

Passage 18 — Flying the Coast

1. **FALSE**

unlike Fiordland to the south, which was and still is an uninhabitable wilderness, the Coast in the 1930s was not only habitable, it was also potentially rich — opposite population situations, not similar. Answer: **FALSE**.

2. **NOT GIVEN**

The passage mentions settlers hunting, fishing, logging, and mining, but never says where they originally came from. Answer: **NOT GIVEN**.

3. **TRUE**

isolation was due to the **topography of the place — the mountains, gorges, glaciers, rivers and headlands**, which necessitated long detours. Answer: **TRUE**.

4. **NOT GIVEN**

The passage says bridges were scarce and ferry crossings were often impossible after heavy rains, but never says bridges were swept away. Answer: **NOT GIVEN**.

5. **FALSE**

he transported across the country by rail, wings off, before reassembling it in a local garage — not flown. Answer: **FALSE**.

6. **FALSE**

Coasters queued up for joyrides — the opposite of unwillingness. Answer: **FALSE**.

7. **luxury**

the Fox Moth was a plane that offered considerable luxury. Answer: **luxury**.

8. **mail**

commenced a regular delivery of mail. Answer: **mail**.

9. **beaches**

using remote beaches such as the one at the mouth of the Paringa River, to collect the whitebait. Answer: **beaches**.

10. **subsidy**

thanks in large part to a government-issued subsidy. Answer: **subsidy**.

11. **pilots**

with so many now joining the Air Force, he no longer had enough pilots. Answer: **pilots**.

12. **cargo**

some of Mercer's newly formed team objected to the amount of cargo they had to carry. Answer: **cargo**.

13. **helicopters**

Helicopters were soon found to be ideal machines for the Coast terrain, and quickly took over the vast majority of the local air transport business. Answer: **helicopters**.

Passage 19 — Gobekli Tepe: A Remarkable Discovery

1. **FALSE**

the builders 'lacked even basic technology such as metal tools.' Answer: **FALSE**.

2. **TRUE**

'they shaped blocks of limestone into pillars at a stone quarry. They then moved each pillar...to the Gobekli Tepe site.' Answer: **TRUE**.

3. **NOT GIVEN**

the passage describes the ring arrangement in detail but never links it to the seasons. Answer: **NOT GIVEN**.

4. **NOT GIVEN**

the passage describes the carvings themselves but never says who made them or whether they were specially trained. Answer: **NOT GIVEN**.

5. **FALSE**

the opposite: other sites depict animals in herds, but 'at Gobekli Tepe, **the images are of individual fiercely-looking beasts**.' Answer: **FALSE**.

6. **TRUE**

'these were for years generally **dismissed as nothing more than a jumble of broken stones**...and the site was essentially ignored.' Answer: **TRUE**.

7. **FALSE**

the opposite: he realised the hill's shape was something that '**only man could have created**.' Answer: **FALSE**.

8. **NOT GIVEN**

the passage never mentions any need for permission to study the site. Answer: **NOT GIVEN**.

9. **cemetery**

'it is possible that Gobekli Tepe was built as an ancient **cemetery**.' Answer: **cemetery**.

10. **writing**

'the lack of any **writing** on them make interpreting the builders' motivations virtually impossible.' Answer: **writing**.

11. **settlement**

'archaeologists have found no evidence of a well-established ancient **settlement** such as houses, hearths, cooking sites, or rubbish dumps.' Answer: **settlement**.

12. **bones**

'100,000 fragments of the **bones** of wild game animals like boar, gazelle and deer.' Answer: **bones**.

13. **nuts**

'It would have been full of fruit and **nuts**, all readily available to be collected by gatherers.' Answer: **nuts**.

14. **construction**

‘the process of **construction** itself eventually caused the builders to abandon hunting and gathering.’
Answer: **construction**.

15. **wheat**

‘the original strains of the modern **wheat** that people farm today have been traced to an ancient village only 32 kilometres from Gobekli Tepe.’ Answer: **wheat**.

16. **pigs**

‘the first **pigs** raised in domestic conditions were kept at a village about 100 kilometres away.’ Answer: **pigs**.

Passage 20 — Going Nowhere Fast

1. **TRUE**

the writer contrasts fast communication and space travel with the fact that **when it comes to getting around our cities, we depend on systems that have scarcely changed since the days of gottlieb daimler**. Answer: **TRUE**.

2. **FALSE**

on the contrary, **the pollution belched out by millions of vehicles has dominated the debate about transport**. Answer: **FALSE**.

3. **NOT GIVEN**

the passage only mentions **California** curbing traffic growth — nothing about most US states doing the same. Answer: **NOT GIVEN**.

4. **NOT GIVEN**

there is no mention of hills or gradients in connection with buses or trains anywhere in the passage.
Answer: **NOT GIVEN**.

5. **TRUE**

cars, after all, are popular for very good reasons, as anyone with small children or heavy shopping knows. Answer: **TRUE**.

6. **FALSE**

the writer argues the opposite: **politicians should be trying to lure people out of their cars, not forcing them out**. Answer: **FALSE**.

7. **A**

PRT: **any transport system that threatened the car's dominance would be up against all those with a stake in maintaining the status quo, from private car owners to manufacturers and oil multinationals.** Answer: **A.**

8. **C**

Both: PRT guideways **carry far more traffic, nonstop, than any inner-city road**, and the RUF plan is to **'build a fast network of guideways in a busy city centre'**. Answer: **C.**

9. **A**

PRT: **'you wouldn't have to share your space with strangers.'** Answer: **A.**

10. **A**

PRT: **'hop into a computer-controlled car'** — whereas with RUF **'once on a road, the occupant would take over from the computer'**. Answer: **A.**

11. **B**

RUF: vehicles **'have wheels as well as a slot allowing them to travel on a monorail, so they can drive off the rail onto a normal road.'** Answer: **B.**

12. **B**

RUF: people would have **'a strong incentive not just to use public RUF vehicles, but also to buy their own dual-mode vehicle.'** Answer: **B.**

13. **"C**

Correct three are **C, E & F** — C: the RUF vehicle would become an **electric car** (low pollution); E: **'build a fast network of guideways'** (speed); F: the passage closes by noting **more than 30 million people have died in road accidents** and asks **'surely, we can find a better way to get around?'** (safety). A, B, D and G are never discussed as advantages of PRT/RUF.

Passage 21 — How Consumers Decide

1. **TRUE**

understanding how consumers make decisions, and the crucial role of packaging in this process has been a neglected area of research so far. Answer: **TRUE.**

2. **TRUE**

one heuristic rule is to **choose what a trusted source suggests.** Answer: **TRUE.**

3. **FALSE**

heuristic processing **requires comparatively little effort**, while systematic processing involves much more mental effort — the opposite of the statement. Answer: **FALSE.**

4. **NOT GIVEN**

the passage says the research team's work **focused on** heuristic and systematic processing, but never claims they originated the concept. Answer: **NOT GIVEN**.

5. **TRUE**

systematic processing involves **taking account of the product information, including its price, its perceived quality and so on**. Answer: **TRUE**.

6. **FALSE**

under heuristic processing, consumers **may simply need to be able to distinguish the pack from those of competitors** — packaging needs to stand out, not be similar. Answer: **FALSE**.

7. **D**

testing is ineffective **if the methods you are employing concern one form of thinking... but your consumers are purchasing in the other mode**. Answer: **D**.

8. **A**

changing brands needs packaging **that includes information that does stimulate systematic processing**. Answer: **A**.

9. **advantages**

packaging should communicate the product's characteristics, **highlighting its advantages over possible competitors**. Answer: **advantages**.

10. **situation**

people only engage in systematic processing **when the situation justifies it**. Answer: **situation**.

11. **information**

if we present too much, therefore, they will become confused, leading consumers to choose another product. Answer: **information**.

12. **ingredients**

people often lack the knowledge needed to understand things **such as the ingredients of food products**. Answer: **ingredients**.

13. **simple**

those low in need for cognition **prefer to use very simple forms of thinking**. Answer: **simple**.

Passage 22 — How Tennis Rackets Have Changed

1. **FALSE**

yet five years previously, he had been regarded as a talented outsider who entered but never won the major tournaments. Answer: **FALSE**.

2. **FALSE**

another change was so subtle as to pass more or less unnoticed. Answer: **FALSE**.

3. **NOT GIVEN**

touring professionals have their rackets customised to their specific needs is mentioned, but nothing is said about a stringer travelling with players. Answer: **NOT GIVEN**.

4. **FALSE**

the rackets they use now weigh more than the average model. Answer: **FALSE**.

5. **NOT GIVEN**

the amateur german tennis player werner fischer started playing with the so-called spaghetti-strung racket — no mention of who designed it. Answer: **NOT GIVEN**.

6. **TRUE**

they will continually change it depending on various factors including the court surface, climatic conditions, and game styles. Answer: **TRUE**.

7. **TRUE**

much of the serving power of US professional player Pete Sampras was attributed to the addition of four to five lead weights onto his rackets. Answer: **TRUE**.

8. **paint**

they even experimented with different kinds of paint. Answer: **paint**.

9. **topspin**

it created a string bed that generated so much topspin that it was quickly banned. Answer: **topspin**.

10. **training**

an aspect of the game that is equal in significance to nutrition or training. Answer: **training**.

11. **layer**

natural gut made from the outer layer of sheep or cow intestines. Answer: **layer**.

12. **weights**

addition of four to five lead weights onto his rackets. Answer: **weights**.

13. **grips**

replaced the original grips of his rackets with something thinner. Answer: **grips**.

Passage 23 — Katherine Mansfield

1. **FALSE**

her origin name was **Katherine Mansfield Beauchamp Murry**, but she published **using the pen name of Katherine Mansfield** — not her full origin name. Answer: **FALSE**.

2. **NOT GIVEN**

the passage only says **her first published stories appeared in the High School Reporter** — nothing about winning a prize. Answer: **NOT GIVEN**.

3. **TRUE**

the Maori people **were often portrayed in a sympathetic light in her later stories, such as How Pearl Button Was Kidnapped**. Answer: **TRUE**.

4. **FALSE**

at Queen's, it was the **cello, an occupation that she believed, during her time at Queen's, she would take up professionally** — not writing. Her ambition to become a professional writer came later, in 1906. Answer: **FALSE**.

5. **FALSE**

she was **appreciated amongst fellow students at Queen's for her lively and charismatic approach to life and work**. Answer: **FALSE**.

6. **TRUE**

Mansfield did not actively support the suffragette movement in the UK, a leading political cause of the day. Answer: **TRUE**.

7. **1906**

she returned to her New Zealand home in **1906**. Answer: **1906**.

8. **Australia**

several works published in Australia in a magazine called Native Companion, which was her first paid writing work. Answer: **Australia**.

9. family

Mansfield with the provincial New Zealand lifestyle, and with her family. Answer: family.

10. bankruptcy

their attempt to set up as writers in Paris was cut short by Murry's bankruptcy. Answer: bankruptcy.

11. writers

she and Murry developed close contact with other well-known writers of the time such as DH Lawrence, Bertrand Russell and Aldous Huxley. Answer: writers.

12. reputation

it was the publication of Bliss and Other Stories in 1920 that was to solidify Mansfield's reputation as a writer. Answer: reputation.

13. husband

after her death, her husband, Murry, took on the task of editing and publishing her works. Answer: husband.

Passage 24 — Living in a Noisy World

1. 85 dBA

his agency's definition of hazardous noise is sound that exceeds 85 dba. Answer: 85 dBA.

2. high-frequency

continued exposure to excessive noise can lead to difficulties with high-frequency sound discrimination. Answer: high-frequency.

3. stomach problems

the results showed that exposure to noise caused stomach problems in healthy human beings. Answer: stomach problems.

4. noise maps

European cities are developing noise maps of those cities to help leaders determine noise pollution strategies. Answer: noise maps.

5. E

researchers from the center for disease control and prevention reported that... an estimated 12.5% of american children have noise-induced hearing threshold shifts. Answer: E.

6. **D**

for many people in the united states, noise has drastically affected the quality of their lives, says Arline Bronzaft. Answer: **D**.

7. **C**

osha does require that any employer with staff overexposed to noise safeguard those employees against the harmful impact of noise, says Craig Moulton. Answer: **C**.

8. **D**

bronzafst stresses that governments worldwide need to direct more financial resources towards studies investigating the effects of noise. Answer: **D**.

9. **A**

WHO declared: noise-induced hearing impairment is the most prevalent irreversible occupational hazard. Answer: **A**.

10. **B**

niosh studies... show that 90% of coal miners have hearing impairment by age 52. Answer: **B**.

11. **B**

acoustical tiles installed in the classrooms... the transit authority also undertook work on the track near the school in order to reduce noise levels. Answer: **B**.

12. **C**

the map data will be fed into computer models that will help test the sound impact of new road layouts or buildings before construction begins. Answer: **C**.

13. **D**

the passage covers noise's effects on hearing, health, children's learning, and city planning worldwide — far broader than just workplace performance, city rankings, or worker education. Answer: **D**.

Passage 25 — Manhattan's Skyscrapers Race to the Top

1. **FALSE**

‘the first tall buildings to rise in New York City appeared in the late 1800s’ — the 19th century, not the 20th. Answer: **FALSE**.

2. **TRUE**

‘other essential factors included an increasing population, soaring property values, rivalry, geology, vanity, and greed.’ Answer: **TRUE**.

3. **FALSE**

it was the opposite: New York had the advantage — ‘while Chicago was built on muddy soil, **Manhattan stood on solid granite.**’ Answer: **FALSE.**

4. **NOT GIVEN**

the passage says it was surpassed ‘four years later’ — this refers to being overtaken, not the building’s construction time. Answer: **NOT GIVEN.**

5. **NOT GIVEN**

the passage says it became ‘as much a **symbol** of New York City as the...Statue of Liberty’ — a symbol, not a claim about visitor numbers. Answer: **NOT GIVEN.**

6. **TRUE**

‘unable to find tenants, Raskob’s magnificent structure was **mockingly called the ‘Empty State Building.’**’ Answer: **TRUE.**

7. **NOT GIVEN**

the passage mentions its ‘appearance in the 1933 movie King Kong’ but never claims it was the **FIRST** skyscraper ever shown in a film. Answer: **NOT GIVEN.**

8. **TRUE**

‘Le Corbusier is **often cited as the villain** behind this move towards less elegant architecture.’ Answer: **TRUE.**

9. **brake**

‘Elisha Otis demonstrated his elevator safety **brake.**’ Answer: **brake.**

10. **steel**

‘Henry Bessemer patented his process for producing a new kind of strong and low-cost **steel.**’ Answer: **steel.**

11. **laws**

‘city authorities issued zoning **laws** that stipulated that tall buildings must become narrower at higher levels.’ Answer: **laws.**

12. **automobile**

‘the building’s distinctive ornamentation resembled features of the **automobile** from which Chrysler made his fortune.’ Answer: **automobile.**

13. **fog**

‘a B-25 bomber almost brought it down, crashing...between the 78th and 80th floors in dense **fog.**’ Answer: **fog.**

14. surfaces

'buildings started to favor a modern look with a sleek design and also gleaming **surfaces**.' Answer: **surfaces**.

15. model

'it became a **model** for designing other office towers built from the 1960s onward.' Answer: **model**.

16. walking

'New York City remains one of the world's great **walking** cities.' Answer: **walking**.

Passage 26 — Moving Continents

1. earthquakes and floods

Para A: Ortelius suggested the Americas were torn away from Europe and Africa **by earthquakes and floods**. Answer: **earthquakes and floods**.

2. a supercontinent

Para A: all the present continents used to **form one supercontinent, which he called Pangaea**. Answer: **a supercontinent**.

3. fossils

Para B: **the occurrences of unusual geologic structures and of fossils** — of both plants and animals — found on the matching coastlines. Answer: **fossils**.

4. climate change

Para C: the drifting of continents explained not only the fossil occurrences but also the **evidence of dramatic climate change on some continents**. Answer: **climate change**.

5. plates

Para E: **the Earth's outermost layer is fragmented into a dozen or more plates** of various sizes. Answer: **plates**.

6. earthquakes / volcanic eruptions

Para F: **why earthquakes and volcanic eruptions occur in very specific areas** around the world. Answer: **earthquakes / volcanic eruptions**.

7. E

Para E: **new evidence from ocean floor exploration and other studies rekindled interest in his theory**. Answer: **E**.

8. C

Para C: **indications that there were once glaciers in regions of South Africa led to the conclusion** this land was once far from the tropics; coal in Spitzbergen shows the opposite. Answer: C.

9. D

Para D: it **could not satisfactorily answer the most fundamental question raised by his critics**. Answer: D.

10. B

Para B: **physically impossible for most of these creatures to have swum or have been transported across the vast ocean**. Answer: B.

11. D

Para D: his proposal **was not well received, even though it seemed to agree with the scientific information available at the time**. Answer: D.

12. E

Para E: leading to the **development of the theory of plate tectonics**, which the paragraph goes on to outline. Answer: E.

Passage 27 — Museum Blockbuster

1. C

Para C: **requires continued product development** — regular exhibition changes and innovations. Answer: C.

2. A

Para A: people **will stand in line for hours to see** blockbuster exhibitions. Answer: A.

3. B

Para B: words that appear in descriptions of blockbuster are '**less scholarly**', '**non-elitist**' and '**popularist**'. Answer: B.

4. B

Para B: detractors argue **blockbusters are designed to appeal to the lowest common denominator**, while others extol the virtues of catering for a broad selection of the community rather than an elite sector — a dispute over who blockbusters should target. Answer: B.

5. customers

Para C: **the visiting publics have become customers rather than visitors**. Answer: customers.

6. public relation skills

Para C: as succinctly summarised in the Economist **'business nous and public relation skills'** were essential requirements for a director. Answer: **public relation skills**.

7. New museology

Para D: **the new museology has resulted in** the convergence of museums, the heritage industry, and tourism. Answer: **New museology**.

8. tourist attractions

Para D: whether it is appropriate to see **museums primarily as tourist attractions**. Answer: **tourist attractions**.

9. A

Correct two are **A & D** — para E: **money that they need to update parts of their collections or to repair buildings** (A); para G: **crowds are good for the local economy... shops, hotels, restaurants, the transport industry and retailers** (D). B and C describe general museology trends rather than advantages of blockbusters specifically, and E is never linked to directors' personal benefit.

10. D

11. B

Correct three are **B, C & E** — para G: **creating, mounting or hiring blockbusters is exhausting for staff** (B); para G: museums will be **seen as part of the entertainment and tourism industry, rather than as cultural icons** (C); para F: **blockbusters require large capital expenditure** (E). A softens "choose between" into "hesitated", D and F are close paraphrases already covered by B, and F swaps the passage's "vagaries" (uncertainties) for "fantasies" — a mismatched word.

12. C

13. E

Passage 28 — New Zealand's Early Crafts and Traditions

1. TRUE

during the 13th century that their canoes eventually landed on New Zealand's shores. Answer: **TRUE**.

2. NOT GIVEN

The passage only says Europeans arrived "in search of new opportunities" — no mention of a specific interest in trade. Answer: **NOT GIVEN**.

3. NOT GIVEN

The same craftsmen made both tools and weapons — the passage never describes a division where individuals specialised in either one or the other. Answer: **NOT GIVEN**.

4. FALSE

Pottery is an example of this, despite the fact that the clay needed to make pots and bowls could easily be found in the new country — pottery was NOT practised in New Zealand. Answer: **FALSE**.

5. TRUE

Whakairo means 'decorative work', and the same term can also apply to weaving. Answer: **TRUE**.

6. FALSE

it was the few people of high status rather than low-ranking members of a tribe who would possess such objects. Answer: **FALSE**.

7. feathers

feathers from kiwi, pigeons or other native birds might be attached. Answer: **feathers**.

8. hood

neither was a hood a feature of this garment. Answer: **hood**.

9. shoulder

pins allowed longer cloaks to be secured at the shoulder. Answer: **shoulder**.

10. insulation

A cloak made from fur or wool could provide insulation from the cold, but not so a cloak made of flax. Answer: **insulation**.

11. water

the fibres were left to soak in water over a period of time in order to soften them. Answer: **water**.

12. iron

the mud was rich in iron due to New Zealand's volcanic landscape. Answer: **iron**.

13. spear tips

the tough fibres could reduce the impact of spear tips during a fight with enemy tribes. Answer: **spear tips**.

Passage 29 — Obesity: Do Citizens of Developed Countries Really Have a Choice?

1. **B**

Nestle: the food industry **needs to promote the consumption of ever greater quantities to remain competitive, increase sales, and keep shareholders happy.** Answer: B.

2. **G**

Lakdawalla: **agricultural technology has brought down the cost of food, and because it is less costly, americans eat more.** Answer: G.

3. **H**

Philipson: **technological advances in the work place have made americans' lives less strenuous, causing people to burn fewer calories than they once did.** Answer: H.

4. **E**

Gortmaker: **cardiovascular disease and hypertension are just two of the conditions that researchers have connected with obesity, and people need to be aware of the health risk.** Answer: E.

5. **F**

NIH: **only 15 per cent of americans get the recommended amount of vigorous activity each week, and a quarter reported no physical activity at all outside of work.** Answer: F.

6. **E**

Gortmaker: asthma sufferers seek help, 'while **those suffering from obesity rarely do unless they are experiencing serious problems.**' Answer: E.

7. **A**

OECD: **us citizens are more likely to be obese than those in any other nation.** Answer: A.

8. **D**

CDCP: **that results in more than 30 per cent of men and women in america being classified as obese, more than twice as many as 40 years ago.** Answer: D.

9. **YES**

america produces far more food than its citizens need. the us food industry provides a daily average of 3,800 calories per person, twice an individual's requirement. Answer: YES.

10. **YES**

soft-drink companies use their economic power to get vending machines into schools. Answer: YES.

11. **NO**

on the contrary, **the average american uses 68kg of sweeteners such as refined sugar and corn syrup each year, up from 5.5kg in 1910** — a large increase, not a decrease. Answer: **NO**.

12. **NOT GIVEN**

the passage gives a single combined figure ('close to \$100 billion a year' for direct + indirect expenses together) but never states the two are equal in size. Answer: **NOT GIVEN**.

13. **YES**

exercise... is now a luxury that must be paid for either in money or time or both. Answer: **YES**.

Passage 30 — Stonehenge

1. **deer antlers**

'primitive tools, which may have been fashioned out of **deer antlers**.' Answer: **deer antlers**.

2. **timber posts**

'deep pits...may have once held a ring of **timber posts**.' Answer: **timber posts**.

3. **tree trunks**

'fashioned sledges and rollers out of **tree trunks**.' Answer: **tree trunks**.

4. **oxen**

'wicker baskets...hailed by **oxen**.' Answer: **oxen**.

5. **glaciers**

'it was **glaciers**, not humans, that carried the bluestones to Salisbury Plain.' Answer: **glaciers**.

6. **druids**

'Stonehenge was the work of **druids**.' Answer: **druids**.

7. **burial ground**

'Stonehenge once served the function of **burial ground**.' Answer: **burial ground**.

8. **calendar**

'operated as a form of **calendar**.' Answer: **calendar**.

9. **TRUE**

the ditch and henge were dug first (Stage 1); '**several hundred years later**...builders hoisted...bluestones...into standing positions' (Stage 2). Answer: **TRUE**.

10. **TRUE**

‘sandstone slabs...were arranged into an **outer crescent or ring**; some were assembled into...trilithons that stand tall **in the centre** of Stonehenge.’ Answer: **TRUE**.

11. **FALSE**

the opposite is true: ‘work continued at Stonehenge until roughly 1600 BCE, with the bluestones in particular being **repositioned multiple times**.’ Answer: **FALSE**.

12. **FALSE**

the opposite: ‘in the mid-twentieth century, radiocarbon dating **demonstrated that Stonehenge stood more than 1,000 years before the Celts** inhabited the region’ — disproving Aubrey’s druid theory. Answer: **FALSE**.

13. **TRUE**

‘Bones, tools and other artefacts found on the site seem to **support this hypothesis**’ (that several distinct tribes contributed). Answer: **TRUE**.

14. **NOT GIVEN**

the passage presents two competing theories — ‘some believe that they were immigrants from the European continent, while others maintain that they were probably native Britons’ — with no single accepted answer. Answer: **NOT GIVEN**.

15. **NOT GIVEN**

the passage says ‘critics maintain...’ without ever specifying that these critics were astronomers. Answer: **NOT GIVEN**.

16. **NOT GIVEN**

the passage only says this led ‘a group of British archaeologists to speculate’ — nothing about worldwide acceptance. Answer: **NOT GIVEN**.

Passage 31 — Sydney Opera House

1. **FALSE**

Para B: **won by Jørn Utzon, a relatively little-known architect from Denmark** — the opposite of “famous”. Answer: **FALSE**.

2. **FALSE**

Para B: Saarinen **looked through their rejected entries and stopped at the Utzon design**, meaning the other three judges had initially rejected it. Answer: **FALSE**.

3. **TRUE**

Para C: **his early exposure to shipbuilding provided the inspiration for the design of the roof.**
Answer: **TRUE.**

4. **NOT GIVEN**

opera houses he had seen in Mexico — para C only says "From his travels to Mexico, he had the idea of placing his building on a wide horizontal platform"; no opera houses in Mexico are mentioned at all. Answer: **NOT GIVEN.**

5. **TRUE**

Para D: **Construction of the platform began in 1959, and throughout the early 1960s Utzon amended his original designs.** Answer: **TRUE.**

6. **TRUE**

Para E: **there were protests through the streets led by prominent architect Harry Seidler and others, demanding Utzon be reinstated as architect.** Answer: **TRUE.**

7. **FALSE**

Para E: **He never returned and new architects were appointed to complete the building in his absence.** Answer: **FALSE.**

8. **102 million**

Para E: **the Opera House was not formally completed until 1973, having cost \$102 million.**
Answer: **102 million.**

9. **Sweden**

Para G: **1,056,000 white and cream-coloured tiles manufactured in a factory in Sweden.**
Answer: **Sweden.**

10. **granite**

Para G: **the building's exterior is mostly clad with granite quarried in Australia.** Answer: **granite.**

11. **four**

Para H: **including the four resident companies.** Answer: **four.**

12. **forecourt**

Para I: **There is a sizeable outdoor forecourt from which people ascend to the main entrance.**
Answer: **forecourt.**

13. harbour

Para J: **enabled theatre patrons to see the harbour for the first time from the theatre foyers.**
Answer: **harbour.**

Passage 32 — Temperature Rising

1. TRUE

in scotland higher winter temperatures have led several ski areas to **diversify into summer leisure pursuits such as golf, and paragliding in the winter.** Answer: **TRUE.**

2. TRUE

intense forest fires in recent years have limited public access to national and state parks.
Answer: **TRUE.**

3. NOT GIVEN

\$578 billion is given as the tourism industry's **total revenue in 2002**, not the cost/consequence of global warming specifically. Answer: **NOT GIVEN.**

4. NOT GIVEN

the passage discusses ski-resort closures in the **United States** (Michael Berry) and low-altitude resorts in **Austria** (Kitzbuhel) — no Australian ski resort is mentioned anywhere. Answer: **NOT GIVEN.**

5. diving industry

'one of the most impacted areas is the diving industry, whose very survival depends on the health of coral reefs.' Answer: **diving industry.**

6. colourless

bleaching 'essentially heats coral until it sheds the algae that feed it, **leaving it colourless.**' Answer: **colourless.**

7. two

the IPCC '**estimates a rise of between two- and 10.4-degrees Fahrenheit by 2100.**' Answer: **two.**

8. polar bears

'scientists feel polar bears will die out within the next hundred years or so.' Answer: **polar bears.**

9. F

Michael Berry: **'the reason why some small ski resorts aren't around anymore is because they decided they could not afford to install snow-making systems.'** Answer: **F.**

10. **A**

Bill MacCracken: 'we're slowly going to be getting more extreme weather, but **we can't say that any particular type of weather is due to the changing climate.**' Answer: **A**.

11. **D**

John Rumney: '**we hear tourists say, "this doesn't look like the postcard or as it did three years ago".**' Answer: **D**.

12. **G**

James Wilson believes that **the low-lying Maldiv Islands of the Indian Ocean** will have gone within 20 to 30 years. Answer: **G**.

13. **B**

Robert Mills reports that cooler summers and warmer winters have resulted in **resort owners having to restrict outdoor sports such as swimming in lakes and ice fishing.** Answer: **B**.

Passage 33 — The Ascension Island Experiment

1. **FALSE**

the island 'is certainly difficult for an ordinary commercial traveler to reach even today.' Answer: **FALSE**.

2. **TRUE**

'its volcano is especially active to this day.' Answer: **TRUE**.

3. **FALSE**

the island 'was given the name Ascension Island two years later by a second Portuguese explorer, **Alfonso de Albuquerque**' — not Castella. Answer: **FALSE**.

4. **FALSE**

it was the opposite: '**coastal regions...supported a few small, underdeveloped plants, while the central peaks were essentially bald.**' Answer: **FALSE**.

5. **NOT GIVEN**

the passage never mentions the Portuguese military wanting to use Ascension as a base — only the **BRITISH** military later did. Answer: **NOT GIVEN**.

6. **TRUE**

'it was a thriving waystation providing much needed provisions for **British ships.**' Answer: **TRUE**.

7. NOT GIVEN

the passage only says Ascension ‘experienced strong trade winds from southern Africa’ — there is no mention of water being imported from there. Answer: **NOT GIVEN**.

8. TRUE

‘the little rain that did fall swiftly disappeared in the hot environment.’ Answer: **TRUE**.

9. expectations

‘Darwin later admitted that his **expectations** of Ascension were low.’ Answer: **expectations**.

10. potential

‘Darwin became excited by what he considered to be its **potential**.’ Answer: **potential**.

11. Antarctica

‘an ambitious journey around the icy polar region of **Antarctica** aboard HMS Erebus and HMS Terror.’
Answer: **Antarctica**.

12. evaporation

‘These would provide shade, helping to reduce **evaporation** after rain.’ Answer: **evaporation**.

13. gardens

‘each depositing trees selected from special **gardens** maintained by scientists in Europe, South Africa and South America.’ Answer: **gardens**.

14. mist

‘The trees capture sea **mist**, creating moisture in the centre of an essentially dry island.’ Answer: **mist**.

15. ecosystem

‘This forest is a fully functioning, but in fact totally artificial **ecosystem**.’ Answer: **ecosystem**.

16. colonies

‘could one day be used to transform the planet Mars to make it sustainable for **colonies** from Earth.’
Answer: **colonies**.

Passage 34 — A review of Nigel Townson's The British at Play

1. three million

an estimated three million britons take part in some sort of sporting activity every week.
Answer: **three million**.

2. strangers

sport is **acceptable** when initiating social interaction with **strangers**. Answer: **strangers**.

3. expressions

expressions from sport have passed into general use: we talk about 'team players' in situations that have nothing to do with sport. Answer: **expressions**.

4. loyalty

informal associations **define themselves by their loyalty to their own group**. Answer: **loyalty**.

5. opposition

and opposition to others, the 'out groups', and in an extreme form, this opposition leads to the phenomenon of football hooliganism. Answer: **opposition**.

6. stereotypes

the author shows **what is wrong with the well-known stereotypes of soccer hooligans**. Answer: **stereotypes**.

7. fan behaviour

he argues that **media coverage of fan behaviour helps to create a climate in which hooliganism occurs**. Answer: **fan behaviour**.

8. panic

the media sensationalises and exaggerates it, with the result that an atmosphere of **panic builds up in the country**. Answer: **panic**.

9. C

9: this greater visibility of women highlights the weakening of the traditional view that sport is mainly for men. 10: that desire has been encouraged, if not created, by the emphasis in sport on images of 'ideal' male and female bodies. 11: sport fits in well with the global tv world of beautiful and perfectly muscled young people, exercising or playing sports, dressed in the latest fashionable sports gear. 12: football clubs turn themselves into businesses, raising money by selling merchandise and by selling their shares on the stock market. 13: the book did not, for me, go far enough in transmitting the power, the energy, the passion, the emotion and the joy of sport. Answers: C / F / D / G / B.

10. **F**

9: this greater visibility of women highlights the weakening of the traditional view that sport is mainly for men. 10: that desire has been encouraged, if not created, by the emphasis in sport on images of 'ideal' male and female bodies. 11: sport fits in well with the global tv world of beautiful and perfectly muscled young people, exercising or playing sports, dressed in the latest fashionable sports gear. 12: football clubs turn themselves into businesses, raising money by selling merchandise and by selling their shares on the stock market. 13: the book did not, for me, go far enough in transmitting the power, the energy, the passion, the emotion and the joy of sport. Answers: C / F / D / G / B.

11. **D**

9: this greater visibility of women highlights the weakening of the traditional view that sport is mainly for men. 10: that desire has been encouraged, if not created, by the emphasis in sport on images of 'ideal' male and female bodies. 11: sport fits in well with the global tv world of beautiful and perfectly muscled young people, exercising or playing sports, dressed in the latest fashionable sports gear. 12: football clubs turn themselves into businesses, raising money by selling merchandise and by selling their shares on the stock market. 13: the book did not, for me, go far enough in transmitting the power, the energy, the passion, the emotion and the joy of sport. Answers: C / F / D / G / B.

12. **G**

9: this greater visibility of women highlights the weakening of the traditional view that sport is mainly for men. 10: that desire has been encouraged, if not created, by the emphasis in sport on images of 'ideal' male and female bodies. 11: sport fits in well with the global tv world of beautiful and perfectly muscled young people, exercising or playing sports, dressed in the latest fashionable sports gear. 12: football clubs turn themselves into businesses, raising money by selling merchandise and by selling their shares on the stock market. 13: the book did not, for me, go far enough in transmitting the power, the energy, the passion, the emotion and the joy of sport. Answers: C / F / D / G / B.

13. **B**

9: this greater visibility of women highlights the weakening of the traditional view that sport is mainly for men. 10: that desire has been encouraged, if not created, by the emphasis in sport on images of 'ideal' male and female bodies. 11: sport fits in well with the global tv world of beautiful and perfectly muscled young people, exercising or playing sports, dressed in the latest fashionable sports gear. 12: football clubs turn themselves into businesses, raising money by selling merchandise and by selling their shares on the stock market. 13: the book did not, for me, go far enough in transmitting the power, the energy, the passion, the emotion and the joy of sport. Answers: C / F / D / G / B.

Passage 35 — The Burgess Shale Fossils

1. **TRUE**

popularised in 1989 in a book, wonderful life, by stephen jay gould. Answer: TRUE.

2. **NOT GIVEN**

the passage only says **he is said to have tripped over a block of shale in 1909** — nothing about needing official permission. Answer: **NOT GIVEN**.

3. **TRUE**

it is one of the earliest fossil beds to contain impressions of the soft body parts alongside the remains of bones and shells. Answer: **TRUE**.

4. **FALSE**

these animals originally existed below an ocean. Answer: **FALSE**.

5. **FALSE**

Hallucigenia is now thought to be an ancestor of the modern group of arthropods. Answer: **FALSE**.

6. **Cambrian explosion**

the burgess shale began to form soon after a period of time known as the cambrian explosion. Answer: **Cambrian explosion**.

7. **railway**

following reports of fabulous fossils finds by construction workers on canadian railways. Answer: **railway**.

8. **1979**

described by palaeontologist Simon Conway Morris when he re-examined walcott's specimens in 1979. Answer: **1979**.

9. **evolution**

this suggests that the evolution of such complex life went on uninterrupted. Answer: **evolution**.

10. **fan**

a tail in the shape of a fan. Answer: **fan**.

11. **food**

the proboscis had a set of grasping claws on the end, with which it grabbed food and stuffed it into its mouth. Answer: **food**.

12. **leech**

nectocaris, meanwhile, could be mistaken for a leech. Answer: **leech**.

13. walk

⚠ Not directly stated in this excerpt of the passage — based on the well-known real interpretation of Hallucigenia's spines (originally mistaken for walking legs before later research revised this view). Flagged as an inferred answer, not a verbatim citation. Answer: **walk** (unconfirmed).

Passage 36 — The Clipper Races

1. FALSE

the british east india company had the monopoly on trade with china and india... no rival could legally import tea or other goods from these countries at this time. A monopoly means no competition. Answer: **FALSE**.

2. TRUE

its priority was to minimise costs by carrying as much as possible on each ship. Answer: **TRUE**.

3. TRUE

even with favourable sailing conditions, the round trip lasted almost two years. Answer: **TRUE**.

4. NOT GIVEN

the passage discusses the length and cost of voyages, but never their level of danger or risk. Answer: **NOT GIVEN**.

5. FALSE

many were bought by merchants or their captains, who continued to plough the seas between britain and china. Answer: **FALSE**.

6. FALSE

consumers believed the fresher and earlier-picked the tea, the better the resulting drink — younger, not mature, leaves were preferred. Answer: **FALSE**.

7. sails

clippers had masses of sails on tall masts. Answer: **sails**.

8. storms

the narrower beam made them less powerful during storms, but faster in calmer weather. Answer: **storms**.

9. Challenger

richard green, built the aptly named clipper challenger... the British ship beat its rival to London by two days. Answer: **Challenger**.

10. **ports**

competition was really ignited when new ports were opened up for trade in china. Answer: **ports**.

11. **paperwork**

the clippers set off sometimes not even taking time to fill out the official paperwork. Answer: **paperwork**.

12. **tugs**

they would be towed by tugs, up the river and into the docks. Answer: **tugs**.

13. **money**

the races were about more than just money: the crews... were expert sailors, proud of their ships, and they delighted in competing against each other. Answer: **money**.

Passage 37 — The Development of Plastics

1. **photographic film**

the best-known celluloid product was photographic film. Answer: **photographic film**.

2. **Bakelite**

Baekeland called it Bakelite. Answer: **Bakelite**.

3. **electrical switches**

used in the manufacture of electrical switches. Answer: **electrical switches**.

4. **Britain**

chemists in Britain discovered that the gas ethylene would polymerize under heat and pressure to form a thermoplastic they called polythene. Answer: **Britain**.

5. **fireproof**

a fireproof plastic suitable for drains and gutters. Answer: **fireproof**.

6. **"hard"**

a hard, clear material like glass. Answer: **hard, clear material**.

7. **commonly produced as**

was commonly produced as a foam. Answer: **commonly produced as (foams)**.

8. **FALSE**

the same bonding principle, polymerization, is the basis of the creation of a huge range of plastics — rubber and plastics share the same underlying structure. Answer: **FALSE**.

9. **NOT GIVEN**

The passage only says Hyatt won the prize with celluloid — his profession is never mentioned. Answer: **NOT GIVEN**.

10. **FALSE**

Celluloid can be repeatedly softened and reshaped by heat, but Bakelite **cannot be softened by heat and reshaped once they have set** — opposite reactions. Answer: **FALSE**.

11. **TRUE**

different types of plastic are often found in the same items and call for different treatments. Answer: **TRUE**.

12. **FALSE**

starch is attacked by bacteria and causes the plastic to fall apart — the opposite of more durable. Answer: **FALSE**.

13. **TRUE**

bottles made of such materials have to be stored in the dark. Answer: **TRUE**.

Passage 38 — The Early History of Scott and Bowne's Cod Liver Oil

1. **FALSE**

the oil grew darker according to how long the livers were left — the longest time produced the DARKEST oil, not the lightest. Answer: **FALSE**.

2. **TRUE**

de Jongh, of the Netherlands, **concluded that the light-brown oil was the healthiest** and attributed this to its chemical content. Answer: **TRUE**.

3. **TRUE**

he **produced the first extensive chemical analysis of cod liver oil** (researcher), and 'Dr. de Jongh's Light Brown Cod Liver Oil' was marketed throughout Europe (supplier). Answer: **TRUE**.

4. **NOT GIVEN**

only de Jongh's own suggestions for dealing with the taste/smell are mentioned — the passage never says "many scientists" worked on this. Answer: **NOT GIVEN**.

5. **NOT GIVEN**

the passage never directly compares the two companies' experimental methods. Answer: **NOT GIVEN**.

6. **FALSE**

'**though not a doctor or pharmacist by training, Scott had the eye for opportunity** that was necessary for achievement in business.' Answer: **FALSE**.

7. **NOT GIVEN**

Scott sourced oil from the Lofoten Islands, already described as 'the world center of cod fishery' — the passage never says whether Scott himself discovered or was the first to use this location. Answer: **NOT GIVEN**.

8. **TRUE**

the codfish streamed to the islands in early january to lay their eggs, and by the end of april were gone — January to April is the first four months of the year. Answer: **TRUE**.

9. **wine**

de Jongh recommended overcoming the taste 'by following it with some fruit or biscuit, or **glass of wine**.' Answer: **wine**.

10. **coffee**

'it was often combined with **coffee**, although a few people recommended taking the oil with tomato ketchup.' Answer: **coffee**.

11. **milk**

'**Palatable as milk**' became a key phrase in Scott's advertising. Answer: **milk**.

12. **rope**

'a thick **rope**, wrapped round his waist, shoulders and hands, secures the load on his back.' Answer: **rope**.

13. **tail**

'its **tail** sweeping the floor.' Answer: **tail**.

14. **spots**

'the brown and amber **spots** all over its body.' Answer: **spots**.

15. **quality**

the realistic image 'served as a reassurance of **quality** in a market that contained some impure, unsafe products.' Answer: **quality**.

16. **building**

‘in one instance painted several stories high on the side of a **building**.’ Answer: **building**.

Passage 39 — The Extinction of Cave Bear

1. **FALSE**

He dissolves 30,000-year-old animal bones in hydrochloric acid, which is strong enough to burn through metal, until what's left is a speck of powder — not measuring intact bones. Answer: **FALSE**.

2. **NOT GIVEN**

The passage says Bocherens is "at the forefront" of cave bear research, but never claims he was the first to study them in a lab. Answer: **NOT GIVEN**.

3. **TRUE**

a European species that died out 25,000 years ago. Answer: **TRUE**.

4. **TRUE**

in the Middle Ages, the massive skulls were attributed to dragons. Answer: **TRUE**.

5. **FALSE**

Males weighed up to 1,500 pounds, 50 per cent more than the largest modern grizzly bears — a big difference, not similar. Answer: **FALSE**.

6. **NOT GIVEN**

The passage says Neanderthals shared the continent with cave bears, but never compares their understanding of bear behaviour to that of modern humans. Answer: **NOT GIVEN**.

7. **skeletons**

the floor is covered with 150 cave bear skeletons. Answer: **skeletons**.

8. **skull**

a cave bear skull was perched on a stone slab in the center of one chamber. Answer: **skull**.

9. **shelter**

cave bears and human beings sought shelter here from winter weather. Answer: **shelter**.

10. **plants**

Bocherens has found that cave bears ate mainly plants. Answer: **plants**.

11. Europe

altered the distributions of plant species across Europe. Answer: Europe.

12. teeth

according to a DNA analysis of teeth found near the Danube River. Answer: teeth.

13. hibernate

cave bears had fewer safe places to hibernate. Answer: hibernate.

Passage 40 — The Giant Panda

1. A

the giant panda **has evolved a sixth digit, a prehensile elongated wrist bone called the radial sesamoid** — its ‘false thumb’, not mentioned for the spectacled bear. Answer: A.

2. B

spectacled bears **construct tree nests to act as a bed**. Answer: B.

3. C

the panda and the spectacled bear **are survivors of an ancient lineage dating back 18 million years**. Answer: C.

4. B

spectacled bears are highly adaptable and are found in a wide range of habitats, unlike the giant panda, which lives only in mountain bamboo forests. Answer: B.

5. A

the giant panda (compared here with the red panda) **has a distinctive cry, which they use to communicate with others in their group** — no such trait is mentioned for the spectacled bear. Answer: A.

6. C

the giant panda is **capable of climbing trees as well**, and the spectacled bear is an even more frequent climber. Answer: C.

7. C

the giant panda **supplements its diet with meat, which it scavenges**, and spectacled bears eat carrion too. Answer: C.

8. **A**

zoologist Alphonse Milne Edwards **concluded that it had more in common with the red panda, a member of the raccoon family** — this raccoon-family link is made only for the giant panda; the passage never connects the spectacled bear to raccoons. Answer: **A**.

9. **carnivore**

its physiology is typical of a **carnivore**. Answer: **carnivore**.

10. **digest**

a panda manages to **digest only about 17% of the bamboo it eats**. Answer: **digest**.

11. **hibernate**

unable to store fat effectively, it continues eating in the bitterly cold winter, at a time when many other bears **hibernate**. Answer: **hibernate**.

12. **tree nests**

they construct tree nests to act as a bed as well as **a platform to feed from fruit-laden branches**. Answer: **tree nests**.

13. **eye patches**

'in pandas, a stare is a threat,' Schaller says. **the eye patches enlarge the panda's small, dark eyes tenfold, making the stare more powerful**. Answer: **eye patches**.

Passage 41 — The History and Production of Chocolate

1. **FALSE**

the climate of tropical rain forests combines great quantities of rainfall with high year-round temperatures and humidity — constant, not seasonally varying, temperatures. Answer: **FALSE**.

2. **TRUE**

the olmec indians... are believed to have been the first to grow cocoa beans as a domestic crop. Answer: **TRUE**.

3. **TRUE**

because they lived further north at more arid, higher altitudes not suitable for the cultivation of the cacao tree, they had to acquire the beans through trade or by force. Answer: **TRUE**.

4. **NOT GIVEN**

the passage only says **taxes were paid in the beans to their rulers** — nothing about the taxes being heavy. Answer: **NOT GIVEN**.

5. **FALSE**

the beverage **featured prominently in rituals and as a luxury available only to the very wealthy.**
Answer: **FALSE.**

6. **bitterness**

they **found the aztec drink distasteful because of its bitterness. however, they learned to improve its taste by mixing the ground roasted beans with sugar and vanilla.** Answer: **bitterness.**

7. **business**

these establishments were used as clubs where wealthy people could meet to talk to their friends and to discuss business over a cup of chocolate. Answer: **business.**

8. **baker**

the first recorded instance of chocolate being eaten in solid form occurred when a baker in england began adding cocoa powder to cakes in the mid-1600s. Answer: **baker.**

9. **bar**

unknowingly he **paved the way for the chocolate bar that we consume today.** Answer: **bar.**

10. **employees**

many of these chocolate factory owners were noticed for their enlightened treatment of employees, providing not just jobs but facilities such as housing and schools. Answer: **employees.**

11. **dried**

the beans are allowed to ferment and then dried either in the sun... or artificially, using specially designed machines. Answer: **dried.**

12. **roasted**

after that the beans are roasted according to various formulas which affect the flavour of the final product. Answer: **roasted.**

13. **liquor**

in the grinding process heat is produced which liquefies the product into a substance called the liquor. Answer: **liquor.**

Passage 42 — The History of Lighting

1. **cheap**

Tallow candles were **plentiful and cheap.** Answer: **cheap.**

2. **smelled**

Tallow candles **smelled very unpleasant**. Answer: **smelled**.

3. **steadier**

Beeswax candles **gave a steadier light**. Answer: **steadier**.

4. **special occasions**

Beeswax candles **tended to be used only on special occasions**. Answer: **special occasions**.

5. **metal**

Gas lighting, if installed indoors, would **corrode metal**. Answer: **metal**.

6. **FALSE**

A candle **provides barely a hundredth of the illumination of a single 100 watt light bulb**.
Answer: **FALSE**.

7. **FALSE**

Beeswax candles **tended to be used only on special occasions**, not everyday. Answer: **FALSE**.

8. **NOT GIVEN**

Gas was used in businesses and streets first, and **didn't become common in homes until closer to the middle of the century** — but no explicit ordering against theatres is given. Answer: **NOT GIVEN**.

9. **TRUE**

Thomas Edison's successful development of a long-lasting incandescent light bulb in 1879.
Answer: **TRUE**.

10. **FALSE**

its adoption was gradual, not immediate. Answer: **FALSE**.

11. **B**

Tallow **could be made at home as a by-product of any animal which also supplied food to eat, it was plentiful and cheap**. Answer: **B**.

12. **B**

To minimise the health problem, **gas lights were sometimes erected outside windows** to light the interior. Answer: **B**.

13. **C**

the fundamental human desire that drove the history of lighting – to push back the darkness – remains unchanged. Answer: **C**.

Passage 43 — The History of Pencil

1. tree

found under a ____ ≈ para A: the black substance was "**spotted at the root of the unlucky tree**" that was struck by lightning. Answer: **tree**.

2. sheep

possession and number of ____ ≈ para A: graphite was used "**in marking their sheep for signs of ownership and calculation**". Answer: **sheep**.

3. string

wrapped in ____ or animal skin ≈ para B: "**graphite sticks were initially wrapped in string or sheepskin for stability**". Answer: **string**.

4. government

____ protected the mines ≈ para D: "**the mines were taken over by the government and guarded**". Answer: **government**.

5. steal

____ or received the "wad" ≈ para D: it was "**an offence to steal or receive 'wad'**". Answer: **steal**.

6. grease

____ pencils ≈ para E: "**astronauts in Russia used grease pencils, which don't have breakage problems**". Answer: **grease**.

7. TRUE

Italy... first country ≈ para B: Italy's wrapped graphite sticks are described as "**becoming perhaps the very first pencil in the world**". Answer: **TRUE**.

8. NOT GIVEN

various kinds of wood — para B only says the German technique carved "**two wooden halves**" and glued them; no wood type is mentioned for Germany at all (juniper wood belongs to the earlier Italian version). Answer: **NOT GIVEN**.

9. FALSE

graphite makes... harder contradicts para C: it's the **clay**, not the graphite, that controls hardness — "**the more clay you put in, the higher hardness the core has**". Answer: **FALSE**.

10. FALSE

not produced any more contradicts para D: pencils "**are getting drastically more popular in many countries with each passing decade**" — clearly still made. Answer: **FALSE**.

11. **NOT GIVEN**

first American space expedition — para E only says Americans "**used pencils in the outer space before real zero-gravity pencil was invented**", without saying this was specifically their first expedition. Answer: **NOT GIVEN**.

12. **TRUE**

did not replace... immediately ≈ para E: "**despite the fact that the Americans did invent zero-gravity pencils later, they stuck to mechanical pencils for many years**". Answer: **TRUE**.

13. **FALSE**

unlikely to be used in the future contradicts para F: "**a spectrum of users are likely to continue to use it into the future**". Answer: **FALSE**.

Passage 44 — The History of the Biro

1. **vii**

Para A: **five thousand shoppers crowded the pavements... the new writing instrument was heralded as "fantastic ... miraculous"... within six hours, gimbels had sold its entire stock** — a bold advertisement drew huge crowds. Answer: **vii**.

2. **ix**

Para B: **over the next five decades, 350 additional patents were issued for similar ball-type pens... each had their own faults** — many patents, all with problems. Answer: **ix**.

3. **vi**

Para C: **he found himself becoming more and more frustrated... biro came up with the idea of fitting his pen with a tiny ball bearing** — frustration led directly to the new design. Answer: **vi**.

4. **iv**

Para D: **the biro brothers had a rethink and eventually devised a new design... the biro brothers fled to argentina** — a second design plus a move to a new country. Answer: **iv**.

5. **viii**

Para E: **the united states department of war contacted several american companies... eversharp paid them \$500,000 for the exclusive manufacturing and marketing rights** — a government request leads to the rights being sold. Answer: **viii**.

6. **iii**

Para F: **milton reynolds stumbled upon the biro pen while on vacation in buenos aires** — a holiday trip is how the biro reached America. Answer: **iii**.

7. **D**

the major difficulty was the ink: if the ink was thin, the pens leaked, and if it was too thick, they clogged. depending on the climate or air temperature, sometimes the pens would do both. Answer: **D**.

8. **A**

the first biro pen, like the designs that had gone before it, relied on gravity for the ink to flow to the ball bearing at the tip. Answer: **A**.

9. **C**

eversharp challenged reynolds in the law courts, but lost the case because the biro brothers had failed to secure a u.s. patent on their invention. Answer: **C**.

10. **leather**

John Loud devised and patented **the first "rolling-pointed marker pen" for marking leather**. Answer: **leather**.

11. **Argentina**

the biro brothers fled to argentina, where they applied for a patent for their pen and established their first factory. Answer: **Argentina**.

12. **1945**

Reynolds' factory **began production on 6th october 1945**. Answer: **1945**.

Passage 45 — The History of the Picnic

1. **FALSE**

‘they were initially **purely indoor affairs, held at home or in hired rooms.**’ Answer: **FALSE**.

2. **TRUE**

‘**it was assumed that invited guests would contribute - they could either bring a dish or pay a share of the overall cost.**’ Answer: **TRUE**.

3. **NOT GIVEN**

music is mentioned as part of larger gatherings, but the passage never says HOSTS asked GUESTS to play it themselves. Answer: **NOT GIVEN**.

4. **TRUE**

‘many wealthy French aristocrats fleeing abroad...the majority chose to go to England...and in doing so, **introduced the picnic to England.**’ Answer: **TRUE**.

5. **NOT GIVEN**

the passage says the picnic 'became decidedly less refined and noisier', but never mentions complaints being received. Answer: **NOT GIVEN**.

6. **NOT GIVEN**

the passage describes what members had to bring, but never mentions any approval process for new members. Answer: **NOT GIVEN**.

7. **TRUE**

'each member tried to outdo the others in luxury and expense.' Answer: **TRUE**.

8. **FALSE**

held in hired rooms; 'the stage was small and crudely built, unlike in a theatre.' Answer: **FALSE**.

9. **outside**

'picnics were adopted by the emerging middle classes and moved **outside**.' Answer: **outside**.

10. **music**

'**music** ceased to be provided at picnics.' Answer: **music**.

11. **cities**

'it remained a refined activity most popular with the middle classes in **cities**.' Answer: **cities**.

12. **paintings**

'**paintings** of picnics by well-known American artists.' Answer: **paintings**.

13. **aristocracy**

'their indoor version still predominated, especially among the **aristocracy**.' Answer: **aristocracy**.

14. **transport**

'the development of different forms of **transport**...made the countryside accessible.' Answer: **transport**.

15. **baskets**

'specialist picnic **baskets** were being produced for the mass market.' Answer: **baskets**.

16. **variety**

'as a result of the international availability of a huge **variety** of food.' Answer: **variety**.

Passage 46 — The Life of Louis Braille

1. FALSE

oldest child contradicts para 1: he was "**the only child of Louis and Constance Braille**" — being an only child rules out having any siblings, older or younger. Answer: **FALSE**.

2. FALSE

immediately after contradicts para 1: "**Within weeks of the accident, the infection that had developed in his eyes took away his sight completely**" — the blindness came from a later infection, not the injury itself. Answer: **FALSE**.

3. FALSE

school for blind children contradicts para 1: "**He attended school in their village with sighted children**" — a regular village school, not a blind school, before Paris. Answer: **FALSE**.

4. TRUE

designed for military purposes ≈ para 3: Barbier invented 'night writing' so "**soldiers could use to read and send messages in complete darkness**". Answer: **TRUE**.

5. FALSE

before Braille died contradicts para 6: adoption came "**one year after his passing**" (1854, following his 1852 death), i.e. after, not before. Answer: **FALSE**.

6. harnesses

leather ____ for local farmers ≈ para 1: his father "**made leather saddles and harnesses for farmers in the area**". Answer: **harnesses** (saddles also accepted).

7. standard alphabet

raised letters of the ____ ≈ para 2: books had "**raised letters of the standard alphabet pressed into the paper**". Answer: **standard alphabet**.

8. twelve

grids of ____ dots ≈ para 3: "**The system was based on grids of twelve dots**". Answer: **twelve**.

9. fifteen

only ____ years old ≈ para 4: "**By 1824, when Braille was just fifteen years old, he had completed the basic framework of his system**". Answer: **fifteen**.

10. B

succeed as a student ≈ para 1: "**He was an excellent student, mostly because of his exceptional memory**". Answer: **B**.

11. C

disadvantage of raised-print books ≈ para 2: "Reading them was slow and laborious... the books were extremely expensive to produce". Answer: C.

12. B

reject Braille's system ≈ para 4: "Many sighted teachers at the institute feared that Braille's code would isolate blind students from the sighted world". Answer: B.

13. C

author's main purpose — the whole passage traces Braille's birth, education, invention, struggle for acceptance, and lasting legacy — a biographical account of **his life and achievements**, not a narrow history or comparison. Answer: C.

Passage 47 — The National Parks of America

1. FALSE

the concept of the national park originated in america, though it was quickly replicated by similar parks in canada, finland, new zealand, australia and elsewhere — it was the other way round. Answer: FALSE.

2. NOT GIVEN

the passage only says Yellowstone **attracts more than four million visitors a year** — no comparison is made with visitor numbers at other parks. Answer: NOT GIVEN.

3. TRUE

the NPS **also manages significant monuments and historic structures such as the statue of liberty, the gettysburg museum and ford's theatre**. Answer: TRUE.

4. NOT GIVEN

the passage only says **catlin failed to see such a park established in his lifetime, but his vision lived on** — nothing about criticism. Answer: NOT GIVEN.

5. FALSE

Emerson argued **that natural landscapes should be protected rather than exploited for agriculture and commerce** — the opposite view. Answer: FALSE.

6. TRUE

the creation of a public park in the yosemite valley was directly the result of his work. Answer: TRUE.

7. hunting

Yellowstone provided a sanctuary for endangered species **by imposing a ban on hunting in all its forms**. Answer: **hunting**.

8. military

their management was for many years placed in the hands of the military, rather than the federal or state government. Answer: **military**.

9. tourism

railroads increasingly ran through remote areas, **making tourism possible in these locations for the very first time**. Answer: **tourism**.

10. education

Langford saw the park as an opportunity for **the education of ordinary citizens in all aspects of the natural world and environmental conservation, which was his number one priority**. Answer: **education**.

11. finances

mather's transformation of the hitherto poorly managed finances of the parks. Answer: **finances**.

12. dams

the controversial decision to permit the construction of dams in national parks. Answer: **dams**.

13. automobiles

the number of visitors expanded dramatically **largely because mass-produced automobiles became affordable**. Answer: **automobiles**.

Passage 48 — The Origins of Laughter

1. B

Zimmerman **discovered that chimp and human baby laughter follow broadly the same pattern**. Answer: **B**.

2. C

Panksepp researched **the ultrasonic chirps produced by rats during play and in response to tickling**. Answer: **C**.

3. D

according to Tom Flamson, **laughing is what makes it a reliable signal of trust**. Answer: **D**.

4. **A**

Provine found that most laughter comes as a polite reaction to everyday remarks such as “see you later”, rather than anything particularly funny. Answer: A.

5. **B**

Zimmerman believes the closeness of baby laughter to chimp laughter supports the idea that laughter was around long before humans arrived on the scene. Answer: B.

6. **A**

Provine: ‘laughter evolved as a signal to others — it almost disappears when we are alone.’ Answer: A.

7. **play**

7: provine believes we need to look at play. 8: ‘does this pant laughter have the same **source** as our own laughter?’ (source ≈ origins). 9: **more distantly related primates, including gorillas, laugh, and anecdotal evidence suggests that other social mammals can do too.** 10: laughter in response to tickling is a way for two individuals to **signal and test their trust** in one another (trust ≈ confidence). Answers: E / D / G / H.

8. **origins**

7: provine believes we need to look at play. 8: ‘does this pant laughter have the same **source** as our own laughter?’ (source ≈ origins). 9: **more distantly related primates, including gorillas, laugh, and anecdotal evidence suggests that other social mammals can do too.** 10: laughter in response to tickling is a way for two individuals to **signal and test their trust** in one another (trust ≈ confidence). Answers: E / D / G / H.

9. **primates**

7: provine believes we need to look at play. 8: ‘does this pant laughter have the same **source** as our own laughter?’ (source ≈ origins). 9: **more distantly related primates, including gorillas, laugh, and anecdotal evidence suggests that other social mammals can do too.** 10: laughter in response to tickling is a way for two individuals to **signal and test their trust** in one another (trust ≈ confidence). Answers: E / D / G / H.

10. **confidence**

7: provine believes we need to look at play. 8: ‘does this pant laughter have the same **source** as our own laughter?’ (source ≈ origins). 9: **more distantly related primates, including gorillas, laugh, and anecdotal evidence suggests that other social mammals can do too.** 10: laughter in response to tickling is a way for two individuals to **signal and test their trust** in one another (trust ≈ confidence). Answers: E / D / G / H.

11. **FALSE**

men laugh more with other men, but **women tend to laugh more and at a higher pitch when men are present** — the opposite sex, not the same sex. Answer: FALSE.

12. **TRUE**

‘while other animals pant, **we alone can control our breath well enough to produce the sound of laughter.**’ Answer: **TRUE**.

13. **NOT GIVEN**

the passage discusses chimp laughter (in play) and rat laughter (in play and tickling) but never directly compares the **RANGE** of situations between the two species. Answer: **NOT GIVEN**.

Passage 49 — The Origins of Weather Forecasting

1. **FALSE**

Para 2: **predicting the weather had gained a reputation similar to astrology or fortune-telling** — the opposite of "respected". Answer: **FALSE**.

2. **TRUE**

Para 3: **his captaining of the ship HMS Beagle on Charles Darwin's voyage around the world, then "retired from active service as an admiral"** — a faultless background as a naval captain and admiral confirms a successful career at sea. Answer: **TRUE**.

3. **NOT GIVEN**

other applicants — para 3 only says he "**secured a job as chief statistician**"; no other candidates or a competitive process are mentioned at all. Answer: **NOT GIVEN**.

4. **FALSE**

Para 3: the 1854 funding was "**to chart the safest sailing routes**", not for forecasts — and **the British Government was not impressed with the idea of collecting weather reports and establishing a weather forecast**, with an MP's suggestion "met with laughter". Answer: **FALSE**.

5. **TRUE**

Para 5: "**This was just the right moment for a national forecasting service**" because **the invention of the electric telegraph and Morse code in 1844 gave instant communication for the first time**, alongside improving barometer readings. Answer: **TRUE**.

6. **instruments**

Para 6: **issue ships with standardised instruments which allowed for accurate weather reports**. Answer: **instruments**.

7. **ports**

Para 6: **FitzRoy could telegraph warnings to ports on the coast**. Answer: **ports**.

8. flags

Para 6: **an ingenious system of flags was hoisted up high for passing ships to read.** Answer: **flags.**

9. lifeboat

Para 6: **this represented the biggest advance in shipping safety after the introduction of the lifeboat.** Answer: **lifeboat.**

10. island

Para 7: Queen Victoria wanted forecasts for **the sea crossing over the Solent to an island which she liked to visit.** Answer: **island.**

11. owners

Para 7: **the owners of fishing vessels, however, were not supportive of FitzRoy's forecasts.** Answer: **owners.**

12. name

Para 9: **with the honour of an area of sea being given his name in the shipping forecast.** Answer: **name.**

13. book

Para 9: **he published a book about weather forecasting which was recognized by other scientists to be well ahead of its time.** Answer: **book.**

Passage 50 — The Pyramid of Cestius

1. FALSE

the opposite: 'one notable monument there has **never attracted nearly as much interest.**' Answer: **FALSE.**

2. NOT GIVEN

the passage says Cestius 'ordered that the building work be completed within a period of 330 days', but never confirms whether he was still alive when it was finished. Answer: **NOT GIVEN.**

3. TRUE

'popular myth had developed that it might be a tomb for one of the twin brothers **Romulus and Remus, who were regarded as the men who had established the city of Rome.**' Answer: **TRUE.**

4. FALSE

the opposite: it rests below street level so that 'passing tourists and residents could **easily fail to notice its full height.**' Answer: **FALSE.**

5. NOT GIVEN

the passage only mentions his gratitude to Italy ('our company has grown thanks to Italy'); nothing about Egyptian architecture. Answer: **NOT GIVEN**.

6. TRUE

'Work began at the site...and was **completed ahead of schedule** thanks to his 2-million-euro contribution.' Answer: **TRUE**.

7. FALSE

the opposite: 'there used to be more here, but the **majority have disappeared over time**.' Answer: **FALSE**.

8. white

'Cestius' pyramid had a layer of **white** Carrara marble on the outside.' Answer: **white**.

9. shape

'the **shape** of Cestius' pyramid is quite unlike that of typical Egyptian ones.' Answer: **shape**.

10. countryside

'there was a strict Roman law prohibiting the placement of tombs within the city itself, the pyramid of Cestius would have stood in the **countryside**.' Answer: **countryside**.

11. statues

'two marble bases were found in front of the pyramid, as well as fragments of bronze **statues** that had once stood on them.' Answer: **statues**.

12. tunnel

'they did come across a **tunnel**. It was quite possible, therefore, that robbers had at some earlier time removed the contents of the tomb.' Answer: **tunnel**.

13. pollution

'The white exterior of the pyramid of Cestius will have to be cleaned every few months to remove the layer of urban **pollution**.' Answer: **pollution**.

Passage 51 — The Seed Hunters

1. extinction

'working to halt the sad slide into **extinction** facing so many plant species.' Answer: **extinction**.

2. drugs or crops

'species that will yield the **drugs or crops** of the future.' Answer: **drugs or crops**.

3. **pioneers**

‘among the **pioneers** of this botanical treasure hunt was John Tradescant.’ Answer: **pioneers**.

4. **Joseph Banks**

Sir **Joseph Banks**...‘was so driven to expand his collections that he sent botanists around the world at his own expense.’ Answer: **Joseph Banks**.

5. **vaults**

‘within its underground **vaults** are 260 million dry seeds.’ Answer: **vaults**.

6. **environments**

‘including the restoration of damaged **environments**.’ Answer: **environments**.

7. **TRUE**

‘those heady days of exploration and discovery may be over, but they have been replaced by **a pressing need to preserve our natural history for the future**.’ Answer: **TRUE**.

8. **NOT GIVEN**

the passage calls it the world’s **LARGEST** depository, but never claims it was the **FIRST** one ever established. Answer: **NOT GIVEN**.

9. **TRUE**

‘**a further one billion hectares of wilderness will be converted to farmland in developing countries alone**.’ Answer: **TRUE**.

10. **NOT GIVEN**

the passage mentions only 4% of plants have been evaluated, but never says this is due to the Union lacking **RESOURCES**. Answer: **NOT GIVEN**.

11. **TRUE**

‘**seed conservation techniques were originally developed by farmers**...just as in farming.’ Answer: **TRUE**.

12. **FALSE**

Smith says ‘**there’s no reason why any plant species should become extinct, given today’s technology**’ — current technology is already sufficient. Answer: **FALSE**.

13. **B & D**

Correct two are **B & D** — ‘besides **providing staple crops for human consumption** (D), plants are a source of many medicines, and **the principal supply of fuel** and building materials (B) in many parts of the world.’ A, C and E are never mentioned as uses of plants in the passage.

Passage 52 — The Story of Silk

1. tea

‘one of the cocoons that she had collected landed in the hot **tea**.’ Answer: **tea**.

2. reel

‘She also devised a special **reel** to draw the fibres from the cocoon into a single thread.’ Answer: **reel**.

3. women

‘silkworm farming was solely restricted to **women**.’ Answer: **women**.

4. royalty

‘originally, only **royalty** were entitled to have clothes made of silk.’ Answer: **royalty**.

5. currency

‘silk was so prized that it was also used as a unit of **currency**.’ Answer: **currency**.

6. paper

‘The earliest indication of silk **paper** being used was discovered in the tomb of a noble...around 168 AD.’
Answer: **paper**.

7. wool

‘taking silk westward and bringing gold, silver and **wool** to the East.’ Answer: **wool**.

8. monks

‘**monks** working for the Byzantine emperor Justinian smuggled silkworm eggs to Constantinople...concealed inside hollow bamboo walking canes.’ Answer: **monks**.

9. nylon

‘new manmade fibres, such as **nylon**, started to be used in what had traditionally been silk products.’
Answer: **nylon**.

10. FALSE

the opposite: ‘It was named the Silk Road after its most precious commodity, which was considered to be **worth more than gold**.’ Answer: **FALSE**.

11. TRUE

‘Few **merchants** travelled the entire route; goods were handled mostly by a series of **middlemen**.’ Answer: **TRUE**.

12. **FALSE**

the opposite: 'The Byzantines were as secretive as the Chinese...and for many centuries the weaving and trading of silk fabric was a **strict imperial monopoly**' — it was the Arabs who later spread production westward. Answer: **FALSE**.

13. **NOT GIVEN**

the passage only says China is 'the world's biggest producer and exporter of **raw silk and silk yarn**' together — it never breaks down what share is yarn specifically. Answer: **NOT GIVEN**.

Passage 53 — The Terracotta Army

1. **well**

'forced farmer Yang Zhifa and his five brothers to start digging a **well**.' Answer: **well**.

2. **arrowhead**

'along with a heavily-corroded bronze **arrowhead**.' Answer: **arrowhead**.

3. **truck**

'loaded them onto a **truck** to transport them to the Lingtong County Museum.' Answer: **truck**.

4. **journalist**

'when a **journalist** visiting the area heard about his work and published an article about the site.' Answer: **journalist**.

5. **swords**

'scattered around their feet a huge collection of **swords**.' Answer: **swords**.

6. **cavalry**

'alongside **cavalry** units and chariots pulled by teams of terracotta horses.' Answer: **cavalry**.

7. **workshops**

'Built on an industrial scale in various **workshops** in the late 3rd century BC.' Answer: **workshops**.

8. **face**

'each soldier's **face** had been individually finished off by hand to give it a unique look.' Answer: **face**.

9. **FALSE**

the opposite: '**they did not understand the significance** of what they had found at the time.' Answer: **FALSE**.

10. **TRUE**

‘Once there, **his worst fears were realized.**’ Answer: **TRUE.**

11. **FALSE**

the opposite: ‘**they were reluctant to agree,** but finally did so.’ Answer: **FALSE.**

12. **TRUE**

‘two imposing, 1.78-metre-tall figures stood before him...he **assigned a small team to start work on excavating the site.**’ Answer: **TRUE.**

13. **NOT GIVEN**

the passage only says ‘the dig was teeming with archaeologists’ — it never specifies they came from every part of China. Answer: **NOT GIVEN.**

14. **TRUE**

‘the third pit revealed **high-ranking officers** and an elaborate bronze war chariot.’ Answer: **TRUE.**

15. **FALSE**

the opposite: ‘**insufficient environmental research** had been undertaken before the figures were exposed to the elements.’ Answer: **FALSE.**

16. **NOT GIVEN**

the passage describes tourists visiting freely and figures going on display abroad, but never mentions any permission requirement for large groups. Answer: **NOT GIVEN.**

Passage 54 — The Trans-Australian Railway

1. **sea**

the only form of transport between the west coast and the eastern cities was a long and often rough sea voyage across the great australian bight. Answer: **sea.**

2. **guide**

the first east-west crossing of the nullarbor plain on foot was achieved by the english explorer edward eyre and his indigenous australian guide wylie in 1841. Answer: **guide.**

3. **telegraph**

in 1877 a telegraph line was built across the plain. Answer: **telegraph.**

4. **pipeline**

this line was built alongside a pipeline carrying drinking water to the arid goldfields around kalgoorlie. Answer: **pipeline.**

5. **federation**

in 1901, the self-governing colonies of queensland, new south wales, victoria, tasmania, south australia and western australia got together to form a federation. Answer: **federation**.

6. **desert**

the two lines finally converged, at a site in an area of desert, in 1917. Answer: **desert**.

7. **straight**

the longest stretch - 297 miles- of straight track in the world. Answer: **straight**.

8. **standard**

the new 1,052 mile stretch was built to the standard gauge of 4ft 8in. Answer: **standard**.

9. **FALSE**

the tar line's original chief engineer had requested diesel engines, but his request was not granted. Answer: **FALSE**.

10. **NOT GIVEN**

the passage only says the service **commenced running in 1917 to serve isolated communities and settlements along its route** — nothing about its popularity with locals or tourists. Answer: **NOT GIVEN**.

11. **TRUE**

facing increased competition from air travel it last ran in june 1991. Answer: **TRUE**.

12. **TRUE**

the train brought to communities along the route not only the necessities for everyday life, but also a butcher's shop, medical facilities and a cinema. Answer: **TRUE**.

13. **FALSE**

the indian pacific service has operated between sydney on the east coast and perth since 1970, the year that the tar route was completely converted to standard gauge — the service began the SAME year the conversion was completed, not before. Answer: **FALSE**.

Passage 55 — Thomas Cole: American Nature Painter

1. **TRUE**

later making reproductions of well-known paintings — this was part of his early work as a print shop apprentice. Answer: **TRUE**.

2. **FALSE**

stayed behind on the east coast as his parents traveled westward — it was his parents who went west; the teenage Cole stayed east. Answer: **FALSE**.

3. **NOT GIVEN**

The passage says he sketched in the Caribbean and drew The Tree there, but never says he sold anything there. Answer: **NOT GIVEN**.

4. **TRUE**

where he made lasting contacts in the still small art world. Answer: **TRUE**.

5. **NOT GIVEN**

The passage mentions selling paintings in New York and, separately, the 1825 Hudson River trip — but never links the two. Answer: **NOT GIVEN**.

6. **NOT GIVEN**

Italy was "a more pleasant stay" — about enjoyment, not length of time. Answer: **NOT GIVEN**.

7. **FALSE**

left new york city with his wife and children for the countryside — his family lived with him outside the city. Answer: **FALSE**.

8. **animal**

it looks bizarre, almost resembling some kind of animal. Answer: **animal**.

9. **ruins**

but on the ruins of a long-dead culture. Answer: **ruins**.

10. **stages**

it narrates the stages that an unnamed fictional civilization passes through. Answer: **stages**.

11. **mountain**

depicts thin clouds in the cool fresh air, a forested mountain. Answer: **mountain**.

12. **fields**

treeless, square-cut fields which are cultivated and fertile. Answer: **fields**.

13. **canvas**

he turns away from his canvas and looks right at us. Answer: **canvas**.

Passage 56 — Tickling and Laughter

1. E

Para E: **'the soles of the feet, the underarm area, the stomach and the neck are most commonly mentioned.'** Answer: E.

2. B

Para B: **'it is a subject that has intrigued philosophers and scientists since antiquity.'** Answer: B.

3. D

Para D: **'when someone tells you a joke, they take you along a path of expectation to a punchline...the same applies to tickling.'** Answer: D.

4. F

Para F: Blakemore **constructed a robotic tickle machine** to compare tickling oneself vs. being tickled. Answer: F.

5. E

Para E: **'people who do not consider themselves ticklish may be inhibited about laughing and exposing their vulnerability.'** Answer: E.

6. B

Ramachandran: **'laughter is a signal that the tickling is a false alarm, that there is nothing to really worry about.'** Answer: B.

7. A

Darwin **came to the conclusion that tickling was an ingredient in forming social bonds.** Answer: A.

8. C

Christine Harris believes there are **two types of tickling. The lighter pressure results in the urge to scratch or rub, while the heavier provokes laughter.** Answer: C.

9. D

Blakemore: **the part of the brain that registers touch reacted more strongly when the machine tickled them than when they tickled themselves.** Answer: D.

10. A

Darwin noted that **'the precise point to be tickled must not be known.'** Answer: A.

11. **B**

For Ramachandran, 'laughter is the essential key to unlocking the mystery of tickling.' Answer: **B**.

12. **cerebellum**

'the **cerebellum**, a more primitive part of the brain, dampens the tickle sensation if you try to tickle yourself.' Answer: **cerebellum**.

13. **blood flow**

'Blakemore used MRI scans which measure **blood flow** in the brain to compare cerebral activity.' Answer: **blood flow**.

Passage 57 — Tunnelling under the Thames

1. **NOT GIVEN**

the passage only says **the port of London was the busiest in the world** — nothing about its relative safety. Answer: **NOT GIVEN**.

2. **FALSE**

the bridge was **as impractical as its early date implies...** the centre of the world's largest traffic **jam**. Answer: **FALSE**.

3. **TRUE**

if private enterprise could build another crossing closer to the docks, there would be good money to be made in tolls paid by users. Answer: **TRUE**.

4. **FALSE**

another bridge was out of the question, as this would deny sailing ships access to the city centre. Answer: **FALSE**.

5. **NOT GIVEN**

coal mining is mentioned only as background context for tunnelling experience in general — **increasing demand for coal had meant a great many tunnels had been dug in mines in Britain** — nothing about coal existing under the Thames itself. Answer: **NOT GIVEN**.

6. **TRUE**

their ambition was to tunnel below the Thames, but there was little to guide them as there had been no previous attempt to do this. Answer: **TRUE**.

7. **NOT GIVEN**

the passage says **things began to go disastrously wrong, with muddy soil pouring into the tunnel**, but never mentions any injuries to the men. Answer: **NOT GIVEN**.

8. **TRUE**

its funds were exhausted. Answer: **TRUE**.

9. **worm**

he observed the wood was infested with something that looked like a worm. Answer: **worm**.

10. **clay**

he decided to dig close to the river bottom, where he could expect to find clay. Answer: **clay**.

11. **headaches**

they also complained of frequent headaches and dizziness, caused by the poor air quality.
Answer: **headaches** (or dizziness).

12. **accidents**

there were a number of accidents because the miners could not see what they were doing.
Answer: **accidents**.

13. **government**

the government loan he had required to complete the project had to be paid back with interest. Answer: **government**.

Passage 58 — What Exactly Is a Hoard?

1. **TRUE**

a typical image of a hoard is a pot full of gold coins. Answer: **TRUE**.

2. **NOT GIVEN**

fallen trees are mentioned as ONE way hoards were exposed, but the passage never says most hoards were found in wooded areas specifically. Answer: **NOT GIVEN**.

3. **FALSE**

the romans...seem to have found ancient artefacts intriguing: they buried fossils and prehistoric weapons alongside their own items. Answer: **FALSE**.

4. **FALSE**

in urban areas hoards are usually found by archaeologists excavating sites, whereas metal-detector users find hoards mainly in RURAL areas, not towns/cities. Answer: **FALSE**.

5. **TRUE**

the act of hiding a group of objects in a pot in the ground or behind a wall often keeps them in good condition, unlike single dropped objects. Answer: **TRUE**.

6. **NOT GIVEN**

the passage discusses soil as a source of CONTEXTUAL INFORMATION for understanding a hoard, not as something that reduces damage to buried objects. Answer: **NOT GIVEN**.

7. **NOT GIVEN**

the passage describes the Beau Street hoard's size ('over 17,500 coins') but never claims it is the LARGEST in the UK. Answer: **NOT GIVEN**.

8. **FALSE**

there are no records to allow us ever to know the exact significance of prehistoric rituals.
Answer: **FALSE**.

9. **bags**

'sometimes still with the **bags** that had contained them.' Answer: **bags**.

10. **silver**

'the official coinage decreased in **silver** content.' Answer: **silver**.

11. **sorted**

'composed of eight lots of money that appear to have been **sorted** and then stored under the floor.' Answer: **sorted**.

12. **walls**

'many hoards from Roman villas were hidden within **walls** that later collapsed.' Answer: **walls**.

13. **water**

'putting something underground or under **water**.' Answer: **water**.

14. **inscriptions**

'some later offerings are accompanied by legible **inscriptions**.' Answer: **inscriptions**.

15. **boundaries**

objects were placed 'or defining significant **boundaries**.' Answer: **boundaries**.

16. **leader**

'these ceremonies may be carried out when there is a change of **leader**.' Answer: **leader**.

Passage 59 — What if Everything Had a Barcode?

1. 1.7 million

scientists have classified about **1.7 million organisms**, a small fraction of the total. Answer: **1.7 million**.

2. flies

such as the different kinds of **flies**, look remarkably similar. Answer: **flies**.

3. mammals

the International Union for the Conservation of Nature **estimates that a quarter of the world's mammals are threatened with extinction**. Answer: **mammals**.

4. DNA barcode reader

He envisaged a '**DNA barcode reader**', similar to the scanners at retail checkouts. Answer: **DNA barcode reader**.

5. gene

He selected a **gene** that gives rise to an enzyme known as CO1. Answer: **gene**.

6. reference database

This information would be instantly relayed to a **reference database**, a public library of DNA barcodes. Answer: **reference database**.

7. 98 per cent

results indicate that **animals can now be identified by their barcodes in 98 per cent of cases**. Answer: **98 per cent**.

8. Astraptes fulgerator

caterpillars of the tropical butterfly Astraptes fulgerator... barcoding has shown there are 10 different kinds. Answer: **Astraptes fulgerator**.

9. TRUE

It won't be long before everyone, from **experts to amateurs**, will be able to scan. Answer: **TRUE**.

10. TRUE

such a device would let you scan fish at the fishmonger's to check if it's been labelled properly. Answer: **TRUE**.

11. FALSE

the inspiration came while he was walking up and down the aisles of a supermarket — the idea came to Hebert himself, not from an employee. Answer: **FALSE**.

12. NOT GIVEN

the passage gives the current figure, **supported by 150 institutions in 45 countries**, but says nothing about growth over time. Answer: **NOT GIVEN**.

13. FALSE

he estimates that out of the world's 10,000 bird species, DNA barcoding will distinguish at least 1,000 new ones — a future prediction, not already achieved. Answer: **FALSE**.

Passage 60 — William Gilbert and Magnetism

1. v

Para A: **"The 16th and 17th centuries saw two great pioneers of modern science: Galileo and Gilbert"** — introduces both as early scientific pioneers. Answer: **v**.

2. i

Para B: **"went to grammar school, and then studied medicine at St John's College, Cambridge"** — covers his birth and education. Answer: **i**.

3. vi

Para C covers **"he was a very successful and eminent doctor"**, presidency of the Royal Science Society, royal physician, and knighthood together — broader than any single achievement. Answer: **vi** (not vii, which names only one clause).

4. x

Para D: **"Gilbert was first interested in chemistry but later changed his focus due to the large portion of mysticism of alchemy involved"** — the paragraph's arc is his shift from chemistry to physics. Answer: **x** (not viii, which only fits the closing sentence about his books).

5. ix

Para E: **"Gilbert's discovery was so important to modern physics. He investigated the nature of magnetism and electricity"**. Answer: **ix**.

6. iv

Para F: **"He also questioned the traditional astronomical beliefs"**. Answer: **iv**.

7. ii

Para G: **"His research method was revolutionary in that he used experiments rather than pure logic and reasoning... It was a new attitude towards scientific investigation"**. Answer: **ii**.

8. TRUE

Para A: **"However, he is less well known than he deserves"**. Answer: **TRUE**.

9. **TRUE**

Para C: "He was a very successful and eminent doctor. All this culminated in his election to the president..." then "He was also appointed personal physician to the queen" — the sequence shows **his doctor fame preceded the royal appointment**. Answer: **TRUE**.

10. **NOT GIVEN**

medical theories of his time — the passage only discusses his shift away from alchemy/chemistry (para D) and his views on astronomy (para F); nothing about losing faith in medicine specifically. Answer: **NOT GIVEN**.

11. **C**

Correct three are **C, D & E** ≈ para E: "**Gilbert also found that metals can be magnetised by rubbing materials such as fur, plastic or the like on them**" (C); para F: "**he believed that stars are not equidistant from the earth**" (D); para F: "**the sun and other stars wobble just like the earth does**", implying the earth itself wobbles (E). A is alchemy (which Gilbert moved away from), B is a superstition explicitly labelled as such, and F is credited to du Fay, not Gilbert.

12. **D**

13. **E**